

11.3 The Dot Product

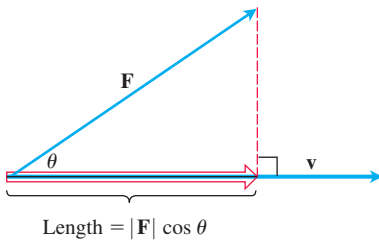


FIGURE 11.20 The magnitude of the force $\mathbf{F}$ in the direction of vector $\mathbf{v}$ is the length $|\mathbf{F}|\cos\theta$ of the projection of $\mathbf{F}$ onto $\mathbf{v}$.

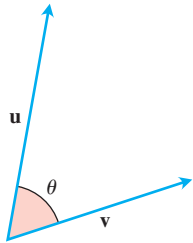


FIGURE 11.21 The angle between $\mathbf{u}$ and $\mathbf{v}$ given by Theorem 1 lies in the interval $[0, \pi]$.

If a force $\mathbf{F}$ is applied to a particle moving along a path, we often need to know the magnitude of the force in the direction of motion. If $\mathbf{v}$ is parallel to the tangent line to the path at the point where $\mathbf{F}$ is applied, then we want the magnitude of $\mathbf{F}$ in the direction of $\mathbf{v}$. Figure 11.20 shows that the scalar quantity we seek is the length $|\mathbf{F}|\cos\theta$, where θ is the angle between the two vectors $\mathbf{F}$ and $\mathbf{v}$.

In this section we show how to calculate easily the angle between two vectors directly from their components. A key part of the calculation is an expression called the *dot product*. Dot products are also called *inner* or *scalar* products because the product results in a scalar, not a vector. After investigating the dot product, we apply it to finding the projection of one vector onto another (as displayed in Figure 11.20) and to finding the work done by a constant force acting through a displacement.

Angle Between Vectors

When two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are placed so their initial points coincide, they form an angle θ of measure $0 \leq \theta \leq \pi$ (Figure 11.21). If the vectors do not lie along the same line, the angle θ is measured in the plane containing both of them. If they do lie along the same line, the angle between them is 0 if they point in the same direction and π if they point in opposite directions. The angle θ is the **angle between $\mathbf{u}$ and $\mathbf{v}$** . Theorem 1 gives a formula to determine this angle.

THEOREM 1—Angle Between Two Vectors

The angle θ between two nonzero vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is given by

$$\theta = \arccos \left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{|\mathbf{u}||\mathbf{v}|} \right).$$

We use the law of cosines to prove Theorem 1, but before doing so, we focus attention on the expression $u_1v_1 + u_2v_2 + u_3v_3$ in the calculation for θ . This expression is the sum of the products of the corresponding components of the vectors $\mathbf{u}$ and $\mathbf{v}$.

DEFINITION The **dot product $\mathbf{u} \cdot \mathbf{v}$** (“**u dot v**”) of vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is the scalar

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

EXAMPLE 1 We illustrate the definition.

- (a) $\langle 1, -2, -1 \rangle \cdot \langle -6, 2, -3 \rangle = (1)(-6) + (-2)(2) + (-1)(-3)$
 $= -6 - 4 + 3 = -7$
- (b) $\left(\frac{1}{2}\mathbf{i} + 3\mathbf{j} + \mathbf{k}\right) \cdot (4\mathbf{i} - \mathbf{j} + 2\mathbf{k}) = \left(\frac{1}{2}\right)(4) + (3)(-1) + (1)(2) = 1$ ■

The dot product of a pair of two-dimensional vectors is defined in a similar fashion:

$$\langle u_1, u_2 \rangle \cdot \langle v_1, v_2 \rangle = u_1v_1 + u_2v_2.$$

We will see throughout the remainder of this text that the dot product is a key tool for many important geometric and physical calculations in space (and the plane).

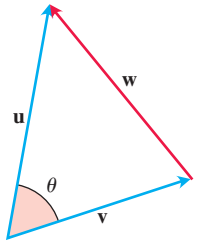


FIGURE 11.22 The parallelogram law of addition of vectors gives $\mathbf{w} = \mathbf{u} - \mathbf{v}$.

Proof of Theorem 1 Applying the law of cosines (Equation (8), Section 1.3) to the triangle in Figure 11.22, we find that

$$|\mathbf{w}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2|\mathbf{u}||\mathbf{v}|\cos \theta \quad \text{Law of cosines}$$

$$2|\mathbf{u}||\mathbf{v}|\cos \theta = |\mathbf{u}|^2 + |\mathbf{v}|^2 - |\mathbf{w}|^2.$$

Because $\mathbf{w} = \mathbf{u} - \mathbf{v}$, the component form of $\mathbf{w}$ is $\langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle$. So

$$|\mathbf{u}|^2 = (\sqrt{u_1^2 + u_2^2 + u_3^2})^2 = u_1^2 + u_2^2 + u_3^2$$

$$|\mathbf{v}|^2 = (\sqrt{v_1^2 + v_2^2 + v_3^2})^2 = v_1^2 + v_2^2 + v_3^2$$

$$\begin{aligned} |\mathbf{w}|^2 &= (\sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2})^2 \\ &= (u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2 \\ &= u_1^2 - 2u_1v_1 + v_1^2 + u_2^2 - 2u_2v_2 + v_2^2 + u_3^2 - 2u_3v_3 + v_3^2 \end{aligned}$$

and

$$|\mathbf{u}|^2 + |\mathbf{v}|^2 - |\mathbf{w}|^2 = 2(u_1v_1 + u_2v_2 + u_3v_3).$$

Therefore,

$$2|\mathbf{u}||\mathbf{v}|\cos \theta = |\mathbf{u}|^2 + |\mathbf{v}|^2 - |\mathbf{w}|^2 = 2(u_1v_1 + u_2v_2 + u_3v_3)$$

$$|\mathbf{u}||\mathbf{v}|\cos \theta = u_1v_1 + u_2v_2 + u_3v_3$$

$$\cos \theta = \frac{u_1v_1 + u_2v_2 + u_3v_3}{|\mathbf{u}||\mathbf{v}|}.$$

Thus, for $0 \leq \theta \leq \pi$, we have $\theta = \arccos \left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{|\mathbf{u}||\mathbf{v}|} \right)$. ■

Dot Product and Angles

The angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ is $\theta = \arccos \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} \right)$.

The dot product of two vectors $\mathbf{u}$ and $\mathbf{v}$ is given by $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos \theta$.

EXAMPLE 2 Find the angle between $\mathbf{u} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{v} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$.

Solution We use the formula above:

$$\mathbf{u} \cdot \mathbf{v} = (1)(6) + (-2)(3) + (-2)(2) = 6 - 6 - 4 = -4$$

$$|\mathbf{u}| = \sqrt{(1)^2 + (-2)^2 + (-2)^2} = \sqrt{9} = 3$$

$$|\mathbf{v}| = \sqrt{(6)^2 + (3)^2 + (2)^2} = \sqrt{49} = 7$$

$$\theta = \arccos \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} \right) = \arccos \left(\frac{-4}{(3)(7)} \right) \approx 1.76 \text{ radians or } 100.98^\circ. \quad \text{■}$$

The angle formula applies to two-dimensional vectors as well. Note that the angle θ is acute if $\mathbf{u} \cdot \mathbf{v} > 0$ and obtuse if $\mathbf{u} \cdot \mathbf{v} < 0$.

EXAMPLE 3 Find the angle θ in the triangle ABC determined by the vertices $A = (0, 0)$, $B = (3, 5)$, and $C = (5, 2)$ (Figure 11.23).

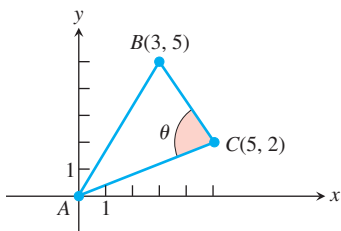


FIGURE 11.23 The triangle in Example 3.

Solution The angle θ is the angle between the vectors $\overrightarrow{CA}$ and $\overrightarrow{CB}$. The component forms of these two vectors are

$$\overrightarrow{CA} = \langle -5, -2 \rangle \quad \text{and} \quad \overrightarrow{CB} = \langle -2, 3 \rangle.$$

First we calculate the dot product and magnitudes of these two vectors.

$$\overrightarrow{CA} \cdot \overrightarrow{CB} = (-5)(-2) + (-2)(3) = 4$$

$$|\overrightarrow{CA}| = \sqrt{(-5)^2 + (-2)^2} = \sqrt{29}$$

$$|\overrightarrow{CB}| = \sqrt{(-2)^2 + (3)^2} = \sqrt{13}$$

Then, applying the angle formula, we have

$$\begin{aligned} \theta &= \arccos \left(\frac{\overrightarrow{CA} \cdot \overrightarrow{CB}}{|\overrightarrow{CA}| |\overrightarrow{CB}|} \right) = \arccos \left(\frac{4}{(\sqrt{29})(\sqrt{13})} \right) \\ &\approx 78.1^\circ \text{ or } 1.36 \text{ radians.} \end{aligned}$$

Orthogonal Vectors

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if the angle between them is $\pi/2$. For such vectors, we have $\mathbf{u} \cdot \mathbf{v} = 0$ because $\cos(\pi/2) = 0$. The converse is also true. If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors with $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta = 0$, then $\cos \theta = 0$ and $\theta = \arccos 0 = \pi/2$. The following definition also allows for one or both of the vectors to be the zero vector.

DEFINITION Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$.

EXAMPLE 4 To determine if two vectors are orthogonal, calculate their dot product.

(a) $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle 4, 6 \rangle$ are orthogonal because $\mathbf{u} \cdot \mathbf{v} = (3)(4) + (-2)(6) = 0$.

(b) $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 2\mathbf{j} + 4\mathbf{k}$ are orthogonal because

$$\mathbf{u} \cdot \mathbf{v} = (3)(0) + (-2)(2) + (1)(4) = 0.$$

(c) $\mathbf{0}$ is orthogonal to every vector $\mathbf{u}$ because

$$\begin{aligned} \mathbf{0} \cdot \mathbf{u} &= \langle 0, 0, 0 \rangle \cdot \langle u_1, u_2, u_3 \rangle \\ &= (0)(u_1) + (0)(u_2) + (0)(u_3) = 0. \end{aligned}$$

Dot Product Properties and Vector Projections

The dot product obeys many of the laws that hold for ordinary products of real numbers (scalars).

Properties of the Dot Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and c is a scalar, then

$$1. \mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$2. (c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$$

$$3. \mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$$

$$4. \mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$$

$$5. \mathbf{0} \cdot \mathbf{u} = 0.$$

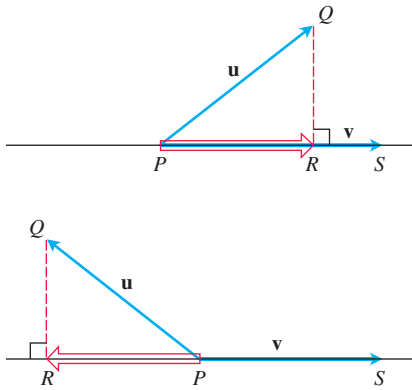


FIGURE 11.24 The vector projection of $\mathbf{u}$ onto $\mathbf{v}$.

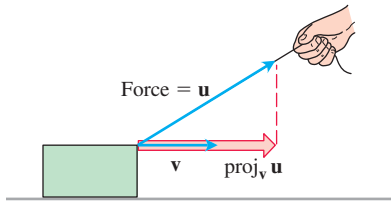


FIGURE 11.25 If we pull on the box with force $\mathbf{u}$, the effective force moving the box forward in the direction $\mathbf{v}$ is the projection of $\mathbf{u}$ onto $\mathbf{v}$.

Proofs of Properties 1 and 3 The properties are easy to prove using the definition. For instance, here are the proofs of Properties 1 and 3.

$$1. \mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = v_1u_1 + v_2u_2 + v_3u_3 = \mathbf{v} \cdot \mathbf{u}$$

$$\begin{aligned} 3. \mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) &= \langle u_1, u_2, u_3 \rangle \cdot \langle v_1 + w_1, v_2 + w_2, v_3 + w_3 \rangle \\ &= u_1(v_1 + w_1) + u_2(v_2 + w_2) + u_3(v_3 + w_3) \\ &= u_1v_1 + u_1w_1 + u_2v_2 + u_2w_2 + u_3v_3 + u_3w_3 \\ &= (u_1v_1 + u_2v_2 + u_3v_3) + (u_1w_1 + u_2w_2 + u_3w_3) \\ &= \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w} \end{aligned}$$

We now return to the problem of projecting one vector onto another, posed in the opening to this section. The **vector projection** of $\mathbf{u} = \overrightarrow{PQ}$ onto a nonzero vector $\mathbf{v} = \overrightarrow{PS}$ (Figure 11.24) is the vector $\overrightarrow{PR}$ determined by dropping a perpendicular from Q to the line PS . The notation for this vector is

$$\text{proj}_{\mathbf{v}} \mathbf{u} \quad (\text{“the vector projection of } \mathbf{u} \text{ onto } \mathbf{v} \text{”}).$$

If $\mathbf{u}$ represents a force, then $\text{proj}_{\mathbf{v}} \mathbf{u}$ represents the effective force in the direction of $\mathbf{v}$ (Figure 11.25).

If the angle θ between $\mathbf{u}$ and $\mathbf{v}$ is acute, $\text{proj}_{\mathbf{v}} \mathbf{u}$ has length $|\mathbf{u}| \cos \theta$ and direction $\mathbf{v}/|\mathbf{v}|}$ (Figure 11.26). If θ is obtuse, $\cos \theta < 0$ and $\text{proj}_{\mathbf{v}} \mathbf{u}$ has length $-|\mathbf{u}| \cos \theta$ and direction $-\mathbf{v}/|\mathbf{v}|}$. In both cases,

$$\begin{aligned} \text{proj}_{\mathbf{v}} \mathbf{u} &= (|\mathbf{u}| \cos \theta) \frac{\mathbf{v}}{|\mathbf{v}|} \\ &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \right) \frac{\mathbf{v}}{|\mathbf{v}|} & |\mathbf{u}| \cos \theta &= \frac{|\mathbf{u}| |\mathbf{v}| \cos \theta}{|\mathbf{v}|} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \\ &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}. \end{aligned}$$

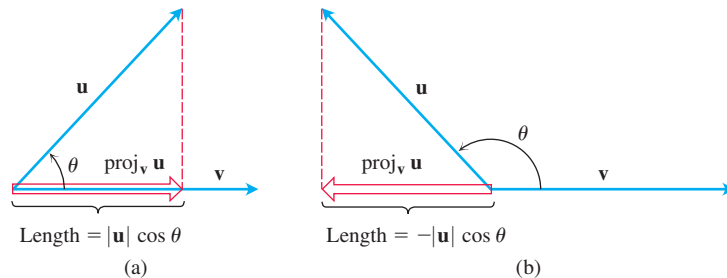


FIGURE 11.26 The length of $\text{proj}_{\mathbf{v}} \mathbf{u}$ is (a) $|\mathbf{u}| \cos \theta$ if $\cos \theta \geq 0$ and (b) $-|\mathbf{u}| \cos \theta$ if $\cos \theta < 0$.

The number $|\mathbf{u}| \cos \theta$ is called the **scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$** . To summarize,

The vector projection of $\mathbf{u}$ onto $\mathbf{v}$ is the vector

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} \right) \frac{\mathbf{v}}{|\mathbf{v}|}. \quad (1)$$

The scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$ is the scalar

$$|\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} = \mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|}. \quad (2)$$

Note that both the vector projection of $\mathbf{u}$ onto $\mathbf{v}$ and the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$ depend only on the direction of the vector $\mathbf{v}$, not on its length. This is because in both cases we take the dot product of $\mathbf{u}$ with the direction vector $\mathbf{v}/|\mathbf{v}|$, which is the direction of $\mathbf{v}$, and for the projection we go on to multiply the result by the direction vector.

EXAMPLE 5 Find the vector projection of $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$.

Solution We find $\text{proj}_{\mathbf{v}} \mathbf{u}$ from Equation (1):

$$\begin{aligned}\text{proj}_{\mathbf{v}} \mathbf{u} &= \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v} = \frac{6 - 6 - 4}{1 + 4 + 4} (\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}) \\ &= -\frac{4}{9} (\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}) = -\frac{4}{9} \mathbf{i} + \frac{8}{9} \mathbf{j} + \frac{8}{9} \mathbf{k}.\end{aligned}$$

We find the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$ from Equation (2):

$$\begin{aligned}|\mathbf{u}| \cos \theta &= \mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|} = (6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}) \cdot \left(\frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}\right) \\ &= 2 - 2 - \frac{4}{3} = -\frac{4}{3}.\end{aligned}$$

Equations (1) and (2) also apply to two-dimensional vectors. We demonstrate this in the next example.

EXAMPLE 6 Find the vector projection of a force $\mathbf{F} = 5\mathbf{i} + 2\mathbf{j}$ onto $\mathbf{v} = \mathbf{i} - 3\mathbf{j}$ and the scalar component of $\mathbf{F}$ in the direction of $\mathbf{v}$.

Solution The vector projection is

$$\begin{aligned}\text{proj}_{\mathbf{v}} \mathbf{F} &= \left(\frac{\mathbf{F} \cdot \mathbf{v}}{|\mathbf{v}|^2}\right) \mathbf{v} = \left(\frac{\mathbf{F} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\right) \mathbf{v} \\ &= \frac{5 - 6}{1 + 9} (\mathbf{i} - 3\mathbf{j}) = -\frac{1}{10} (\mathbf{i} - 3\mathbf{j}) \\ &= -\frac{1}{10} \mathbf{i} + \frac{3}{10} \mathbf{j}.\end{aligned}$$

The scalar component of $\mathbf{F}$ in the direction of $\mathbf{v}$ is

$$|\mathbf{F}| \cos \theta = \frac{\mathbf{F} \cdot \mathbf{v}}{|\mathbf{v}|} = \frac{5 - 6}{\sqrt{1 + 9}} = -\frac{1}{\sqrt{10}}.$$

EXAMPLE 7 Verify that the vector $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$ is orthogonal to the projection vector $\text{proj}_{\mathbf{v}} \mathbf{u}$.

Solution The vector $\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\right) \mathbf{v}$ is parallel to $\mathbf{v}$. So it suffices to show that the vector $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$ is orthogonal to $\mathbf{v}$. We verify orthogonality by showing that the dot product of $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$ with $\mathbf{v}$ is zero:

$$\begin{aligned}(\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}) \cdot \mathbf{v} &= \mathbf{u} \cdot \mathbf{v} - \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v}\right) \cdot \mathbf{v} && \text{Definition of } \text{proj}_{\mathbf{v}} \mathbf{u} \\ &= \mathbf{u} \cdot \mathbf{v} - \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} (\mathbf{v} \cdot \mathbf{v}) && \text{Dot product property (2)} \\ &= \mathbf{u} \cdot \mathbf{v} - \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} |\mathbf{v}|^2 && \mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2 \\ &= \mathbf{u} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{v} = 0.\end{aligned}$$

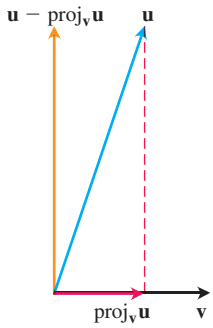


FIGURE 11.27 The vector $\mathbf{u}$ is the sum of two perpendicular vectors: a vector $\text{proj}_{\mathbf{v}} \mathbf{u}$, parallel to $\mathbf{v}$, and a vector $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$, perpendicular to $\mathbf{v}$.

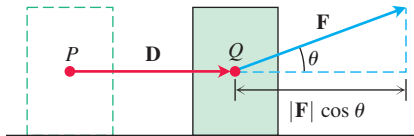


FIGURE 11.28 The work done by a constant force $\mathbf{F}$ during a displacement $\mathbf{D}$ is $(|\mathbf{F}|\cos\theta)|\mathbf{D}|$, which is the dot product $\mathbf{F} \cdot \mathbf{D}$.

Example 7 verifies that the vector $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$ is orthogonal to the projection vector $\text{proj}_{\mathbf{v}} \mathbf{u}$ (which has the same direction as $\mathbf{v}$). So the equation

$$\mathbf{u} = \text{proj}_{\mathbf{v}} \mathbf{u} + (\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}) = \underbrace{\left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}}_{\text{Parallel to } \mathbf{v}} + \underbrace{\left(\mathbf{u} - \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v} \right)}_{\text{Orthogonal to } \mathbf{v}}$$

expresses $\mathbf{u}$ as a sum of orthogonal vectors (see Figure 11.27).

Work

In Chapter 6, we calculated the work done by a constant force of magnitude F in moving an object through a distance d as $W = Fd$. That formula holds only if the force is directed along the line of motion. If a force $\mathbf{F}$ moving an object through a displacement $\mathbf{D} = \overrightarrow{PQ}$ has some other direction, the work is performed by the component of $\mathbf{F}$ in the direction of $\mathbf{D}$. If θ is the angle between $\mathbf{F}$ and $\mathbf{D}$ (Figure 11.28), then

$$\begin{aligned} \text{Work} &= \left(\begin{array}{l} \text{scalar component of } \mathbf{F} \\ \text{in the direction of } \mathbf{D} \end{array} \right) (\text{length of } \mathbf{D}) \\ &= (|\mathbf{F}|\cos\theta)|\mathbf{D}| \\ &= \mathbf{F} \cdot \mathbf{D}. \end{aligned}$$

DEFINITION The work done by a constant force $\mathbf{F}$ acting through a displacement $\mathbf{D} = \overrightarrow{PQ}$ is

$$W = \mathbf{F} \cdot \mathbf{D}.$$

EXAMPLE 8 If $|\mathbf{F}| = 40$ N (newtons), $|\mathbf{D}| = 3$ m, and $\theta = 60^\circ$, the work done by $\mathbf{F}$ in acting from P to Q is

$$\begin{aligned} \text{Work} &= \mathbf{F} \cdot \mathbf{D} && \text{Definition} \\ &= |\mathbf{F}||\mathbf{D}|\cos\theta \\ &= (40)(3)\cos 60^\circ && \text{Given values} \\ &= (120)(1/2) = 60 \text{ J (joules)}. \end{aligned}$$

We encounter more challenging work problems in Chapter 15 when we learn to find the work done by a variable force along a more general *path* in space.

The Dot Product of Two n -Dimensional Vectors

If $\mathbf{u} = \langle u_1, u_2, \dots, u_n \rangle$ and $\mathbf{v} = \langle v_1, v_2, \dots, v_n \rangle$ are n -dimensional vectors, then we define the dot product to be

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

As for two- and three-dimensional vectors, the dot product is calculated by adding the products of the corresponding components of the two vectors.

This generalized dot product can be shown to satisfy the Properties of the Dot Product that were introduced earlier in this section, and similar terminology is used. If $\mathbf{u}$ and $\mathbf{v}$ are n -dimensional vectors, then

1. $\mathbf{u}$ and $\mathbf{v}$ are said to be orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$,
2. the vector projection of $\mathbf{u}$ onto $\mathbf{v}$ is $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v}$, and

3. the angle between the vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as $\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right)$. (The Cauchy-Schwarz inequality, $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$, stated in Exercise 27 can be extended to n -dimensional vectors. This guarantees that $\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$ is within the interval $[-1, 1]$.)

EXAMPLE 9 An automobile assembly plant makes four different car models. The components of the vector $\mathbf{u} = \langle 36, 50, 24, 10 \rangle$ indicate the plant's output of each model per hour, whereas the revenue per vehicle (in US dollars) of each model is represented by the vector $\mathbf{v} = \langle 24,000, 31,000, 39,000, 52,000 \rangle$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ and explain the significance of the value that was obtained.

Solution

$$\mathbf{u} \cdot \mathbf{v} = (36)(24,000) + (50)(31,000) + (24)(39,000) + (10)(52,000) = 3,870,000.$$

The value \$3,870,000 represents the total hourly revenue. ■

EXERCISES 11.3

For some exercises, a calculator may be helpful when expressing answers in decimal form.

Dot Product and Projections

In Exercises 1–8, find

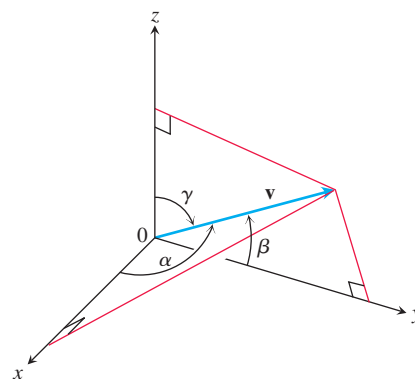
- $\mathbf{v} \cdot \mathbf{u}$, $|\mathbf{v}|$, $|\mathbf{u}|$
 - the cosine of the angle between $\mathbf{v}$ and $\mathbf{u}$
 - the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$
 - the vector $\text{proj}_{\mathbf{u}} \mathbf{v}$.
- $\mathbf{v} = 2\mathbf{i} - 4\mathbf{j} + \sqrt{5}\mathbf{k}$, $\mathbf{u} = -2\mathbf{i} + 4\mathbf{j} - \sqrt{5}\mathbf{k}$
 - $\mathbf{v} = (3/5)\mathbf{i} + (4/5)\mathbf{k}$, $\mathbf{u} = 5\mathbf{i} + 12\mathbf{j}$
 - $\mathbf{v} = 10\mathbf{i} + 11\mathbf{j} - 2\mathbf{k}$, $\mathbf{u} = 3\mathbf{j} + 4\mathbf{k}$
 - $\mathbf{v} = 2\mathbf{i} + 10\mathbf{j} - 11\mathbf{k}$, $\mathbf{u} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
 - $\mathbf{v} = 5\mathbf{j} - 3\mathbf{k}$, $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
 - $\mathbf{v} = -\mathbf{i} + \mathbf{j}$, $\mathbf{u} = \sqrt{2}\mathbf{i} + \sqrt{3}\mathbf{j} + 2\mathbf{k}$
 - $\mathbf{v} = 5\mathbf{i} + \mathbf{j}$, $\mathbf{u} = 2\mathbf{i} + \sqrt{17}\mathbf{j}$
 - $\mathbf{v} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}} \right\rangle$, $\mathbf{u} = \left\langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{3}} \right\rangle$

Angle Between Vectors

Find the angles between the vectors in Exercises 9–12 to the nearest hundredth of a radian.

- $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$
 - $\mathbf{u} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 3\mathbf{i} + 4\mathbf{k}$
 - $\mathbf{u} = \sqrt{3}\mathbf{i} - 7\mathbf{j}$, $\mathbf{v} = \sqrt{3}\mathbf{i} + \mathbf{j} - 2\mathbf{k}$
 - $\mathbf{u} = \mathbf{i} + \sqrt{2}\mathbf{j} - \sqrt{2}\mathbf{k}$, $\mathbf{v} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$
13. **Triangle** Find the measures of the angles of the triangle whose vertices are $A = (-1, 0)$, $B = (2, 1)$, and $C = (1, -2)$.
14. **Rectangle** Find the measures of the angles between the diagonals of the rectangle whose vertices are $A = (1, 0)$, $B = (0, 3)$, $C = (3, 4)$, and $D = (4, 1)$.

15. **Direction angles and direction cosines** The *direction angles* α , β , and γ of a vector $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ are defined as follows:
- α is the angle between $\mathbf{v}$ and the positive x -axis ($0 \leq \alpha \leq \pi$).
 - β is the angle between $\mathbf{v}$ and the positive y -axis ($0 \leq \beta \leq \pi$).
 - γ is the angle between $\mathbf{v}$ and the positive z -axis ($0 \leq \gamma \leq \pi$).



- a. Show that

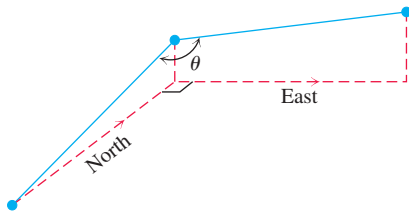
$$\cos \alpha = \frac{a}{|\mathbf{v}|}, \quad \cos \beta = \frac{b}{|\mathbf{v}|}, \quad \cos \gamma = \frac{c}{|\mathbf{v}|},$$

and $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$. These cosines are called the *direction cosines* of $\mathbf{v}$.

- b. **Unit vectors are built from direction cosines** Show that if $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$ is a unit vector, then a , b , and c are the direction cosines of $\mathbf{v}$.

16. **Water main construction** A water main is to be constructed with a 20% grade in the north direction and a 10% grade in the

east direction. Determine the angle θ required in the water main for the turn from north to east.



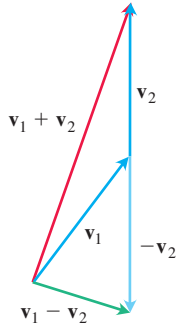
For Exercises 17 and 18, find the acute angle between the given lines by using vectors parallel to the lines.

17. $y = x$, $y = 2x + 3$

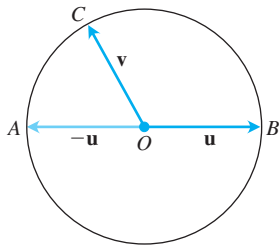
18. $2 - x + 2y = 0$, $3x - 4y = -12$

Theory and Examples

19. Sums and differences In the accompanying figure, it looks as if $\mathbf{v}_1 + \mathbf{v}_2$ and $\mathbf{v}_1 - \mathbf{v}_2$ are orthogonal. Is this mere coincidence, or are there circumstances under which we may expect the sum of two vectors to be orthogonal to their difference? Give reasons for your answer.



20. Orthogonality on a circle Suppose that AB is the diameter of a circle with center O and that C is a point on one of the two arcs joining A and B . Show that $\overline{CA}$ and $\overline{CB}$ are orthogonal.

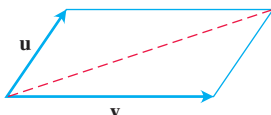


21. Diagonals of a rhombus Show that the diagonals of a rhombus (parallelogram with sides of equal length) are perpendicular.

22. Perpendicular diagonals Show that squares are the only rectangles with perpendicular diagonals.

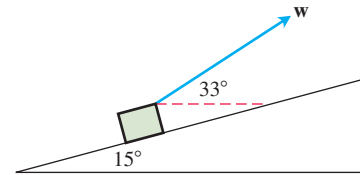
23. When parallelograms are rectangles Prove that a parallelogram is a rectangle if and only if its diagonals are equal in length. (This fact is often exploited by carpenters.)

24. Diagonal of parallelogram Show that the indicated diagonal of the parallelogram determined by vectors $\mathbf{u}$ and $\mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$ if $|\mathbf{u}| = |\mathbf{v}|$.



25. Projectile motion A gun with muzzle velocity of 400 m/s is fired at an angle of 8° above the horizontal. Find the horizontal and vertical components of the velocity.

26. Inclined plane Suppose that a box is being towed up an inclined plane as shown in the figure. Find the force $\mathbf{w}$ needed to make the component of the force parallel to the inclined plane equal to 2.5 N.



27. a. Cauchy-Schwarz inequality Since $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$, show that the inequality $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$ holds for any vectors $\mathbf{u}$ and $\mathbf{v}$.

b. Under what circumstances, if any, does $|\mathbf{u} \cdot \mathbf{v}|$ equal $|\mathbf{u}||\mathbf{v}|$? Give reasons for your answer.

28. Dot multiplication is positive definite Show that dot multiplication of vectors is *positive definite*; that is, show that $\mathbf{u} \cdot \mathbf{u} \geq 0$ for every vector $\mathbf{u}$ and that $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

29. Orthogonal unit vectors If $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthogonal unit vectors and $\mathbf{v} = a\mathbf{u}_1 + b\mathbf{u}_2$, find $\mathbf{v} \cdot \mathbf{u}_1$.

30. Cancellation in dot products In real-number multiplication, if $uv_1 = uv_2$ and $u \neq 0$, we can cancel the u and conclude that $v_1 = v_2$. Does the same rule hold for the dot product? That is, if $\mathbf{u} \cdot \mathbf{v}_1 = \mathbf{u} \cdot \mathbf{v}_2$ and $\mathbf{u} \neq \mathbf{0}$, can you conclude that $\mathbf{v}_1 = \mathbf{v}_2$? Give reasons for your answer.

31. If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, show that $\text{proj}_{\mathbf{v}} \mathbf{u} = \mathbf{0}$.

32. A force $\mathbf{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ is applied to a spacecraft with velocity vector $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$. Express $\mathbf{F}$ as a sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

Equations for Lines in the Plane

33. Line perpendicular to a vector Show that $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$ is perpendicular to the line $ax + by = c$. (Hint: For a and b nonzero, establish that the slope of the vector $\mathbf{v}$ is the negative reciprocal of the slope of the given line. Also verify the statement when $a = 0$ or $b = 0$.)

34. Line parallel to a vector Show that the vector $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$ is parallel to the line $bx - ay = c$. (Hint: For a and b nonzero, establish that the slope of the line segment representing $\mathbf{v}$ is the same as the slope of the given line. Also verify the statement when $a = 0$ or $b = 0$.)

In Exercises 35–38, use the result of Exercise 33 to find an equation for the line through P perpendicular to $\mathbf{v}$. Then sketch the line. Include $\mathbf{v}$ in your sketch as a vector starting at the origin.

35. $P(2, 1)$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$

36. $P(-1, 2)$, $\mathbf{v} = -2\mathbf{i} - \mathbf{j}$

37. $P(-2, -7)$, $\mathbf{v} = -2\mathbf{i} + \mathbf{j}$

38. $P(11, 10)$, $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j}$

In Exercises 39–42, use the result of Exercise 34 to find an equation for the line through P parallel to $\mathbf{v}$. Then sketch the line. Include $\mathbf{v}$ in your sketch as a vector starting at the origin.

39. $P(-2, 1)$, $\mathbf{v} = \mathbf{i} - \mathbf{j}$

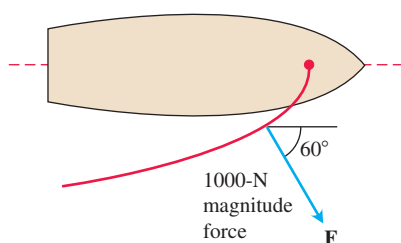
40. $P(0, -2)$, $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j}$

41. $P(1, 2)$, $\mathbf{v} = -\mathbf{i} - 2\mathbf{j}$

42. $P(1, 3)$, $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j}$

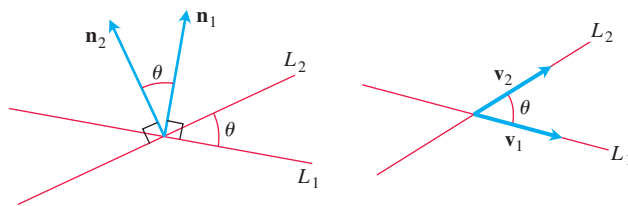
Work

- 43. Work along a line** Find the work done by a force $\mathbf{F} = 5\mathbf{i}$ (magnitude 5 N) in moving an object along the line from the origin to the point $(1, 1)$ (distance in meters).
- 44. Locomotive** The Union Pacific's *Big Boy* locomotive could pull 6000-tonne trains with a tractive effort (pull) of 602,148 N. At this level of effort, about how much work did *Big Boy* do on the (approximately straight) 605-km journey from San Francisco to Los Angeles?
- 45. Inclined plane** How much work does it take to slide a crate 20 m along a loading dock by pulling on it with a 200-N force at an angle of 30° from the horizontal?
- 46. Sailboat** The wind passing over a boat's sail exerted a 1000 N magnitude force $\mathbf{F}$ as shown here. How much work did the wind perform in moving the boat forward 1 km? Answer in joules.



Angles Between Lines in the Plane

The **acute angle between intersecting lines** that do not cross at right angles is the same as the angle determined by vectors normal to the lines or by vectors parallel to the lines.



Use this fact and the results of Exercise 33 or 34 to find the acute angles between the lines in Exercises 47–52.

- 47.** $3x + y = 5$, $2x - y = 4$
48. $y = \sqrt{3}x - 1$, $y = -\sqrt{3}x + 2$
49. $\sqrt{3}x - y = -2$, $x - \sqrt{3}y = 1$
50. $x + \sqrt{3}y = 1$, $(1 - \sqrt{3})x + (1 + \sqrt{3})y = 8$
51. $3x - 4y = 3$, $x - y = 7$
52. $12x + 5y = 1$, $2x - 2y = 3$

Dot Products of n -Dimensional Vectors

In Exercises 53–56, (a) find $\mathbf{u} \cdot \mathbf{v}$ and (b) determine whether the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

- 53.** $\mathbf{u} = \langle 3, 2, -4, 0 \rangle$, $\mathbf{v} = \langle 1, 0, 0, 2 \rangle$
54. $\mathbf{u} = \langle -2, 1, 1, 2 \rangle$, $\mathbf{v} = \langle -1, 2, -2, -1 \rangle$
55. $\mathbf{u} = \langle 6, 3, 0, 1, -2 \rangle$, $\mathbf{v} = \langle 0, 2, -7, 0, 3 \rangle$
56. $\mathbf{u} = \langle 4, 2, -3, -2, 1, 5 \rangle$, $\mathbf{v} = \langle 3, -3, 2, -2, 1, -1 \rangle$

11.4 The Cross Product

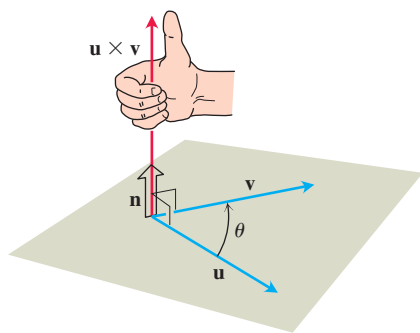


FIGURE 11.29 The construction of $\mathbf{u} \times \mathbf{v}$.

In studying lines in the plane, when we needed to describe how a line was tilting, we used the notions of slope and angle of inclination. In space, we want a way to describe how a *plane* is tilting. We accomplish this by multiplying two vectors in the plane together to get a third vector perpendicular to the plane. The direction of this third vector tells us the “inclination” of the plane. The product we use to multiply the vectors together is the *vector* or *cross product*, the second of the two vector multiplication methods. The cross product gives us a simple way to find a variety of geometric quantities, including volumes, areas, and perpendicular vectors. We study the cross product in this section.

The Cross Product of Two Vectors in Space

We start with two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in space. Two vectors are *parallel* if one is a nonzero multiple of the other. If $\mathbf{u}$ and $\mathbf{v}$ are not parallel, they determine a plane. The vectors in this plane are linear combinations of $\mathbf{u}$ and $\mathbf{v}$, so they can be written as a sum $a\mathbf{u} + b\mathbf{v}$. We select the unit vector $\mathbf{n}$ perpendicular to the plane by the **right-hand rule**. This means that we choose $\mathbf{n}$ to be the unit normal vector that points the way your right thumb points when your fingers curl through the angle θ from $\mathbf{u}$ to $\mathbf{v}$ (Figure 11.29). Then we define a new vector as follows.

DEFINITION The **cross product** $\mathbf{u} \times \mathbf{v}$ (“**u cross v**”) is the vector

$$\mathbf{u} \times \mathbf{v} = (|\mathbf{u}||\mathbf{v}| \sin \theta) \mathbf{n}.$$

Unlike the dot product, the cross product is a vector. For this reason it is also called the **vector product** of $\mathbf{u}$ and $\mathbf{v}$, and can be applied *only* to vectors in space. The vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$ because it is a scalar multiple of $\mathbf{n}$.

There is a straightforward way to calculate the cross product of two vectors from their components. The method does not require that we know the angle between them (as suggested by the definition), but we postpone that calculation momentarily so we can focus first on the properties of the cross product.

Because the sines of 0 and π are both zero, it makes sense to define the cross product of two parallel nonzero vectors to be $\mathbf{0}$. If one or both of $\mathbf{u}$ and $\mathbf{v}$ are zero, we also define $\mathbf{u} \times \mathbf{v}$ to be zero. This way, the cross product of two vectors $\mathbf{u}$ and $\mathbf{v}$ is zero if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel or one or both of them are zero.

Parallel Vectors

Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.

The cross product obeys the following laws.

Properties of the Cross Product

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are any vectors and r , s are scalars, then

- | | |
|--|---|
| 1. $(r\mathbf{u}) \times (s\mathbf{v}) = (rs)(\mathbf{u} \times \mathbf{v})$ | 2. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$ |
| 3. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$ | 4. $(\mathbf{v} + \mathbf{w}) \times \mathbf{u} = \mathbf{v} \times \mathbf{u} + \mathbf{w} \times \mathbf{u}$ |
| 5. $\mathbf{0} \times \mathbf{u} = \mathbf{0}$ | 6. $\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$ |

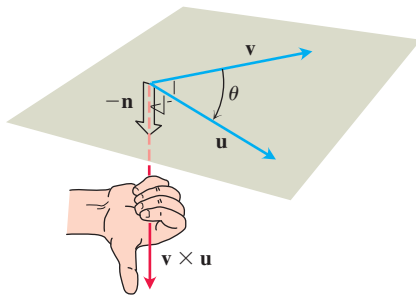


FIGURE 11.30 The construction of $\mathbf{v} \times \mathbf{u}$.

To visualize Property 3, for example, notice that when the fingers of your right hand curl through the angle θ from $\mathbf{v}$ to $\mathbf{u}$, your thumb points the opposite way; the unit vector we choose in forming $\mathbf{v} \times \mathbf{u}$ is the negative of the one we choose in forming $\mathbf{u} \times \mathbf{v}$ (Figure 11.30).

Property 1 can be verified by applying the definition of cross product to both sides of the equation and comparing the results. Property 2 is proved in Appendix A.9. Property 4 follows by multiplying both sides of the equation in Property 2 by -1 and reversing the order of the products using Property 3. Property 5 is a definition. As a rule, cross product multiplication is *not associative* so $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ does not generally equal $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$. (See Additional Exercise 17.)

When we apply the definition and Property 3 to calculate the pairwise cross products of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, we find (Figure 11.31)

$$\mathbf{i} \times \mathbf{j} = -(\mathbf{j} \times \mathbf{i}) = \mathbf{k}$$

$$\mathbf{j} \times \mathbf{k} = -(\mathbf{k} \times \mathbf{j}) = \mathbf{i}$$

$$\mathbf{k} \times \mathbf{i} = -(\mathbf{i} \times \mathbf{k}) = \mathbf{j}$$

and

$$\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}.$$

$|\mathbf{u} \times \mathbf{v}|$ Is the Area of a Parallelogram

Because $\mathbf{n}$ is a unit vector, the magnitude of $\mathbf{u} \times \mathbf{v}$ is

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}||\sin \theta||\mathbf{n}| = |\mathbf{u}||\mathbf{v}| \sin \theta.$$

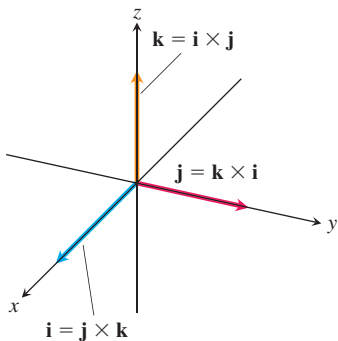


FIGURE 11.31 The pairwise cross products of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

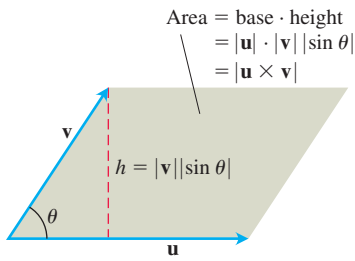


FIGURE 11.32 The parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$.

Determinants

2×2 and 3×3 determinants are evaluated as follows:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

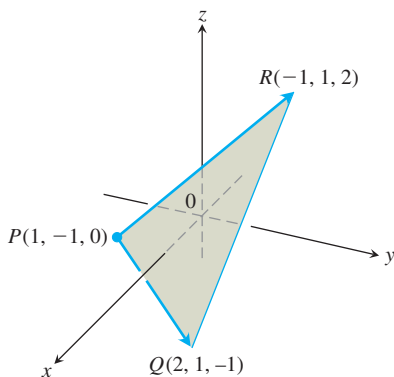


FIGURE 11.33 The vector $\overrightarrow{PQ} \times \overrightarrow{PR}$ is perpendicular to the plane of triangle PQR (Example 2). The area of triangle PQR is half of $|\overrightarrow{PQ} \times \overrightarrow{PR}|$ (Example 3).

This is the area of the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$ (Figure 11.32), $|\mathbf{u}|$ being the base of the parallelogram and $|\mathbf{v}|\sin\theta$ being the height.

Determinant Formula for $\mathbf{u} \times \mathbf{v}$

Our next objective is to calculate $\mathbf{u} \times \mathbf{v}$ from the components of $\mathbf{u}$ and $\mathbf{v}$ relative to a Cartesian coordinate system.

Suppose that

$$\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k} \quad \text{and} \quad \mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}.$$

Then the distributive laws and the rules for multiplying $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ tell us that

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= (u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}) \times (v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}) \\ &= u_1v_1\mathbf{i} \times \mathbf{i} + u_1v_2\mathbf{i} \times \mathbf{j} + u_1v_3\mathbf{i} \times \mathbf{k} \\ &\quad + u_2v_1\mathbf{j} \times \mathbf{i} + u_2v_2\mathbf{j} \times \mathbf{j} + u_2v_3\mathbf{j} \times \mathbf{k} \\ &\quad + u_3v_1\mathbf{k} \times \mathbf{i} + u_3v_2\mathbf{k} \times \mathbf{j} + u_3v_3\mathbf{k} \times \mathbf{k} \\ &= (u_2v_3 - u_3v_2)\mathbf{i} - (u_1v_3 - u_3v_1)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k}. \end{aligned}$$

The component terms in the last line are hard to remember, but they are the same as the terms in the expansion of the symbolic determinant

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

So we restate the calculation in the following easy-to-remember form.

Calculating the Cross Product as a Determinant

If $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$, then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

EXAMPLE 1 Find $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$ if $\mathbf{u} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{v} = -4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.

Solution We expand the symbolic determinant.

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} \mathbf{k} \\ &= -2\mathbf{i} - 6\mathbf{j} + 10\mathbf{k} \\ \mathbf{v} \times \mathbf{u} &= -(\mathbf{u} \times \mathbf{v}) = 2\mathbf{i} + 6\mathbf{j} - 10\mathbf{k} \end{aligned}$$

Property 3

EXAMPLE 2 Find a vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$ (Figure 11.33).

Solution The vector $\overrightarrow{PQ} \times \overrightarrow{PR}$ is perpendicular to the plane because it is perpendicular to both vectors. In terms of components,

$$\begin{aligned}\overrightarrow{PQ} &= (2 - 1)\mathbf{i} + (1 + 1)\mathbf{j} + (-1 - 0)\mathbf{k} = \mathbf{i} + 2\mathbf{j} - \mathbf{k} \\ \overrightarrow{PR} &= (-1 - 1)\mathbf{i} + (1 + 1)\mathbf{j} + (2 - 0)\mathbf{k} = -2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k} \\ \overrightarrow{PQ} \times \overrightarrow{PR} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 2 & 2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & -1 \\ -2 & 2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ -2 & 2 \end{vmatrix} \mathbf{k} \\ &= 6\mathbf{i} + 6\mathbf{k}. \end{aligned}$$

EXAMPLE 3 Find the area of the triangle with vertices $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$ (Figure 11.33).

Solution The area of the parallelogram determined by P , Q , and R is

$$\begin{aligned}|\overrightarrow{PQ} \times \overrightarrow{PR}| &= |6\mathbf{i} + 6\mathbf{k}| && \text{Values from Example 2} \\ &= \sqrt{(6)^2 + (6)^2} = \sqrt{2 \cdot 36} = 6\sqrt{2}.\end{aligned}$$

The triangle's area is half of this, or $3\sqrt{2}$.

EXAMPLE 4 Find a unit vector perpendicular to the plane of $P(1, -1, 0)$, $Q(2, 1, -1)$, and $R(-1, 1, 2)$.

Solution Since $\overrightarrow{PQ} \times \overrightarrow{PR}$ is perpendicular to the plane, its direction $\mathbf{n}$ is a unit vector perpendicular to the plane. Taking values from Examples 2 and 3, we have

$$\mathbf{n} = \frac{\overrightarrow{PQ} \times \overrightarrow{PR}}{|\overrightarrow{PQ} \times \overrightarrow{PR}|} = \frac{6\mathbf{i} + 6\mathbf{k}}{6\sqrt{2}} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{k}.$$

For ease in calculating the cross product using determinants, we usually write vectors in the form $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ rather than as ordered triples $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$.

Torque

When we turn a bolt by applying a force $\mathbf{F}$ to a wrench (Figure 11.34), we produce a torque that causes the bolt to rotate. The **torque vector** points in the direction of the axis of the bolt according to the right-hand rule (so the rotation is counterclockwise when viewed from the *tip* of the vector). The magnitude of the torque depends on how far out on the wrench the force is applied and on how much of the force is perpendicular to the wrench at the point of application. The number we use to measure the torque's magnitude is the product of the length of the lever arm $\mathbf{r}$ and the scalar component of $\mathbf{F}$ perpendicular to $\mathbf{r}$. In the notation of Figure 11.34,

$$\text{Magnitude of torque vector} = |\mathbf{r}||\mathbf{F}| \sin \theta,$$

or $|\mathbf{r} \times \mathbf{F}|$. If we let $\mathbf{n}$ be a unit vector along the axis of the bolt in the direction of the torque, then a complete description of the torque vector is $\mathbf{r} \times \mathbf{F}$, or

$$\text{Torque vector} = \mathbf{r} \times \mathbf{F} = (|\mathbf{r}||\mathbf{F}| \sin \theta) \mathbf{n}.$$

Recall that we defined $\mathbf{u} \times \mathbf{v}$ to be $\mathbf{0}$ when $\mathbf{u}$ and $\mathbf{v}$ are parallel. This is consistent with the torque interpretation as well. If the force $\mathbf{F}$ in Figure 11.34 is parallel to the wrench, meaning that we are trying to turn the bolt by pushing or pulling along the line of the wrench's handle, the torque produced is zero.

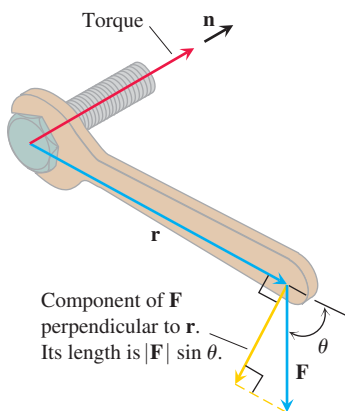


FIGURE 11.34 The torque vector describes the tendency of the force $\mathbf{F}$ to drive the bolt forward.

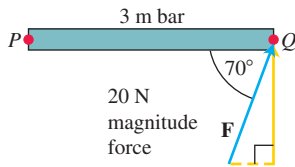


FIGURE 11.35 The magnitude of the torque exerted by $\mathbf{F}$ at P is about $56.4 \text{ N} \cdot \text{m}$ (Example 5). The bar rotates counter-clockwise around P .

EXAMPLE 5 The magnitude of the torque generated by force $\mathbf{F}$ at the pivot point P in Figure 11.35 is

$$|\overrightarrow{PQ} \times \mathbf{F}| = |\overrightarrow{PQ}| |\mathbf{F}| \sin 70^\circ \approx (3)(20)(0.94) \approx 56.4 \text{ N} \cdot \text{m}.$$

In this example, the torque vector is pointing out of the page toward you. ■

Triple Scalar or Box Product

The product $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ is called the **triple scalar product** of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ (in that order). As you can see from the formula

$$|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = |\mathbf{u} \times \mathbf{v}| |\mathbf{w}| |\cos \theta|,$$

the absolute value of this product is the volume of the parallelepiped (parallelogram-sided box) determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ (Figure 11.36). The number $|\mathbf{u} \times \mathbf{v}|$ is the area of the base parallelogram. The number $|\mathbf{w}| |\cos \theta|$ is the parallelepiped's height. Because of this geometry, $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ is also called the **box product** of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

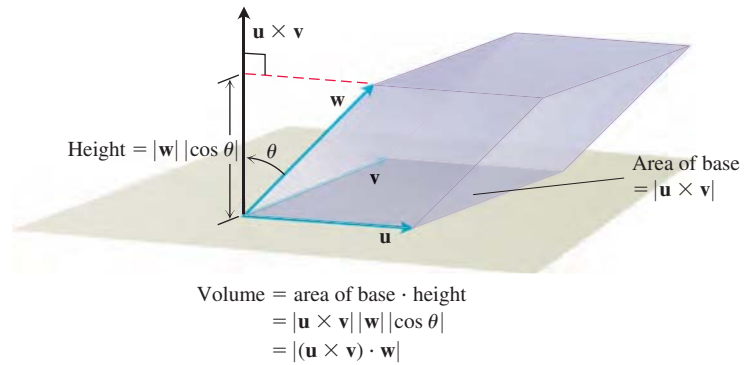


FIGURE 11.36 The number $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$ is the volume of a parallelepiped.

By treating the planes of $\mathbf{v}$ and $\mathbf{w}$ and of $\mathbf{w}$ and $\mathbf{u}$ as the base planes of the parallelepiped determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, we see that

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u} = (\mathbf{w} \times \mathbf{u}) \cdot \mathbf{v}.$$

Since the dot product is commutative, we also have

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}).$$

The triple scalar product can be evaluated as a determinant:

$$\begin{aligned}
 (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} &= \left(\begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \right) \cdot \mathbf{w} \\
 &= w_1 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - w_2 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + w_3 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \\
 &= \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.
 \end{aligned}$$

The dot and cross may be interchanged in a triple scalar product without altering its value.

Calculating the Triple Scalar Product as a Determinant

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Any two rows of a matrix can be interchanged without changing the *absolute value* of the determinant. So we can take the vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in any order when calculating the absolute value of the triple product.

EXAMPLE 6 Find the volume of the box (parallelepiped) that is determined by $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = -2\mathbf{i} + 3\mathbf{k}$, and $\mathbf{w} = 7\mathbf{j} - 4\mathbf{k}$.

Solution Using the rule for calculating a 3×3 determinant, we find

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix} = (1) \begin{vmatrix} 0 & 3 \\ 7 & -4 \end{vmatrix} - (2) \begin{vmatrix} -2 & 3 \\ 0 & -4 \end{vmatrix} + (-1) \begin{vmatrix} -2 & 0 \\ 0 & 7 \end{vmatrix} = -23.$$

The volume is $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = 23$ units cubed. ■

EXERCISES 11.4

Cross Product Calculations

In Exercises 1–8, find the length and direction (when defined) of $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$.

1. $\mathbf{u} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = \mathbf{i} - \mathbf{k}$
2. $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j}$, $\mathbf{v} = -\mathbf{i} + \mathbf{j}$
3. $\mathbf{u} = 2\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$, $\mathbf{v} = -\mathbf{i} + \mathbf{j} - 2\mathbf{k}$
4. $\mathbf{u} = \mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{v} = \mathbf{0}$
5. $\mathbf{u} = 2\mathbf{i}$, $\mathbf{v} = -3\mathbf{j}$
6. $\mathbf{u} = \mathbf{i} \times \mathbf{j}$, $\mathbf{v} = \mathbf{j} \times \mathbf{k}$
7. $\mathbf{u} = -8\mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$, $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$
8. $\mathbf{u} = \frac{3}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + \mathbf{k}$, $\mathbf{v} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$

In Exercises 9–14, sketch the coordinate axes and then include the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$ as vectors starting at the origin.

9. $\mathbf{u} = \mathbf{i}$, $\mathbf{v} = \mathbf{j}$
10. $\mathbf{u} = \mathbf{i} - \mathbf{k}$, $\mathbf{v} = \mathbf{j}$
11. $\mathbf{u} = \mathbf{i} - \mathbf{k}$, $\mathbf{v} = \mathbf{j} + \mathbf{k}$
12. $\mathbf{u} = 2\mathbf{i} - \mathbf{j}$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j}$
13. $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{i} - \mathbf{j}$
14. $\mathbf{u} = \mathbf{j} + 2\mathbf{k}$, $\mathbf{v} = \mathbf{i}$

Triangles in Space

In Exercises 15–18,

- a. Find the area of the triangle determined by the points P , Q , and R .
 - b. Find a unit vector perpendicular to plane PQR .
15. $P(1, -1, 2)$, $Q(2, 0, -1)$, $R(0, 2, 1)$
 16. $P(1, 1, 1)$, $Q(2, 1, 3)$, $R(3, -1, 1)$
 17. $P(2, -2, 1)$, $Q(3, -1, 2)$, $R(3, -1, 1)$
 18. $P(-2, 2, 0)$, $Q(0, 1, -1)$, $R(-1, 2, -2)$

Triple Scalar Products

In Exercises 19–22, verify that

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u} = (\mathbf{w} \times \mathbf{u}) \cdot \mathbf{v}$$

and find the volume of the parallelepiped (box) determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

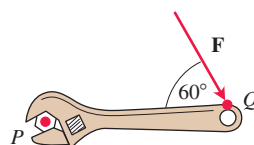
$\mathbf{u}$	$\mathbf{v}$	$\mathbf{w}$
19. $2\mathbf{i}$	$2\mathbf{j}$	$2\mathbf{k}$
20. $\mathbf{i} - \mathbf{j} + \mathbf{k}$	$2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$	$-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
21. $2\mathbf{i} + \mathbf{j}$	$2\mathbf{i} - \mathbf{j} + \mathbf{k}$	$\mathbf{i} + 2\mathbf{k}$
22. $\mathbf{i} + \mathbf{j} - 2\mathbf{k}$	$-\mathbf{i} - \mathbf{k}$	$2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$

Theory and Examples

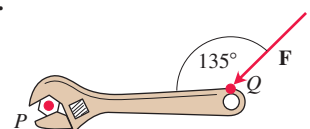
23. **Parallel and perpendicular vectors** Let $\mathbf{u} = 5\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{v} = \mathbf{j} - 5\mathbf{k}$, $\mathbf{w} = -15\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$. Which vectors, if any, are (a) perpendicular? (b) Parallel? Give reasons for your answers.
24. **Parallel and perpendicular vectors** Let $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{w} = \mathbf{i} + \mathbf{k}$, $\mathbf{r} = -(\pi/2)\mathbf{i} - \pi\mathbf{j} + (\pi/2)\mathbf{k}$. Which vectors, if any, are (a) perpendicular? (b) Parallel? Give reasons for your answers.

In Exercises 25 and 26, find the magnitude of the torque exerted by $\mathbf{F}$ on the bolt at P if $|\overrightarrow{PQ}| = 20$ cm and $|\mathbf{F}| = 15$ N. Answer in newton-meters.

25.



26.



27. Which of the following are *always true*, and which are *not always true*? Give reasons for your answers.
- $|\mathbf{u}| = \sqrt{\mathbf{u} \cdot \mathbf{u}}$
 - $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|$
 - $\mathbf{u} \times \mathbf{0} = \mathbf{0} \times \mathbf{u} = \mathbf{0}$
 - $\mathbf{u} \times (-\mathbf{u}) = \mathbf{0}$
 - $\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{u}$
 - $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
 - $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0$
 - $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$
28. Which of the following are *always true*, and which are *not always true*? Give reasons for your answers.
- $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
 - $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$
 - $(-\mathbf{u}) \times \mathbf{v} = -(\mathbf{u} \times \mathbf{v})$
 - $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$ (any number c)
 - $c(\mathbf{u} \times \mathbf{v}) = (c\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (c\mathbf{v})$ (any number c)
 - $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$
 - $(\mathbf{u} \times \mathbf{u}) \cdot \mathbf{u} = 0$
 - $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = \mathbf{v} \cdot (\mathbf{u} \times \mathbf{v})$
29. Given nonzero vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, use dot product and cross product notation, as appropriate, to describe the following.
- The vector projection of $\mathbf{u}$ onto $\mathbf{v}$
 - A vector orthogonal to $\mathbf{u}$ and $\mathbf{v}$
 - A vector orthogonal to $\mathbf{u} \times \mathbf{v}$ and $\mathbf{w}$
 - The volume of the parallelepiped determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$
 - A vector orthogonal to $\mathbf{u} \times \mathbf{v}$ and $\mathbf{u} \times \mathbf{w}$
 - A vector of length $|\mathbf{u}|$ in the direction of $\mathbf{v}$
30. Compute $(\mathbf{i} \times \mathbf{j}) \times \mathbf{j}$ and $\mathbf{i} \times (\mathbf{j} \times \mathbf{j})$. What can you conclude about the associativity of the cross product?
31. Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors. Which of the following make sense, and which do not? Give reasons for your answers.
- $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$
 - $\mathbf{u} \times (\mathbf{v} \cdot \mathbf{w})$
 - $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$
 - $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w})$
32. **Cross products of three vectors** Show that except in degenerate cases, $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ lies in the plane of $\mathbf{u}$ and $\mathbf{v}$, whereas $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ lies in the plane of $\mathbf{v}$ and $\mathbf{w}$. What *are* the degenerate cases?
33. **Cancellation in cross products** If $\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \mathbf{w}$ and $\mathbf{u} \neq \mathbf{0}$, then does $\mathbf{v} = \mathbf{w}$? Give reasons for your answer.
34. **Double cancellation** If $\mathbf{u} \neq \mathbf{0}$ and if $\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \mathbf{w}$ and $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$, then does $\mathbf{v} = \mathbf{w}$? Give reasons for your answer.

Area of a Parallelogram

Find the areas of the parallelograms whose vertices are given in Exercises 35–40.

- $A(1, 0)$, $B(0, 1)$, $C(-1, 0)$, $D(0, -1)$
- $A(0, 0)$, $B(7, 3)$, $C(9, 8)$, $D(2, 5)$
- $A(-1, 2)$, $B(2, 0)$, $C(7, 1)$, $D(4, 3)$
- $A(-6, 0)$, $B(1, -4)$, $C(3, 1)$, $D(-4, 5)$
- $A(0, 0, 0)$, $B(3, 2, 4)$, $C(5, 1, 4)$, $D(2, -1, 0)$
- $A(1, 0, -1)$, $B(1, 7, 2)$, $C(2, 4, -1)$, $D(0, 3, 2)$

Area of a Triangle

Find the areas of the triangles whose vertices are given in Exercises 41–47.

- $A(0, 0)$, $B(-2, 3)$, $C(3, 1)$
- $A(-1, -1)$, $B(3, 3)$, $C(2, 1)$
- $A(-5, 3)$, $B(1, -2)$, $C(6, -2)$
- $A(-6, 0)$, $B(10, -5)$, $C(-2, 4)$
- $A(1, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, -1)$
- $A(0, 0, 0)$, $B(-1, 1, -1)$, $C(3, 0, 3)$
- $A(1, -1, 1)$, $B(0, 1, 1)$, $C(1, 0, -1)$
- Find the volume of a parallelepiped with one of its eight vertices at $A(0, 0, 0)$ and three adjacent vertices at $B(1, 2, 0)$, $C(0, -3, 2)$, and $D(3, -4, 5)$.
- Triangle area** Find a 2×2 determinant formula for the area of the triangle in the xy -plane with vertices at $(0, 0)$, (a_1, a_2) , and (b_1, b_2) . Explain your work.
- Triangle area** Find a concise 3×3 determinant formula that gives the area of a triangle in the xy -plane having vertices (a_1, a_2) , (b_1, b_2) , and (c_1, c_2) .

Volume of a Tetrahedron

Using the methods of Section 6.1, where volume is computed by integrating cross-sectional area, it can be shown that the volume of a tetrahedron formed by three vectors is equal to $\frac{1}{6}$ the volume of the parallelepiped formed by the three vectors. Find the volumes of the tetrahedra whose vertices are given in Exercises 51–54.

- $A(0, 0, 0)$, $B(2, 0, 0)$, $C(0, 3, 0)$, $D(0, 0, 4)$
- $A(0, 0, 0)$, $B(1, 0, 2)$, $C(0, 2, 1)$, $D(3, 4, 0)$
- $A(1, -1, 0)$, $B(0, 2, -2)$, $C(-3, 0, 3)$, $D(0, 4, 4)$
- $A(-1, 2, 3)$, $B(2, 0, 1)$, $C(1, -3, 2)$, $D(-2, 1, -1)$

In Exercises 55–57, determine whether the given points are coplanar.

- $A(1, 1, 1)$, $B(-1, 0, 4)$, $C(0, 2, 1)$, $D(2, -2, 3)$
- $A(0, 0, 4)$, $B(6, 2, 0)$, $C(2, -1, 1)$, $D(-3, -4, 3)$
- $A(0, 1, 2)$, $B(-1, 1, 0)$, $C(2, 0, -1)$, $D(1, -1, 1)$

11.5 Lines and Planes in Space

This section shows how to use scalar and vector products to write equations for lines, line segments, and planes in space. We will use these representations throughout the rest of the text in studying the calculus of curves and surfaces in space.

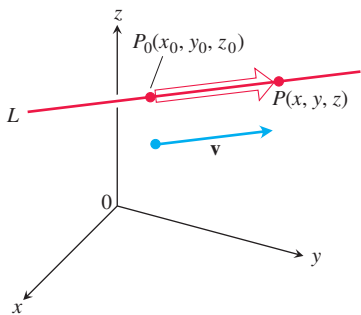


FIGURE 11.37 A point P lies on L through P_0 parallel to $\mathbf{v}$ if and only if $\overrightarrow{P_0P}$ is a scalar multiple of $\mathbf{v}$.

Lines and Line Segments in Space

In the plane, a line is determined by a point and a number giving the slope of the line. In space a line is determined by a point and a *vector* giving the direction of the line.

Suppose that L is a line in space passing through a point $P_0(x_0, y_0, z_0)$ parallel to a vector $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$. Then L is the set of all points $P(x, y, z)$ for which $\overrightarrow{P_0P}$ is parallel to $\mathbf{v}$ (Figure 11.37). Thus, $\overrightarrow{P_0P} = t\mathbf{v}$ for some scalar parameter t . The value of t depends on the location of the point P along the line, and the domain of t is $(-\infty, \infty)$. The expanded form of the equation $\overrightarrow{P_0P} = t\mathbf{v}$ is

$$(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k} = t(v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}),$$

which can be rewritten as

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k} + t(v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}). \quad (1)$$

If $\mathbf{r}(t)$ is the position vector of a point $P(x, y, z)$ on the line and $\mathbf{r}_0$ is the position vector of the point $P_0(x_0, y_0, z_0)$, then Equation (1) gives the following vector form for the equation of a line in space.

Vector Equation for a Line

A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\mathbf{v}$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \quad -\infty < t < \infty, \quad (2)$$

where $\mathbf{r}$ is the position vector of a point $P(x, y, z)$ on L and $\mathbf{r}_0$ is the position vector of $P_0(x_0, y_0, z_0)$.

Equating the corresponding components of the two sides of Equation (1) gives three scalar equations involving the parameter t :

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3.$$

These equations give us the standard parametrization of the line for the parameter interval $-\infty < t < \infty$.

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to a nonzero vector $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1, \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad -\infty < t < \infty. \quad (3)$$

EXAMPLE 1 Find parametric equations for the line through $(-2, 0, 4)$ parallel to $\mathbf{v} = 2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ (Figure 11.38).

Solution With $P_0(x_0, y_0, z_0)$ equal to $(-2, 0, 4)$ and $v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ equal to $2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, Equations (3) become

$$x = -2 + 2t, \quad y = 4t, \quad z = 4 - 2t. \quad \blacksquare$$

EXAMPLE 2 Find parametric equations for the line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$.

Solution The vector

$$\overrightarrow{PQ} = (1 - (-3))\mathbf{i} + (-1 - 2)\mathbf{j} + (4 - (-3))\mathbf{k} = 4\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$$

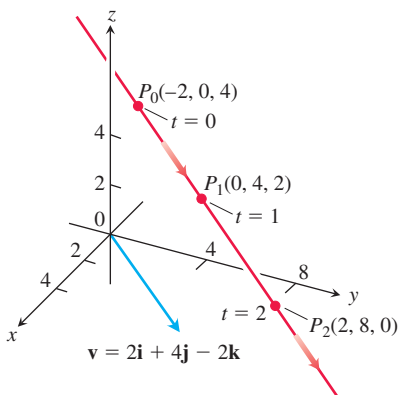


FIGURE 11.38 Selected points and parameter values on the line in Example 1. The arrows show the direction of increasing t .

is parallel to the line, and Equations (3) with $(x_0, y_0, z_0) = (-3, 2, -3)$ give

$$x = -3 + 4t, \quad y = 2 - 3t, \quad z = -3 + 7t.$$

We could have chosen $Q(1, -1, 4)$ as the “base point” and written

$$x = 1 + 4t, \quad y = -1 - 3t, \quad z = 4 + 7t.$$

These equations serve as well as the first; they simply place you at a different point on the line for a given value of t . ■

Notice that parametrizations are not unique. Not only can the “base point” change, but so can the parameter. The equations $x = -3 + 4t^3$, $y = 2 - 3t^3$, and $z = -3 + 7t^3$ also parametrize the line in Example 2.

To parametrize a line segment joining two points, we first parametrize the line through the points. We then find the t -values for the endpoints and restrict t to lie in the closed interval bounded by these values. The line equations, together with this added restriction, parametrize the segment.

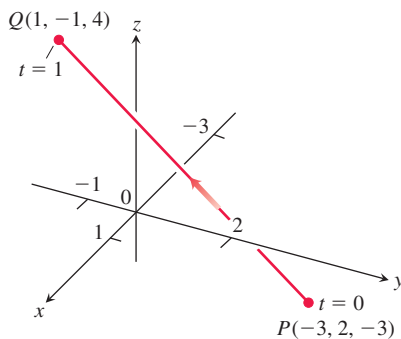


FIGURE 11.39 Example 3 derives a parametrization of line segment PQ . The arrow shows the direction of increasing t .

EXAMPLE 3 Parametrize the line segment joining the points $P(-3, 2, -3)$ and $Q(1, -1, 4)$ (Figure 11.39).

Solution We begin with equations for the line through P and Q , taking them, in this case, from Example 2:

$$x = -3 + 4t, \quad y = 2 - 3t, \quad z = -3 + 7t.$$

We observe that the point

$$(x, y, z) = (-3 + 4t, 2 - 3t, -3 + 7t)$$

on the line passes through $P(-3, 2, -3)$ at $t = 0$ and $Q(1, -1, 4)$ at $t = 1$. We add the restriction $0 \leq t \leq 1$ to parametrize the segment:

$$x = -3 + 4t, \quad y = 2 - 3t, \quad z = -3 + 7t, \quad 0 \leq t \leq 1. \quad \text{■}$$

The vector form (Equation (2)) for a line in space is more revealing if we think of a line as the path of a particle starting at position $P_0(x_0, y_0, z_0)$ and moving in the direction of vector $\mathbf{v}$. Rewriting Equation (2), we have

$$\begin{aligned} \mathbf{r}(t) &= \mathbf{r}_0 + t\mathbf{v} \\ &= \mathbf{r}_0 + \underbrace{t}_{\text{Time}} \underbrace{|\mathbf{v}|}_{\text{Speed}} \underbrace{\frac{\mathbf{v}}{|\mathbf{v}|}}_{\text{Direction}}. \end{aligned} \quad (4)$$

Initial position
Time
Speed
Direction

In other words, the position of the particle at time t is its initial position plus its distance moved (speed $\times$ time) in the direction $\mathbf{v}/|\mathbf{v}|$ of its straight-line motion.

EXAMPLE 4 A helicopter is to fly directly from a helipad at the origin in the direction of the point $(1, 1, 1)$ at a speed of 60 m/s. What is the position of the helicopter after 10 s?

Solution We place the origin at the starting position (helipad) of the helicopter. Then the unit vector

$$\mathbf{u} = \frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$$

gives the flight direction of the helicopter. From Equation (4), the position of the helicopter at any time t is

$$\begin{aligned}\mathbf{r}(t) &= \mathbf{r}_0 + t(\text{speed})\mathbf{u} \\ &= \mathbf{0} + t(60)\left(\frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}\right) \\ &= 20\sqrt{3}t(\mathbf{i} + \mathbf{j} + \mathbf{k}).\end{aligned}$$

When $t = 10$ s,

$$\begin{aligned}\mathbf{r}(10) &= 200\sqrt{3}(\mathbf{i} + \mathbf{j} + \mathbf{k}) \\ &= \langle 200\sqrt{3}, 200\sqrt{3}, 200\sqrt{3} \rangle.\end{aligned}$$

After 10 s of flight from the origin toward $(1, 1, 1)$, the helicopter is located at the point $(200\sqrt{3}, 200\sqrt{3}, 200\sqrt{3})$ in space. It has traveled a distance of $(60 \text{ m/s})(10 \text{ s}) = 600 \text{ m}$, which is the length of the vector $\mathbf{r}(10)$. ■

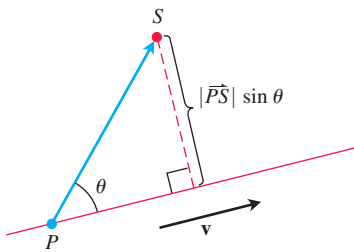


FIGURE 11.40 The distance from S to the line through P parallel to $\mathbf{v}$ is $|\overrightarrow{PS}|\sin\theta$, where θ is the angle between $\overrightarrow{PS}$ and $\mathbf{v}$.

The Distance from a Point to a Line in Space

To find the distance from a point S to a line that passes through a point P parallel to a vector $\mathbf{v}$, we find the absolute value of the scalar component of $\overrightarrow{PS}$ in the direction of a vector normal to the line (Figure 11.40). In the notation of the figure, the absolute value of the scalar component is $|\overrightarrow{PS}|\sin\theta$, which is $\frac{|\overrightarrow{PS}||\mathbf{v}|\sin\theta}{|\mathbf{v}|} = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$.

Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|} \quad (5)$$

EXAMPLE 5 Find the distance from the point $S(1, 1, 5)$ to the line

$$L: \quad x = 1 + t, \quad y = 3 - t, \quad z = 2t.$$

Solution We see from the equations for L that L passes through $P(1, 3, 0)$ parallel to $\mathbf{v} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$. With

$$\overrightarrow{PS} = (1 - 1)\mathbf{i} + (1 - 3)\mathbf{j} + (5 - 0)\mathbf{k} = -2\mathbf{j} + 5\mathbf{k}$$

and

$$\overrightarrow{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = \mathbf{i} + 5\mathbf{j} + 2\mathbf{k},$$

Equation (5) gives

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|} = \frac{\sqrt{1 + 25 + 4}}{\sqrt{1 + 1 + 4}} = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}. \quad \blacksquare$$

An Equation for a Plane in Space

A plane in space is determined by knowing a point on the plane and its “tilt” or orientation. This “tilt” is defined by specifying a vector that is perpendicular, or normal, to the plane.

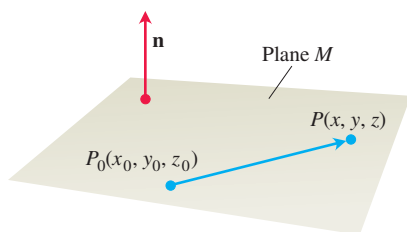


FIGURE 11.41 The standard equation for a plane in space is defined in terms of a vector normal to the plane: A point P lies in the plane through P_0 normal to $\mathbf{n}$ if and only if $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$.

Suppose that plane M passes through a point $P_0(x_0, y_0, z_0)$ and is normal to the nonzero vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$. A vector from P_0 to any point P on the plane is orthogonal to $\mathbf{n}$. Then M is the set of all points $P(x, y, z)$ for which $\overrightarrow{P_0P}$ is orthogonal to $\mathbf{n}$ (Figure 11.41). Thus, the dot product $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$. This equation is equivalent to

$$(A\mathbf{i} + B\mathbf{j} + C\mathbf{k}) \cdot [(x - x_0)\mathbf{i} + (y - y_0)\mathbf{j} + (z - z_0)\mathbf{k}] = 0,$$

so the plane M consists of the points (x, y, z) satisfying

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0.$$

Equation for a Plane

The plane through $P_0(x_0, y_0, z_0)$ normal to a nonzero vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ has

Vector equation: $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$

Component equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Component equation simplified: $Ax + By + Cz = D$, where
 $D = Ax_0 + By_0 + Cz_0$

EXAMPLE 6 Find an equation for the plane through $P_0(-3, 0, 7)$ perpendicular to $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

Solution The component equation is

$$5(x - (-3)) + 2(y - 0) + (-1)(z - 7) = 0.$$

Simplifying, we obtain

$$\begin{aligned} 5x + 15 + 2y - z + 7 &= 0 \\ 5x + 2y - z &= -22. \end{aligned}$$

Notice in Example 6 how the components of $\mathbf{n} = 5\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ became the coefficients of x , y , and z in the equation $5x + 2y - z = -22$. The vector $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is normal to the plane $Ax + By + Cz = D$.

EXAMPLE 7 Find an equation for the plane through $A(0, 0, 1)$, $B(2, 0, 0)$, and $C(0, 3, 0)$.

Solution We find a vector normal to the plane and use it with one of the points (it does not matter which) to write an equation for the plane.

The cross product

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -1 \\ 0 & 3 & -1 \end{vmatrix} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$$

is normal to the plane. We substitute the components of this vector and the coordinates of $A(0, 0, 1)$ into the component form of the equation to obtain

$$\begin{aligned} 3(x - 0) + 2(y - 0) + 6(z - 1) &= 0 \\ 3x + 2y + 6z &= 6. \end{aligned}$$

Lines of Intersection

Just as lines are parallel if and only if they have the same direction, two planes are **parallel** if and only if their normals are parallel, or $\mathbf{n}_1 = k\mathbf{n}_2$ for some scalar k . Two planes that are not parallel intersect in a line.

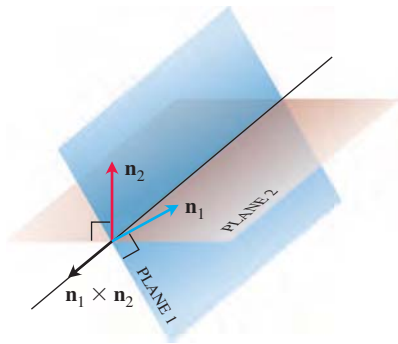


FIGURE 11.42 How the line of intersection of two planes is related to the planes' normal vectors (Example 8).

EXAMPLE 8 Find a vector parallel to the line of intersection of the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution The line of intersection of two planes is perpendicular to both planes' normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ (Figure 11.42) and therefore parallel to $\mathbf{n}_1 \times \mathbf{n}_2$. Turning this around, $\mathbf{n}_1 \times \mathbf{n}_2$ is a vector parallel to the planes' line of intersection. In our case,

$$\mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -6 & -2 \\ 2 & 1 & -2 \end{vmatrix} = 14\mathbf{i} + 2\mathbf{j} + 15\mathbf{k}.$$

Any nonzero scalar multiple of $\mathbf{n}_1 \times \mathbf{n}_2$ will do as well. ■

EXAMPLE 9 Find parametric equations for the line in which the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$ intersect.

Solution We find a vector parallel to the line and a point on the line and use Equations (3).

Example 8 identifies $\mathbf{v} = 14\mathbf{i} + 2\mathbf{j} + 15\mathbf{k}$ as a vector parallel to the line. To find a point on the line, we can take any point common to the two planes. Substituting $z = 0$ in the plane equations and solving for x and y simultaneously identifies one of these points as $(3, -1, 0)$. The line is

$$x = 3 + 14t, \quad y = -1 + 2t, \quad z = 15t.$$

The choice $z = 0$ is arbitrary, and we could have chosen $z = 1$ or $z = -1$ just as well. Or we could have let $x = 0$ and solved for y and z . The different choices would simply give different parametrizations of the same line. ■

Sometimes we want to know where a line and a plane intersect. For example, if we are looking at a flat plate and a line segment passes through it, we may be interested in knowing what portion of the line segment is hidden from our view by the plate. This application is used in computer graphics (Exercise 78).

EXAMPLE 10 Find the point where the line

$$x = \frac{8}{3} + 2t, \quad y = -2t, \quad z = 1 + t$$

intersects the plane $3x + 2y + 6z = 6$.

Solution The point

$$\left(\frac{8}{3} + 2t, -2t, 1 + t\right)$$

lies in the plane if its coordinates satisfy the equation of the plane—that is, if

$$\begin{aligned} 3\left(\frac{8}{3} + 2t\right) + 2(-2t) + 6(1 + t) &= 6 \\ 8 + 6t - 4t + 6 + 6t &= 6 \\ 8t &= -8 \\ t &= -1. \end{aligned}$$

The point of intersection is

$$(x, y, z)|_{t=-1} = \left(\frac{8}{3} - 2, 2, 1 - 1\right) = \left(\frac{2}{3}, 2, 0\right). \quad \blacksquare$$

The Distance from a Point to a Plane

If P is a point on a plane with a normal $\mathbf{n}$, then the distance from any point S to the plane is the length of the vector projection of $\overrightarrow{PS}$ onto $\mathbf{n}$, as given in the following formula.

Distance from a Point S to a Plane Through a Point P with a Normal $\mathbf{n}$

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| \quad (6)$$

EXAMPLE 11 Find the distance from $S(1, 1, 3)$ to the plane $3x + 2y + 6z = 6$.

Solution We find a point P in the plane and calculate the length of the vector projection of $\overrightarrow{PS}$ onto a vector $\mathbf{n}$ normal to the plane (Figure 11.43). The coefficients in the equation $3x + 2y + 6z = 6$ give

$$\mathbf{n} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}.$$

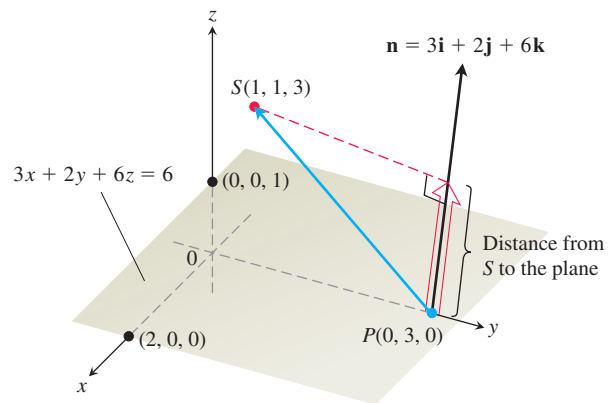


FIGURE 11.43 The distance from S to the plane is the length of the vector projection of $\overrightarrow{PS}$ onto $\mathbf{n}$ (Example 11).

The points on the plane easiest to find from the plane's equation are the intercepts. If we take P to be the y -intercept $(0, 3, 0)$, then

$$\overrightarrow{PS} = (1 - 0)\mathbf{i} + (1 - 3)\mathbf{j} + (3 - 0)\mathbf{k} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k},$$

$$|\mathbf{n}| = \sqrt{(3)^2 + (2)^2 + (6)^2} = \sqrt{49} = 7.$$

Therefore, the distance from S to the plane is

$$\begin{aligned} d &= \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| && \text{Length of } \text{proj}_{\mathbf{n}} \overrightarrow{PS} \\ &= \left| (\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) \cdot \left(\frac{3}{7}\mathbf{i} + \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k} \right) \right| \\ &= \left| \frac{3}{7} - \frac{4}{7} + \frac{18}{7} \right| = \frac{17}{7}. \end{aligned}$$

Angles Between Planes

The angle between two intersecting planes is defined to be the acute angle between their normal vectors (Figure 11.44).

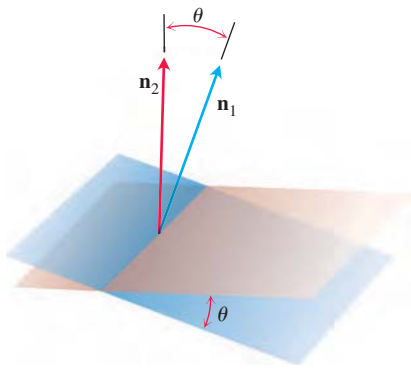


FIGURE 11.44 The angle between two planes is obtained from the angle between their normals.

EXAMPLE 11 Find the angle between the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Solution The vectors

$$\mathbf{n}_1 = 3\mathbf{i} - 6\mathbf{j} - 2\mathbf{k}, \quad \mathbf{n}_2 = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$$

are normals to the planes. The angle between them is

$$\begin{aligned} \theta &= \arccos \left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} \right) \\ &= \arccos \left(\frac{4}{21} \right) \approx 1.38 \text{ radians.} \quad \text{About 79 degrees} \end{aligned}$$

EXERCISES 11.5

Lines and Line Segments

Find parametric equations for the lines in Exercises 1–12.

- The line through the point $P(3, -4, -1)$ parallel to the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$
- The line through $P(1, 2, -1)$ and $Q(-1, 0, 1)$
- The line through $P(-2, 0, 3)$ and $Q(3, 5, -2)$
- The line through $P(1, 2, 0)$ and $Q(1, 1, -1)$
- The line through the origin parallel to the vector $2\mathbf{j} + \mathbf{k}$
- The line through the point $(3, -2, 1)$ parallel to the line $x = 1 + 2t$, $y = 2 - t$, $z = 3t$
- The line through $(1, 1, 1)$ parallel to the z -axis
- The line through $(2, 4, 5)$ perpendicular to the plane $3x + 7y - 5z = 21$
- The line through $(0, -7, 0)$ perpendicular to the plane $x + 2y + 2z = 13$
- The line through $(2, 3, 0)$ perpendicular to the vectors $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$
- The x -axis
- The z -axis

Find parametrizations for the line segments joining the points in Exercises 13–20. Draw coordinate axes and sketch each segment, indicating the direction of increasing t for your parametrization.

- $(0, 0, 0)$, $(1, 1, 3/2)$
- $(0, 0, 0)$, $(1, 0, 0)$
- $(1, 0, 0)$, $(1, 1, 0)$
- $(1, 1, 0)$, $(1, 1, 1)$
- $(0, 1, 1)$, $(0, -1, 1)$
- $(0, 2, 0)$, $(3, 0, 0)$
- $(2, 0, 2)$, $(0, 2, 0)$
- $(1, 0, -1)$, $(0, 3, 0)$

Planes

Find equations for the planes in Exercises 21–26.

- The plane through $P_0(0, 2, -1)$ normal to $\mathbf{n} = 3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$
- The plane through $(1, -1, 3)$ parallel to the plane $3x + y + z = 7$
- The plane through $(1, 1, -1)$, $(2, 0, 2)$, and $(0, -2, 1)$
- The plane through $(2, 4, 5)$, $(1, 5, 7)$, and $(-1, 6, 8)$

- The plane through $P_0(2, 4, 5)$ perpendicular to the line

$$x = 5 + t, \quad y = 1 + 3t, \quad z = 4t$$

- The plane through $A(1, -2, 1)$ perpendicular to the vector from the origin to A .
- Find the point of intersection of the lines $x = 2t + 1$, $y = 3t + 2$, $z = 4t + 3$, and $x = s + 2$, $y = 2s + 4$, $z = -4s - 1$, and then find the plane determined by these lines.
- Find the point of intersection of the lines $x = t$, $y = -t + 2$, $z = t + 1$, and $x = 2s + 2$, $y = s + 3$, $z = 5s + 6$, and then find the plane determined by these lines.

In Exercises 29 and 30, find the plane containing the intersecting lines.

- $L1: x = -1 + t$, $y = 2 + t$, $z = 1 - t$; $-\infty < t < \infty$
 $L2: x = 1 - 4s$, $y = 1 + 2s$, $z = 2 - 2s$; $-\infty < s < \infty$
- $L1: x = t$, $y = 3 - 3t$, $z = -2 - t$; $-\infty < t < \infty$
 $L2: x = 1 + s$, $y = 4 + s$, $z = -1 + s$; $-\infty < s < \infty$
- Find a plane through $P_0(2, 1, -1)$ and perpendicular to the line of intersection of the planes $2x + y - z = 3$, $x + 2y + z = 2$.
- Find a plane through the points $P_1(1, 2, 3)$, and $P_2(3, 2, 1)$ and perpendicular to the plane $4x - y + 2z = 7$.

Distances

In Exercises 33–38, find the distance from the point to the line.

- $(0, 0, 12)$; $x = 4t$, $y = -2t$, $z = 2t$
- $(0, 0, 0)$; $x = 5 + 3t$, $y = 5 + 4t$, $z = -3 - 5t$
- $(2, 1, 3)$; $x = 2 + 2t$, $y = 1 + 6t$, $z = 3$
- $(2, 1, -1)$; $x = 2t$, $y = 1 + 2t$, $z = 2t$
- $(3, -1, 4)$; $x = 4 - t$, $y = 3 + 2t$, $z = -5 + 3t$
- $(-1, 4, 3)$; $x = 10 + 4t$, $y = -3$, $z = 4t$

In Exercises 39–44, find the distance from the point to the plane.

- $(2, -3, 4)$, $x + 2y + 2z = 13$
- $(0, 0, 0)$, $3x + 2y + 6z = 6$
- $(0, 1, 1)$, $4y + 3z = -12$

42. $(2, 2, 3)$, $2x + y + 2z = 4$
 43. $(0, -1, 0)$, $2x + y + 2z = 4$
 44. $(1, 0, -1)$, $-4x + y + z = 4$
 45. Find the distance from the plane $x + 2y + 6z = 1$ to the plane $x + 2y + 6z = 10$.
 46. Find the distance from the line $x = 2 + t$, $y = 1 + t$, $z = -(1/2) - (1/2)t$ to the plane $x + 2y + 6z = 10$.

Angles

In Exercises 47 and 48, find the angles between the planes.

47. $x + y = 1$, $2x + y - 2z = 2$
 48. $5x + y - z = 10$, $x - 2y + 3z = -1$

In Exercises 49 and 50, find the acute angles between the intersecting lines.

49. $x = t$, $y = 2t$, $z = -t$ and $x = 1 - t$, $y = 5 + t$, $z = 2t$
 50. $x = 2 + t$, $y = 4t + 2$, $z = 1 + t$ and $x = 3t - 2$, $y = -2$, $z = 2 - 2t$

In Exercises 51 and 52, find the acute angles between the lines and planes.

51. $x = 1 - t$, $y = 3t$, $z = 1 + t$; $2x - y + 3z = 6$
 52. $x = 2$, $y = 3 + 2t$, $z = 1 - 2t$; $x - y + z = 0$

T Use a calculator to find the acute angles between the planes in Exercises 53–56 to the nearest hundredth of a radian.

53. $2x + 2y + 2z = 3$, $2x - 2y - z = 5$
 54. $x + y + z = 1$, $z = 0$ (the xy -plane)
 55. $2x + 2y - z = 3$, $x + 2y + z = 2$
 56. $4y + 3z = -12$, $3x + 2y + 6z = 6$

Intersecting Lines and Planes

In Exercises 57–60, find the point in which the line meets the plane.

57. $x = 1 - t$, $y = 3t$, $z = 1 + t$; $2x - y + 3z = 6$
 58. $x = 2$, $y = 3 + 2t$, $z = -2 - 2t$; $6x + 3y - 4z = -12$
 59. $x = 1 + 2t$, $y = 1 + 5t$, $z = 3t$; $x + y + z = 2$
 60. $x = -1 + 3t$, $y = -2$, $z = 5t$; $2x - 3z = 7$

Find parametrizations for the lines in which the planes in Exercises 61–64 intersect.

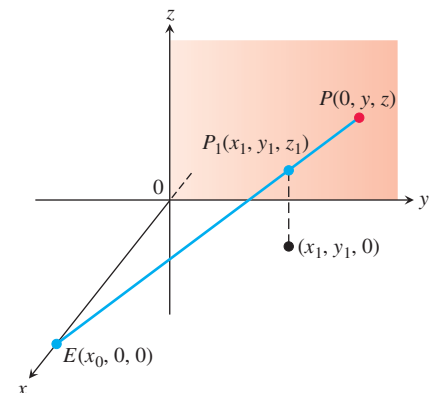
61. $x + y + z = 1$, $x + y = 2$
 62. $3x - 6y - 2z = 3$, $2x + y - 2z = 2$
 63. $x - 2y + 4z = 2$, $x + y - 2z = 5$
 64. $5x - 2y = 11$, $4y - 5z = -17$

Given two lines in space, either they are parallel, they intersect, or they are skew (lie in parallel planes). In Exercises 65 and 66, determine whether the lines, taken two at a time, are parallel, intersect, or are skew. If they intersect, find the point of intersection. Otherwise, find the distance between the two lines.

65. $L1: x = 3 + 2t$, $y = -1 + 4t$, $z = 2 - t$; $-\infty < t < \infty$
 $L2: x = 1 + 4s$, $y = 1 + 2s$, $z = -3 + 4s$; $-\infty < s < \infty$
 $L3: x = 3 + 2r$, $y = 2 + r$, $z = -2 + 2r$; $-\infty < r < \infty$
 66. $L1: x = 1 + 2t$, $y = -1 - t$, $z = 3t$; $-\infty < t < \infty$
 $L2: x = 2 - s$, $y = 3s$, $z = 1 + s$; $-\infty < s < \infty$
 $L3: x = 5 + 2r$, $y = 1 - r$, $z = 8 + 3r$; $-\infty < r < \infty$

Theory and Examples

67. Use Equations (3) to generate a parametrization of the line through $P(2, -4, 7)$ parallel to $\mathbf{v}_1 = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. Then generate another parametrization of the line using the point $P_2(-2, -2, 1)$ and the vector $\mathbf{v}_2 = -\mathbf{i} + (1/2)\mathbf{j} - (3/2)\mathbf{k}$.
 68. Use the component form to generate an equation for the plane through $P_1(4, 1, 5)$ normal to $\mathbf{n}_1 = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$. Then generate another equation for the same plane using the point $P_2(3, -2, 0)$ and the normal vector $\mathbf{n}_2 = -\sqrt{2}\mathbf{i} + 2\sqrt{2}\mathbf{j} - \sqrt{2}\mathbf{k}$.
 69. Find the points in which the line $x = 1 + 2t$, $y = -1 - t$, $z = 3t$ meets the coordinate planes. Describe the reasoning behind your answer.
 70. Find equations for the line in the plane $z = 3$ that makes an angle of $\pi/6$ rad with $\mathbf{i}$ and an angle of $\pi/3$ rad with $\mathbf{j}$. Describe the reasoning behind your answer.
 71. Is the line $x = 1 - 2t$, $y = 2 + 5t$, $z = -3t$ parallel to the plane $2x + y - z = 8$? Give reasons for your answer.
 72. How can you tell when two planes $A_1x + B_1y + C_1z = D_1$ and $A_2x + B_2y + C_2z = D_2$ are parallel? Perpendicular? Give reasons for your answer.
 73. Find two different planes whose intersection is the line $x = 1 + t$, $y = 2 - t$, $z = 3 + 2t$. Write equations for each plane in the form $Ax + By + Cz = D$.
 74. Find a plane through the origin that is perpendicular to the plane $M: 2x + 3y + z = 12$ in a right angle. How do you know that your plane is perpendicular to M ?
 75. The graph of $(x/a) + (y/b) + (z/c) = 1$ is a plane for any non-zero numbers a, b , and c . Which planes have an equation of this form?
 76. Suppose L_1 and L_2 are disjoint (nonintersecting) nonparallel lines. Is it possible for a nonzero vector to be perpendicular to both L_1 and L_2 ? Give reasons for your answer.
 77. **Perspective in computer graphics** In computer graphics and perspective drawing, we need to represent objects seen by the eye in space as images on a two-dimensional plane. Suppose that the eye is at $E(x_0, 0, 0)$ as shown here and that we want to represent a point $P_1(x_1, y_1, z_1)$ as a point on the yz -plane. We do this by projecting P_1 onto the plane with a ray from E . The point P_1 will be portrayed as the point $P(0, y, z)$. The problem for us as graphics designers is to find y and z given E and P_1 .
 a. Write a vector equation that holds between $\overrightarrow{EP}$ and $\overrightarrow{EP_1}$. Use the equation to express y and z in terms of x_0 , x_1 , y_1 , and z_1 .
 b. Test the formulas obtained for y and z in part (a) by investigating their behavior at $x_1 = 0$ and $x_1 = x_0$ and by seeing what happens as $x_0 \rightarrow \infty$. What do you find?



78. Hidden lines in computer graphics Here is another typical problem in computer graphics. Your eye is at $(4, 0, 0)$. You are looking at a triangular plate whose vertices are at $(1, 0, 1)$, $(1, 1, 0)$, and $(-2, 2, 2)$. The line segment from $(1, 0, 0)$ to $(0, 2, 2)$

passes through the plate. What portion of the line segment is hidden from your view by the plate? (This is an exercise in finding intersections of lines and planes.)

11.6 Cylinders and Quadric Surfaces

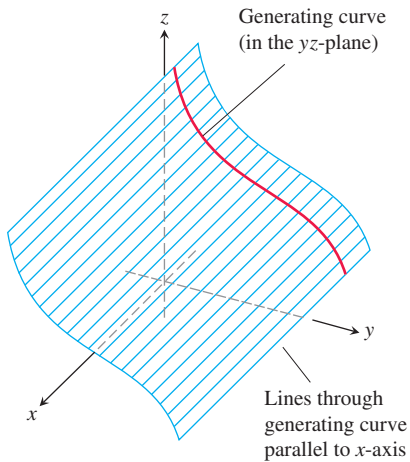


FIGURE 11.45 A cylinder and generating curve.

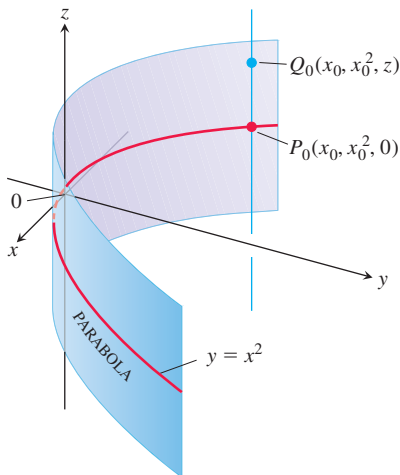


FIGURE 11.46 Every point of the cylinder in Example 1 has coordinates of the form (x_0, x_0^2, z) .

Up to now, we have studied two special types of surfaces: spheres and planes. In this section, we extend our inventory to include a variety of cylinders and quadric surfaces. Quadric surfaces are surfaces defined by second-degree equations in x , y , and z . Spheres are quadric surfaces, but there are others of equal interest that will be needed in Chapters 13–15.

Cylinders

Suppose we are given a plane in space that contains a curve, and in addition we are given a line that is not parallel to this plane. A **cylinder** is a surface that is generated by moving a line that is parallel to the given line along the curve, while keeping it parallel to the given line. The curve is called a **generating curve** for the cylinder (Figure 11.45 illustrates this when the given plane is the yz -plane and the given line is the x -axis). In solid geometry, where *cylinder* means *circular cylinder*, the generating curves are circles, but now we allow generating curves of any kind. The cylinder in our first example is generated by a parabola.

EXAMPLE 1 Find an equation for the cylinder made by the lines parallel to the z -axis that pass through the parabola $y = x^2$, $z = 0$ (Figure 11.46).

Solution The point $P_0(x_0, x_0^2, 0)$ lies on the parabola $y = x^2$ in the xy -plane. Then, for any value of z , the point $Q(x_0, x_0^2, z)$ lies on the cylinder because it lies on the line $x = x_0$, $y = x_0^2$ through P_0 parallel to the z -axis. Conversely, any point $Q(x_0, x_0^2, z)$ whose y -coordinate is the square of its x -coordinate lies on the cylinder because it lies on the line $x = x_0$, $y = x_0^2$ through P_0 parallel to the z -axis (Figure 11.46).

Regardless of the value of z , therefore, the points on the surface are the points whose coordinates satisfy the equation $y = x^2$. This makes $y = x^2$ an equation for the cylinder. ■

As Example 1 suggests, any curve $f(x, y) = c$ in the xy -plane generates a cylinder parallel to the z -axis whose equation is also $f(x, y) = c$. For instance, the equation $x^2 + y^2 = 1$ corresponds to the circular cylinder made by the lines parallel to the z -axis that pass through the circle $x^2 + y^2 = 1$ in the xy -plane.

In a similar way, any curve $g(x, z) = c$ in the xz -plane generates a cylinder parallel to the y -axis whose space equation is also $g(x, z) = c$. Any curve $h(y, z) = c$ generates a cylinder parallel to the x -axis whose space equation is also $h(y, z) = c$. The axis of a cylinder need not be parallel to a coordinate axis, however.

Quadric Surfaces

A **quadric surface** is the graph in space of a second-degree equation in x , y , and z . We first focus on quadric surfaces given by the equation

$$Ax^2 + By^2 + Cz^2 + Dz = E,$$

where A , B , C , D , and E are constants. The basic quadric surfaces are **ellipsoids**, **paraboloids**, **elliptical cones**, and **hyperboloids**. Spheres are special cases of ellipsoids. We present a few examples illustrating how to sketch a quadric surface, and then we give a summary table of graphs of the basic types.

CHAPTER 11 Questions to Guide Your Review

- When do directed line segments in the plane represent the same vector?
- How are vectors added and subtracted geometrically? How are they added and subtracted algebraically?
- How do you find a vector's magnitude and direction?
- If a vector is multiplied by a positive scalar, how is the result related to the original vector? What if the scalar is zero? Negative?
- Define the *dot product* (*scalar product*) of two vectors. Which algebraic laws are satisfied by dot products? Give examples. When is the dot product of two vectors equal to zero?
- What geometric interpretation does the dot product have? Give examples.
- What is the vector projection of a vector $\mathbf{u}$ onto a vector $\mathbf{v}$? Give an example of a useful application of a vector projection.
- Define the *cross product* (*vector product*) of two vectors. Which algebraic laws are satisfied by cross products, and which are not? Give examples. When is the cross product of two vectors equal to zero?
- What geometric or physical interpretations do cross products have? Give examples.
- What is the determinant formula for calculating the cross product of two vectors relative to the Cartesian $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ -coordinate system? Use it in an example.
- How do you find equations for lines, line segments, and planes in space? Give examples. Can you express a line in space by a single equation? A plane?
- How do you find the distance from a point to a line in space? From a point to a plane? Give examples.
- What are box products? What significance do they have? How are they evaluated? Give an example.
- How do you find equations for spheres in space? Give examples.
- How do you find the intersection of two lines in space? A line and a plane? Two planes? Give examples.
- What is a cylinder? Give examples of equations that define cylinders in Cartesian coordinates.
- What are quadric surfaces? Give examples of different kinds of ellipsoids, paraboloids, cones, and hyperboloids (equations and sketches).

CHAPTER 11 Practice Exercises

Vector Calculations in Two Dimensions

In Exercises 1–4, let $\mathbf{u} = \langle -3, 4 \rangle$ and $\mathbf{v} = \langle 2, -5 \rangle$. Find (a) the component form of the vector and (b) its magnitude.

- $3\mathbf{u} - 4\mathbf{v}$
- $\mathbf{u} + \mathbf{v}$
- $-2\mathbf{u}$
- $5\mathbf{v}$

In Exercises 5–8, find the component form of the vector.

- The vector obtained by rotating $\langle 0, 1 \rangle$ through an angle of $2\pi/3$ radians
- The unit vector that makes an angle of $\pi/6$ radian with the positive x -axis
- The vector 2 units long in the direction $4\mathbf{i} - \mathbf{j}$
- The vector 5 units long in the direction opposite to the direction of $(3/5)\mathbf{i} + (4/5)\mathbf{j}$

Express the vectors in Exercises 9–12 in terms of their lengths and directions.

- $\sqrt{2}\mathbf{i} + \sqrt{2}\mathbf{j}$
- $-\mathbf{i} - \mathbf{j}$
- Velocity vector $\mathbf{v} = (-2 \sin t)\mathbf{i} + (2 \cos t)\mathbf{j}$ when $t = \pi/2$.
- Velocity vector $\mathbf{v} = (e^t \cos t - e^t \sin t)\mathbf{i} + (e^t \sin t + e^t \cos t)\mathbf{j}$ when $t = \ln 2$.

Vector Calculations in Three Dimensions

Express the vectors in Exercises 13 and 14 in terms of their lengths and directions.

- $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$
- $\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
- Find a vector 2 units long in the direction of $\mathbf{v} = 4\mathbf{i} - \mathbf{j} + 4\mathbf{k}$.
- Find a vector 5 units long in the direction opposite to the direction of $\mathbf{v} = (3/5)\mathbf{i} + (4/5)\mathbf{k}$.

In Exercises 17 and 18, find $|\mathbf{v}|$, $|\mathbf{u}|$, $\mathbf{v} \cdot \mathbf{u}$, $\mathbf{u} \cdot \mathbf{v}$, $\mathbf{v} \times \mathbf{u}$, $\mathbf{u} \times \mathbf{v}$, $|\mathbf{v} \times \mathbf{u}|$, the angle between $\mathbf{v}$ and $\mathbf{u}$, the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$, and the vector projection of $\mathbf{u}$ onto $\mathbf{v}$.

- $\mathbf{v} = \mathbf{i} + \mathbf{j}$
 $\mathbf{u} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$
- $\mathbf{v} = \mathbf{i} + \mathbf{j} + 2\mathbf{k}$
 $\mathbf{u} = -\mathbf{i} - \mathbf{k}$

In Exercises 19 and 20, find $\text{proj}_{\mathbf{v}} \mathbf{u}$.

- $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$
 $\mathbf{u} = \mathbf{i} + \mathbf{j} - 5\mathbf{k}$
- $\mathbf{u} = \mathbf{i} - 2\mathbf{j}$
 $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

In Exercises 21 and 22, draw coordinate axes and then sketch $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$ as vectors at the origin.

- $\mathbf{u} = \mathbf{i}$, $\mathbf{v} = \mathbf{i} + \mathbf{j}$
- $\mathbf{u} = \mathbf{i} - \mathbf{j}$, $\mathbf{v} = \mathbf{i} + \mathbf{j}$
- If $|\mathbf{v}| = 2$, $|\mathbf{w}| = 3$, and the angle between $\mathbf{v}$ and $\mathbf{w}$ is $\pi/3$, find $|\mathbf{v} - 2\mathbf{w}|$.
- For what value or values of a will the vectors $\mathbf{u} = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ and $\mathbf{v} = -4\mathbf{i} - 8\mathbf{j} + a\mathbf{k}$ be parallel?

In Exercises 25 and 26, find (a) the area of the parallelogram determined by vectors $\mathbf{u}$ and $\mathbf{v}$ and (b) the volume of the parallelepiped determined by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

25. $\mathbf{u} = \mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{w} = -\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

26. $\mathbf{u} = \mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{j}$, $\mathbf{w} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

Lines, Planes, and Distances

27. Suppose that $\mathbf{n}$ is normal to a plane and that $\mathbf{v}$ is parallel to the plane. Describe how you would find a vector $\mathbf{n}$ that is both perpendicular to $\mathbf{v}$ and parallel to the plane.

28. Find a vector in the plane parallel to the line $ax + by = c$.

In Exercises 29 and 30, find the distance from the point to the line.

29. $(2, 2, 0)$; $x = -t$, $y = t$, $z = -1 + t$

30. $(0, 4, 1)$; $x = 2 + t$, $y = 2 + t$, $z = t$

31. Parametrize the line that passes through the point $(1, 2, 3)$ parallel to the vector $\mathbf{v} = -3\mathbf{i} + 7\mathbf{k}$.

32. Parametrize the line segment joining the points $P(1, 2, 0)$ and $Q(1, 3, -1)$.

In Exercises 33 and 34, find the distance from the point to the plane.

33. $(6, 0, -6)$, $x - y = 4$

34. $(3, 0, 10)$, $2x + 3y + z = 2$

35. Find an equation for the plane that passes through the point $(3, -2, 1)$ normal to the vector $\mathbf{n} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$.

36. Find an equation for the plane that passes through the point $(-1, 6, 0)$ perpendicular to the line $x = -1 + t$, $y = 6 - 2t$, $z = 3t$.

In Exercises 37 and 38, find an equation for the plane through points P , Q , and R .

37. $P(1, -1, 2)$, $Q(2, 1, 3)$, $R(-1, 2, -1)$

38. $P(1, 0, 0)$, $Q(0, 1, 0)$, $R(0, 0, 1)$

39. Find the points in which the line $x = 1 + 2t$, $y = -1 - t$, $z = 3t$ meets the three coordinate planes.

40. Find the point in which the line through the origin perpendicular to the plane $2x - y - z = 4$ meets the plane $3x - 5y + 2z = 6$.

41. Find the acute angle between the planes $x = 7$ and $x + y + \sqrt{2}z = -3$.

42. Find the acute angle between the planes $x + y = 1$ and $y + z = 1$.

43. Find parametric equations for the line in which the planes $x + 2y + z = 1$ and $x - y + 2z = -8$ intersect.

44. Show that the line in which the planes

$$x + 2y - 2z = 5 \quad \text{and} \quad 5x - 2y - z = 0$$

intersect is parallel to the line

$$x = -3 + 2t, \quad y = 3t, \quad z = 1 + 4t.$$

45. The planes $3x + 6z = 1$ and $2x + 2y - z = 3$ intersect in a line.

a. Show that the planes are orthogonal.

b. Find equations for the line of intersection.

46. Find an equation for the plane that passes through the point $(1, 2, 3)$ parallel to $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

47. Is $\mathbf{v} = 2\mathbf{i} - 4\mathbf{j} + \mathbf{k}$ related in any special way to the plane $2x + y = 5$? Give reasons for your answer.

48. The equation $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$ represents the plane through P_0 normal to $\mathbf{n}$. What set does the inequality $\mathbf{n} \cdot \overrightarrow{P_0P} > 0$ represent?

49. Find the distance from the point $P(1, 4, 0)$ to the plane through $A(0, 0, 0)$, $B(2, 0, -1)$, and $C(2, -1, 0)$.

50. Find the distance from the point $(2, 2, 3)$ to the plane $2x + 3y + 5z = 0$.

51. Find a vector parallel to the plane $2x - y - z = 4$ and orthogonal to $\mathbf{i} + \mathbf{j} + \mathbf{k}$.

52. Find a unit vector orthogonal to $\mathbf{A}$ in the plane of $\mathbf{B}$ and $\mathbf{C}$ if $\mathbf{A} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{B} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, and $\mathbf{C} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

53. Find a vector of magnitude 2 parallel to the line of intersection of the planes $x + 2y + z - 1 = 0$ and $x - y + 2z + 7 = 0$.

54. Find the point in which the line through the origin perpendicular to the plane $2x - y - z = 4$ meets the plane $3x - 5y + 2z = 6$.

55. Find the point in which the line through $P(3, 2, 1)$ normal to the plane $2x - y + 2z = -2$ meets the plane.

56. What angle does the line of intersection of the planes $2x + y - z = 0$ and $x + y + 2z = 0$ make with the positive x -axis?

57. The line

$$L: x = 3 + 2t, \quad y = 2t, \quad z = t$$

intersects the plane $x + 3y - z = -4$ in a point P . Find the coordinates of P and find equations for the line in the plane through P perpendicular to L .

58. Show that for every real number k , the plane

$$x - 2y + z + 3 + k(2x - y - z + 1) = 0$$

contains the line of intersection of the planes

$$x - 2y + z + 3 = 0 \quad \text{and} \quad 2x - y - z + 1 = 0.$$

59. Find an equation for the plane through $A(-2, 0, -3)$ and $B(1, -2, 1)$ that lies parallel to the line through $C(-2, -13/5, 26/5)$ and $D(16/5, -13/5, 0)$.

60. Is the line $x = 1 + 2t$, $y = -2 + 3t$, $z = -5t$ related in any way to the plane $-4x - 6y + 10z = 9$? Give reasons for your answer.

61. Which of the following are equations for the plane through the points $P(1, 1, -1)$, $Q(3, 0, 2)$, and $R(-2, 1, 0)$?

a. $(2\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}) \cdot ((x + 2)\mathbf{i} + (y - 1)\mathbf{j} + z\mathbf{k}) = 0$

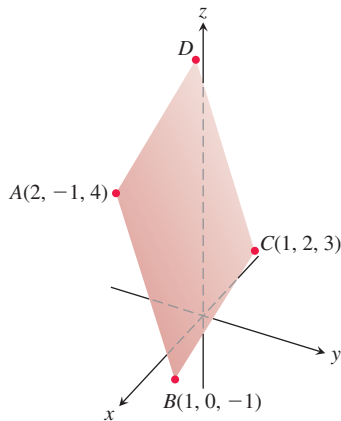
b. $x = 3 - t$, $y = -11t$, $z = 2 - 3t$

c. $(x + 2) + 11(y - 1) = 3z$

d. $(2\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}) \times ((x + 2)\mathbf{i} + (y - 1)\mathbf{j} + z\mathbf{k}) = \mathbf{0}$

e. $(2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) \times (-3\mathbf{i} + \mathbf{k}) \cdot ((x + 2)\mathbf{i} + (y - 1)\mathbf{j} + z\mathbf{k}) = 0$

62. The parallelogram shown here has vertices at $A(2, -1, 4)$, $B(1, 0, -1)$, $C(1, 2, 3)$, and D . Find



- the coordinates of D .
- the cosine of the interior angle at B .
- the vector projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.
- the area of the parallelogram.
- an equation for the plane of the parallelogram.
- the areas of the orthogonal projections of the parallelogram on the three coordinate planes.

63. **Distance between skew lines** Find the distance between the line L_1 through the points $A(1, 0, -1)$ and $B(-1, 1, 0)$ and the line L_2 through the points $C(3, 1, -1)$ and $D(4, 5, -2)$. The distance is to be measured along the line perpendicular to the two lines. First find a vector $\mathbf{n}$ perpendicular to both lines. Then project $\overrightarrow{AC}$ onto $\mathbf{n}$.
64. (Continuation of Exercise 63.) Find the distance between the line through $A(4, 0, 2)$ and $B(2, 4, 1)$ and the line through $C(1, 3, 2)$ and $D(2, 2, 4)$.

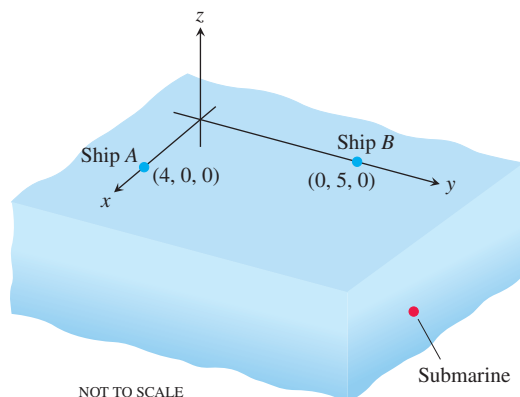
Quadric Surfaces

Identify and sketch the surfaces in Exercises 65–76.

- $x^2 + y^2 + z^2 = 4$
- $x^2 + (y - 1)^2 + z^2 = 1$
- $4x^2 + 4y^2 + z^2 = 4$
- $36x^2 + 9y^2 + 4z^2 = 36$
- $z = -(x^2 + y^2)$
- $y = -(x^2 + z^2)$
- $x^2 + y^2 = z^2$
- $x^2 + z^2 = y^2$
- $x^2 + y^2 - z^2 = 4$
- $4y^2 + z^2 - 4x^2 = 4$
- $y^2 - x^2 - z^2 = 1$
- $z^2 - x^2 - y^2 = 1$

CHAPTER 11 Additional and Advanced Exercises

1. **Submarine hunting** Two surface ships on maneuvers are trying to determine a submarine's course and speed to prepare for an aircraft intercept. As shown here, ship A is located at $(4, 0, 0)$, whereas ship B is located at $(0, 5, 0)$. All coordinates are given in thousands of meters. Ship A locates the submarine in the direction of the vector $2\mathbf{i} + 3\mathbf{j} - (1/3)\mathbf{k}$, and ship B locates it in the direction of the vector $18\mathbf{i} - 6\mathbf{j} - \mathbf{k}$. Four minutes ago, the submarine was located at $(2, -1, -1/3)$. The aircraft is due in 20 min. Assuming that the submarine moves in a straight line at a constant speed, to what position should the surface ships direct the aircraft?



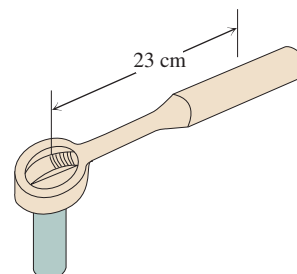
2. **A helicopter rescue** Two helicopters, H_1 and H_2 , are traveling together. At time $t = 0$, they separate and follow different straight-line paths given by

$$H_1: x = 6 + 40t, \quad y = -3 + 10t, \quad z = -3 + 2t$$

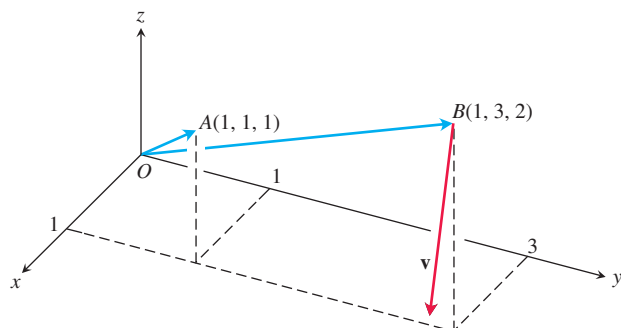
$$H_2: x = 6 + 110t, \quad y = -3 + 4t, \quad z = -3 + t.$$

Time t is measured in hours, and all coordinates are measured in kilometers. Due to system malfunctions, H_2 stops its flight at $(446, 13, 1)$ and, in a negligible amount of time, lands at $(446, 13, 0)$. Two hours later, H_1 is advised of this fact and heads toward H_2 at 150 km/h. How long will it take H_1 to reach H_2 ?

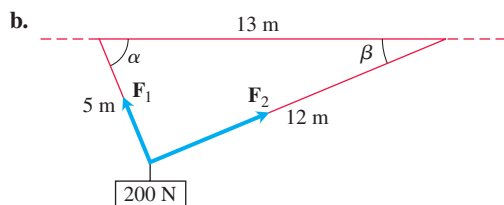
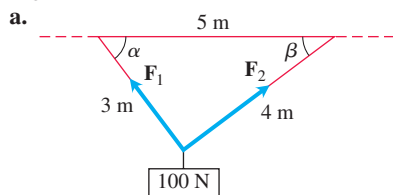
3. **Torque** The operator's manual for the Toro® 53-cm lawnmower says, "tighten the spark plug to $20.4 \text{ N} \cdot \text{m}$." If you are installing the plug with a 26.5-cm socket wrench that places the center of your hand 23 cm from the axis of the spark plug, about how hard should you pull? Answer in newtons.



4. **Rotating body** The line through the origin and the point $A(1, 1, 1)$ is the axis of rotation of a rigid body rotating with a constant angular speed of $3/2$ rad/s. The rotation appears to be clockwise when we look toward the origin from A . Find the velocity $\mathbf{v}$ of the point of the body that is at the position $B(1, 3, 2)$.

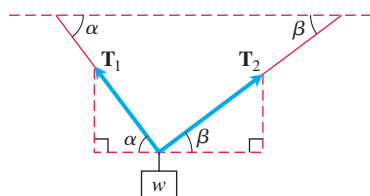


5. Consider the weight suspended by two wires in each diagram. Find the magnitudes and components of vectors $\mathbf{F}_1$ and $\mathbf{F}_2$, and angles α and β .



(Hint: This triangle is a right triangle.)

6. Consider a weight of w N suspended by two wires in the diagram, where $\mathbf{T}_1$ and $\mathbf{T}_2$ are force vectors directed along the wires.



- a. Find the vectors $\mathbf{T}_1$ and $\mathbf{T}_2$ and show that their magnitudes are

$$|\mathbf{T}_1| = \frac{w \cos \beta}{\sin(\alpha + \beta)}$$

and

$$|\mathbf{T}_2| = \frac{w \cos \alpha}{\sin(\alpha + \beta)}.$$

- b. For a fixed β , determine the value of α that minimizes the magnitude $|\mathbf{T}_1|$.
c. For a fixed α , determine the value of β that minimizes the magnitude $|\mathbf{T}_2|$.

7. Determinants and planes

- a. Show that

$$\begin{vmatrix} x_1 - x & y_1 - y & z_1 - z \\ x_2 - x & y_2 - y & z_2 - z \\ x_3 - x & y_3 - y & z_3 - z \end{vmatrix} = 0$$

is an equation for the plane through the three noncollinear points $P_1(x_1, y_1, z_1)$, $P_2(x_2, y_2, z_2)$, and $P_3(x_3, y_3, z_3)$.

- b. What set of points in space is described by the equation

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0?$$

8. Determinants and lines

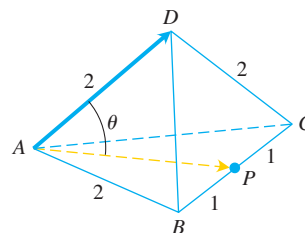
Show that the lines $x = a_1s + b_1$, $y = a_2s + b_2$, $z = a_3s + b_3$, $-\infty < s < \infty$ and

$x = c_1t + d_1$, $y = c_2t + d_2$, $z = c_3t + d_3$, $-\infty < t < \infty$ intersect or are parallel if and only if

$$\begin{vmatrix} a_1 & c_1 & b_1 - d_1 \\ a_2 & c_2 & b_2 - d_2 \\ a_3 & c_3 & b_3 - d_3 \end{vmatrix} = 0.$$

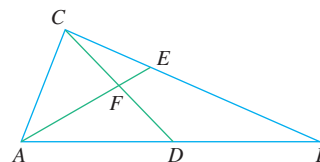
9. Consider a regular tetrahedron of side length 2.

- a. Use vectors to find the angle θ formed by the base of the tetrahedron and any one of its other edges.



- b. Use vectors to find the angle θ formed by any two adjacent faces of the tetrahedron. This angle is commonly referred to as a dihedral angle.

10. In the figure here, D is the midpoint of side AB of triangle ABC , and E is one-third of the way between C and B . Use vectors to prove that F is the midpoint of line segment CD .



11. Use vectors to show that the distance from $P_1(x_1, y_1)$ to the line $ax + by = c$ is

$$d = \frac{|ax_1 + by_1 - c|}{\sqrt{a^2 + b^2}}.$$

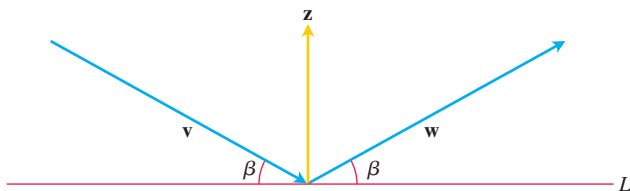
12. a. Use vectors to show that the distance from $P_1(x_1, y_1, z_1)$ to the plane $Ax + By + Cz = D$ is

$$d = \frac{|Ax_1 + By_1 + Cz_1 - D|}{\sqrt{A^2 + B^2 + C^2}}.$$

- b. Find an equation for the sphere that is tangent to the planes $x + y + z = 3$ and $x + y + z = 9$ if the planes $2x - y = 0$ and $3x - z = 0$ pass through the center of the sphere.
13. a. **Distance between parallel planes** Show that the distance between the parallel planes $Ax + By + Cz = D_1$ and $Ax + By + Cz = D_2$ is

$$d = \frac{|D_1 - D_2|}{|A\mathbf{i} + B\mathbf{j} + C\mathbf{k}|}.$$

- b. Find the distance between the planes $2x + 3y - z = 6$ and $2x + 3y - z = 12$.
- c. Find an equation for the plane parallel to the plane $2x - y + 2z = -4$ if the point $(3, 2, -1)$ is equidistant from the two planes.
- d. Write equations for the planes that lie parallel to, and 5 units away from, the plane $x - 2y + z = 3$.
14. Prove that four points A, B, C , and D are coplanar (lie in a common plane) if and only if $\overrightarrow{AD} \cdot (\overrightarrow{AB} \times \overrightarrow{BC}) = 0$.
15. **The projection of a vector on a plane** Let P be a plane in space and let $\mathbf{v}$ be a vector. The vector projection of $\mathbf{v}$ onto the plane P , $\text{proj}_P \mathbf{v}$, can be defined informally as follows. Suppose the sun is shining so that its rays are normal to the plane P . Then $\text{proj}_P \mathbf{v}$ is the “shadow” of $\mathbf{v}$ onto P . If P is the plane $x + 2y + 6z = 6$ and $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, find $\text{proj}_P \mathbf{v}$.
16. The accompanying figure shows nonzero vectors $\mathbf{v}, \mathbf{w}$, and $\mathbf{z}$, with $\mathbf{z}$ orthogonal to the line L , and $\mathbf{v}$ and $\mathbf{w}$ making equal angles β with L . Assuming $|\mathbf{v}| = |\mathbf{w}|$, find $\mathbf{w}$ in terms of $\mathbf{v}$ and $\mathbf{z}$.



17. **Triple vector products** The *triple vector products* $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$ and $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ are usually not equal, although the formulas for evaluating them from components are similar:

$$(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{v} \cdot \mathbf{w})\mathbf{u}.$$

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}.$$

Verify each formula for the following vectors by evaluating its two sides and comparing the results.

$\mathbf{u}$	$\mathbf{v}$	$\mathbf{w}$
a. $2\mathbf{i}$	$2\mathbf{j}$	$2\mathbf{k}$
b. $\mathbf{i} - \mathbf{j} + \mathbf{k}$	$2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$	$-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$
c. $2\mathbf{i} + \mathbf{j}$	$2\mathbf{i} - \mathbf{j} + \mathbf{k}$	$\mathbf{i} + 2\mathbf{k}$
d. $\mathbf{i} + \mathbf{j} - 2\mathbf{k}$	$-\mathbf{i} - \mathbf{k}$	$2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$

18. **Cross and dot products** Show that if $\mathbf{u}, \mathbf{v}, \mathbf{w}$, and $\mathbf{r}$ are any vectors, then

$$\mathbf{a}. \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) + \mathbf{v} \times (\mathbf{w} \times \mathbf{u}) + \mathbf{w} \times (\mathbf{u} \times \mathbf{v}) = \mathbf{0}$$

$$\mathbf{b}. \mathbf{u} \times \mathbf{v} = (\mathbf{u} \cdot \mathbf{v} \times \mathbf{i})\mathbf{i} + (\mathbf{u} \cdot \mathbf{v} \times \mathbf{j})\mathbf{j} + (\mathbf{u} \cdot \mathbf{v} \times \mathbf{k})\mathbf{k}$$

$$\mathbf{c}. (\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{w} \times \mathbf{r}) = \begin{vmatrix} \mathbf{u} \cdot \mathbf{w} & \mathbf{v} \cdot \mathbf{w} \\ \mathbf{u} \cdot \mathbf{r} & \mathbf{v} \cdot \mathbf{r} \end{vmatrix}.$$

19. **Cross and dot products** Prove or disprove the formula

$$\mathbf{u} \times (\mathbf{u} \times (\mathbf{u} \times \mathbf{v})) \cdot \mathbf{w} = -|\mathbf{u}|^2 \mathbf{u} \cdot \mathbf{v} \times \mathbf{w}.$$

20. By forming the cross product of two appropriate vectors, derive the trigonometric identity

$$\sin(A - B) = \sin A \cos B - \cos A \sin B.$$

21. Use vectors to prove that

$$(a^2 + b^2)(c^2 + d^2) \geq (ac + bd)^2$$

for any four numbers a, b, c , and d . (Hint: Let $\mathbf{u} = a\mathbf{i} + b\mathbf{j}$ and $\mathbf{v} = c\mathbf{i} + d\mathbf{j}$.)

22. **Dot multiplication is positive definite** Show that dot multiplication of vectors is *positive definite*; that is, show that $\mathbf{u} \cdot \mathbf{u} \geq 0$ for every vector $\mathbf{u}$ and that $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

23. Show that $|\mathbf{u} + \mathbf{v}| \leq |\mathbf{u}| + |\mathbf{v}|$ for any vectors $\mathbf{u}$ and $\mathbf{v}$.

24. Show that $\mathbf{w} = |\mathbf{v}|\mathbf{u} + |\mathbf{u}|\mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$.

25. Show that $|\mathbf{v}|\mathbf{u} + |\mathbf{u}|\mathbf{v}$ and $|\mathbf{v}|\mathbf{u} - |\mathbf{u}|\mathbf{v}$ are orthogonal.

CHAPTER 11 Technology Application Projects

Mathematica/Maple Projects

Projects can be found within MyLab Math.

Using Vectors to Represent Lines and Find Distances

Parts I and II: Learn the advantages of interpreting lines as vectors.

Part III: Use vectors to find the distance from a point to a line.

Putting a Scene in Three Dimensions onto a Two-Dimensional Canvas

Use the concept of planes in space to obtain a two-dimensional image.

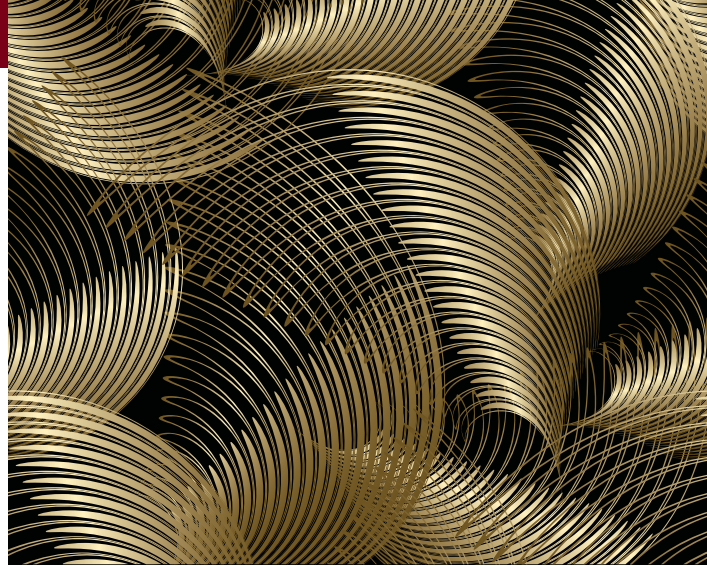
Getting Started in Plotting in 3D

Part I: Use the vector definition of lines and planes to generate graphs and equations, and to compare different forms for the equations of a single line.

Part II: Plot functions that are defined implicitly.

12

Vector-Valued Functions and Motion in Space



OVERVIEW In this chapter we introduce the calculus of vector-valued functions. The domains of these functions are sets of real numbers, as before, but their ranges consist of vectors instead of scalars. When a vector-valued function changes, the change can occur in both magnitude and direction, so the derivative is itself a vector. The integral of a vector-valued function is also a vector. We use the calculus of these functions to describe the paths and motions of objects moving in a plane or in space, so their velocities and accelerations are given by vectors.

12.1 Curves in Space and Their Tangents

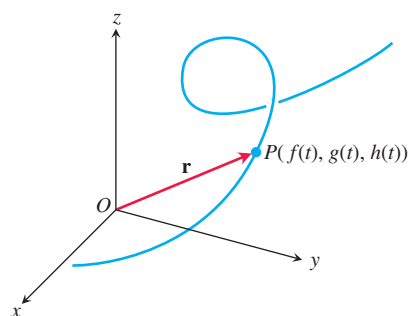


FIGURE 12.1 The position vector $\mathbf{r} = \overrightarrow{OP}$ of a particle moving through space is a function of time.

When a particle moves through space during a time interval I , we think of the particle's coordinates as functions defined on I :

$$x = f(t), \quad y = g(t), \quad z = h(t), \quad t \in I. \quad (1)$$

The points $(x, y, z) = (f(t), g(t), h(t))$, $t \in I$, make up the **curve** in space that we call the particle's **path**. The equations and interval in Equation (1) parametrize the curve.

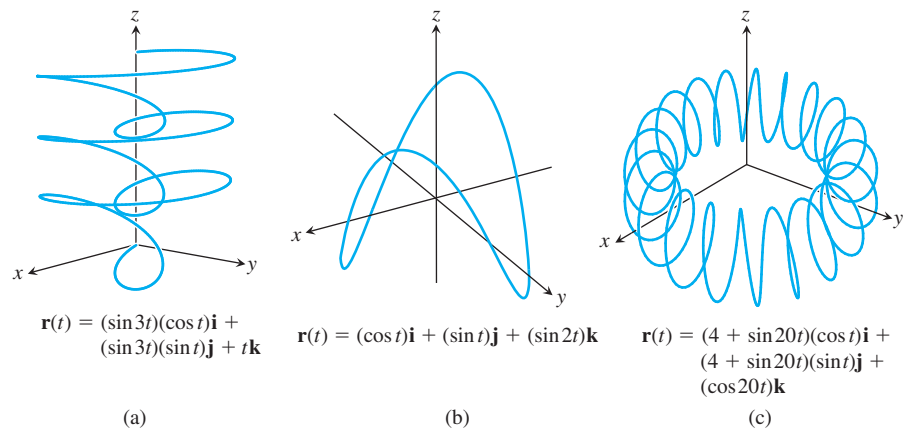
A curve in space can also be represented in vector form. The vector

$$\mathbf{r}(t) = \overrightarrow{OP} = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} \quad (2)$$

from the origin to the particle's position $P(f(t), g(t), h(t))$ at time t is the particle's position vector (Figure 12.1). The functions f , g , and h are the **component functions** (or components) of the position vector. We think of the particle's path as the curve traced by $\mathbf{r}$ during the time interval I . Figure 12.2 displays several space curves generated by a computer graphing program.

Equation (2) defines $\mathbf{r}$ as a vector function of the real variable t on the interval I . More generally, a **vector-valued function** or **vector function** on a domain set D is a rule that assigns a vector in space to each element in D . For now, the domains will be intervals of real numbers, and the graph of the function represents a curve in space. Vector functions on a domain in the plane or in space give rise to “vector fields,” which are important to the study of fluid flows, gravitational fields, and electromagnetic phenomena. We investigate vector fields and their applications in Chapter 15.

Real-valued functions are often called **scalar functions** to distinguish them from vector functions. The components of $\mathbf{r}$ in Equation (2) are scalar functions of t . The domain of a vector-valued function is the common domain of its components.


 FIGURE 12.2 Space curves are defined by the position vectors $\mathbf{r}(t)$.

EXAMPLE 1 Graph the vector function

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}.$$

Solution This vector function $\mathbf{r}(t)$ is defined for all real values of t . The curve traced by $\mathbf{r}$ winds around the circular cylinder $x^2 + y^2 = 1$ (Figure 12.3). The curve lies on the cylinder because the $\mathbf{i}$ - and $\mathbf{j}$ -components of $\mathbf{r}$, being the x - and y -coordinates of the tip of $\mathbf{r}$, satisfy the cylinder's equation:

$$x^2 + y^2 = (\cos t)^2 + (\sin t)^2 = 1.$$

The curve rises as the $\mathbf{k}$ -component $z = t$ increases. Each time t increases by 2π , the curve completes one turn around the cylinder. The curve is called a **helix** (from an ancient Greek word for “spiral”). The equations

$$x = \cos t, \quad y = \sin t, \quad z = t$$

parametrize the helix. The domain is the largest set of points t for which all three equations are defined, or $-\infty < t < \infty$ for this example. Figure 12.4 shows more helices. ■

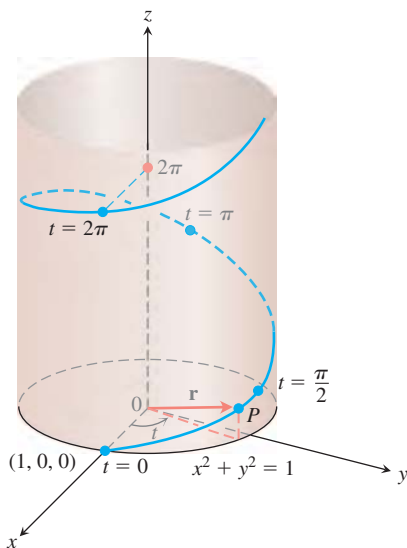


FIGURE 12.3 The helix $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$ (Example 1).

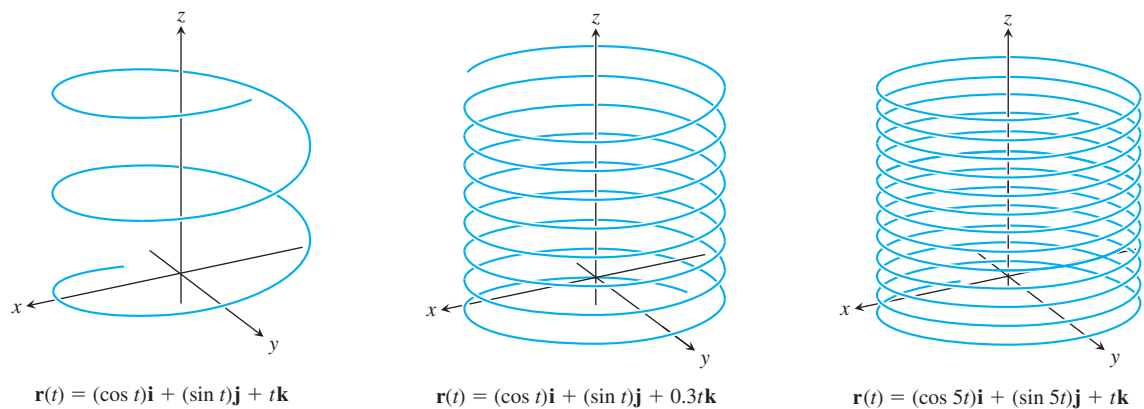


FIGURE 12.4 Helices spiral upward around a cylinder, like coiled springs.

Limits and Continuity

The way we define limits of vector-valued functions is similar to the way we define limits of real-valued functions.

DEFINITION Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function with domain D , and let $\mathbf{L}$ be a vector. We say that $\mathbf{r}$ has **limit** $\mathbf{L}$ as t approaches t_0 and write

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$$

if, for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that, for all $t \in D$,

$$|\mathbf{r}(t) - \mathbf{L}| < \varepsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta.$$

If $\mathbf{L} = L_1\mathbf{i} + L_2\mathbf{j} + L_3\mathbf{k}$, then it can be shown that $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$ precisely when

$$\lim_{t \rightarrow t_0} f(t) = L_1, \quad \lim_{t \rightarrow t_0} g(t) = L_2, \quad \text{and} \quad \lim_{t \rightarrow t_0} h(t) = L_3.$$

We omit the proof. The equation

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \left(\lim_{t \rightarrow t_0} f(t) \right) \mathbf{i} + \left(\lim_{t \rightarrow t_0} g(t) \right) \mathbf{j} + \left(\lim_{t \rightarrow t_0} h(t) \right) \mathbf{k} \quad (3)$$

provides a practical way to calculate limits of vector functions.

EXAMPLE 2 If $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$, then

$$\begin{aligned} \lim_{t \rightarrow \pi/4} \mathbf{r}(t) &= \left(\lim_{t \rightarrow \pi/4} \cos t \right) \mathbf{i} + \left(\lim_{t \rightarrow \pi/4} \sin t \right) \mathbf{j} + \left(\lim_{t \rightarrow \pi/4} t \right) \mathbf{k} \\ &= \frac{\sqrt{2}}{2} \mathbf{i} + \frac{\sqrt{2}}{2} \mathbf{j} + \frac{\pi}{4} \mathbf{k}. \end{aligned}$$

We define continuity for vector functions the same way we define continuity for scalar functions defined over an interval.

DEFINITION A vector function $\mathbf{r}(t)$ is **continuous at a point** $t = t_0$ in its domain if $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$. The function is **continuous** if it is continuous at every point in its domain.

From Equation (3), we see that $\mathbf{r}(t)$ is continuous at $t = t_0$ if and only if each component function is continuous there (Exercise 45).

EXAMPLE 3

- (a) All the space curves shown in Figures 12.2 and 12.4 are continuous because their component functions are continuous at every value of t in $(-\infty, \infty)$.
 (b) The function

$$\mathbf{g}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + \lfloor t \rfloor \mathbf{k}$$

is discontinuous at every integer, because the greatest integer function $\lfloor t \rfloor$ is discontinuous at every integer.

Derivatives and Motion

Suppose that $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ is the position vector of a particle moving along a curve in space and that f , g , and h are differentiable functions of t . Then the difference between the particle's positions at time t and time $t + \Delta t$ is the vector

$$\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$$

To calculate the limit of a vector function, we find the limit of each component scalar function.

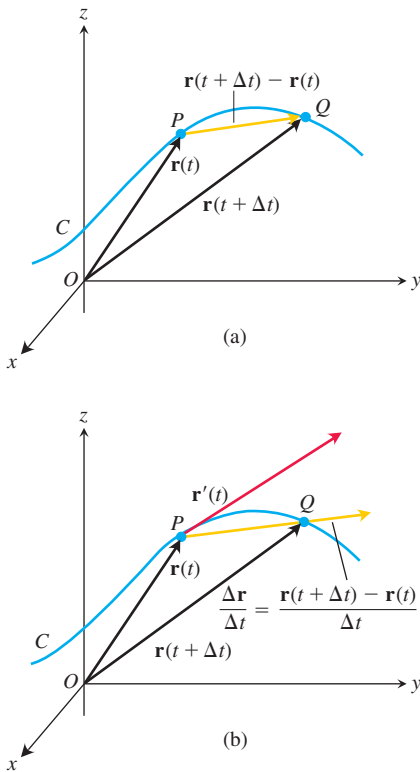


FIGURE 12.5 As $\Delta t \rightarrow 0$, the point Q approaches the point P along the curve C . In the limit, the vector $\overrightarrow{PQ}/\Delta t$ becomes the tangent vector $\mathbf{r}'(t)$.

(Figure 12.5a). In terms of components,

$$\begin{aligned}\Delta \mathbf{r} &= \mathbf{r}(t + \Delta t) - \mathbf{r}(t) \\ &= [f(t + \Delta t)\mathbf{i} + g(t + \Delta t)\mathbf{j} + h(t + \Delta t)\mathbf{k}] - [f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}] \\ &= [f(t + \Delta t) - f(t)]\mathbf{i} + [g(t + \Delta t) - g(t)]\mathbf{j} + [h(t + \Delta t) - h(t)]\mathbf{k}.\end{aligned}$$

As Δt approaches zero, three things seem to happen simultaneously. First, Q approaches P along the curve. Second, the secant line PQ seems to approach a limiting position tangent to the curve at P . Third, the quotient $\Delta \mathbf{r}/\Delta t$ (Figure 12.5b) approaches the limit

$$\begin{aligned}\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} &= \left[\lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \right] \mathbf{i} + \left[\lim_{\Delta t \rightarrow 0} \frac{g(t + \Delta t) - g(t)}{\Delta t} \right] \mathbf{j} \\ &\quad + \left[\lim_{\Delta t \rightarrow 0} \frac{h(t + \Delta t) - h(t)}{\Delta t} \right] \mathbf{k} \\ &= \left[\frac{df}{dt} \right] \mathbf{i} + \left[\frac{dg}{dt} \right] \mathbf{j} + \left[\frac{dh}{dt} \right] \mathbf{k}.\end{aligned}$$

These observations lead us to the following definition.

DEFINITION The vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ has a **derivative (is differentiable)** at t if f , g , and h have derivatives at t . The derivative is the vector function

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{df}{dt} \mathbf{i} + \frac{dg}{dt} \mathbf{j} + \frac{dh}{dt} \mathbf{k}.$$

A vector function $\mathbf{r}$ is **differentiable** if it is differentiable at every point of its domain.

The geometric significance of the definition of derivative is shown in Figure 12.5. The points P and Q have position vectors $\mathbf{r}(t)$ and $\mathbf{r}(t + \Delta t)$, and the vector $\overrightarrow{PQ}$ is represented by $\mathbf{r}(t + \Delta t) - \mathbf{r}(t)$. For $\Delta t > 0$, the scalar multiple $(1/\Delta t)(\mathbf{r}(t + \Delta t) - \mathbf{r}(t))$ points in the same direction as the vector $\overrightarrow{PQ}$. As $\Delta t \rightarrow 0$, this vector approaches the vector $\mathbf{r}'(t)$, which is a vector tangent to the curve at P , as long as it is different from the zero vector $\mathbf{0}$ (Figure 12.5b).

The curve traced by $\mathbf{r}$ is **smooth** if $d\mathbf{r}/dt$ is continuous and never $\mathbf{0}$, that is, if f , g , and h have continuous first derivatives that are not simultaneously 0. We require $d\mathbf{r}/dt \neq \mathbf{0}$ for a smooth curve to make sure the curve has a continuously turning tangent at each point. On a smooth curve, there are no sharp corners or cusps.

A curve that is made up of a finite number of smooth curves pieced together in a continuous fashion is called **piecewise smooth** (Figure 12.6).

Look once again at Figure 12.5. We drew the figure for Δt positive, so $\Delta \mathbf{r}$ points forward, in the direction of the motion. The vector $\Delta \mathbf{r}/\Delta t$, having the same direction as $\Delta \mathbf{r}$, points forward too. Had Δt been negative, $\Delta \mathbf{r}$ would have pointed backward, against the direction of motion. The quotient $\Delta \mathbf{r}/\Delta t$, however, being a negative scalar multiple of $\Delta \mathbf{r}$, would once again have pointed forward. No matter how $\Delta \mathbf{r}$ points, $\Delta \mathbf{r}/\Delta t$ points forward, and we expect the vector $d\mathbf{r}/dt = \lim_{\Delta t \rightarrow 0} \Delta \mathbf{r}/\Delta t$, when different from $\mathbf{0}$, to do the same.

This means that the derivative $d\mathbf{r}/dt$, which is the rate of change of position with respect to time, always points in the direction of motion. For a smooth curve, $d\mathbf{r}/dt$ is never zero; the particle does not stop or reverse direction.

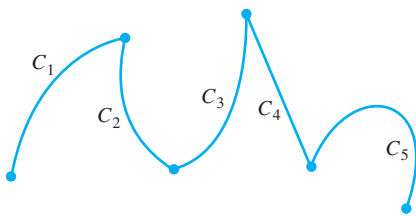


FIGURE 12.6 A piecewise smooth curve made up of five smooth curves connected end to end in a continuous fashion. The curve here is not smooth at the points joining the five smooth curves.

DEFINITIONS If $\mathbf{r}$ is the position vector of a particle moving along a smooth curve in space, then

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

is the particle's **velocity vector**. If $\mathbf{v}$ is a nonzero vector, then it is tangent to the curve, and its direction is the **direction of motion**. The magnitude of $\mathbf{v}$ is the particle's **speed**, and the derivative $\mathbf{a} = d\mathbf{v}/dt$, when it exists, is the particle's **acceleration vector**. In summary,

1. Velocity is the derivative of position: $\mathbf{v} = \frac{d\mathbf{r}}{dt}$.
2. Speed is the magnitude of velocity: $\text{Speed} = |\mathbf{v}|$.
3. Acceleration is the derivative of velocity: $\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$.
4. The unit vector $\mathbf{v}/|\mathbf{v}|$ is the direction of motion at time t .

EXAMPLE 4 Find the velocity, speed, and acceleration of a particle whose motion in space is given by the position vector $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + 5 \cos^2 t \mathbf{k}$. Sketch the velocity vector $\mathbf{v}(7\pi/4)$.

Solution The velocity and acceleration vectors at time t are

$$\begin{aligned}\mathbf{v}(t) &= \mathbf{r}'(t) = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} - 10 \cos t \sin t \mathbf{k} \\ &= -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} - 5 \sin 2t \mathbf{k}, \\ \mathbf{a}(t) &= \mathbf{r}''(t) = -2 \cos t \mathbf{i} - 2 \sin t \mathbf{j} - 10 \cos 2t \mathbf{k},\end{aligned}$$

and the speed is

$$|\mathbf{v}(t)| = \sqrt{(-2 \sin t)^2 + (2 \cos t)^2 + (-5 \sin 2t)^2} = \sqrt{4 + 25 \sin^2 2t}.$$

When $t = 7\pi/4$, we have

$$\mathbf{v}\left(\frac{7\pi}{4}\right) = \sqrt{2}\mathbf{i} + \sqrt{2}\mathbf{j} + 5\mathbf{k}, \quad \mathbf{a}\left(\frac{7\pi}{4}\right) = -\sqrt{2}\mathbf{i} + \sqrt{2}\mathbf{j}, \quad \left|\mathbf{v}\left(\frac{7\pi}{4}\right)\right| = \sqrt{29}.$$

A sketch of the curve of motion, and the velocity vector when $t = 7\pi/4$, can be seen in Figure 12.7. ■

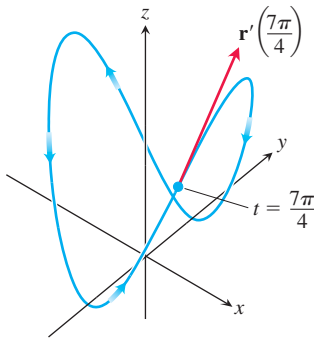


FIGURE 12.7 The curve and the velocity vector when $t = 7\pi/4$ for the motion given in Example 4.

We can express the velocity of a moving particle as the product of its speed and direction:

$$\text{Velocity} = |\mathbf{v}| \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right) = (\text{speed})(\text{direction}).$$

Differentiation Rules

Because the derivatives of vector functions may be computed component by component, the rules for differentiating vector functions have the same form as the rules for differentiating scalar functions.

Differentiation Rules for Vector Functions

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions of t , $\mathbf{C}$ a constant vector, c any scalar, and f any differentiable scalar function.

1. *Constant Function Rule:* $\frac{d}{dt}\mathbf{C} = \mathbf{0}$

2. *Scalar Multiple Rules:* $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$

$$\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$$

3. *Sum Rule:* $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$

4. *Difference Rule:* $\frac{d}{dt}[\mathbf{u}(t) - \mathbf{v}(t)] = \mathbf{u}'(t) - \mathbf{v}'(t)$

5. *Dot Product Rule:* $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

6. *Cross Product Rule:* $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$

7. *Chain Rule:* $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

When you use the Cross Product Rule, remember to preserve the order of the factors. If $\mathbf{u}$ comes first on the left side of the equation, it must also come first on the right, or the signs will be wrong.

We will prove the product rules and the Chain Rule but will leave the rules for constants, scalar multiples, sums, and differences as exercises.

Proof of the Dot Product Rule Suppose that

$$\mathbf{u} = u_1(t)\mathbf{i} + u_2(t)\mathbf{j} + u_3(t)\mathbf{k}$$

and

$$\mathbf{v} = v_1(t)\mathbf{i} + v_2(t)\mathbf{j} + v_3(t)\mathbf{k}.$$

Then

$$\begin{aligned}\frac{d}{dt}(\mathbf{u} \cdot \mathbf{v}) &= \frac{d}{dt}(u_1v_1 + u_2v_2 + u_3v_3) \\ &= \underbrace{u_1'v_1 + u_2'v_2 + u_3'v_3}_{\mathbf{u}' \cdot \mathbf{v}} + \underbrace{u_1v_1' + u_2v_2' + u_3v_3'}_{\mathbf{u} \cdot \mathbf{v}'}.\end{aligned}$$

Proof of the Cross Product Rule We model the proof after the proof of the Product Rule for scalar functions. According to the definition of derivative,

$$\frac{d}{dt}(\mathbf{u} \times \mathbf{v}) = \lim_{h \rightarrow 0} \frac{\mathbf{u}(t+h) \times \mathbf{v}(t+h) - \mathbf{u}(t) \times \mathbf{v}(t)}{h}.$$

To change this fraction into an equivalent one that contains the difference quotients for the derivatives of $\mathbf{u}$ and $\mathbf{v}$, we subtract and add $\mathbf{u}(t) \times \mathbf{v}(t+h)$ in the numerator. Then

$$\begin{aligned}\frac{d}{dt}(\mathbf{u} \times \mathbf{v}) &= \lim_{h \rightarrow 0} \frac{\mathbf{u}(t+h) \times \mathbf{v}(t+h) - \mathbf{u}(t) \times \mathbf{v}(t+h) + \mathbf{u}(t) \times \mathbf{v}(t+h) - \mathbf{u}(t) \times \mathbf{v}(t)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{\mathbf{u}(t+h) - \mathbf{u}(t)}{h} \times \mathbf{v}(t+h) + \mathbf{u}(t) \times \frac{\mathbf{v}(t+h) - \mathbf{v}(t)}{h} \right] \\ &= \lim_{h \rightarrow 0} \frac{\mathbf{u}(t+h) - \mathbf{u}(t)}{h} \times \lim_{h \rightarrow 0} \mathbf{v}(t+h) + \lim_{h \rightarrow 0} \mathbf{u}(t) \times \lim_{h \rightarrow 0} \frac{\mathbf{v}(t+h) - \mathbf{v}(t)}{h}.\end{aligned}$$

The last of these equalities holds because the limit of the cross product of two vector functions is the cross product of their limits if the latter exist (Exercise 46). As h approaches zero, $\mathbf{v}(t+h)$ approaches $\mathbf{v}(t)$ because $\mathbf{v}$, being differentiable at t , is continuous at t (Exercise 47). The two fractions approach the values of $d\mathbf{u}/dt$ and $d\mathbf{v}/dt$ at t . In short,

$$\frac{d}{dt}(\mathbf{u} \times \mathbf{v}) = \left(\frac{d\mathbf{u}}{dt} \times \mathbf{v} \right) + \left(\mathbf{u} \times \frac{d\mathbf{v}}{dt} \right). \quad \blacksquare$$

Proof of the Chain Rule Suppose that $\mathbf{u}(s) = a(s)\mathbf{i} + b(s)\mathbf{j} + c(s)\mathbf{k}$ is a differentiable vector function of s and that $s = f(t)$ is a differentiable scalar function of t . Then a , b , and c are differentiable functions of t , and the Chain Rule for differentiable real-valued functions gives

$$\begin{aligned} \frac{d}{dt}[\mathbf{u}(s)] &= \frac{da}{dt}\mathbf{i} + \frac{db}{dt}\mathbf{j} + \frac{dc}{dt}\mathbf{k} \\ &= \frac{da}{ds}\frac{ds}{dt}\mathbf{i} + \frac{db}{ds}\frac{ds}{dt}\mathbf{j} + \frac{dc}{ds}\frac{ds}{dt}\mathbf{k} \\ &= \frac{ds}{dt}\left(\frac{da}{ds}\mathbf{i} + \frac{db}{ds}\mathbf{j} + \frac{dc}{ds}\mathbf{k}\right) \\ &= \frac{ds}{dt}\frac{d\mathbf{u}}{ds} \\ &= f'(t)\mathbf{u}'(f(t)). \end{aligned} \quad s = f(t) \quad \blacksquare$$

As an algebraic convenience, we sometimes write the product of a scalar c and a vector $\mathbf{v}$ as $\mathbf{v}c$ instead of $c\mathbf{v}$. This permits us, for instance, to write the Chain Rule in a familiar form:

$$\frac{d\mathbf{u}}{dt} = \frac{d\mathbf{u}}{ds} \frac{ds}{dt},$$

where $s = f(t)$.

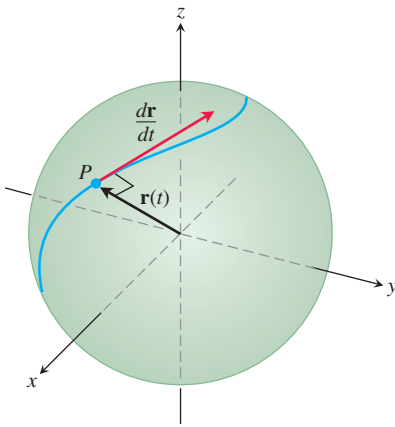


FIGURE 12.8 If a particle moves on a sphere in such a way that its position $\mathbf{r}$ is a differentiable function of time, then $\mathbf{r} \cdot (d\mathbf{r}/dt) = 0$.

Vector Functions of Constant Length

When we track a particle moving on a sphere centered at the origin (Figure 12.8), the position vector has a constant length equal to the radius of the sphere. The velocity vector $d\mathbf{r}/dt$, tangent to the path of motion, is tangent to the sphere and hence perpendicular to $\mathbf{r}$. This is always the case for a differentiable vector function of constant length: The vector and its first derivative are orthogonal. By direct calculation,

$$\begin{aligned} \mathbf{r}(t) \cdot \mathbf{r}(t) &= |\mathbf{r}(t)|^2 = c^2 && |\mathbf{r}(t)| = c \text{ is constant.} \\ \frac{d}{dt}[\mathbf{r}(t) \cdot \mathbf{r}(t)] &= 0 && \text{Differentiate both sides.} \\ \mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) &= 0 && \text{Rule 5 with } \mathbf{r}(t) = \mathbf{u}(t) = \mathbf{v}(t) \\ 2\mathbf{r}'(t) \cdot \mathbf{r}(t) &= 0. \end{aligned}$$

Thus the vectors $\mathbf{r}'(t)$ and $\mathbf{r}(t)$ are orthogonal because their dot product is 0. In summary, the following holds.

If $\mathbf{r}$ is a differentiable vector function of t and the length of $\mathbf{r}(t)$ is constant, then

$$\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0. \quad (4)$$

We will use this observation repeatedly in Section 12.4. The converse is also true (see Exercise 41).

EXERCISES 12.1

In Exercises 1–4, find the given limits.

$$1. \lim_{t \rightarrow \pi} \left[\left(\sin \frac{t}{2} \right) \mathbf{i} + \left(\cos \frac{2}{3} t \right) \mathbf{j} + \left(\tan \frac{5}{4} t \right) \mathbf{k} \right]$$

$$2. \lim_{t \rightarrow -1} \left[t^3 \mathbf{i} + \left(\sin \frac{\pi}{2} t \right) \mathbf{j} + (\ln(t+2)) \mathbf{k} \right]$$

$$3. \lim_{t \rightarrow 1} \left[\left(\frac{t^2 - 1}{\ln t} \right) \mathbf{i} - \left(\frac{\sqrt{t} - 1}{1 - t} \right) \mathbf{j} + (\arctan t) \mathbf{k} \right]$$

$$4. \lim_{t \rightarrow 0} \left[\left(\frac{\sin t}{t} \right) \mathbf{i} + \left(\frac{\tan^2 t}{\sin 2t} \right) \mathbf{j} - \left(\frac{t^3 - 8}{t + 2} \right) \mathbf{k} \right]$$

Motion in the Plane

In Exercises 5–8, $\mathbf{r}(t)$ is the position of a particle in the xy -plane at time t . Find an equation in x and y whose graph is the path of the particle. Then find the particle's velocity and acceleration vectors at the given value of t .

$$5. \mathbf{r}(t) = (t+1)\mathbf{i} + (t^2 - 1)\mathbf{j}, \quad t = 1$$

$$6. \mathbf{r}(t) = \frac{t}{t+1}\mathbf{i} + \frac{1}{t}\mathbf{j}, \quad t = -\frac{1}{2}$$

$$7. \mathbf{r}(t) = e^t \mathbf{i} + \frac{2}{9} e^{2t} \mathbf{j}, \quad t = \ln 3$$

$$8. \mathbf{r}(t) = (\cos 2t)\mathbf{i} + (3 \sin 2t)\mathbf{j}, \quad t = 0$$

Exercises 9–12 give the position vectors of particles moving along various curves in the xy -plane. In each case, find the particle's velocity and acceleration vectors at the stated times and sketch them as vectors on the curve.

9. Motion on the circle $x^2 + y^2 = 1$

$$\mathbf{r}(t) = (\sin t)\mathbf{i} + (\cos t)\mathbf{j}; \quad t = \pi/4 \text{ and } \pi/2$$

10. Motion on the circle $x^2 + y^2 = 16$

$$\mathbf{r}(t) = \left(4 \cos \frac{t}{2} \right) \mathbf{i} + \left(4 \sin \frac{t}{2} \right) \mathbf{j}; \quad t = \pi \text{ and } 3\pi/2$$

11. Motion on the cycloid $x = t - \sin t$, $y = 1 - \cos t$

$$\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}; \quad t = \pi \text{ and } 3\pi/2$$

12. Motion on the parabola $y = x^2 + 1$

$$\mathbf{r}(t) = t\mathbf{i} + (t^2 + 1)\mathbf{j}; \quad t = -1, 0, \text{ and } 1$$

Motion in Space

In Exercises 13–18, $\mathbf{r}(t)$ is the position of a particle in space at time t . Find the particle's velocity and acceleration vectors. Then find the particle's speed and direction of motion at the given value of t . Write the particle's velocity at that time as the product of its speed and direction.

$$13. \mathbf{r}(t) = (t+1)\mathbf{i} + (t^2 - 1)\mathbf{j} + 2t\mathbf{k}, \quad t = 1$$

$$14. \mathbf{r}(t) = (1+t)\mathbf{i} + \frac{t^2}{\sqrt{2}}\mathbf{j} + \frac{t^3}{3}\mathbf{k}, \quad t = 1$$

$$15. \mathbf{r}(t) = (2 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j} + 4t\mathbf{k}, \quad t = \pi/2$$

$$16. \mathbf{r}(t) = (\sec t)\mathbf{i} + (\tan t)\mathbf{j} + \frac{4}{3}t\mathbf{k}, \quad t = \pi/6$$

$$17. \mathbf{r}(t) = (2 \ln(t+1))\mathbf{i} + t^2\mathbf{j} + \frac{t^2}{2}\mathbf{k}, \quad t = 1$$

$$18. \mathbf{r}(t) = e^{-t}\mathbf{i} + (2 \cos 3t)\mathbf{j} + (2 \sin 3t)\mathbf{k}, \quad t = 0$$

In Exercises 19–22, $\mathbf{r}(t)$ is the position of a particle in space at time t . Find the angle between the velocity and acceleration vectors at time $t = 0$.

$$19. \mathbf{r}(t) = (3t+1)\mathbf{i} + \sqrt{3}t\mathbf{j} + t^2\mathbf{k}$$

$$20. \mathbf{r}(t) = \left(\frac{\sqrt{2}}{2}t \right) \mathbf{i} + \left(\frac{\sqrt{2}}{2}t - 16t^2 \right) \mathbf{j}$$

$$21. \mathbf{r}(t) = (\ln(t^2 + 1))\mathbf{i} + (\arctan t)\mathbf{j} + \sqrt{t^2 + 1}\mathbf{k}$$

$$22. \mathbf{r}(t) = \frac{4}{9}(1+t)^{3/2}\mathbf{i} + \frac{4}{9}(1-t)^{3/2}\mathbf{j} + \frac{1}{3}t\mathbf{k}$$

Tangents to Curves

As mentioned in the text, the **tangent line** to a smooth curve $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ at $t = t_0$ is the line that passes through the point $(f(t_0), g(t_0), h(t_0))$ parallel to $\mathbf{v}(t_0)$, the curve's velocity vector at t_0 . In Exercises 23–26, find parametric equations for the line that is tangent to the given curve at the given parameter value $t = t_0$.

$$23. \mathbf{r}(t) = (\sin t)\mathbf{i} + (t^2 - \cos t)\mathbf{j} + e^t\mathbf{k}, \quad t_0 = 0$$

$$24. \mathbf{r}(t) = t^2\mathbf{i} + (2t-1)\mathbf{j} + t^3\mathbf{k}, \quad t_0 = 2$$

$$25. \mathbf{r}(t) = \ln t\mathbf{i} + \frac{t-1}{t+2}\mathbf{j} + t \ln t\mathbf{k}, \quad t_0 = 1$$

$$26. \mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + (\sin 2t)\mathbf{k}, \quad t_0 = \frac{\pi}{2}$$

In Exercises 27–30, find the value(s) of t so that the tangent line to the given curve contains the given point.

$$27. \mathbf{r}(t) = t^2\mathbf{i} + (1+t)\mathbf{j} + (2t-3)\mathbf{k}; \quad (-8, 2, -1)$$

$$28. \mathbf{r}(t) = t\mathbf{i} + 3\mathbf{j} + \left(\frac{2}{3}t^{3/2} \right) \mathbf{k}; \quad (0, 3, -8/3)$$

$$29. \mathbf{r}(t) = 2t\mathbf{i} + t^2\mathbf{j} - t^2\mathbf{k}; \quad (0, -4, 4)$$

$$30. \mathbf{r}(t) = -t\mathbf{i} + t^2\mathbf{j} + (\ln t)\mathbf{k}; \quad (2, -5, -3)$$

In Exercises 31–36, $\mathbf{r}(t)$ is the position of a particle in space at time t . Match each position function with one of the graphs A–F.

$$31. \mathbf{r}(t) = (t \cos t)\mathbf{i} + (t \sin t)\mathbf{j} + t\mathbf{k}$$

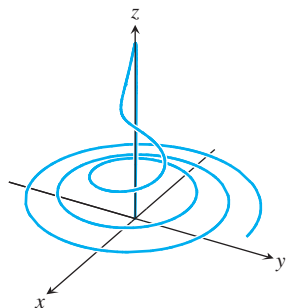
$$32. \mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + (\sin 2t)\mathbf{k}$$

$$33. \mathbf{r}(t) = t^2\mathbf{i} + (t^2 + 1)\mathbf{j} + t^4\mathbf{k}$$

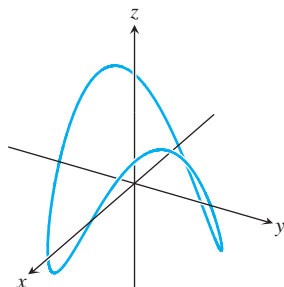
$$34. \mathbf{r}(t) = t\mathbf{i} + (\ln t)\mathbf{j} + (\sin t)\mathbf{k}$$

35. $\mathbf{r}(t) = t\mathbf{i} + (\cos t)\mathbf{j} + (\sin t)\mathbf{k}$

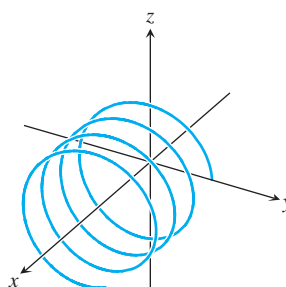
36. $\mathbf{r}(t) = (t \sin t)\mathbf{i} + (t \cos t)\mathbf{j} + \left(\frac{t}{t^2 + 1}\right)\mathbf{k}$



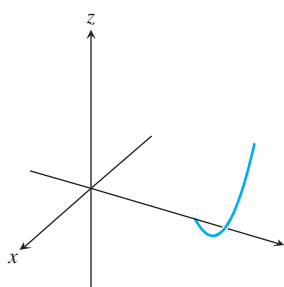
A.



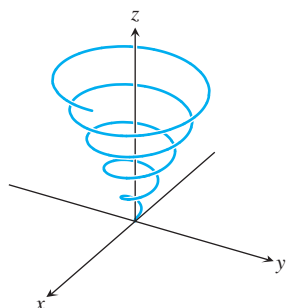
B.



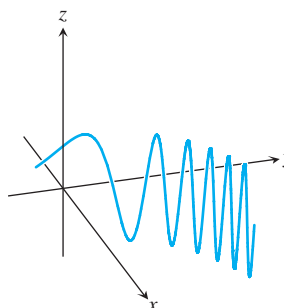
C.



D.



E.



F.

Theory and Examples

37. Motion along a circle Each of the following equations in parts (a)–(e) describes the motion of a particle having the same path, namely the unit circle $x^2 + y^2 = 1$. Although the path of each particle in parts (a)–(e) is the same, the behavior, or “dynamics,” of each particle is different. For each particle, answer the following questions.

- Does the particle have constant speed? If so, what is its constant speed?
- Is the particle's acceleration vector always orthogonal to its velocity vector?
- Does the particle move clockwise or counterclockwise around the circle?
- Is the particle initially located at the point $(1, 0)$?

a. $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}, \quad t \geq 0$

b. $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \sin(2t)\mathbf{j}, \quad t \geq 0$

c. $\mathbf{r}(t) = \cos(t - \pi/2)\mathbf{i} + \sin(t - \pi/2)\mathbf{j}, \quad t \geq 0$

d. $\mathbf{r}(t) = (\cos t)\mathbf{i} - (\sin t)\mathbf{j}, \quad t \geq 0$

e. $\mathbf{r}(t) = \cos(t^2)\mathbf{i} + \sin(t^2)\mathbf{j}, \quad t \geq 0$

38. Motion along a circle Show that the vector-valued function

$$\mathbf{r}(t) = (2\mathbf{i} + 2\mathbf{j} + \mathbf{k}) + \cos t\left(\frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}\right) + \sin t\left(\frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}\right)$$

describes the motion of a particle moving in the circle of radius 1 centered at the point $(2, 2, 1)$ and lying in the plane $x + y - 2z = 2$.

39. Motion along a parabola A particle moves along the top of the parabola $y^2 = 2x$ from left to right at a constant speed of 5 units per second. Find the velocity of the particle as it moves through the point $(2, 2)$.

40. Motion along a cycloid A particle moves in the xy -plane in such a way that its position at time t is

$$\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}.$$

- T** a. Graph $\mathbf{r}(t)$. The resulting curve is a cycloid.
 b. Find the maximum and minimum values of $|\mathbf{v}|$ and $|\mathbf{a}|$.
 (Hint: Find the extreme values of $|\mathbf{v}|^2$ and $|\mathbf{a}|^2$ first, and take square roots later.)

41. Let $\mathbf{r}$ be a differentiable vector function of t . Show that if $\mathbf{r} \cdot (d\mathbf{r}/dt) = 0$ for all t , then $|\mathbf{r}|$ is constant.

42. Derivatives of triple scalar products

a. Show that if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are differentiable vector functions of t , then

$$\frac{d}{dt}(\mathbf{u} \cdot \mathbf{v} \times \mathbf{w}) = \frac{d\mathbf{u}}{dt} \cdot \mathbf{v} \times \mathbf{w} + \mathbf{u} \cdot \frac{d\mathbf{v}}{dt} \times \mathbf{w} + \mathbf{u} \cdot \mathbf{v} \times \frac{d\mathbf{w}}{dt}.$$

b. Show that

$$\frac{d}{dt}\left(\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} \times \frac{d^2\mathbf{r}}{dt^2}\right) = \mathbf{r} \cdot \left(\frac{d\mathbf{r}}{dt} \times \frac{d^3\mathbf{r}}{dt^3}\right).$$

(Hint: Differentiate on the left and look for vectors whose products are zero.)

43. Prove the two Scalar Multiple Rules for vector functions.

44. Prove the Sum and Difference Rules for vector functions.

45. Component test for continuity at a point Show that the vector function $\mathbf{r}$ defined by $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ is continuous at $t = t_0$ if and only if f , g , and h are continuous at t_0 .

46. Limits of cross products of vector functions Suppose that $\mathbf{r}_1(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$, $\mathbf{r}_2(t) = g_1(t)\mathbf{i} + g_2(t)\mathbf{j} + g_3(t)\mathbf{k}$, $\lim_{t \rightarrow t_0} \mathbf{r}_1(t) = \mathbf{A}$, and $\lim_{t \rightarrow t_0} \mathbf{r}_2(t) = \mathbf{B}$. Use the determinant formula for cross products and the Limit Product Rule for scalar functions to show that

$$\lim_{t \rightarrow t_0} (\mathbf{r}_1(t) \times \mathbf{r}_2(t)) = \mathbf{A} \times \mathbf{B}.$$

47. Differentiable vector functions are continuous Show that if $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ is differentiable at $t = t_0$, then it is continuous at t_0 as well.

48. Constant Function Rule Prove that if $\mathbf{u}$ is the vector function with the constant value $\mathbf{C}$, then $d\mathbf{u}/dt = \mathbf{0}$.

COMPUTER EXPLORATIONS

Use a CAS to perform the following steps in Exercises 49–52.

- Plot the space curve traced out by the position vector $\mathbf{r}$.
 - Find the components of the velocity vector $d\mathbf{r}/dt$.
 - Evaluate $d\mathbf{r}/dt$ at the given point t_0 and determine the equation of the tangent line to the curve at $\mathbf{r}(t_0)$.
 - Plot the tangent line together with the curve over the given interval.
- 49.** $\mathbf{r}(t) = (\sin t - t \cos t)\mathbf{i} + (\cos t + t \sin t)\mathbf{j} + t^2\mathbf{k}$,
 $0 \leq t \leq 6\pi$, $t_0 = 3\pi/2$
- 50.** $\mathbf{r}(t) = \sqrt{2}t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$, $-2 \leq t \leq 3$, $t_0 = 1$
- 51.** $\mathbf{r}(t) = (\sin 2t)\mathbf{i} + (\ln(1+t))\mathbf{j} + t\mathbf{k}$, $0 \leq t \leq 4\pi$,
 $t_0 = \pi/4$

52. $\mathbf{r}(t) = (\ln(t^2 + 2))\mathbf{i} + (\arctan 3t)\mathbf{j} + \sqrt{t^2 + 1}\mathbf{k}$,
 $-3 \leq t \leq 5$, $t_0 = 3$

In Exercises 53 and 54, you will explore graphically the behavior of the helix

$$\mathbf{r}(t) = (\cos at)\mathbf{i} + (\sin at)\mathbf{j} + btk$$

as you change the values of the constants a and b . Use a CAS to perform the steps in each exercise.

- 53.** Set $b = 1$. Plot the helix $\mathbf{r}(t)$ together with the tangent line to the curve at $t = 3\pi/2$ for $a = 1, 2, 4$, and 6 over the interval $0 \leq t \leq 4\pi$. Describe in your own words what happens to the graph of the helix and the position of the tangent line as a increases through these positive values.
- 54.** Set $a = 1$. Plot the helix $\mathbf{r}(t)$ together with the tangent line to the curve at $t = 3\pi/2$ for $b = 1/4, 1/2, 2$, and 4 over the interval $0 \leq t \leq 4\pi$. Describe in your own words what happens to the graph of the helix and the position of the tangent line as b increases through these positive values.

12.2 Integrals of Vector Functions; Projectile Motion

In this section we investigate integrals of vector functions and their application to motion along a path in space or in the plane.

Integrals of Vector Functions

A differentiable vector function $\mathbf{R}(t)$ is an **antiderivative** of a vector function $\mathbf{r}(t)$ on an interval I if $d\mathbf{R}/dt = \mathbf{r}$ at each point of I . If $\mathbf{R}$ is an antiderivative of $\mathbf{r}$ on I , it can be shown, working one component at a time, that every antiderivative of $\mathbf{r}$ on I has the form $\mathbf{R} + \mathbf{C}$ for some constant vector $\mathbf{C}$ (Exercise 45). The set of all antiderivatives of $\mathbf{r}$ on I is the **indefinite integral** of $\mathbf{r}$ on I .

DEFINITION The **indefinite integral** of $\mathbf{r}$ with respect to t is the set of all antiderivatives of $\mathbf{r}$, denoted by

$$\int \mathbf{r}(t) dt.$$

The usual arithmetic rules for indefinite integrals apply.

EXAMPLE 1 To integrate a vector function, we integrate each of its components.

$$\int ((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k}) dt = \left(\int \cos t dt \right) \mathbf{i} + \left(\int dt \right) \mathbf{j} - \left(\int 2t dt \right) \mathbf{k} \quad (1)$$

$$= (\sin t + C_1)\mathbf{i} + (t + C_2)\mathbf{j} - (t^2 + C_3)\mathbf{k} \quad (2)$$

$$= (\sin t)\mathbf{i} + t\mathbf{j} - t^2\mathbf{k} + \mathbf{C} \quad \mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} - C_3\mathbf{k}$$

As in the integration of scalar functions, we recommend that you skip the steps in Equations (1) and (2) and go directly to the final form. Find an antiderivative for each component and add a *constant vector* at the end. ■

Definite integrals of vector functions are best defined in terms of components. The definition is consistent with how we compute limits and derivatives of vector functions.

DEFINITION If the components of $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ are integrable over $[a, b]$, then so is $\mathbf{r}$, and the **definite integral** of $\mathbf{r}$ from a to b is

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}.$$

EXAMPLE 2 As in Example 1, we integrate each component.

$$\begin{aligned} \int_0^\pi ((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k}) dt &= \left(\int_0^\pi \cos t dt \right) \mathbf{i} + \left(\int_0^\pi dt \right) \mathbf{j} - \left(\int_0^\pi 2t dt \right) \mathbf{k} \\ &= \left[\sin t \right]_0^\pi \mathbf{i} + \left[t \right]_0^\pi \mathbf{j} - \left[t^2 \right]_0^\pi \mathbf{k} \\ &= [0 - 0]\mathbf{i} + [\pi - 0]\mathbf{j} - [\pi^2 - 0^2]\mathbf{k} \\ &= \pi\mathbf{j} - \pi^2\mathbf{k} \end{aligned}$$

The Fundamental Theorem of Calculus for continuous vector functions says that

$$\int_a^b \mathbf{r}(t) dt = \mathbf{R}(t) \Big|_a^b = \mathbf{R}(b) - \mathbf{R}(a),$$

where $\mathbf{R}$ is any antiderivative of $\mathbf{r}$, so that $\mathbf{R}'(t) = \mathbf{r}(t)$ (Exercise 46). Notice that an antiderivative of a vector function is also a vector function, whereas a definite integral of a vector function is a single constant vector.

EXAMPLE 3 Suppose we do not know the path of a hang glider, but only its acceleration vector $\mathbf{a}(t) = -(3 \cos t)\mathbf{i} - (3 \sin t)\mathbf{j} + 2\mathbf{k}$. We also know that initially (at time $t = 0$) the glider departed from the point $(4, 0, 0)$ with velocity $\mathbf{v}(0) = 3\mathbf{j}$. Find the glider's position as a function of t .

Solution Our goal is to find $\mathbf{r}(t)$ knowing

The differential equation: $\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2} = -(3 \cos t)\mathbf{i} - (3 \sin t)\mathbf{j} + 2\mathbf{k}$

The initial conditions: $\mathbf{v}(0) = 3\mathbf{j}$ and $\mathbf{r}(0) = 4\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$.

Integrating both sides of the differential equation with respect to t gives

$$\mathbf{v}(t) = -(3 \sin t)\mathbf{i} + (3 \cos t)\mathbf{j} + 2t\mathbf{k} + \mathbf{C}_1.$$

We use $\mathbf{v}(0) = 3\mathbf{j}$ to find $\mathbf{C}_1$:

$$3\mathbf{j} = -(3 \sin 0)\mathbf{i} + (3 \cos 0)\mathbf{j} + (0)\mathbf{k} + \mathbf{C}_1$$

$$3\mathbf{j} = 3\mathbf{j} + \mathbf{C}_1$$

$$\mathbf{C}_1 = \mathbf{0}.$$

The glider's velocity as a function of time is

$$\frac{d\mathbf{r}}{dt} = \mathbf{v}(t) = -(3 \sin t)\mathbf{i} + (3 \cos t)\mathbf{j} + 2t\mathbf{k}.$$

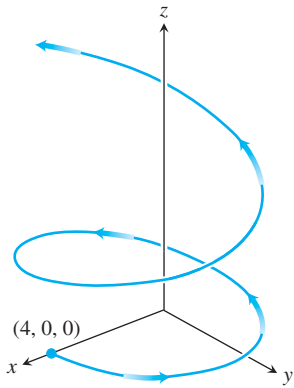


FIGURE 12.9 The path of the hang glider in Example 3. Although the path spirals around the z -axis, it is not a helix.

Integrating both sides of this last differential equation gives

$$\mathbf{r}(t) = (3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j} + t^2\mathbf{k} + \mathbf{C}_2.$$

We then use the initial condition $\mathbf{r}(0) = 4\mathbf{i}$ to find $\mathbf{C}_2$:

$$4\mathbf{i} = (3 \cos 0)\mathbf{i} + (3 \sin 0)\mathbf{j} + (0^2)\mathbf{k} + \mathbf{C}_2$$

$$4\mathbf{i} = 3\mathbf{i} + (0)\mathbf{j} + (0)\mathbf{k} + \mathbf{C}_2$$

$$\mathbf{C}_2 = \mathbf{i}.$$

The glider's position as a function of t is

$$\mathbf{r}(t) = (1 + 3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j} + t^2\mathbf{k}.$$

This is the path of the glider shown in Figure 12.9. Although the path resembles that of a helix due to its spiraling nature around the z -axis, it is not a helix because of the way it is rising. (We say more about this in Section 12.5.)

The Vector and Parametric Equations for Ideal Projectile Motion

A classic example of integrating vector functions is the derivation of the equations for the motion of a projectile. In physics, projectile motion describes how an object fired at some angle from an initial position, and acted upon by only the force of gravity, moves in a vertical coordinate plane. In the classic example, we ignore the effects of any frictional drag on the object, which may vary with its speed and altitude, and also the fact that the force of gravity changes slightly with the projectile's changing height. In addition, we ignore the long-distance effects of Earth turning beneath the projectile, such as in a rocket launch or the firing of a projectile from a cannon. Ignoring these effects gives us a reasonable approximation of the motion in most cases.

To derive equations for projectile motion, we assume that the projectile behaves like a particle moving in a vertical coordinate plane and that the only force acting on the projectile during its flight is the constant force of gravity, which always points straight down. The magnitude of the gravitational acceleration g is approximately 9.8 m/s^2 at sea level. We assume that the projectile is launched from the origin at time $t = 0$ into the first quadrant with an initial velocity $\mathbf{v}_0$ (Figure 12.10). If $\mathbf{v}_0$ makes an angle α with the horizontal, then

$$\mathbf{v}_0 = (|\mathbf{v}_0| \cos \alpha)\mathbf{i} + (|\mathbf{v}_0| \sin \alpha)\mathbf{j}.$$

If we use the simpler notation v_0 for the initial speed $|\mathbf{v}_0|$, then

$$\mathbf{v}_0 = (v_0 \cos \alpha)\mathbf{i} + (v_0 \sin \alpha)\mathbf{j}. \quad (3)$$

The projectile's initial position is

$$\mathbf{r}_0 = 0\mathbf{i} + 0\mathbf{j} = \mathbf{0}. \quad (4)$$

Newton's second law of motion says that the force acting on the projectile is equal to the projectile's mass m times its acceleration, or $m(d^2\mathbf{r}/dt^2)$, if $\mathbf{r}$ is the projectile's position vector and t is time. If the force is solely the gravitational force $-mg\mathbf{j}$, then

$$m \frac{d^2\mathbf{r}}{dt^2} = -mg\mathbf{j} \quad \text{and} \quad \frac{d^2\mathbf{r}}{dt^2} = -g\mathbf{j},$$

where g is the acceleration due to gravity. We find $\mathbf{r}$ as a function of t by solving the following initial value problem.

$$\text{Differential equation:} \quad \frac{d^2\mathbf{r}}{dt^2} = -g\mathbf{j}$$

$$\text{Initial conditions:} \quad \mathbf{r} = \mathbf{r}_0 \quad \text{and} \quad \frac{d\mathbf{r}}{dt} = \mathbf{v}_0 \quad \text{when } t = 0$$

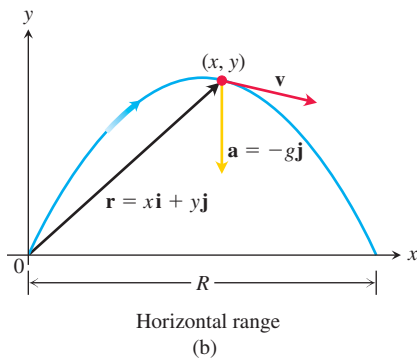
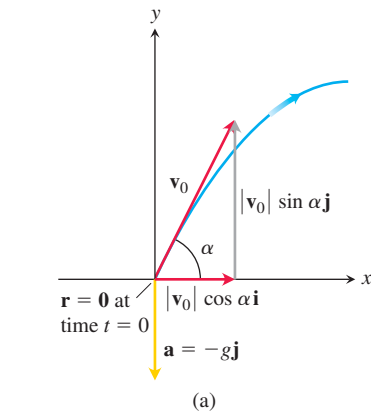


FIGURE 12.10 (a) Position, velocity, acceleration, and launch angle at $t = 0$. (b) Position, velocity, and acceleration at a later time t .

The first integration gives

$$\frac{d\mathbf{r}}{dt} = -(gt)\mathbf{j} + \mathbf{v}_0.$$

A second integration gives

$$\mathbf{r} = -\frac{1}{2}gt^2\mathbf{j} + \mathbf{v}_0t + \mathbf{r}_0.$$

Substituting the values of $\mathbf{v}_0$ and $\mathbf{r}_0$ from Equations (3) and (4) gives

$$\mathbf{r} = -\frac{1}{2}gt^2\mathbf{j} + \underbrace{(v_0 \cos \alpha)t\mathbf{i} + (v_0 \sin \alpha)t\mathbf{j}}_{\mathbf{v}_0t} + \mathbf{0}.$$

Collecting terms, we obtain the following.

Ideal Projectile Motion Equation

$$\mathbf{r} = (v_0 \cos \alpha)t\mathbf{i} + \left((v_0 \sin \alpha)t - \frac{1}{2}gt^2\right)\mathbf{j}. \quad (5)$$

Equation (5) is the *vector equation* of the path for ideal projectile motion. The angle α is the projectile's **launch angle (firing angle, angle of elevation)**, and v_0 , as we said before, is the projectile's **initial speed**. The components of $\mathbf{r}$ give the parametric equations

$$x = (v_0 \cos \alpha)t \quad \text{and} \quad y = (v_0 \sin \alpha)t - \frac{1}{2}gt^2, \quad (6)$$

where x is the distance downrange and y is the height of the projectile at time $t \geq 0$.

EXAMPLE 4 A projectile is fired from the origin over horizontal ground at an initial speed of 500 m/s and a launch angle of 60° . Where will the projectile be 10 s later?

Solution We use Equation (5) with $v_0 = 500$, $\alpha = 60^\circ$, $g = 9.8$, and $t = 10$ to find the projectile's components 10 s after firing.

$$\begin{aligned} \mathbf{r} &= (v_0 \cos \alpha)t\mathbf{i} + \left((v_0 \sin \alpha)t - \frac{1}{2}gt^2\right)\mathbf{j} \\ &= (500)\left(\frac{1}{2}\right)(10)\mathbf{i} + \left((500)\left(\frac{\sqrt{3}}{2}\right)10 - \left(\frac{1}{2}\right)(9.8)(100)\right)\mathbf{j} \\ &\approx 2500\mathbf{i} + 3840\mathbf{j} \end{aligned}$$

Ten seconds after firing, the projectile is about 3840 m above ground and 2500 m downrange from the origin. ■

Ideal projectiles move along parabolas, as we now deduce from Equations (6). If we substitute $t = x/(v_0 \cos \alpha)$ from the first equation into the second, we obtain the Cartesian coordinate equation

$$y = -\left(\frac{g}{2v_0^2 \cos^2 \alpha}\right)x^2 + (\tan \alpha)x.$$

This equation has the form $y = ax^2 + bx$, so its graph is a parabola.

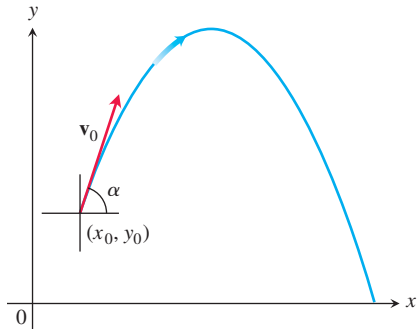


FIGURE 12.11 The path of a projectile fired from (x_0, y_0) with an initial velocity $\mathbf{v}_0$ at an angle of α degrees with the horizontal.

A projectile reaches its highest point when its vertical velocity component is zero. When fired over horizontal ground, the projectile lands when its vertical component equals zero in Equation (5), and the **range** R is the distance from the origin to the point of impact. We summarize the results here, which you are asked to verify in Exercise 31.

Height, Flight Time, and Range for Ideal Projectile Motion

For ideal projectile motion when an object is launched from the origin over a horizontal surface with initial speed v_0 and launch angle α :

$$\text{Maximum height:} \quad y_{\max} = \frac{(v_0 \sin \alpha)^2}{2g}$$

$$\text{Flight time:} \quad t = \frac{2v_0 \sin \alpha}{g}$$

$$\text{Range:} \quad R = \frac{v_0^2}{g} \sin 2\alpha$$

If we fire our ideal projectile from the point (x_0, y_0) instead of the origin (Figure 12.11), the position vector for the path of motion is

$$\mathbf{r} = (x_0 + (v_0 \cos \alpha)t)\mathbf{i} + \left(y_0 + (v_0 \sin \alpha)t - \frac{1}{2}gt^2\right)\mathbf{j}, \quad (7)$$

as you are asked to show in Exercise 33.

Projectile Motion with Wind Gusts

The next example shows how to account for another force acting on a projectile due to a gust of wind. We assume that the path of the baseball in Example 5 lies in a vertical plane.

EXAMPLE 5 A baseball is hit when it is 1 m above the ground. It leaves the bat with initial speed of 50 m/s, making an angle of 20° with the horizontal. At the instant the ball is hit, an instantaneous gust of wind blows in the horizontal direction directly opposite the direction the ball is taking toward the outfield, adding a component of $-2.5\mathbf{i}$ (m/s) to the ball's initial velocity ($2.5 \text{ m/s} = 9 \text{ km/h}$).

- Find a vector equation (position vector) for the path of the baseball.
- How high does the baseball go, and when does it reach maximum height?
- Assuming that the ball is not caught, find its range and flight time.

Solution

- Using Equation (3) and accounting for the gust of wind, the initial velocity of the baseball is

$$\begin{aligned} \mathbf{v}_0 &= (v_0 \cos \alpha)\mathbf{i} + (v_0 \sin \alpha)\mathbf{j} - 2.5\mathbf{i} \\ &= (50 \cos 20^\circ)\mathbf{i} + (50 \sin 20^\circ)\mathbf{j} - (2.5)\mathbf{i} \\ &= (50 \cos 20^\circ - 2.5)\mathbf{i} + (50 \sin 20^\circ)\mathbf{j}. \end{aligned}$$

The initial position is $\mathbf{r}_0 = 0\mathbf{i} + 1\mathbf{j}$. Integration of $d^2\mathbf{r}/dt^2 = -g\mathbf{j}$ gives

$$\frac{d\mathbf{r}}{dt} = -(gt)\mathbf{j} + \mathbf{v}_0.$$

A second integration gives

$$\mathbf{r} = -\frac{1}{2}gt^2\mathbf{j} + \mathbf{v}_0t + \mathbf{r}_0.$$

Substituting the values of $\mathbf{v}_0$ and $\mathbf{r}_0$ into the last equation gives the position vector of the baseball.

$$\begin{aligned}\mathbf{r} &= -\frac{1}{2}gt^2\mathbf{j} + \mathbf{v}_0t + \mathbf{r}_0 \\ &= -4.9t^2\mathbf{j} + (50 \cos 20^\circ - 2.5)t\mathbf{i} + (50 \sin 20^\circ)t\mathbf{j} + 1\mathbf{j} \\ &= (50 \cos 20^\circ - 2.5)t\mathbf{i} + (1 + (50 \sin 20^\circ)t - 4.9t^2)\mathbf{j}.\end{aligned}$$

- (b) The baseball reaches its highest point when the vertical component of velocity is zero, or

$$\frac{dy}{dt} = 50 \sin 20^\circ - 9.8t = 0.$$

Solving for t we find

$$t = \frac{50 \sin 20^\circ}{9.8} \approx 1.75 \text{ s}.$$

Substituting this time into the vertical component for $\mathbf{r}$ gives the maximum height

$$y_{\max} = 1 + (50 \sin 20^\circ)(1.75) - 4.9(1.75)^2 \approx 15.9 \text{ m}.$$

That is, the maximum height of the baseball is about 15.9 m, reached about 1.75 s after leaving the bat.

- (c) To find when the baseball lands, we set the vertical component for $\mathbf{r}$ equal to 0 and solve for t :

$$\begin{aligned}1 + (50 \sin 20^\circ)t - 4.9t^2 &= 0 \\ 1 + (17.1)t - 4.9t^2 &= 0.\end{aligned}$$

The solution values are about $t = 3.55 \text{ s}$ and $t = -0.06 \text{ s}$. Substituting the positive time into the horizontal component for $\mathbf{r}$, we find the range

$$R = (50 \cos 20^\circ - 2.5)(3.55) \approx 157.8 \text{ m}.$$

Thus, the horizontal range is about 157.8 m, and the flight time is about 3.55 s. ■

In Exercises 41 and 42, we consider projectile motion when there is air resistance slowing down the flight.

EXERCISES 12.2

Integrating Vector-Valued Functions

Evaluate the integrals in Exercises 1–10.

1. $\int_0^1 [t^3\mathbf{i} + 7t\mathbf{j} + (t+1)\mathbf{k}] dt$

2. $\int_1^2 \left[(6-6t)\mathbf{i} + 3\sqrt{t}\mathbf{j} + \left(\frac{4}{t^2}\right)\mathbf{k} \right] dt$

3. $\int_{-\pi/4}^{\pi/4} [(\sin t)\mathbf{i} + (1+\cos t)\mathbf{j} + (\sec^2 t)\mathbf{k}] dt$

4. $\int_0^{\pi/3} [(\sec t \tan t)\mathbf{i} + (\tan t)\mathbf{j} + (2 \sin t \cos t)\mathbf{k}] dt$

5. $\int_1^4 \left[\frac{1}{t}\mathbf{i} + \frac{1}{5-t}\mathbf{j} + \frac{1}{2t}\mathbf{k} \right] dt$

6. $\int_0^1 \left[\frac{2}{\sqrt{1-t^2}}\mathbf{i} + \frac{\sqrt{3}}{1+t^2}\mathbf{k} \right] dt$

7. $\int_0^1 [te^{t^2}\mathbf{i} + e^{-t}\mathbf{j} + \mathbf{k}] dt$

8. $\int_1^{\ln 3} [te^t\mathbf{i} + e^t\mathbf{j} + \ln t\mathbf{k}] dt$

9. $\int_0^{\pi/2} [\cos t\mathbf{i} - \sin 2t\mathbf{j} + \sin^2 t\mathbf{k}] dt$

10. $\int_0^{\pi/4} [\sec t\mathbf{i} + \tan^2 t\mathbf{j} - t \sin t\mathbf{k}] dt$

Initial Value Problems

Solve the initial value problems in Exercises 11–20 for $\mathbf{r}$ as a vector function of t .

11. Differential equation: $\frac{d\mathbf{r}}{dt} = -t\mathbf{i} - t\mathbf{j} - t\mathbf{k}$
Initial condition: $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$
12. Differential equation: $\frac{d\mathbf{r}}{dt} = (180t)\mathbf{i} + (180t - 16t^2)\mathbf{j}$
Initial condition: $\mathbf{r}(0) = 100\mathbf{j}$
13. Differential equation: $\frac{d\mathbf{r}}{dt} = \frac{3}{2}(t+1)^{1/2}\mathbf{i} + e^{-t}\mathbf{j} + \frac{1}{t+1}\mathbf{k}$
Initial condition: $\mathbf{r}(0) = \mathbf{k}$
14. Differential equation: $\frac{d\mathbf{r}}{dt} = (t^3 + 4t)\mathbf{i} + t\mathbf{j} + 2t^2\mathbf{k}$
Initial condition: $\mathbf{r}(0) = \mathbf{i} + \mathbf{j}$
15. Differential equation:
 $\frac{d\mathbf{r}}{dt} = (\tan t)\mathbf{i} + \left(\cos\left(\frac{1}{2}t\right)\right)\mathbf{j} - (\sec 2t)\mathbf{k}, -\frac{\pi}{4} < t < \frac{\pi}{4}$
Initial condition: $\mathbf{r}(0) = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$
16. Differential equation:
 $\frac{d\mathbf{r}}{dt} = \left(\frac{t}{t^2 + 2}\right)\mathbf{i} - \left(\frac{t^2 + 1}{t - 2}\right)\mathbf{j} + \left(\frac{t^2 + 4}{t^2 + 3}\right)\mathbf{k}, t < 2$
Initial condition: $\mathbf{r}(0) = \mathbf{i} - \mathbf{j} + \mathbf{k}$
17. Differential equation: $\frac{d^2\mathbf{r}}{dt^2} = -32\mathbf{k}$
Initial conditions: $\mathbf{r}(0) = 100\mathbf{k}$ and $\left.\frac{d\mathbf{r}}{dt}\right|_{t=0} = 8\mathbf{i} + 8\mathbf{j}$
18. Differential equation: $\frac{d^2\mathbf{r}}{dt^2} = -(\mathbf{i} + \mathbf{j} + \mathbf{k})$
Initial conditions: $\mathbf{r}(0) = 10\mathbf{i} + 10\mathbf{j} + 10\mathbf{k}$ and $\left.\frac{d\mathbf{r}}{dt}\right|_{t=0} = \mathbf{0}$
19. Differential equation: $\frac{d^2\mathbf{r}}{dt^2} = e^t\mathbf{i} - e^{-t}\mathbf{j} + 4e^{2t}\mathbf{k}$
Initial conditions: $\mathbf{r}(0) = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\left.\frac{d\mathbf{r}}{dt}\right|_{t=0} = -\mathbf{i} + 4\mathbf{j}$
20. Differential equation:
 $\frac{d^2\mathbf{r}}{dt^2} = (\sin t)\mathbf{i} - (\cos t)\mathbf{j} + (4 \sin t \cos t)\mathbf{k}$
Initial conditions: $\mathbf{r}(0) = \mathbf{i} - \mathbf{k}$ and $\left.\frac{d\mathbf{r}}{dt}\right|_{t=0} = \mathbf{i}$

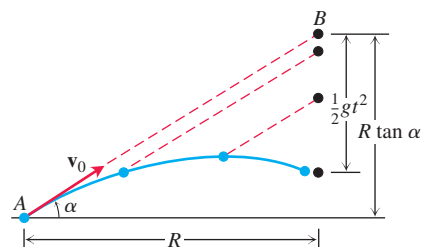
Motion Along a Straight Line

21. At time $t = 0$, a particle is located at the point $(1, 2, 3)$. It travels in a straight line to the point $(4, 1, 4)$, has speed 2 at $(1, 2, 3)$, and has constant acceleration $3\mathbf{i} - \mathbf{j} + \mathbf{k}$. Find an equation for the position vector $\mathbf{r}(t)$ of the particle at time t .
22. A particle traveling in a straight line is located at the point $(1, -1, 2)$ and has speed 2 at time $t = 0$. The particle moves toward the point $(3, 0, 3)$ with constant acceleration $2\mathbf{i} + \mathbf{j} + \mathbf{k}$. Find its position vector $\mathbf{r}(t)$ at time t .

Projectile Motion

Projectile flights in Exercises 23–40 are to be treated as ideal unless stated otherwise. All launch angles are assumed to be measured from the horizontal. All projectiles are assumed to be launched from the origin over a horizontal surface unless stated otherwise. For some exercises, a calculator may be helpful.

23. **Travel time** A projectile is fired at a speed of 840 m/s at an angle of 60° . How long will it take to get 21 km downrange?
24. **Range and height versus speed**
 - a. Show that doubling a projectile's initial speed at a given launch angle multiplies its range by 4.
 - b. By about what percentage should you increase the initial speed to double the height and range?
25. **Flight time and height** A projectile is fired with an initial speed of 500 m/s at an angle of elevation of 45° .
 - a. When and how far away will the projectile strike?
 - b. How high overhead will the projectile be when it is 5 km downrange?
 - c. What is the greatest height reached by the projectile?
26. **Throwing a baseball** A baseball is thrown from the stands 9.8 m above the field at an angle of 30° up from the horizontal. When and how far away will the ball strike the ground if its initial speed is 9.8 m/s?
27. **Firing golf balls** A spring gun at ground level fires a golf ball at an angle of 45° . The ball lands 10 m away.
 - a. What was the ball's initial speed?
 - b. For the same initial speed, find the two firing angles that make the range 6 m.
28. **Beaming electrons** An electron in a cathode-ray tube (CRT) is beamed horizontally at a speed of 5×10^6 m/s toward the face of the tube 40 cm away. About how far will the electron drop before it hits?
29. **Equal-range firing angles** What two angles of elevation will enable a projectile to reach a target 16 km downrange on the same level as the gun if the projectile's initial speed is 400 m/s?
30. **Finding muzzle speed** Find the muzzle speed of a gun whose maximum range is 24.5 km.
31. Verify the results given in the text (following Example 4) for the maximum height, flight time, and range for ideal projectile motion.
32. **Colliding marbles** The accompanying figure shows an experiment with two marbles. Marble A was launched toward marble B with launch angle α and initial speed v_0 . At the same instant, marble B was released to fall from rest at $R \tan \alpha$ units directly above a spot R units downrange from A. The marbles were found to collide regardless of the value of v_0 . Was this mere coincidence, or must this happen? Give reasons for your answer.



33. Firing from (x_0, y_0) Derive the equations

$$x = x_0 + (v_0 \cos \alpha)t,$$

$$y = y_0 + (v_0 \sin \alpha)t - \frac{1}{2}gt^2$$

(see Equation (7) in the text) by solving the following initial value problem for a vector $\mathbf{r}$ in the plane.

Differential equation: $\frac{d^2 \mathbf{r}}{dt^2} = -g\mathbf{j}$

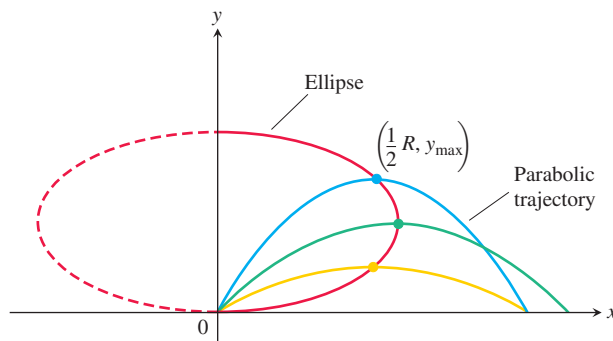
Initial conditions: $\mathbf{r}(0) = x_0\mathbf{i} + y_0\mathbf{j}$

$$\frac{d\mathbf{r}}{dt}(0) = (v_0 \cos \alpha)\mathbf{i} + (v_0 \sin \alpha)\mathbf{j}$$

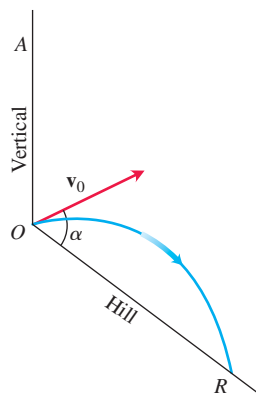
34. Where trajectories crest For a projectile fired from the ground at launch angle α with initial speed v_0 , consider α as a variable and v_0 as a fixed constant. For each α , $0 < \alpha < \pi/2$, we obtain a parabolic trajectory as shown in the accompanying figure. Show that the points in the plane that give the maximum heights of these parabolic trajectories all lie on the ellipse

$$x^2 + 4\left(y - \frac{v_0^2}{4g}\right)^2 = \frac{v_0^4}{4g^2},$$

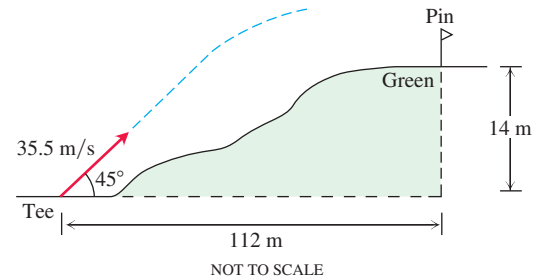
where $x \geq 0$.

**35. Launching downhill** An ideal projectile is launched straight down an inclined plane as shown in the accompanying figure.

- Show that the greatest downhill range is achieved when the initial velocity vector bisects angle AOR .
- If the projectile were fired uphill instead of down, what launch angle would maximize its range? Give reasons for your answer.

**36. Elevated green** A golf ball is hit with an initial speed of 35.5 m/s at an angle of elevation of 45° from the tee to a green

that is elevated 14 m above the tee as shown in the diagram. Assuming that the pin, 112 m downrange, does not get in the way, where will the ball land in relation to the pin?

**37. Volleyball** A volleyball is hit when it is 1.3 m above the ground and 4 m from a 2-m-high net. It leaves the point of impact with an initial velocity of 12 m/s at an angle of 27° and slips by the opposing team untouched.

- Find a vector equation for the path of the volleyball.
- How high does the volleyball go, and when does it reach maximum height?
- Find its range and flight time.
- When is the volleyball 2.3 m above the ground? How far (ground distance) is the volleyball from where it will land?
- Suppose that the net is raised to 2.5 m. Does this change things? Explain.

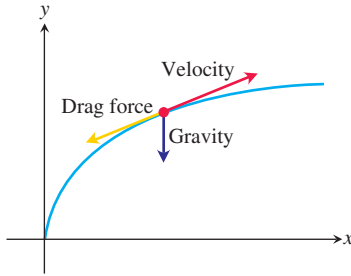
38. Shot put In Moscow in 1987, Natalya Lisouskaya set a women's world record by putting a 4kg shot 22.63 m. Assuming that she launched the shot at a 40° angle to the horizontal from 2 m above the ground, what was the shot's initial speed?**39.** A child throws a ball with an initial speed of 18 m/s at an angle of elevation of 60° toward a tall building that is 7 m from the child. If the child's hand is 1.6 m from the ground, show that the ball hits the building, and find the height above the ground of the point where the ball hits the building.**40. Hitting a baseball under a wind gust** A baseball is hit when it is 0.8 m above the ground. It leaves the bat with an initial velocity of 40 m/s at a launch angle of 23° . At the instant the ball is hit, an instantaneous gust of wind blows against the ball, adding a component of $-4\mathbf{i}$ (m/s) to the ball's initial velocity. A 5-m-high fence lies 90 m from home plate in the direction of the flight.

- Find a vector equation for the path of the baseball.
- How high does the baseball go, and when does it reach maximum height?
- Find the range and flight time of the baseball, assuming that the ball is not caught.
- When is the baseball 6 m high? How far (ground distance) is the baseball from home plate at that height?
- Has the batter hit a home run? Explain.

Projectile Motion with Linear Drag

The main force affecting the motion of a projectile, other than gravity, is air resistance. This slowing down force is **drag force**, and it acts in a direction *opposite* to the velocity of the projectile (see accompanying figure). For projectiles moving through the air at relatively low speeds,

however, the drag force is (very nearly) proportional to the speed (to the first power) and so is called **linear**.



41. Linear drag

Derive the equations

$$x = \frac{v_0}{k}(1 - e^{-kt}) \cos \alpha$$

$$y = \frac{v_0}{k}(1 - e^{-kt})(\sin \alpha) + \frac{g}{k^2}(1 - kt - e^{-kt})$$

by solving the following initial value problem for a vector $\mathbf{r}$ in the plane.

Differential equation: $\frac{d^2 \mathbf{r}}{dt^2} = -g\mathbf{j} - k\mathbf{v} = -g\mathbf{j} - k \frac{d\mathbf{r}}{dt}$

Initial conditions: $\mathbf{r}(0) = \mathbf{0}$

$$\left. \frac{d\mathbf{r}}{dt} \right|_{t=0} = \mathbf{v}_0 = (v_0 \cos \alpha)\mathbf{i} + (v_0 \sin \alpha)\mathbf{j}$$

The **drag coefficient** k is a positive constant representing resistance due to air density, v_0 and α are the projectile's initial speed and launch angle, and g is the acceleration of gravity.

42. Hitting a baseball with linear drag

Consider the baseball problem in Example 5 when there is linear drag (see Exercise 41). Assume a drag coefficient $k = 0.12$, but no gust of wind.

- From Exercise 41, find a vector form for the path of the baseball.
- How high does the baseball go, and when does it reach maximum height?
- Find the range and flight time of the baseball.
- When is the baseball 9 m high? How far (ground distance) is the baseball from home plate at that height?
- A 3-m-high outfield fence is 115 m from home plate in the direction of the flight of the baseball. The outfielder can jump and catch any ball up to 3.3 m off the ground to stop it from going over the fence. Has the batter hit a home run?

Theory and Examples

43. Establish the following properties of integrable vector functions.

- The **Constant Scalar Multiple Rule**:

$$\int_a^b k\mathbf{r}(t) dt = k \int_a^b \mathbf{r}(t) dt \quad (\text{any scalar } k)$$

The **Rule for Negatives**,

$$\int_a^b (-\mathbf{r}(t)) dt = -\int_a^b \mathbf{r}(t) dt,$$

is obtained by taking $k = -1$.

- The **Sum and Difference Rules**:

$$\int_a^b (\mathbf{r}_1(t) \pm \mathbf{r}_2(t)) dt = \int_a^b \mathbf{r}_1(t) dt \pm \int_a^b \mathbf{r}_2(t) dt$$

- The **Constant Vector Multiple Rules**:

$$\int_a^b \mathbf{C} \cdot \mathbf{r}(t) dt = \mathbf{C} \cdot \int_a^b \mathbf{r}(t) dt \quad (\text{any constant vector } \mathbf{C})$$

and

$$\int_a^b \mathbf{C} \times \mathbf{r}(t) dt = \mathbf{C} \times \int_a^b \mathbf{r}(t) dt \quad (\text{any constant vector } \mathbf{C})$$

44. Products of scalar and vector functions

Suppose that the scalar function $u(t)$ and the vector function $\mathbf{r}(t)$ are both defined for $a \leq t \leq b$.

- Show that $u\mathbf{r}$ is continuous on $[a, b]$ if u and $\mathbf{r}$ are continuous on $[a, b]$.
- If u and $\mathbf{r}$ are both differentiable on $[a, b]$, show that $u\mathbf{r}$ is differentiable on $[a, b]$ and that

$$\frac{d}{dt}(u\mathbf{r}) = u \frac{d\mathbf{r}}{dt} + \frac{du}{dt}\mathbf{r}.$$

45. Antiderivatives of vector functions

- Use Corollary 2 of the Mean Value Theorem for scalar functions to show that if two vector functions $\mathbf{R}_1(t)$ and $\mathbf{R}_2(t)$ have identical derivatives on an interval I , then the functions differ by a constant vector value throughout I .
- Use the result in part (a) to show that if $\mathbf{R}(t)$ is any antiderivative of $\mathbf{r}(t)$ on I , then any other antiderivative of $\mathbf{r}$ on I equals $\mathbf{R}(t) + \mathbf{C}$ for some constant vector $\mathbf{C}$.

46. The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus for scalar functions of a real variable holds for vector functions of a real variable as well. Prove this by using the theorem for scalar functions to show first that if a vector function $\mathbf{r}(t)$ is continuous for $a \leq t \leq b$, then

$$\frac{d}{dt} \int_a^t \mathbf{r}(\tau) d\tau = \mathbf{r}(t)$$

at every point t of (a, b) . Then use the conclusion in part (b) of Exercise 45 to show that if $\mathbf{R}$ is any antiderivative of $\mathbf{r}$ on $[a, b]$, then

$$\int_a^b \mathbf{r}(t) dt = \mathbf{R}(b) - \mathbf{R}(a).$$

47. Hitting a baseball with linear drag under a wind gust

Consider again the baseball problem in Example 5. This time, assume a drag coefficient of 0.08 and an instantaneous gust of wind that adds a component of $-5\mathbf{i}$ (m/s) to the initial velocity at the instant the baseball is hit.

- Find a vector equation for the path of the baseball.
- How high does the baseball go, and when does it reach maximum height?
- Find the range and flight time of the baseball.
- When is the baseball 10 m high? How far (ground distance) is the baseball from home plate at that height?
- A 6-m-high outfield fence is 120 m from home plate in the direction of the flight of the baseball. Has the batter hit a home run? If "yes," what change in the horizontal component of the ball's initial velocity would have kept the ball in the park? If "no," what change would have allowed it to be a home run?

48. Height versus time

Show that a projectile attains three-quarters of its maximum height in half the time it takes to reach the maximum height.

12.3 Arc Length in Space

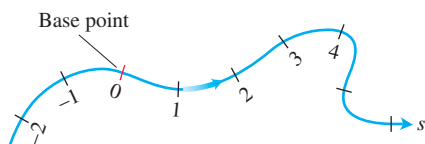


FIGURE 12.12 Smooth curves can be scaled like number lines, the coordinate of each point being its directed distance along the curve from a preselected base point.

In this and the next two sections, we study the mathematical features of a curve's shape that describe the sharpness of its turning and its twisting.

Arc Length Along a Space Curve

One of the features of smooth space and plane curves is that they have a measurable length. This enables us to locate points along these curves by giving their directed distance s along the curve from some base point, the way we locate points on coordinate axes by giving their directed distance from the origin (Figure 12.12). This is what we did for plane curves in Section 10.2.

To measure distance along a smooth curve in space, we add a z -term to the formula we use for curves in the plane.

DEFINITION The **length** of a smooth curve $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \leq t \leq b$, that is traced exactly once as t increases from $t = a$ to $t = b$ is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt. \quad (1)$$

Just as for plane curves, we can calculate the length of a curve in space from any convenient parametrization that meets the stated conditions. We omit the proof.

The square root in Equation (1) is $|\mathbf{v}|$, the length of a velocity vector $d\mathbf{r}/dt$. This enables us to write the formula for length a shorter way.

Arc Length Formula

$$L = \int_a^b |\mathbf{v}| dt \quad (2)$$

EXAMPLE 1 A glider is soaring upward along the helix

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}.$$

How long is the glider's path from $t = 0$ to $t = 2\pi$?

Solution The path segment during this time corresponds to one full turn of the helix (Figure 12.13). The length of this portion of the curve is

$$\begin{aligned} L &= \int_a^b |\mathbf{v}| dt = \int_0^{2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2} dt \\ &= \int_0^{2\pi} \sqrt{2} dt = 2\pi\sqrt{2} \text{ units of length.} \end{aligned}$$

This is $\sqrt{2}$ times the circumference of the circle in the xy -plane over which the helix stands. ■

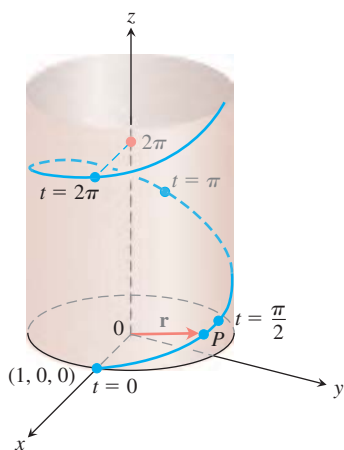


FIGURE 12.13 The helix in Example 1, $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$.

If we choose a base point $P(t_0)$ on a smooth curve C parametrized by t , each value of t determines a point $P(t) = (x(t), y(t), z(t))$ on C and a “directed distance”

$$s(t) = \int_{t_0}^t |\mathbf{v}(\tau)| d\tau,$$

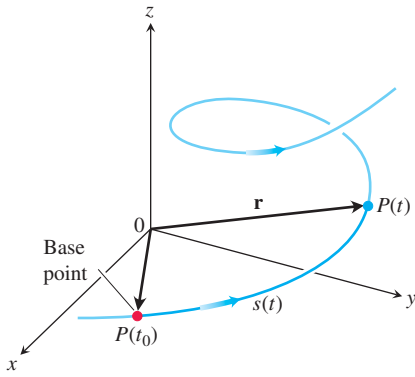


FIGURE 12.14 The directed distance along the curve from $P(t_0)$ to any point $P(t)$ is $s(t) = \int_{t_0}^t |\mathbf{v}(\tau)| d\tau$.

τ is the Greek letter tau (rhymes with “now”)

measured along C from the base point (Figure 12.14). This is the arc length function we defined in Section 10.2 for plane curves that have no z -component. If $t > t_0$, $s(t)$ is the distance along the curve from $P(t_0)$ to $P(t)$. If $t < t_0$, $s(t)$ is the negative of the distance. Each value of s determines a point on C , and this parametrizes C with respect to s . We call s an **arc length parameter** for the curve. The parameter’s value increases in the direction of increasing t . We will see that the arc length parameter is particularly effective for investigating the turning and twisting nature of a space curve.

Arc Length Parameter with Base Point $P(t_0)$

$$s(t) = \int_{t_0}^t \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2} d\tau = \int_{t_0}^t |\mathbf{v}(\tau)| d\tau \quad (3)$$

We use the Greek letter τ (“tau”) as the variable of integration in Equation (3) because the letter t is already in use as the upper limit.

If a curve $\mathbf{r}(t)$ is already given in terms of some parameter t , and $s(t)$ is the arc length function given by Equation (3), then we may be able to solve for t as a function of s : $t = t(s)$. Then the curve can be reparametrized in terms of s by substituting for t : $\mathbf{r} = \mathbf{r}(t(s))$. The new parametrization identifies a point on the curve with its directed distance along the curve from the base point.

EXAMPLE 2 This is an example for which we can actually find the arc length parametrization of a curve. If $t_0 = 0$, then the arc length parameter along the helix

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$$

from t_0 to t is

$$\begin{aligned} s(t) &= \int_{t_0}^t |\mathbf{v}(\tau)| d\tau && \text{Eq. (3)} \\ &= \int_0^t \sqrt{2} d\tau && \text{Value from Example 1} \\ &= \sqrt{2}t. \end{aligned}$$

Solving this equation for t gives $t = s/\sqrt{2}$. Substituting into the position vector $\mathbf{r}$ gives the following arc length parametrization for the helix:

$$\mathbf{r}(t(s)) = \left(\cos \frac{s}{\sqrt{2}}\right)\mathbf{i} + \left(\sin \frac{s}{\sqrt{2}}\right)\mathbf{j} + \frac{s}{\sqrt{2}}\mathbf{k}. \quad \blacksquare$$

Unlike the case that appears in Example 2, the arc length parametrization is generally difficult to find analytically for a curve already given in terms of some other parameter t . Fortunately, however, we rarely need an exact formula for $s(t)$ or its inverse $t(s)$.

HISTORICAL BIOGRAPHY

Josiah Willard Gibbs
(1839–1903)

Gibbs, born in Connecticut, USA, taught at Yale as a professor of mathematics. He made contributions to thermodynamics, electromagnetics, and statistical mechanics. For his foundational work, Gibbs is known as the father of vector analysis.

To know more, visit the companion Website.

Speed on a Smooth Curve

Since the derivatives beneath the radical in Equation (3) are continuous (the curve is smooth), the Fundamental Theorem of Calculus tells us that s is a differentiable function of t with derivative

$$\frac{ds}{dt} = |\mathbf{v}(t)|. \quad (4)$$

Although the base point $P(t_0)$ plays a role in defining s in Equation (3), it plays no role in Equation (4). The rate at which a moving particle covers distance along its path is independent of how far away it is from the base point. Equation (4) says that this rate is the magnitude of $\mathbf{v}$.

Notice that $ds/dt > 0$ since, by definition, $|\mathbf{v}|$ is never zero for a smooth curve. We see once again that s is an increasing function of t .

Unit Tangent Vector

On a smooth curve, we already know that the velocity vector $\mathbf{v} = d\mathbf{r}/dt$ is tangent to the curve $\mathbf{r}(t)$ and that the vector

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

is therefore a unit vector tangent to the curve, called the **unit tangent vector** (Figure 12.15). The unit tangent vector $\mathbf{T}$ for a smooth curve is a differentiable function of t whenever $\mathbf{v}$ is a differentiable function of t . As we will see in Section 12.5, $\mathbf{T}$ is one of three unit vectors in a traveling reference frame that is used to describe the motion of objects traveling in three dimensions.

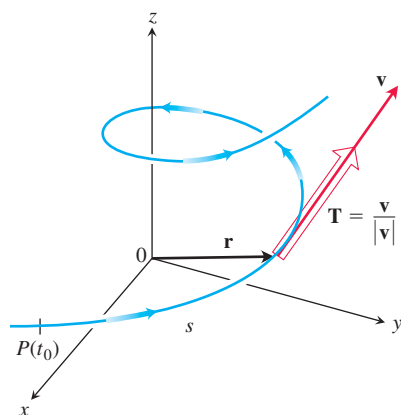


FIGURE 12.15 We find the unit tangent vector $\mathbf{T}$ by dividing $\mathbf{v}$ by its length $|\mathbf{v}|$.

EXAMPLE 3 Find the unit tangent vector of the curve

$$\mathbf{r}(t) = (1 + 3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j} + t^2\mathbf{k}$$

representing the path of the glider in Example 3, Section 12.2.

Solution In that example, we found

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = -(3 \sin t)\mathbf{i} + (3 \cos t)\mathbf{j} + 2t\mathbf{k}$$

and

$$|\mathbf{v}| = \sqrt{9 + 4t^2}.$$

Thus,

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = -\frac{3 \sin t}{\sqrt{9 + 4t^2}}\mathbf{i} + \frac{3 \cos t}{\sqrt{9 + 4t^2}}\mathbf{j} + \frac{2t}{\sqrt{9 + 4t^2}}\mathbf{k}.$$

For the counterclockwise motion

$$\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}$$

around the unit circle, we see that

$$\mathbf{v} = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j}$$

is already a unit vector, so $\mathbf{T} = \mathbf{v}$ and $\mathbf{T}$ is orthogonal to $\mathbf{r}$ (Figure 12.16).

The velocity vector is the change in the position vector $\mathbf{r}$ with respect to time t , but how does the position vector change with respect to arc length? More precisely, what is the derivative $d\mathbf{r}/ds$? Since $ds/dt > 0$ for the curves we are considering, s is one-to-one and has an inverse that gives t as a differentiable function of s (Section 3.8). The derivative of the inverse is

$$\frac{dt}{ds} = \frac{1}{ds/dt} = \frac{1}{|\mathbf{v}|}.$$

This makes $\mathbf{r}$ a differentiable function of s whose derivative can be calculated with the Chain Rule to be

$$\frac{d\mathbf{r}}{ds} = \frac{d\mathbf{r}}{dt} \frac{dt}{ds} = \mathbf{v} \frac{1}{|\mathbf{v}|} = \frac{\mathbf{v}}{|\mathbf{v}|} = \mathbf{T}. \quad (5)$$

This equation says that $d\mathbf{r}/ds$ is the unit tangent vector in the direction of the velocity vector $\mathbf{v}$ (Figure 12.15).

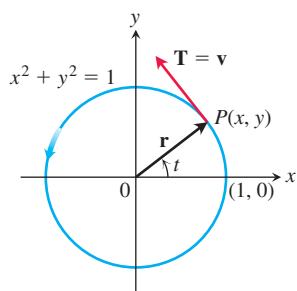


FIGURE 12.16 Counterclockwise motion around the unit circle.

EXERCISES 12.3

Finding Tangent Vectors and Lengths

In Exercises 1–8, find the curve's unit tangent vector. Also, find the length of the indicated portion of the curve.

1. $\mathbf{r}(t) = (2 \cos t)\mathbf{i} + (2 \sin t)\mathbf{j} + \sqrt{5}t\mathbf{k}, \quad 0 \leq t \leq \pi$
2. $\mathbf{r}(t) = (6 \sin 2t)\mathbf{i} + (6 \cos 2t)\mathbf{j} + 5t\mathbf{k}, \quad 0 \leq t \leq \pi$
3. $\mathbf{r}(t) = t\mathbf{i} + (2/3)t^{3/2}\mathbf{k}, \quad 0 \leq t \leq 8$
4. $\mathbf{r}(t) = (2 + t)\mathbf{i} - (t + 1)\mathbf{j} + t\mathbf{k}, \quad 0 \leq t \leq 3$
5. $\mathbf{r}(t) = (\cos^3 t)\mathbf{j} + (\sin^3 t)\mathbf{k}, \quad 0 \leq t \leq \pi/2$
6. $\mathbf{r}(t) = 6t^3\mathbf{i} - 2t^3\mathbf{j} - 3t^3\mathbf{k}, \quad 1 \leq t \leq 2$
7. $\mathbf{r}(t) = (t \cos t)\mathbf{i} + (t \sin t)\mathbf{j} + (2\sqrt{2}/3)t^{3/2}\mathbf{k}, \quad 0 \leq t \leq \pi$
8. $\mathbf{r}(t) = (t \sin t + \cos t)\mathbf{i} + (t \cos t - \sin t)\mathbf{j}, \quad \sqrt{2} \leq t \leq 2$
9. Find the point on the curve

$$\mathbf{r}(t) = (5 \sin t)\mathbf{i} + (5 \cos t)\mathbf{j} + 12t\mathbf{k}$$

at a distance 26π units along the curve from the point $(0, 5, 0)$ in the direction corresponding to increasing t values.

10. Find the point on the curve

$$\mathbf{r}(t) = (12 \sin t)\mathbf{i} - (12 \cos t)\mathbf{j} + 5t\mathbf{k}$$

at a distance 13π units along the curve from the point $(0, -12, 0)$ in the direction corresponding to decreasing t values.

Arc Length Parameter

In Exercises 11–14, find the arc length parameter along the curve from the point where $t = 0$ by evaluating the integral

$$s(t) = \int_0^t |\mathbf{v}(\tau)| d\tau$$

from Equation (3). Then use the formula for $s(t)$ to find the length of the indicated portion of the curve.

11. $\mathbf{r}(t) = (4 \cos t)\mathbf{i} + (4 \sin t)\mathbf{j} + 3t\mathbf{k}, \quad 0 \leq t \leq \pi/2$
12. $\mathbf{r}(t) = (\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}, \quad \pi/2 \leq t \leq \pi$
13. $\mathbf{r}(t) = (e^t \cos t)\mathbf{i} + (e^t \sin t)\mathbf{j} + e^t \mathbf{k}, \quad -\ln 4 \leq t \leq 0$
14. $\mathbf{r}(t) = (1 + 2t)\mathbf{i} + (1 + 3t)\mathbf{j} + (6 - 6t)\mathbf{k}, \quad -1 \leq t \leq 0$

Theory and Examples

15. **Arc length** Find the length of the curve

$$\mathbf{r}(t) = (\sqrt{2}t)\mathbf{i} + (\sqrt{2}t)\mathbf{j} + (1 - t^2)\mathbf{k}$$

from $(0, 0, 1)$ to $(\sqrt{2}, \sqrt{2}, 0)$.

16. **Length of helix** The length $2\pi\sqrt{2}$ of the turn of the helix in Example 1 is also the length of the diagonal of a square 2π units on a side. Show how to obtain this square by cutting away and flattening a portion of the cylinder around which the helix winds.

17. **Ellipse**

- a. Show that the curve $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + (1 - \cos t)\mathbf{k}, \quad 0 \leq t \leq 2\pi$, is an ellipse by showing that it is the intersection of a right circular cylinder and a plane. Find equations for the cylinder and plane.
- b. Sketch the ellipse on the cylinder. Add to your sketch the unit tangent vectors at $t = 0, \pi/2, \pi$, and $3\pi/2$.

- c. Show that the acceleration vector always lies parallel to the plane (orthogonal to a vector normal to the plane). Thus, if you draw the acceleration as a vector attached to the ellipse, it will lie in the plane of the ellipse. Add the acceleration vectors for $t = 0, \pi/2, \pi$, and $3\pi/2$ to your sketch.

- d. Write an integral for the length of the ellipse. Do not try to evaluate the integral; it is nonelementary.

- T** e. **Numerical integrator** Estimate the length of the ellipse to two decimal places.

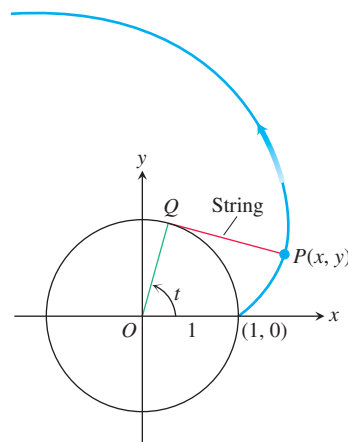
18. **Length is independent of parametrization** To illustrate that the length of a smooth space curve does not depend on the parametrization you use to compute it, calculate the length of one turn of the helix in Example 1 with the following parametrizations.

- a. $\mathbf{r}(t) = (\cos 4t)\mathbf{i} + (\sin 4t)\mathbf{j} + 4t\mathbf{k}, \quad 0 \leq t \leq \pi/2$
- b. $\mathbf{r}(t) = [\cos(t/2)]\mathbf{i} + [\sin(t/2)]\mathbf{j} + (t/2)\mathbf{k}, \quad 0 \leq t \leq 4\pi$
- c. $\mathbf{r}(t) = (\cos t)\mathbf{i} - (\sin t)\mathbf{j} - t\mathbf{k}, \quad -2\pi \leq t \leq 0$

19. **The involute of a circle** If a string wound around a fixed circle is unwound while held taut in the plane of the circle, its end P traces an *involute* of the circle. In the accompanying figure, the circle in question is the circle $x^2 + y^2 = 1$ and the tracing point starts at $(1, 0)$. The unwound portion of the string is tangent to the circle at Q , and t is the radian measure of the angle from the positive x -axis to segment OQ . Derive the parametric equations

$$x = \cos t + t \sin t, \quad y = \sin t - t \cos t, \quad t > 0$$

of the point $P(x, y)$ for the involute.



20. (Continuation of Exercise 19.) Find the unit tangent vector to the involute of the circle at the point $P(x, y)$.
21. **Distance along a line** Show that if $\mathbf{u}$ is a unit vector, then the arc length parameter along the line $\mathbf{r}(t) = P_0 + t\mathbf{u}$ from the point $P_0(x_0, y_0, z_0)$ where $t = 0$, is t itself.
22. Use Simpson's Rule with $n = 10$ to approximate the length of arc of $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ from the origin to the point $(2, 4, 8)$.

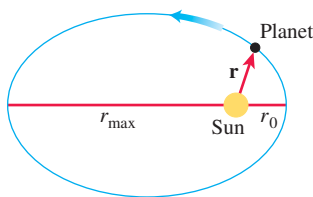


FIGURE 12.36 The length of the major axis of the ellipse is $2a = r_0 + r_{\max}$.

Hence, from Figure 12.36,

$$2a = r_0 + r_{\max} = \frac{2r_0}{1-e} = \frac{2r_0 GM}{2GM - r_0 v_0^2}. \quad (10)$$

Squaring both sides of Equation (9) and substituting the results of Equations (5) and (10) produce Kepler's third law (Exercise 11).

EXERCISES 12.6

In Exercises 1–7, find the velocity and acceleration vectors in terms of $\mathbf{u}_r$ and $\mathbf{u}_\theta$.

- $r = \theta$ and $\frac{d\theta}{dt} = 2$
- $r = \frac{1}{\theta}$ and $\frac{d\theta}{dt} = t^2$
- $r = a(1 - \cos \theta)$ and $\frac{d\theta}{dt} = 3$
- $r = a \sin 2\theta$ and $\frac{d\theta}{dt} = 2t$
- $r = e^{a\theta}$ and $\frac{d\theta}{dt} = 2$
- $r = a(1 + \sin t)$ and $\theta = 1 - e^{-t}$
- $r = 2 \cos 4t$ and $\theta = 2t$
- Type of orbit** For what values of v_0 in Equation (5) is the orbit in Equation (6) a circle? An ellipse? A parabola? A hyperbola?
- Circular orbits** Show that a planet in a circular orbit moves with a constant speed. (*Hint:* This is a consequence of one of Kepler's laws.)
- Suppose that $\mathbf{r}$ is the position vector of a particle moving along a plane curve and dA/dt is the rate at which the vector sweeps out area. Without introducing coordinates, and assuming the necessary derivatives exist, give a geometric argument based on increments and limits for the validity of the equation

$$\frac{dA}{dt} = \frac{1}{2} |\mathbf{r} \times \dot{\mathbf{r}}|.$$

11. Kepler's third law Complete the derivation of Kepler's third law (the part following Equation (10)).

12. Do the data in the accompanying table support Kepler's third law? Give reasons for your answer.

Planet	Semimajor axis	
	a (10^{10} m)	Period T (years)
Mercury	5.79	0.241
Venus	10.81	0.615
Mars	22.78	1.881
Saturn	142.70	29.457

- Earth's major axis** Estimate the length of the major axis of Earth's orbit if its orbital period is 365.256 days.
- Estimate the length of the major axis of the orbit of Uranus if its orbital period is 84 years.
- The eccentricity of Earth's orbit is $e = 0.0167$, so the orbit is nearly circular, with radius approximately 150×10^6 km. Find, in units of km^2/s , the rate dA/dt satisfying Kepler's second law.
- Jupiter's orbital period** Estimate the orbital period of Jupiter, assuming that $a = 77.8 \times 10^{10}$ m.
- Mass of Jupiter** Io is one of the moons of Jupiter. It has a semi-major axis of 0.042×10^{10} m and an orbital period of 1.769 days. Use these data to estimate the mass of Jupiter.
- Distance from Earth to the moon** The period of the moon's rotation around Earth is 2.36055×10^6 s. Estimate the distance to the moon.

CHAPTER 12 Questions to Guide Your Review

- State the rules for differentiating and integrating vector functions. Give examples.
- How do you define and calculate the velocity, speed, direction of motion, and acceleration of a body moving along a sufficiently differentiable space curve? Give an example.
- What is special about the derivatives of vector functions of constant length? Give an example.
- What are the vector and parametric equations for ideal projectile motion? How do you find a projectile's maximum height, flight time, and range? Give examples.
- How do you define and calculate the length of a segment of a smooth space curve? Give an example. What mathematical assumptions are involved in the definition?
- How do you measure distance along a smooth curve in space from a preselected base point? Give an example.
- What is a differentiable curve's unit tangent vector? Give an example.
- Define curvature, circle of curvature (osculating circle), center of curvature, and radius of curvature for twice-differentiable curves in the plane. Give examples. What curves have zero curvature? Constant curvature?

9. What is a plane curve's principal normal vector? When is it defined? Which way does it point? Give an example.
10. How do you define $\mathbf{N}$ and κ for curves in space? How are these quantities related? Give examples.
11. What is a curve's binormal vector? Give an example. How is this vector related to the curve's torsion? Give an example.
12. What formulas are available for writing a moving object's acceleration as a sum of its tangential and normal components? Give an example. Why might one want to write the acceleration this way? What if the object moves at a constant speed? At a constant speed around a circle?
13. State Kepler's laws.

CHAPTER 12 Practice Exercises

Motion in the Plane

In Exercises 1 and 2, graph the curves and sketch their velocity and acceleration vectors at the given values of t . Then write $\mathbf{a}$ in the form $\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N}$ without finding $\mathbf{T}$ and $\mathbf{N}$, and find the value of κ at the given values of t .

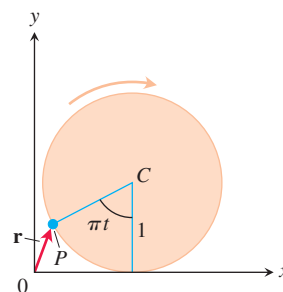
1. $\mathbf{r}(t) = (4 \cos t)\mathbf{i} + (\sqrt{2} \sin t)\mathbf{j}$, $t = 0$ and $\pi/4$
2. $\mathbf{r}(t) = (\sqrt{3} \sec t)\mathbf{i} + (\sqrt{3} \tan t)\mathbf{j}$, $t = 0$
3. The position of a particle in the plane at time t is

$$\mathbf{r} = \frac{1}{\sqrt{1+t^2}}\mathbf{i} + \frac{t}{\sqrt{1+t^2}}\mathbf{j}.$$

Find the particle's highest speed.

4. Suppose $\mathbf{r}(t) = (e^t \cos t)\mathbf{i} + (e^t \sin t)\mathbf{j}$. Show that the angle between $\mathbf{r}$ and $\mathbf{a}$ never changes. What is the angle?
5. **Finding curvature** At point P , the velocity and acceleration of a particle moving in the plane are $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{a} = 5\mathbf{i} + 15\mathbf{j}$. Find the curvature of the particle's path at P .
6. Find the point on the curve $y = e^x$ where the curvature is greatest.
7. A particle moves around the unit circle in the xy -plane. Its position at time t is $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$, where x and y are differentiable functions of t . Find dy/dt if $\mathbf{v} \cdot \mathbf{i} = y$. Is the motion clockwise or counterclockwise?
8. You send a message through a pneumatic tube that follows the curve $9y = x^3$ (distance in meters). At the point $(3, 3)$, $\mathbf{v} \cdot \mathbf{i} = 4$ and $\mathbf{a} \cdot \mathbf{i} = -2$. Find the values of $\mathbf{v} \cdot \mathbf{j}$ and $\mathbf{a} \cdot \mathbf{j}$ at $(3, 3)$.
9. **Characterizing circular motion** A particle moves in the plane so that its velocity and position vectors are always orthogonal. Show that the particle moves in a circle centered at the origin.
10. **Speed along a cycloid** A circular wheel with radius 1 m and center C rolls to the right along the x -axis at a half-turn per second. (See the accompanying figure.) At time t seconds, the position vector of the point P on the wheel's circumference is

$$\mathbf{r} = (\pi t - \sin \pi t)\mathbf{i} + (1 - \cos \pi t)\mathbf{j}.$$
 - a. Sketch the curve traced by P during the interval $0 \leq t \leq 3$.
 - b. Find $\mathbf{v}$ and $\mathbf{a}$ at $t = 0, 1, 2$, and 3 and add these vectors to your sketch.
 - c. At any given time, what is the forward speed of the topmost point of the wheel? Of C ?



Projectile Motion

11. **Shot put** A shot leaves the thrower's hand 2 m above the ground at a 45° angle at 14 m/s. Where is it 3 s later?
12. **Javelin** A javelin leaves the thrower's hand 2.5 m above the ground at a 45° angle at 24 m/s. How high does it go?
13. A golf ball is hit with an initial speed v_0 at an angle α to the horizontal from a point that lies at the foot of a straight-sided hill that is inclined at an angle ϕ to the horizontal, where

$$0 < \phi < \alpha < \frac{\pi}{2}.$$

Show that the ball lands at a distance

$$\frac{2v_0^2 \cos \alpha}{g \cos^2 \phi} \sin(\alpha - \phi),$$

measured up the face of the hill. Hence, show that the greatest range that can be achieved for a given v_0 occurs when $\alpha = (\phi/2) + (\pi/4)$, that is, when the initial velocity vector bisects the angle between the vertical and the hill.

- T 14. Javelin** In Potsdam in 1988, Petra Felke of (then) East Germany set a women's world record by throwing a javelin 80 m.
- a. Assuming that Felke launched the javelin at a 40° angle to the horizontal 2 m above the ground, what was the javelin's initial speed?
 - b. How high did the javelin go?

Motion in Space

Find the lengths of the curves in Exercises 15 and 16.

15. $\mathbf{r}(t) = (2 \cos t)\mathbf{i} + (2 \sin t)\mathbf{j} + t^2\mathbf{k}$, $0 \leq t \leq \pi/4$
16. $\mathbf{r}(t) = (3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j} + 2t^{3/2}\mathbf{k}$, $0 \leq t \leq 3$

In Exercises 17–20, find $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$, κ , and τ at the given value of t .

17. $\mathbf{r}(t) = \frac{4}{9}(1+t)^{3/2}\mathbf{i} + \frac{4}{9}(1-t)^{3/2}\mathbf{j} + \frac{1}{3}t\mathbf{k}, \quad t = 0$

18. $\mathbf{r}(t) = (e^t \sin 2t)\mathbf{i} + (e^t \cos 2t)\mathbf{j} + 2e^t\mathbf{k}, \quad t = 0$

19. $\mathbf{r}(t) = t\mathbf{i} + \frac{1}{2}e^{2t}\mathbf{j}, \quad t = \ln 2$

20. $\mathbf{r}(t) = (3 \cosh 2t)\mathbf{i} + (3 \sinh 2t)\mathbf{j} + 6t\mathbf{k}, \quad t = \ln 2$

In Exercises 21 and 22, write $\mathbf{a}$ in the form $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$ at $t = 0$ without finding $\mathbf{T}$ and $\mathbf{N}$.

21. $\mathbf{r}(t) = (2 + 3t + 3t^2)\mathbf{i} + (4t + 4t^2)\mathbf{j} - (6 \cos t)\mathbf{k}$

22. $\mathbf{r}(t) = (2 + t)\mathbf{i} + (t + 2t^2)\mathbf{j} + (1 + t^2)\mathbf{k}$

23. Find $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$, κ , and τ as functions of t if

$$\mathbf{r}(t) = (\sin t)\mathbf{i} + (\sqrt{2} \cos t)\mathbf{j} + (\sin t)\mathbf{k}.$$

24. At what times in the interval $0 \leq t \leq \pi$ are the velocity and acceleration vectors of the motion $\mathbf{r}(t) = \mathbf{i} + (5 \cos t)\mathbf{j} + (3 \sin t)\mathbf{k}$ orthogonal?

25. The position of a particle moving in space at time $t \geq 0$ is

$$\mathbf{r}(t) = 2\mathbf{i} + \left(4 \sin \frac{t}{2}\right)\mathbf{j} + \left(3 - \frac{t}{\pi}\right)\mathbf{k}.$$

Find the first time $\mathbf{r}$ is orthogonal to the vector $\mathbf{i} - \mathbf{j}$.

26. Find equations for the osculating, normal, and rectifying planes of the curve $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ at the point $(1, 1, 1)$.

27. Find parametric equations for the line that is tangent to the curve $\mathbf{r}(t) = e^t\mathbf{i} + (\sin t)\mathbf{j} + \ln(1-t)\mathbf{k}$ at $t = 0$.

28. Find parametric equations for the line that is tangent to the helix $\mathbf{r}(t) = (\sqrt{2} \cos t)\mathbf{i} + (\sqrt{2} \sin t)\mathbf{j} + t\mathbf{k}$ at the point where $t = \pi/4$.

Theory and Examples

29. **Synchronous curves** By eliminating α from the ideal projectile equations

$$x = (v_0 \cos \alpha)t, \quad y = (v_0 \sin \alpha)t - \frac{1}{2}gt^2,$$

show that $x^2 + (y + gt^2/2)^2 = v_0^2 t^2$. This shows that projectiles launched simultaneously from the origin at the same initial speed will, at any given instant, all lie on the circle of radius $v_0 t$ centered at $(0, -gt^2/2)$, regardless of their launch angle. These circles are the *synchronous curves* of the launching.

30. **Radius of curvature** Show that the radius of curvature of a twice-differentiable plane curve $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$ is given by the formula

$$\rho = \frac{\dot{x}^2 + \dot{y}^2}{\sqrt{\ddot{x}^2 + \ddot{y}^2 - \dot{s}^2}}, \quad \text{where} \quad \dot{s} = \frac{d}{dt} \sqrt{\dot{x}^2 + \dot{y}^2}.$$

31. **An alternative definition of curvature in the plane** An alternative definition gives the curvature of a sufficiently differentiable plane curve to be $|d\phi/ds|$, where ϕ is the angle between $\mathbf{T}$ and $\mathbf{i}$

(Figure 12.37a). Figure 12.37b shows the distance s measured counterclockwise around the circle $x^2 + y^2 = a^2$ from the point $(a, 0)$ to a point P , along with the angle ϕ at P . Calculate the circle's curvature using the alternative definition. (Hint: $\phi = \theta + \pi/2$.)

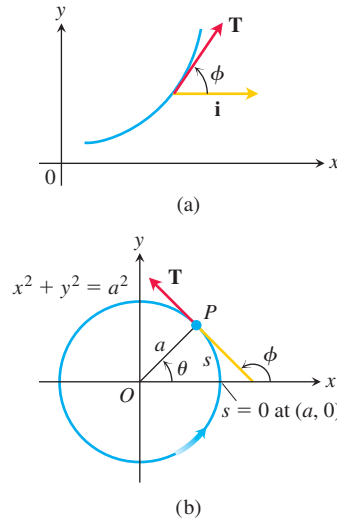


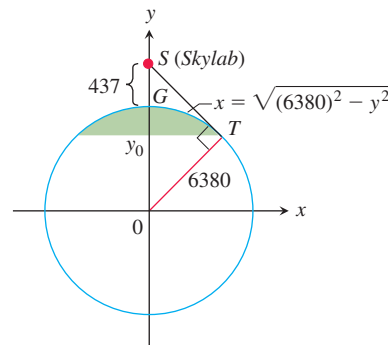
FIGURE 12.37 Figures for Exercise 31.

32. **The view from Skylab 4** What percentage of Earth's surface area could the astronauts see when *Skylab 4* was at its apogee height, 437 km above the surface? To find out, model the visible surface as the surface generated by revolving the circular arc GT , shown here, about the y -axis. Then carry out these steps:

1. Use similar triangles in the figure to show that $y_0/6380 = 6380/(6380 + 437)$. Solve for y_0 .
2. To four significant digits, calculate the visible area as

$$VA = \int_{y_0}^{6380} 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$$

3. Express the result as a percentage of Earth's surface area.



CHAPTER 12 Additional and Advanced Exercises

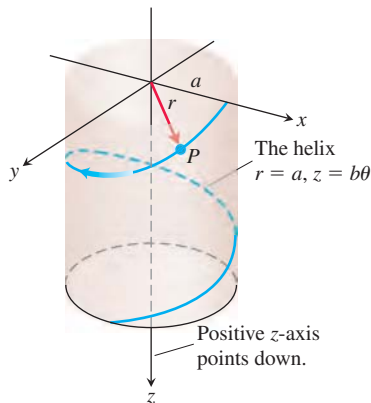
Applications

1. A frictionless particle P , starting from rest at time $t = 0$ at the point $(a, 0, 0)$, slides down the helix

$$\mathbf{r}(\theta) = (a \cos \theta)\mathbf{i} + (a \sin \theta)\mathbf{j} + b\theta\mathbf{k} \quad (a, b > 0)$$

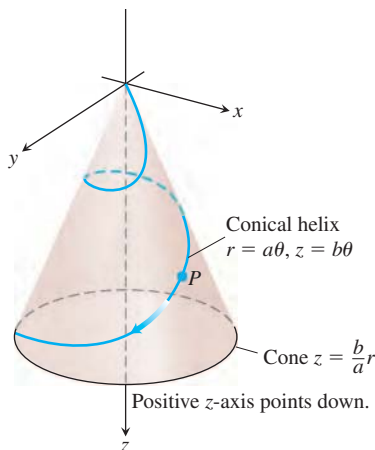
under the influence of gravity, as in the accompanying figure. The θ in this equation is the cylindrical coordinate θ , and the helix is the curve $r = a$, $z = b\theta$, $\theta \geq 0$, in cylindrical coordinates. We assume θ to be a differentiable function of t for the motion. The law of conservation of energy tells us that the particle's speed after it has fallen straight down a distance z is $\sqrt{2gz}$, where g is the constant acceleration of gravity.

- Find the angular velocity $d\theta/dt$ when $\theta = 2\pi$.
- Express the particle's θ - and z -coordinates as functions of t .
- Express the tangential and normal components of the velocity $d\mathbf{r}/dt$ and acceleration $d^2\mathbf{r}/dt^2$ as functions of t . Does the acceleration have any nonzero component in the direction of the binormal vector $\mathbf{B}$?



2. Suppose the curve in Exercise 1 is replaced by the conical helix $r = a\theta$, $z = b\theta$ shown in the accompanying figure.

- Express the angular velocity $d\theta/dt$ as a function of θ .
- Express the distance the particle travels along the helix as a function of θ .



Motion in Polar and Cylindrical Coordinates

3. Deduce from the orbit equation

$$r = \frac{(1+e)r_0}{1+e\cos\theta}$$

that a planet is closest to its sun when $\theta = 0$, and show that $r = r_0$ at that time.

- T** 4. **A Kepler equation** The problem of locating a planet in its orbit at a given time and date eventually leads to solving “Kepler” equations of the form

$$f(x) = x - 1 - \frac{1}{2}\sin x = 0.$$

- Show that this particular equation has a solution between $x = 0$ and $x = 2$.
 - With your computer or calculator in radian mode, use Newton's method to find the solution to as many places as you can.
5. In Section 12.6, we found the velocity of a particle moving in the plane to be

$$\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} = \dot{r}\mathbf{u}_r + r\dot{\theta}\mathbf{u}_\theta.$$

- Express $\dot{x}$ and $\dot{y}$ in terms of $\dot{r}$ and $r\dot{\theta}$ by evaluating the dot products $\mathbf{v} \cdot \mathbf{i}$ and $\mathbf{v} \cdot \mathbf{j}$.
 - Express $\dot{r}$ and $r\dot{\theta}$ in terms of $\dot{x}$ and $\dot{y}$ by evaluating the dot products $\mathbf{v} \cdot \mathbf{u}_r$ and $\mathbf{v} \cdot \mathbf{u}_\theta$.
6. Express the curvature of a twice-differentiable curve $r = f(\theta)$ in the polar coordinate plane in terms of f and its derivatives.
7. A slender rod through the origin of the polar coordinate plane rotates (in the plane) about the origin at the rate of 3 rad/min. A beetle starting from the point $(2, 0)$ crawls along the rod toward the origin at the rate of 1 cm/min.
- Find the beetle's acceleration and velocity in polar form when it is halfway to (1 cm from) the origin.
- T** b. To the nearest millimeter, what will be the length of the path the beetle has traveled by the time it reaches the origin?

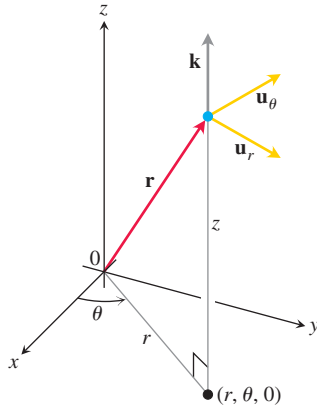
8. Arc length in cylindrical coordinates

- Show that when you express $ds^2 = dx^2 + dy^2 + dz^2$ in terms of cylindrical coordinates, you get $ds^2 = dr^2 + r^2d\theta^2 + dz^2$.
- Interpret this result geometrically in terms of the edges and a diagonal of a box. Sketch the box.
- Use the result in part (a) to find the length of the curve $r = e^\theta$, $z = e^\theta$, $0 \leq \theta \leq \ln 8$.

9. **Unit vectors for position and motion in cylindrical coordinates** When the position of a particle moving in space is given in cylindrical coordinates, the unit vectors we use to describe its position and motion are

$$\mathbf{u}_r = (\cos \theta)\mathbf{i} + (\sin \theta)\mathbf{j}, \quad \mathbf{u}_\theta = -(\sin \theta)\mathbf{i} + (\cos \theta)\mathbf{j}.$$

and $\mathbf{k}$ (see accompanying figure). The particle's position vector is then $\mathbf{r} = r\mathbf{u}_r + z\mathbf{k}$, where r is the positive polar distance coordinate of the particle's position.



- a. Show that $\mathbf{u}_r$, $\mathbf{u}_\theta$, and $\mathbf{k}$, in this order, form a right-handed frame of unit vectors.

- b. Show that

$$\frac{d\mathbf{u}_r}{d\theta} = \mathbf{u}_\theta \quad \text{and} \quad \frac{d\mathbf{u}_\theta}{d\theta} = -\mathbf{u}_r.$$

- c. Assuming that the necessary derivatives with respect to t exist, express $\mathbf{v} = \dot{\mathbf{r}}$ and $\mathbf{a} = \ddot{\mathbf{r}}$ in terms of $\mathbf{u}_r$, $\mathbf{u}_\theta$, $\mathbf{k}$, $\dot{r}$, and $\dot{\theta}$.

- d. **Conservation of angular momentum** Let $\mathbf{r}(t)$ denote the position in space of a moving object at time t . Suppose the force acting on the object at time t is

$$\mathbf{F}(t) = -\frac{c}{|\mathbf{r}(t)|^3} \mathbf{r}(t),$$

where c is a constant. In physics the **angular momentum** of an object at time t is defined to be $\mathbf{L}(t) = \mathbf{r}(t) \times m\mathbf{v}(t)$, where m is the mass of the object and $\mathbf{v}(t)$ is the velocity. Prove that angular momentum is a conserved quantity; that is, prove that $\mathbf{L}(t)$ is a constant vector, independent of time. Remember Newton's law $\mathbf{F} = m\mathbf{a}$. (This is a calculus problem, not a physics problem.)

CHAPTER 12 Technology Application Projects

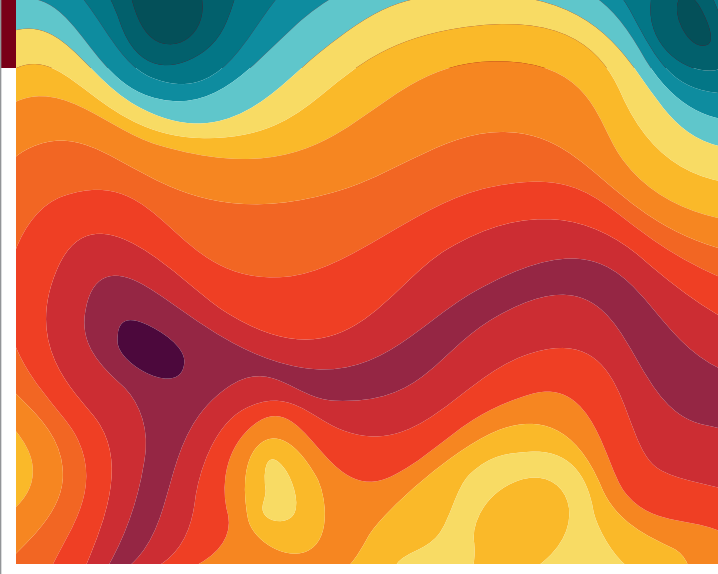
Mathematica/Maple Projects

Projects can be found within MyLab Math.

- **Radar Tracking of a Moving Object**
Visualize position, velocity, and acceleration vectors to analyze motion.
- **Parametric and Polar Equations with a Figure Skater**
Visualize position, velocity, and acceleration vectors to analyze motion.
- **Moving in Three Dimensions**
Compute distance traveled, speed, curvature, and torsion for motion along a space curve. Visualize and compute the tangential, normal, and binormal vectors associated with motion along a space curve.

13

Partial Derivatives



OVERVIEW The volume of a right circular cylinder is a function $V = \pi r^2 h$ of its radius and its height, so it is a function $V(r, h)$ of two variables r and h . The speed of sound through seawater is primarily a function of salinity S and temperature T . The monthly payment on a home mortgage is a function of the principal borrowed P , the interest rate i , and the term t of the loan. These are examples of functions that depend on more than one independent variable. In this chapter we extend the ideas of single-variable differential calculus to functions of several variables.

13.1 Functions of Several Variables

Real-valued functions of several independent real variables are defined analogously to functions of a single variable. Points in the domain are now ordered pairs (or triples, quadruples, n -tuples) of real numbers, and values in the range are real numbers.

DEFINITIONS Suppose D is a set of n -tuples of real numbers $(x_1, x_2, \dots, x_n)$. A **real-valued function** f on D is a rule that assigns a real number

$$w = f(x_1, x_2, \dots, x_n)$$

to each element in D . The set D is the function's **domain**. The set of w -values taken on by f is the function's **range**. The symbol w is the **dependent variable** of f , and f is said to be a function of the n **independent variables** x_1 to x_n . We also call the x_j 's the function's **input variables** and call w the function's **output variable**.

If f is a function of two independent variables, we usually call the independent variables x and y and the dependent variable z , and we picture the domain of f as a region in the xy -plane (Figure 13.1). If f is a function of three independent variables, we call the independent variables x , y , and z and the dependent variable w , and we picture the domain as a region in space.

In applications, we tend to use letters that remind us of what the variables stand for. To say that the volume of a right circular cylinder is a function of its radius and height, we might write $V = f(r, h)$. To be more specific, we might replace the notation $f(r, h)$ by the formula that calculates the value of V from the values of r and h , and write $V = \pi r^2 h$. In either case, r and h would be the independent variables and V the dependent variable of the function.

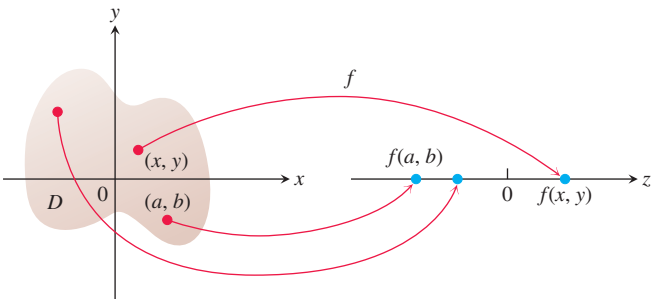


FIGURE 13.1 An arrow diagram for the function $z = f(x, y)$.

As usual, we evaluate functions defined by formulas by substituting the values of the independent variables in the formula and calculating the corresponding value of the dependent variable. For example, the value of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at the point $(3, 0, 4)$ is

$$f(3, 0, 4) = \sqrt{(3)^2 + (0)^2 + (4)^2} = \sqrt{25} = 5.$$

Domains and Ranges

In defining a function of more than one variable, we follow the usual practice of excluding inputs that lead to complex numbers or division by zero. If $f(x, y) = \sqrt{y - x^2}$, then y cannot be less than x^2 . If $f(x, y) = 1/(xy)$, then xy cannot be zero. The domain of a function is assumed to be the largest set for which the defining rule generates real numbers, unless the domain is otherwise specified explicitly. The range consists of the set of output values for the dependent variable.

EXAMPLE 1

(a) These are functions of two variables. Note the restrictions that apply to their domains in order to obtain a real value for the dependent variable z .

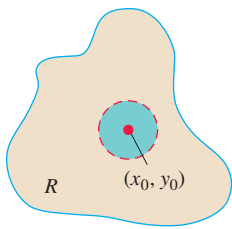
Function	Domain	Range
$z = \sqrt{y - x^2}$	$y \geq x^2$	$[0, \infty)$
$z = \frac{1}{xy}$	$xy \neq 0$	$(-\infty, 0) \cup (0, \infty)$
$z = \sin xy$	Entire plane	$[-1, 1]$

(b) These are functions of three variables with restrictions on some of their domains.

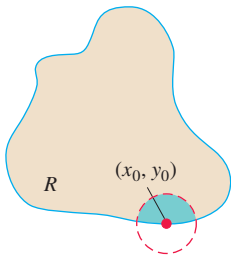
Function	Domain	Range
$w = \sqrt{x^2 + y^2 + z^2}$	Entire space	$[0, \infty)$
$w = \frac{1}{x^2 + y^2 + z^2}$	$(x, y, z) \neq (0, 0, 0)$	$(0, \infty)$
$w = xy \ln z$	Half-space $z > 0$	$(-\infty, \infty)$

Functions of Two Variables

On the real line, closed intervals $[a, b]$ include their boundary points while open intervals (a, b) do not. Intervals such as $[a, b)$, which includes only one of its two boundary points, are neither open nor closed. Regions in the plane can also be open, closed, or neither.



(a) Interior point



(b) Boundary point

FIGURE 13.2 Interior points and boundary points of a plane region R . An interior point is necessarily a point of R . A boundary point of R need not belong to R .

DEFINITIONS A point (x_0, y_0) in a region (set) R in the xy -plane is an **interior point** of R if it is the center of a disk of positive radius that lies entirely in R (Figure 13.2). A point (x_0, y_0) is a **boundary point** of R if every disk centered at (x_0, y_0) contains points that lie outside of R as well as points that lie in R . (The boundary point itself need not belong to R .)

The interior points of a region, as a set, make up the **interior** of the region. The region's boundary points make up its **boundary**. A region is **open** if it consists entirely of interior points. A region is **closed** if it contains all its boundary points (Figure 13.3).

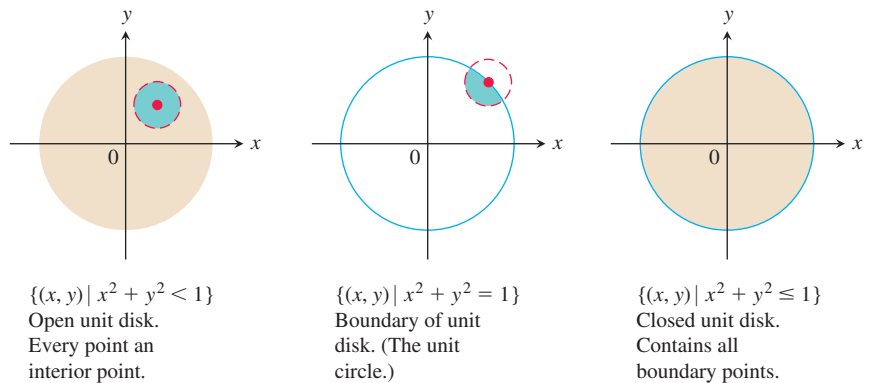


FIGURE 13.3 Interior points and boundary points of the unit disk in the plane.

As with a half-open interval of real numbers $[a, b)$, some regions in the plane are neither open nor closed. If you start with the open disk in Figure 13.3 and add to it some, but not all, of its boundary points, the resulting set is neither open nor closed. The boundary points that *are* there keep the set from being open. The absence of the remaining boundary points keeps the set from being closed. Two interesting examples are the empty set and the entire plane. The empty set has no interior points and no boundary points. This implies that the empty set is open (because it does not contain points that are not interior points), and at the same time it is closed (because there are no boundary points that it fails to contain). The entire xy -plane is also both open and closed: open because every point in the plane is an interior point, and closed because it has no boundary points. The empty set and the entire plane are the only subsets of the plane that are both open and closed. Other sets may be open, or closed, or neither.

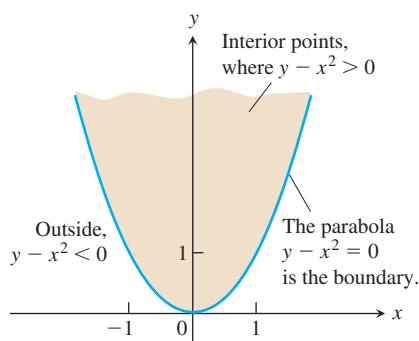


FIGURE 13.4 The domain of $f(x, y)$ in Example 2 consists of the shaded region and its bounding parabola.

DEFINITIONS A region in the plane is **bounded** if it lies inside a disk of finite radius. A region is **unbounded** if it is not bounded.

Examples of *bounded* sets in the plane include line segments, triangles, interiors of triangles, rectangles, circles, and disks. Examples of *unbounded* sets in the plane include lines, coordinate axes, the graphs of functions defined on infinite intervals, quadrants, half-planes, and the plane itself.

EXAMPLE 2 Describe the domain of the function $f(x, y) = \sqrt{y - x^2}$.

Solution Since f is defined only where $y - x^2 \geq 0$, the domain is the closed, unbounded region shown in Figure 13.4. The parabola $y = x^2$ is the boundary of the domain. The points above the parabola make up the domain's interior. ■

Graphs, Level Curves, and Contours of Functions of Two Variables

There are two standard ways to picture the values of a function $f(x, y)$. One is to draw and label curves in the domain on which f has a constant value. The other is to sketch the surface $z = f(x, y)$ in space.

DEFINITIONS The set of points in the plane where a function $f(x, y)$ has a constant value $f(x, y) = c$ is called a **level curve** of f . The set of all points $(x, y, f(x, y))$ in space, for (x, y) in the domain of f , is called the **graph** of f .

The graph of f is often called the **surface** $z = f(x, y)$.

EXAMPLE 3 Graph $f(x, y) = 100 - x^2 - y^2$ and plot the level curves $f(x, y) = 0$, $f(x, y) = 51$, and $f(x, y) = 75$ in the domain of f in the plane.

Solution The domain of f is the entire xy -plane, and the range of f is the set of real numbers less than or equal to 100. The graph is the paraboloid $z = 100 - x^2 - y^2$, the positive portion of which is shown in Figure 13.5.

The level curve $f(x, y) = 0$ is the set of points in the xy -plane at which

$$f(x, y) = 100 - x^2 - y^2 = 0, \quad \text{or} \quad x^2 + y^2 = 100,$$

which is the circle of radius 10 centered at the origin. Similarly, the level curves $f(x, y) = 51$ and $f(x, y) = 75$ (Figure 13.5) are the circles

$$f(x, y) = 100 - x^2 - y^2 = 51, \quad \text{or} \quad x^2 + y^2 = 49$$

$$f(x, y) = 100 - x^2 - y^2 = 75, \quad \text{or} \quad x^2 + y^2 = 25.$$

The level curve $f(x, y) = 100$ consists of the origin alone. (It is still a level curve.)

If $x^2 + y^2 > 100$, then the values of $f(x, y)$ are negative. For example, the circle $x^2 + y^2 = 144$, which is the circle centered at the origin with radius 12, gives the constant value $f(x, y) = -44$ and is a level curve of f . ■

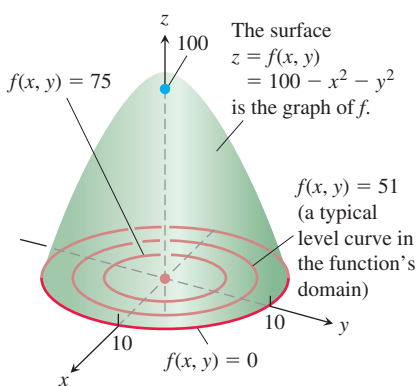


FIGURE 13.5 The graph and selected level curves of the function $f(x, y)$ in Example 3. The level curves lie in the xy -plane, which is the domain of the function $f(x, y)$.

The contour curve $f(x, y) = 100 - x^2 - y^2 = 75$ is the circle $x^2 + y^2 = 25$ in the plane $z = 75$.

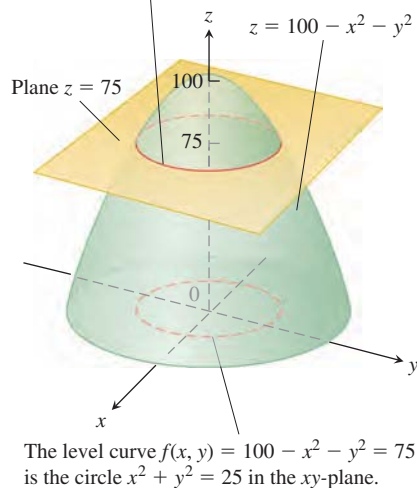


FIGURE 13.6 A plane $z = c$ parallel to the xy -plane intersecting a surface $z = f(x, y)$ produces a contour curve.

The curve in space in which the plane $z = c$ cuts a surface $z = f(x, y)$ is made up of the points that represent the function value $f(x, y) = c$. It is called the **contour curve** $f(x, y) = c$ to distinguish it from the level curve $f(x, y) = c$ in the domain of f . Figure 13.6 shows the contour curve $f(x, y) = 75$ on the surface $z = 100 - x^2 - y^2$ defined by the function $f(x, y) = 100 - x^2 - y^2$. The contour curve lies directly above the circle $x^2 + y^2 = 25$, which is the level curve $f(x, y) = 75$ in the function's domain.

The distinction between level curves and contour curves is often overlooked, and it is common to call both types of curves by the same name, relying on context to make it clear which type of curve is meant. On most maps, for example, the curves that represent constant elevation (height above sea level) are called contours, not level curves (Figure 13.7).

Functions of Three Variables

In the plane, the points where a function of two independent variables has a constant value $f(x, y) = c$ make a curve in the function's domain. In space, the points where a function of three independent variables has a constant value $f(x, y, z) = c$ make a surface in the function's domain.

DEFINITION The set of points (x, y, z) in space where a function of three independent variables has a constant value $f(x, y, z) = c$ is called a **level surface** of f .



FIGURE 13.7 Contours on Mt. Washington in New Hampshire.
(Source: United States Geological Survey)

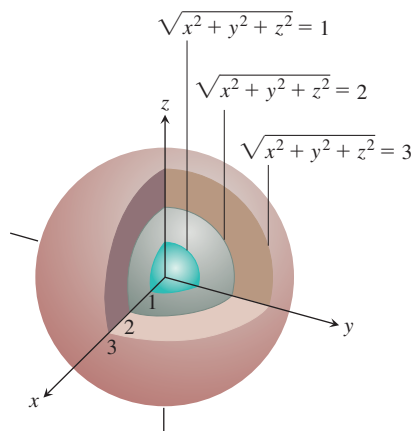


FIGURE 13.8 The level surfaces of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ are concentric spheres (Example 4).

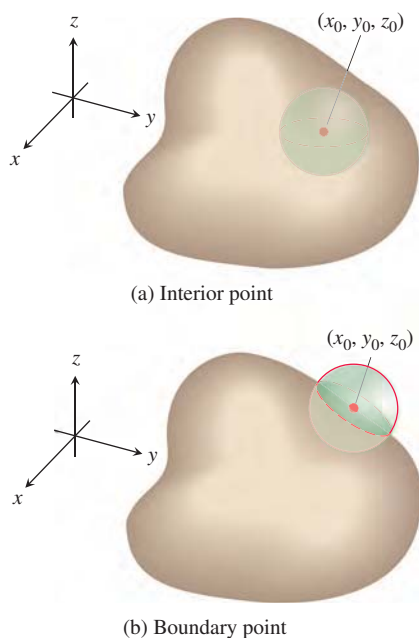


FIGURE 13.9 Interior points and boundary points of a region in space. As with regions in the plane, a boundary point need not belong to the space region R .

Since the graphs of functions of three variables consist of points $(x, y, z, f(x, y, z))$ lying in a four-dimensional space, we cannot sketch them effectively in our three-dimensional frame of reference. We can see how the function behaves, however, by looking at its three-dimensional level surfaces.

EXAMPLE 4 Describe the level surfaces of the function

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}.$$

Solution The value of f is the distance from the origin to the point (x, y, z) . Each level surface $\sqrt{x^2 + y^2 + z^2} = c$, $c > 0$, is a sphere of radius c centered at the origin. Figure 13.8 shows a cutaway view of three of these spheres. The level surface $\sqrt{x^2 + y^2 + z^2} = 0$ consists of the origin alone.

We are not graphing the function here; we are looking at level surfaces in the function's domain. The level surfaces show how the function's values change as we move through its domain. If we remain on a sphere of radius c centered at the origin, the function maintains a constant value, namely c . If we move from a point on one sphere to a point on another, the function's value changes. It increases if we move away from the origin and decreases if we move toward the origin. The way the values change depends on the direction we take. The dependence of change on direction is important. We return to it in Section 13.5. ■

The definitions of interior, boundary, open, closed, bounded, and unbounded for regions in space are similar to those for regions in the plane. To accommodate the extra dimension, we use solid balls of positive radius instead of disks.

DEFINITIONS A point (x_0, y_0, z_0) in a region R in space is an **interior point** of R if it is the center of a solid ball that lies entirely in R (Figure 13.9a). A point (x_0, y_0, z_0) is a **boundary point** of R if every solid ball centered at (x_0, y_0, z_0) contains points that lie outside of R as well as points that lie inside R (Figure 13.9b). The **interior** of R is the set of interior points of R . The **boundary** of R is the set of boundary points of R .

A region is **open** if it consists entirely of interior points. A region is **closed** if it contains its entire boundary.

A region is **bounded** if it lies inside a solid ball of finite radius; otherwise, the region is **unbounded**.

Examples of *open* sets in space include the interior of a sphere, the open half-space $z > 0$, the first octant (where x , y , and z are all positive), and space itself. Examples of *closed* sets in space include lines, planes, and the closed half-space $z \geq 0$. A solid sphere

with part of its boundary removed or a solid cube with a missing face, edge, or corner point is *neither open nor closed*.

Functions of more than three independent variables are also important. For example, a model that measures temperature in the atmosphere may depend not only on the location of the point $P(x, y, z)$ in space, but also on the time t when it is measured, so we would write $T = f(x, y, z, t)$.

Computer Graphing

Three-dimensional graphing software makes it possible to graph functions of two variables. We can often get information more quickly from a graph than from a formula, since the surfaces reveal increasing and decreasing behavior, and high points or low points.

EXAMPLE 5 The temperature w beneath the Earth's surface is a function of the depth x beneath the surface and the time t of the year. If we measure x in meters and t as the number of days elapsed from the expected date of the yearly highest surface temperature, we can model the variation in temperature with the function

$$w = \cos(1.7 \times 10^{-2}t - 0.6x)e^{-0.6x}.$$

(The temperature at 0 m is scaled to vary from +1 to -1, so that the variation at x meters can be interpreted as a fraction of the variation at the surface.)

Figure 13.10 shows a graph of the function. At a depth of 5 m, the variation (change in vertical amplitude in the figure) is about 5% of the surface variation. At 8 m, there is almost no variation during the year.

The graph also shows that the temperature 5 m below the surface is about half a year out of phase with the surface temperature. When the temperature is lowest on the surface (late January, say), it is at its highest 5 m below. Five meters below the ground, the seasons are reversed.

Figure 13.11 shows computer-generated graphs of a number of functions of two variables together with their level curves.

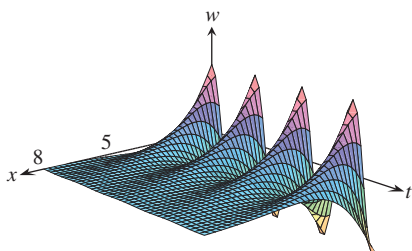


FIGURE 13.10 This graph shows the seasonal variation of the temperature below ground as a fraction of surface temperature (Example 5).

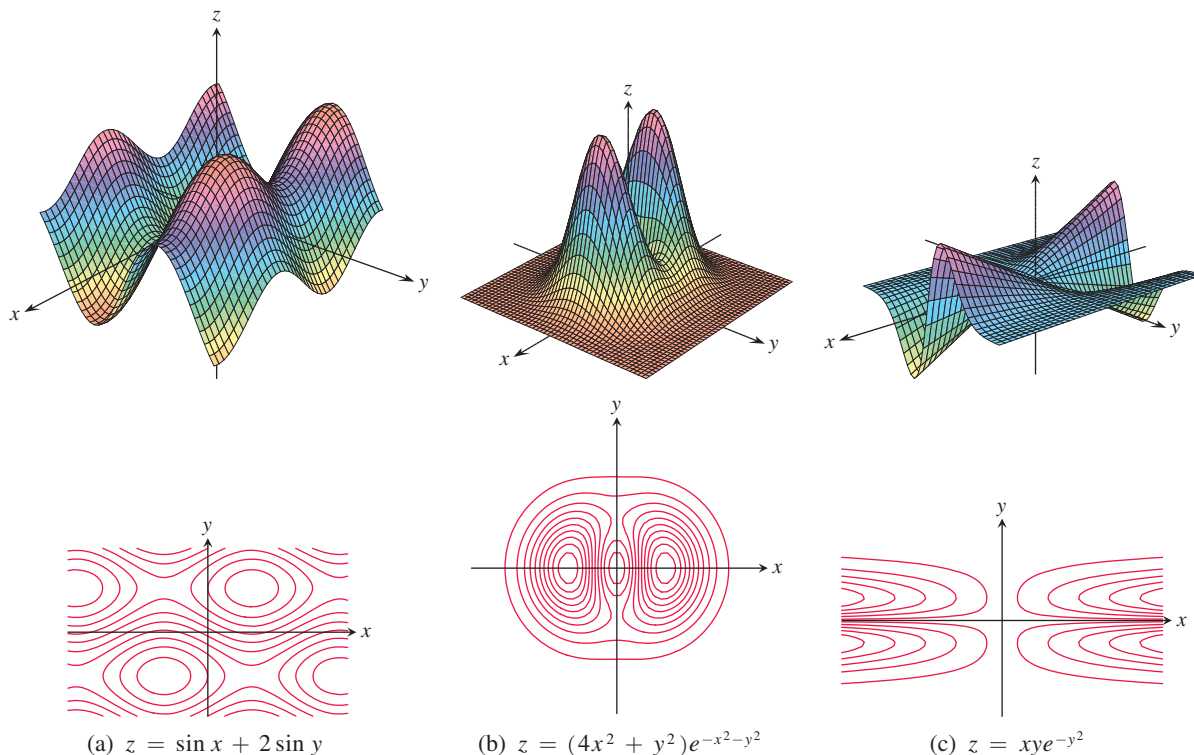


FIGURE 13.11 Computer-generated graphs and level curves of typical functions of two variables.

EXERCISES 13.1

Domain, Range, and Level Curves

In Exercises 1–4, find the specific function values.

1. $f(x, y) = x^2 + xy^3$

a. $f(0, 0)$

b. $f(-1, 1)$

c. $f(2, 3)$

d. $f(-3, -2)$

2. $f(x, y) = \sin(xy)$

a. $f\left(2, \frac{\pi}{6}\right)$

b. $f\left(-3, \frac{\pi}{12}\right)$

c. $f\left(\pi, \frac{1}{4}\right)$

d. $f\left(-\frac{\pi}{2}, -7\right)$

3. $f(x, y, z) = \frac{x - y}{y^2 + z^2}$

a. $f(3, -1, 2)$

b. $f\left(1, \frac{1}{2}, -\frac{1}{4}\right)$

c. $f\left(0, -\frac{1}{3}, 0\right)$

d. $f(2, 2, 100)$

4. $f(x, y, z) = \sqrt{49 - x^2 - y^2 - z^2}$

a. $f(0, 0, 0)$

b. $f(2, -3, 6)$

c. $f(-1, 2, 3)$

d. $f\left(\frac{4}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{6}{\sqrt{2}}\right)$

In Exercises 5–12, find and sketch the domain for each function.

5. $f(x, y) = \sqrt{y - x - 2}$

6. $f(x, y) = \ln(x^2 + y^2 - 4)$

7. $f(x, y) = \frac{(x - 1)(y + 2)}{(y - x)(y - x^3)}$

8. $f(x, y) = \frac{\sin(xy)}{x^2 + y^2 - 25}$

9. $f(x, y) = \cos^{-1}(y - x^2)$

10. $f(x, y) = \ln(xy + x - y - 1)$

11. $f(x, y) = \sqrt{(x^2 - 4)(y^2 - 9)}$

12. $f(x, y) = \frac{1}{\ln(4 - x^2 - y^2)}$

In Exercises 13–16, find and sketch the level curves $f(x, y) = c$ on the same set of coordinate axes for the given values of c . We refer to these level curves as a contour map.

13. $f(x, y) = x + y - 1$, $c = -3, -2, -1, 0, 1, 2, 3$

14. $f(x, y) = x^2 + y^2$, $c = 0, 1, 4, 9, 16, 25$

15. $f(x, y) = xy$, $c = -9, -4, -1, 0, 1, 4, 9$

16. $f(x, y) = \sqrt{25 - x^2 - y^2}$, $c = 0, 1, 2, 3, 4$

In Exercises 17–30, (a) find the function's domain, (b) find the function's range, (c) describe the function's level curves, (d) find the boundary of the function's domain, (e) determine whether the domain is an open region, a closed region, or neither, and (f) decide whether the domain is bounded or unbounded.

17. $f(x, y) = y - x$

19. $f(x, y) = 4x^2 + 9y^2$

21. $f(x, y) = xy$

23. $f(x, y) = \frac{1}{\sqrt{16 - x^2 - y^2}}$

25. $f(x, y) = \ln(x^2 + y^2)$

27. $f(x, y) = \sin^{-1}(y - x)$

29. $f(x, y) = \ln(x^2 + y^2 - 1)$

18. $f(x, y) = \sqrt{y - x}$

20. $f(x, y) = x^2 - y^2$

22. $f(x, y) = y/x^2$

24. $f(x, y) = \sqrt{9 - x^2 - y^2}$

26. $f(x, y) = e^{-(x^2 + y^2)}$

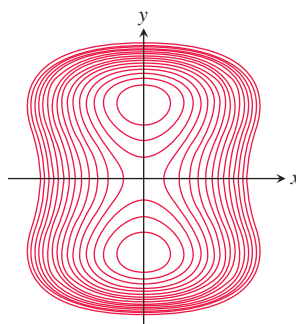
28. $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$

30. $f(x, y) = \ln(9 - x^2 - y^2)$

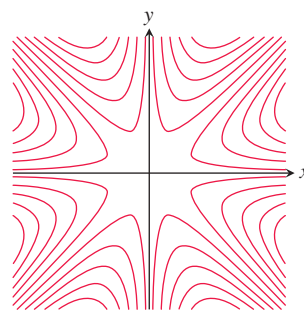
Matching Surfaces with Level Curves

Exercises 31–36 show level curves for six functions. The graphs of these functions are given on the next page (items a–f), as are their equations (items g–l). Match each set of level curves with the appropriate graph and the appropriate equation.

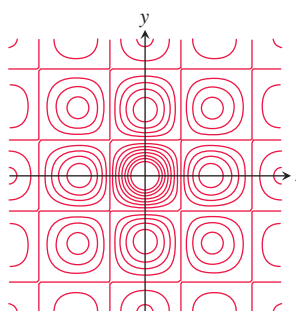
31.



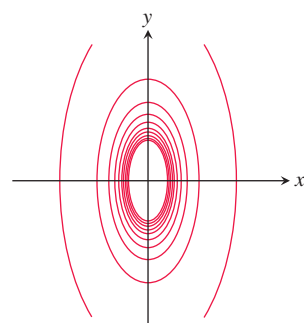
32.



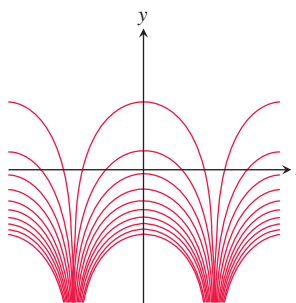
33.



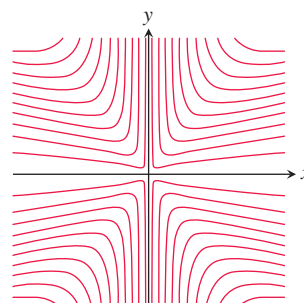
34.

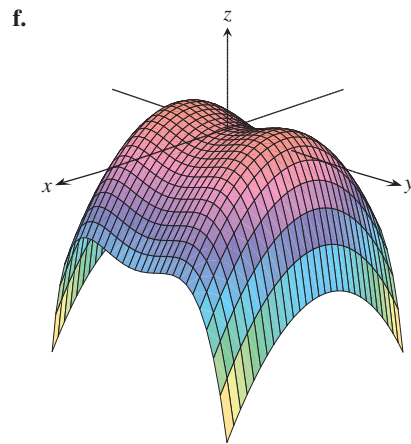
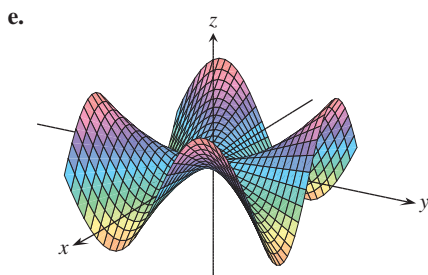
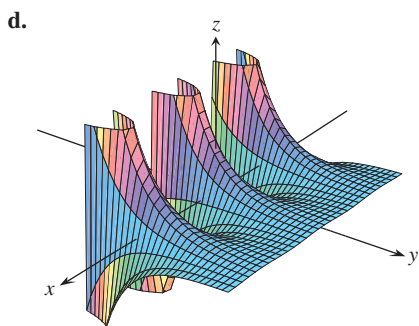
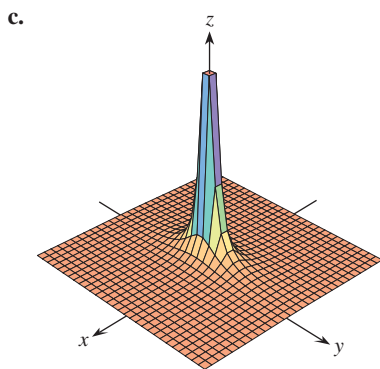
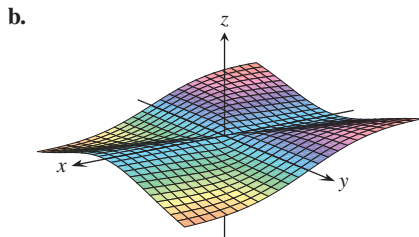
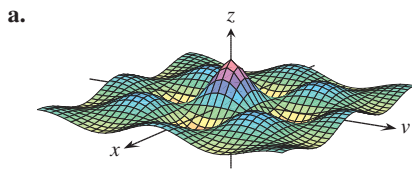


35.



36.





g. $z = -\frac{xy^2}{x^2 + y^2}$

h. $z = y^2 - y^4 - x^2$

i. $z = (\cos x)(\cos y)e^{-\sqrt{x^2+y^2}/4}$

j. $z = e^{-y} \cos x$

k. $z = \frac{1}{4x^2 + y^2}$

l. $z = \frac{xy(x^2 - y^2)}{x^2 + y^2}$

Functions of Two Variables

Display the values of the functions in Exercises 37–48 in two ways: (a) by sketching the surface $z = f(x, y)$ and (b) by drawing an assortment of level curves in the function's domain. Label each level curve with its function value.

37. $f(x, y) = y^2$

38. $f(x, y) = \sqrt{x}$

39. $f(x, y) = x^2 + y^2$

40. $f(x, y) = \sqrt{x^2 + y^2}$

41. $f(x, y) = x^2 - y$

42. $f(x, y) = 4 - x^2 - y^2$

43. $f(x, y) = 4x^2 + y^2$

44. $f(x, y) = 6 - 2x - 3y$

45. $f(x, y) = 1 - |y|$

46. $f(x, y) = 1 - |x| - |y|$

47. $f(x, y) = \sqrt{x^2 + y^2 + 4}$

48. $f(x, y) = \sqrt{x^2 + y^2 - 4}$

Finding Level Curves

In Exercises 49–52, find an equation for, and sketch the graph of, the level curve of the function $f(x, y)$ that passes through the given point.

49. $f(x, y) = 16 - x^2 - y^2$, $(2\sqrt{2}, \sqrt{2})$

50. $f(x, y) = \sqrt{x^2 - 1}$, $(1, 0)$

51. $f(x, y) = \sqrt{x + y^2 - 3}$, $(3, -1)$

52. $f(x, y) = \frac{2y - x}{x + y + 1}$, $(-1, 1)$

Sketching Level Surfaces

In Exercises 53–60, sketch a typical level surface for the function.

53. $f(x, y, z) = x^2 + y^2 + z^2$

54. $f(x, y, z) = \ln(x^2 + y^2 + z^2)$

55. $f(x, y, z) = x + z$

56. $f(x, y, z) = z$

57. $f(x, y, z) = x^2 + y^2$

58. $f(x, y, z) = y^2 + z^2$

59. $f(x, y, z) = z - x^2 - y^2$

60. $f(x, y, z) = (x^2/25) + (y^2/16) + (z^2/9)$

Finding Level Surfaces

In Exercises 61–64, find an equation for the level surface of the function through the given point.

61. $f(x, y, z) = \sqrt{x - y} - \ln z$, $(3, -1, 1)$

$$62. f(x, y, z) = \ln(x^2 + y + z^2), \quad (-1, 2, 1)$$

$$63. g(x, y, z) = \sqrt{x^2 + y^2 + z^2}, \quad (1, -1, \sqrt{2})$$

$$64. g(x, y, z) = \frac{x - y + z}{2x + y - z}, \quad (1, 0, -2)$$

In Exercises 65–68, find and sketch the domain of f . Then find an equation for the level curve or surface of the function passing through the given point.

$$65. f(x, y) = \sum_{n=0}^{\infty} \left(\frac{x}{y}\right)^n, \quad (1, 2)$$

$$66. g(x, y, z) = \sum_{n=0}^{\infty} \frac{(x + y)^n}{n! z^n}, \quad (\ln 4, \ln 9, 2)$$

$$67. f(x, y) = \int_x^y \frac{d\theta}{\sqrt{1 - \theta^2}}, \quad (0, 1)$$

$$68. g(x, y, z) = \int_x^y \frac{dt}{1 + t^2} + \int_0^z \frac{d\theta}{\sqrt{4 - \theta^2}}, \quad (0, 1, \sqrt{3})$$

COMPUTER EXPLORATIONS

Use a CAS to perform the following steps for each of the functions in Exercises 69–72.

a. Plot the surface over the given rectangle.

b. Plot several level curves in the rectangle.

c. Plot the level curve of f through the given point.

$$69. f(x, y) = x \sin \frac{y}{2} + y \sin 2x, \quad 0 \leq x \leq 5\pi, \quad 0 \leq y \leq 5\pi, \\ P(3\pi, 3\pi)$$

$$70. f(x, y) = (\sin x)(\cos y)e^{\sqrt{x^2 + y^2}/8}, \quad 0 \leq x \leq 5\pi, \\ 0 \leq y \leq 5\pi, \quad P(4\pi, 4\pi)$$

$$71. f(x, y) = \sin(x + 2 \cos y), \quad -2\pi \leq x \leq 2\pi, \\ -2\pi \leq y \leq 2\pi, \quad P(\pi, \pi)$$

$$72. f(x, y) = e^{(x^{0.1} - y)} \sin(x^2 + y^2), \quad 0 \leq x \leq 2\pi, \\ -2\pi \leq y \leq \pi, \quad P(\pi, -\pi)$$

Use a CAS to plot the implicitly defined level surfaces in Exercises 73–76.

$$73. 4 \ln(x^2 + y^2 + z^2) = 1 \quad 74. x^2 + z^2 = 1$$

$$75. x + y^2 - 3z^2 = 1$$

$$76. \sin\left(\frac{x}{2}\right) - (\cos y)\sqrt{x^2 + z^2} = 2$$

Parametrized Surfaces Just as you describe curves in the plane parametrically with a pair of equations $x = f(t)$, $y = g(t)$ defined on some parameter interval I , you can sometimes describe surfaces in space with a triple of equations $x = f(u, v)$, $y = g(u, v)$, $z = h(u, v)$ defined on some parameter rectangle $a \leq u \leq b$, $c \leq v \leq d$. Many computer algebra systems permit you to plot such surfaces in *parametric mode*. (Parametrized surfaces are discussed in detail in Section 15.5.) Use a CAS to plot the surfaces in Exercises 77–80. Also plot several level curves in the xy -plane.

$$77. x = u \cos v, \quad y = u \sin v, \quad z = u, \quad 0 \leq u \leq 2, \\ 0 \leq v \leq 2\pi$$

$$78. x = u \cos v, \quad y = u \sin v, \quad z = v, \quad 0 \leq u \leq 2, \\ 0 \leq v \leq 2\pi$$

$$79. x = (2 + \cos u) \cos v, \quad y = (2 + \cos u) \sin v, \quad z = \sin u, \\ 0 \leq u \leq 2\pi, \quad 0 \leq v \leq 2\pi$$

$$80. x = 2 \cos u \cos v, \quad y = 2 \cos u \sin v, \quad z = 2 \sin u, \\ 0 \leq u \leq 2\pi, \quad 0 \leq v \leq \pi$$

13.2 Limits and Continuity in Higher Dimensions

In this section we develop limits and continuity for multivariable functions. The theory is similar to that developed for single-variable functions, but since we now have more than one independent variable, there is additional complexity that requires some new ideas.

Limits for Functions of Two Variables

If the values of $f(x, y)$ lie arbitrarily close to a fixed real number L for all points (x, y) sufficiently close to a point (x_0, y_0) , we say that f approaches the limit L as (x, y) approaches (x_0, y_0) . This is similar to the informal definition for the limit of a function of a single variable. Notice, however, that when (x_0, y_0) lies in the interior of f 's domain, (x, y) can approach (x_0, y_0) from any direction, not just from the left or the right. For the limit to exist, the same limiting value must be obtained whatever direction of approach is taken. We illustrate this issue in several examples following the definition.

DEFINITION Suppose that every open circular disk centered at (x_0, y_0) contains a point in the domain of f other than (x_0, y_0) itself. We say that a function $f(x, y)$ approaches the **limit** L as (x, y) approaches (x_0, y_0) , and write

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L,$$

if, for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f ,

$$|f(x, y) - L| < \varepsilon \quad \text{whenever} \quad 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

The definition of limit says that the distance between $f(x, y)$ and L becomes arbitrarily small whenever the distance from (x, y) to (x_0, y_0) is made sufficiently small (but not 0). The definition applies to interior points (x_0, y_0) as well as boundary points of the domain of f , although a boundary point need not lie within the domain. The points (x, y) that approach (x_0, y_0) are always taken to be in the domain of f . See Figure 13.12.

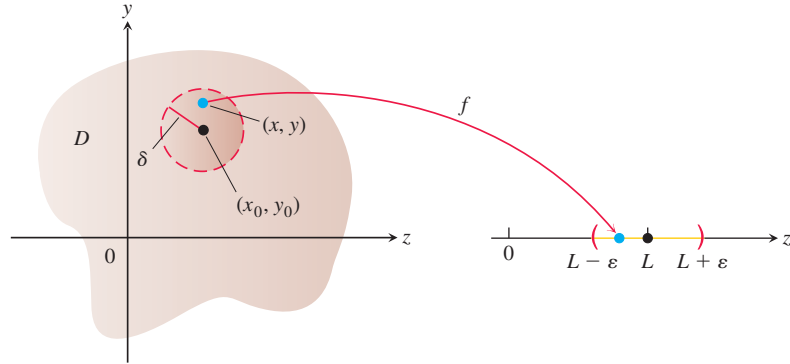


FIGURE 13.12 In the limit definition, δ is the radius of a disk centered at (x_0, y_0) . For all points (x, y) within this disk, the function values $f(x, y)$ lie inside the corresponding interval $(L - \varepsilon, L + \varepsilon)$.

As for functions of a single variable, it can be shown that

$$\lim_{(x, y) \rightarrow (x_0, y_0)} x = x_0 \quad (1)$$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} y = y_0 \quad (2)$$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} k = k \quad (\text{any number } k). \quad (3)$$

For example, in the first limit statement above, $f(x, y) = x$ and $L = x_0$. Using the definition of limit, suppose that $\varepsilon > 0$ is chosen. If we let δ equal this ε , we see that if

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta = \varepsilon,$$

then

$$\begin{aligned} \sqrt{(x - x_0)^2} &< \varepsilon & (x - x_0)^2 &\leq (x - x_0)^2 + (y - y_0)^2 \\ |x - x_0| &< \varepsilon & \sqrt{a^2} &= |a| \\ |f(x, y) - x_0| &< \varepsilon & x &= f(x, y) \end{aligned}$$

That is,

$$|f(x, y) - x_0| < \varepsilon \quad \text{whenever} \quad 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$

So a δ has been found satisfying the requirement of the definition, and therefore we have proved that

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = \lim_{(x, y) \rightarrow (x_0, y_0)} x = x_0.$$

Equation (1) is a special case of the more general formula

$$\lim_{(x, y) \rightarrow (x_0, y_0)} g(x) = \lim_{x \rightarrow x_0} g(x), \quad (4)$$

according to which, if $f(x, y)$ can be expressed as a function g of a single variable x , then $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ depends only on what happens to g as x approaches x_0 . Similarly, the following formula generalizes Equation (2):

$$\lim_{(x, y) \rightarrow (x_0, y_0)} h(y) = \lim_{y \rightarrow y_0} h(y) \quad (5)$$

As with single-variable functions, the limit of the sum of two functions is the sum of their limits (when they both exist), with similar results for the limits of the differences, constant multiples, products, quotients, powers, and roots. These facts are summarized in Theorem 1.

THEOREM 1—Properties of Limits of Functions of Two Variables

The following rules hold if L , M , and k are real numbers and

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M.$$

1. *Sum Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y) + g(x,y)] = L + M$
2. *Difference Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y) - g(x,y)] = L - M$
3. *Constant Multiple Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} kf(x,y) = kL \quad (\text{any number } k)$
4. *Product Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y) \cdot g(x,y)] = L \cdot M$
5. *Quotient Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}, \quad M \neq 0$
6. *Power Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)]^n = L^n, \quad n \text{ a positive integer}$
7. *Root Rule:* $\lim_{(x,y) \rightarrow (x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n},$
 $n \text{ a positive integer, and if } n \text{ is even,}$
 $\text{we assume that } L > 0.$
8. *Composition Rule:* If $h(z)$ is continuous at $z = L$, then
 $\lim_{(x,y) \rightarrow (x_0,y_0)} h(f(x,y)) = h(L).$

Although we will not prove Theorem 1 here, we give an informal discussion of why it is true. If (x, y) is sufficiently close to (x_0, y_0) , then $f(x, y)$ is close to L and $g(x, y)$ is close to M (from the informal interpretation of limits). It is then reasonable that $f(x, y) + g(x, y)$ is close to $L + M$; $f(x, y) - g(x, y)$ is close to $L - M$; $kf(x, y)$ is close to kL ; $f(x, y)g(x, y)$ is close to LM ; and $f(x, y)/g(x, y)$ is close to L/M if $M \neq 0$. Similarly, powers and roots of f are close to those of L , and a continuous function h composed with f has a value close to its value $h(L)$ when applied to L .

When we apply Theorem 1 and Equations (1)–(3) to polynomials and rational functions, we obtain the useful result that the limits of these functions as $(x, y) \rightarrow (x_0, y_0)$ can be calculated by evaluating the functions at (x_0, y_0) . The only requirement is that the rational functions be defined at (x_0, y_0) .

EXAMPLE 1 In this example, we combine Equations (1)–(5) with the results in Theorem 1 to calculate the limits.

$$(a) \quad \lim_{(x,y) \rightarrow (0,1)} \frac{x - xy + 3}{x^2y + 5xy - y^3} = \frac{0 - (0)(1) + 3}{(0)^2(1) + 5(0)(1) - (1)^3} = -3$$

$$(b) \quad \lim_{(x,y) \rightarrow (3,-4)} \sqrt{x^2 + y^2} = \sqrt{\lim_{(x,y) \rightarrow (3,-4)} (x^2 + y^2)} \quad \text{Rule 7}$$

$$= \sqrt{3^2 + (-4)^2} \quad \text{Rules 1 and 6 and Eq. (1) and (2)}$$

$$= \sqrt{25} = 5$$

$$(c) \quad \lim_{(x,y) \rightarrow (\pi/2, 0)} \left(\frac{x}{\sin x} - \frac{\sin y}{y} \right) = \lim_{(x,y) \rightarrow (\pi/2, 0)} \frac{x}{\sin x} - \lim_{(x,y) \rightarrow (\pi/2, 0)} \frac{\sin y}{y} \quad \text{Rule 2}$$

$$= \lim_{x \rightarrow \pi/2} \frac{x}{\sin x} - \lim_{y \rightarrow 0} \frac{\sin y}{y} \quad \text{Eq. (4) and (5)}$$

$$= \frac{\pi}{2} - 1 \quad \text{Theorem 6, Section 2.4} \quad \blacksquare$$

EXAMPLE 2 Find $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$.

Solution Since the denominator $\sqrt{x} - \sqrt{y}$ approaches 0 as $(x, y) \rightarrow (0, 0)$, we cannot use the Quotient Rule from Theorem 1. If we multiply numerator and denominator by $\sqrt{x} + \sqrt{y}$, however, we produce an equivalent fraction whose limit we *can* find:

$$\begin{aligned}
 \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}} &= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 - xy)(\sqrt{x} + \sqrt{y})}{(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y})} && \text{Multiply by a form equal to 1.} \\
 &= \lim_{(x,y) \rightarrow (0,0)} \frac{x(x - y)(\sqrt{x} + \sqrt{y})}{x - y} && \text{Algebra} \\
 &= \lim_{(x,y) \rightarrow (0,0)} x(\sqrt{x} + \sqrt{y}) && \text{Cancel the nonzero factor } (x - y). \\
 &= \left(\lim_{(x,y) \rightarrow (0,0)} x \right) \left[\left(\lim_{(x,y) \rightarrow (0,0)} \sqrt{x} \right) + \left(\lim_{(x,y) \rightarrow (0,0)} \sqrt{y} \right) \right] && \text{Rules 4 and 1} \\
 &= \left(\lim_{(x,y) \rightarrow (0,0)} x \right) \left[\sqrt{\lim_{(x,y) \rightarrow (0,0)} x} + \sqrt{\lim_{(x,y) \rightarrow (0,0)} y} \right] && \text{Rule 7} \\
 &= (0)[\sqrt{0} + \sqrt{0}] = 0 && \text{Eq. (1) and (2)}
 \end{aligned}$$

We can cancel the factor $(x - y)$ because the path $y = x$ (where we would have $x - y = 0$) is *not* in the domain of the function

$$f(x, y) = \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}.$$

EXAMPLE 3 Find $\lim_{(x,y) \rightarrow (0,0)} \frac{4xy^2}{x^2 + y^2}$ if it exists.

Solution We first observe that along the line $x = 0$, the function always has value 0 when $y \neq 0$. Likewise, along the line $y = 0$, the function has value 0 provided $x \neq 0$. So if the limit does exist as (x, y) approaches $(0, 0)$, the value of the limit must be 0 (see Figure 13.13). To see whether this is true, we apply the definition of limit.

Let $\varepsilon > 0$ be given, but arbitrary. We want to find a $\delta > 0$ such that

$$\left| \frac{4xy^2}{x^2 + y^2} - 0 \right| < \varepsilon \quad \text{whenever} \quad 0 < \sqrt{x^2 + y^2} < \delta$$

or

$$\frac{4|x|y^2}{x^2 + y^2} < \varepsilon \quad \text{whenever} \quad 0 < \sqrt{x^2 + y^2} < \delta.$$

Since $y^2 \leq x^2 + y^2$, we have that

$$\frac{4|x|y^2}{x^2 + y^2} \leq 4|x| = 4\sqrt{x^2} \leq 4\sqrt{x^2 + y^2}. \quad \frac{y^2}{x^2 + y^2} \leq 1$$

So if we choose $\delta = \varepsilon/4$ and let $0 < \sqrt{x^2 + y^2} < \delta$, we get

$$\left| \frac{4xy^2}{x^2 + y^2} - 0 \right| \leq 4\sqrt{x^2 + y^2} < 4\delta = 4\left(\frac{\varepsilon}{4}\right) = \varepsilon.$$

It follows from the definition that

$$\lim_{(x,y) \rightarrow (0,0)} \frac{4xy^2}{x^2 + y^2} = 0.$$

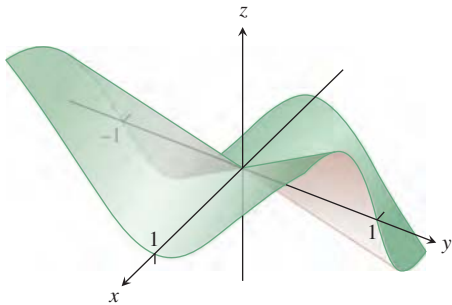


FIGURE 13.13 The surface graph suggests that the limit of the function in Example 3 must be 0, if it exists.

EXAMPLE 4 If $f(x, y) = \frac{y}{x}$, does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist?

Solution The domain of f does not include the y -axis, so we do not consider any points (x, y) where $x = 0$ in the approach toward the origin $(0, 0)$. Along the x -axis, the value of the function is $f(x, 0) = 0$ for all $x \neq 0$. So if the limit does exist as $(x, y) \rightarrow (0, 0)$, the value of the limit must be $L = 0$. On the other hand, along the line $y = x$, the value of the function is $f(x, x) = x/x = 1$ for all $x \neq 0$. That is, the function f approaches the value 1 along the line $y = x$. This means that for every disk of radius δ centered at $(0, 0)$, the disk will contain points $(x, 0)$ on the x -axis where the value of the function is 0, and also points (x, x) along the line $y = x$ where the value of the function is 1. So no matter how small we choose δ as the radius of the disk in Figure 13.12, there will be points within the disk for which the function values differ by 1. Therefore, the limit cannot exist because we can take ε to be any number less than 1 in the limit definition and deny that $L = 0$ or 1, or any other real number. The limit does not exist because we have different limiting values along different paths approaching the point $(0, 0)$. ■

Continuity

As with functions of a single variable, continuity is defined in terms of limits.

DEFINITION Suppose that every open circular disk centered at (x_0, y_0) contains a point in the domain of f other than (x_0, y_0) itself. Then a function $f(x, y)$ is **continuous at the point (x_0, y_0)** if

1. f is defined at (x_0, y_0) ,
2. $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ exists, and
3. $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = f(x_0, y_0)$.

A function is **continuous** if it is continuous at every point of its domain.

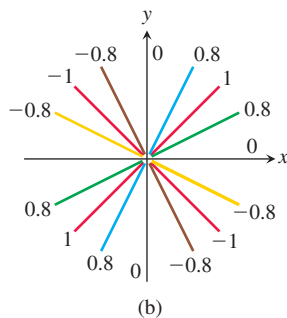
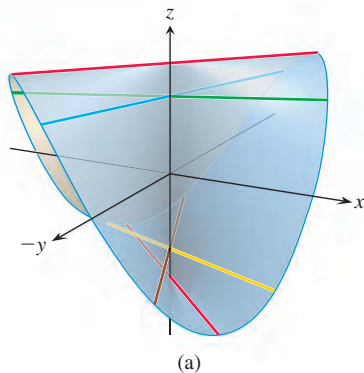


FIGURE 13.14 (a) The graph of

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

The function is continuous at every point except the origin. (b) The value of f along each line $y = mx$, $x \neq 0$, is constant but varies with m (Example 5).

As with the definition of limit, the definition of continuity applies at boundary points as well as interior points of the domain of f .

A consequence of Theorem 1 is that algebraic combinations of continuous functions are continuous at every point at which all the functions involved are defined. This means that sums, differences, constant multiples, products, quotients, and powers of continuous functions are continuous where defined. In particular, polynomials and rational functions of two variables are continuous at every point at which they are defined.

EXAMPLE 5 Show that

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at every point except the origin (Figure 13.14).

Solution The function f is continuous at every point (x, y) except $(0, 0)$ because its values at points other than $(0, 0)$ are given by a rational function of x and y , and therefore at those points the limiting value is simply obtained by substituting the values of x and y into that rational expression.

At $(0, 0)$, the value of f is defined, but f has no limit as $(x, y) \rightarrow (0, 0)$. The reason is that different paths of approach to the origin can lead to different results, as we now see.

For every value of m , the function f has a constant value on the “punctured” line $y = mx, x \neq 0$, because

$$f(x, y) \Big|_{y=mx} = \frac{2xy}{x^2 + y^2} \Big|_{y=mx} = \frac{2x(mx)}{x^2 + (mx)^2} = \frac{2mx^2}{x^2 + m^2x^2} = \frac{2m}{1 + m^2}.$$

Therefore, f has this number as its limit as (x, y) approaches $(0, 0)$ along the line:

$$\lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } y=mx}} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \left[f(x, y) \Big|_{y=mx} \right] = \frac{2m}{1 + m^2}.$$

This limit changes with each value of the slope m . There is therefore no single number we may call the limit of f as (x, y) approaches the origin. The limit fails to exist, and the function is not continuous at the origin. ■

Examples 4 and 5 illustrate an important point about limits of functions of two or more variables. For a limit to exist at a point, the limit must be the same along every approach path. This result is analogous to the single-variable case where both the left- and right-sided limits had to have the same value. For functions of two or more variables, if we ever find paths with different limits, we know the function has no limit at the point they approach.

Two-Path Test for Nonexistence of a Limit

If a function $f(x, y)$ has different limits along two different paths in the domain of f as (x, y) approaches (x_0, y_0) , then $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ does not exist.

EXAMPLE 6 Show that the function

$$f(x, y) = \frac{2x^2y}{x^4 + y^2}$$

(Figure 13.15) has no limit as (x, y) approaches $(0, 0)$.

Solution As (x, y) approaches $(0, 0)$, both the numerator and the denominator approach 0, which gives the indeterminate form $0/0$. We examine the values of f along parabolic curves that end at $(0, 0)$. Along the curve $y = kx^2, x \neq 0$, the function has the constant value

$$f(x, y) \Big|_{y=kx^2} = \frac{2x^2y}{x^4 + y^2} \Big|_{y=kx^2} = \frac{2x^2(kx^2)}{x^4 + (kx^2)^2} = \frac{2kx^4}{x^4 + k^2x^4} = \frac{2k}{1 + k^2}.$$

Therefore,

$$\lim_{\substack{(x, y) \rightarrow (0, 0) \\ \text{along } y=kx^2}} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \left[f(x, y) \Big|_{y=kx^2} \right] = \frac{2k}{1 + k^2}.$$

This limit varies with the path of approach. If (x, y) approaches $(0, 0)$ along the parabola $y = x^2$, for instance, $k = 1$ and the limit is 1. If (x, y) approaches $(0, 0)$ along the x -axis, $k = 0$ and the limit is 0. By the two-path test, f has no limit as (x, y) approaches $(0, 0)$. ■

It can be shown that the function in Example 6 has limit 0 along every straight line path $y = mx$ (Exercise 57). This implies the following observation:

Having the same limit along all straight lines approaching (x_0, y_0) does not imply that a limit exists at (x_0, y_0) .

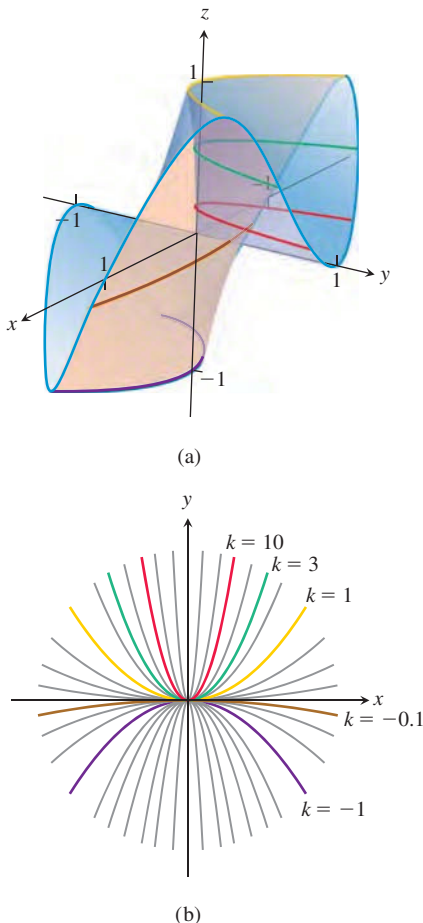


FIGURE 13.15 (a) The graph of $f(x, y) = 2x^2y/(x^4 + y^2)$. (b) Along each path $y = kx^2, x \neq 0$, the value of f is constant, but varies with k (Example 6).

Whenever it is correctly defined, the composition of continuous functions is also continuous. The only requirement is that each function be continuous where it is applied. The proof, omitted here, is similar to that for functions of a single variable (Theorem 9 in Section 2.6).

Continuity of Compositions

If f is continuous at (x_0, y_0) and g is a single-variable function continuous at $f(x_0, y_0)$, then the composition $h = g \circ f$ defined by $h(x, y) = g(f(x, y))$ is also continuous at (x_0, y_0) .

For example, the composite functions

$$e^{x-y}, \quad \cos \frac{xy}{x^2 + 1}, \quad \ln(1 + x^2 y^2)$$

are continuous at every point (x, y) .

Functions of More Than Two Variables

The definitions of limit and continuity for functions of two variables and the conclusions about limits and continuity for sums, products, quotients, powers, and compositions all extend to functions of three or more variables. Functions like

$$\ln(x + y + z) \quad \text{and} \quad \frac{y \sin z}{x - 1}$$

are continuous throughout their domains, and limits like

$$\lim_{P \rightarrow (1, 0, -1)} \frac{e^{x+z}}{z^2 + \cos \sqrt{xy}} = \frac{e^{1-1}}{(-1)^2 + \cos 0} = \frac{1}{2},$$

where P denotes the point (x, y, z) , may be found by direct substitution.

Extreme Values of Continuous Functions on Closed, Bounded Sets

The Extreme Value Theorem (Theorem 1, Section 4.1) states that a function of a single variable that is continuous at every point of a closed, bounded interval $[a, b]$ takes on an absolute maximum value and an absolute minimum value at least once in $[a, b]$. The same holds true of a function $z = f(x, y)$ that is continuous on a closed, bounded set R in the plane (like a line segment, a disk, or a filled-in triangle). The function takes on an absolute maximum value at some point in R and an absolute minimum value at some point in R . The function may take on a maximum or minimum value more than once over R .

Similar results hold for functions of three or more variables. A continuous function $w = f(x, y, z)$ must take on absolute maximum and minimum values on any closed, bounded set (such as a solid ball or cube, spherical shell, or rectangular solid) on which it is defined. We will learn how to find these extreme values in Section 13.7.

EXERCISES 13.2

Limits with Two Variables

Find the limits in Exercises 1–12.

1. $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 - y^2 + 5}{x^2 + y^2 + 2}$

2. $\lim_{(x,y) \rightarrow (0,4)} \frac{x}{\sqrt{y}}$

3. $\lim_{(x,y) \rightarrow (3,4)} \sqrt{x^2 + y^2 - 1}$

4. $\lim_{(x,y) \rightarrow (2,-3)} \left(\frac{1}{x} + \frac{1}{y} \right)^2$

5. $\lim_{(x,y) \rightarrow (0,\pi/4)} \sec x \tan y$

6. $\lim_{(x,y) \rightarrow (0,0)} \cos \frac{x^2 + y^3}{x + y + 1}$

7. $\lim_{(x,y) \rightarrow (0, \ln 2)} e^{x-y}$ 8. $\lim_{(x,y) \rightarrow (1,1)} \ln|1 + x^2y^2|$
9. $\lim_{(x,y) \rightarrow (0,0)} \frac{e^y \sin x}{x}$ 10. $\lim_{(x,y) \rightarrow (1/27, \pi^3)} \cos \sqrt[3]{xy}$
11. $\lim_{(x,y) \rightarrow (1, \pi/6)} \frac{x \sin y}{x^2 + 1}$ 12. $\lim_{(x,y) \rightarrow (\pi/2, 0)} \frac{\cos y + 1}{y - \sin x}$

Limits of Quotients

Find the limits in Exercises 13–24 by rewriting the fractions first.

13. $\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x^2 - 2xy + y^2}{x - y}$ 14. $\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x^2 - y^2}{x - y}$
15. $\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq 1}} \frac{xy - y - 2x + 2}{x - 1}$
16. $\lim_{\substack{(x,y) \rightarrow (2,-4) \\ x \neq -4, x \neq x^2}} \frac{y + 4}{x^2y - xy + 4x^2 - 4x}$
17. $\lim_{\substack{(x,y) \rightarrow (0,0) \\ x \neq y}} \frac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$
18. $\lim_{\substack{(x,y) \rightarrow (2,2) \\ x+y \neq 4}} \frac{x + y - 4}{\sqrt{x} + y - 2}$ 19. $\lim_{\substack{(x,y) \rightarrow (2,0) \\ 2x-y \neq 4}} \frac{\sqrt{2x-y} - 2}{2x - y - 4}$
20. $\lim_{\substack{(x,y) \rightarrow (4,3) \\ x \neq y+1}} \frac{\sqrt{x} - \sqrt{y+1}}{x - y - 1}$
21. $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2}$ 22. $\lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos(xy)}{xy}$
23. $\lim_{(x,y) \rightarrow (1,-1)} \frac{x^3 + y^3}{x + y}$ 24. $\lim_{(x,y) \rightarrow (2,2)} \frac{x - y}{x^4 - y^4}$

Limits with Three Variables

Find the limits in Exercises 25–30.

25. $\lim_{P \rightarrow (1,3,4)} \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$ 26. $\lim_{P \rightarrow (1,-1,-1)} \frac{2xy + yz}{x^2 + z^2}$
27. $\lim_{P \rightarrow (\pi, \pi, 0)} (\sin^2 x + \cos^2 y + \sec^2 z)$
28. $\lim_{P \rightarrow (-1/4, \pi/2, 2)} \tan^{-1} xyz$ 29. $\lim_{P \rightarrow (\pi, 0, 3)} ze^{-2y} \cos 2x$
30. $\lim_{P \rightarrow (2, -3, 6)} \ln \sqrt{x^2 + y^2 + z^2}$

Continuity for Two VariablesAt what points (x, y) in the plane are the functions in Exercises 31–34 continuous?

31. a. $f(x, y) = \sin(x + y)$ b. $f(x, y) = \ln(x^2 + y^2)$
32. a. $f(x, y) = \frac{x + y}{x - y}$ b. $f(x, y) = \frac{y}{x^2 + 1}$
33. a. $g(x, y) = \sin \frac{1}{xy}$ b. $g(x, y) = \frac{x + y}{2 + \cos x}$

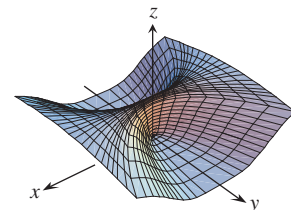
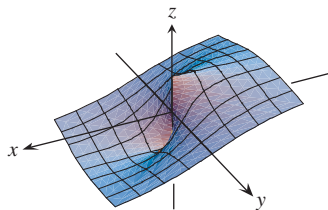
34. a. $g(x, y) = \frac{x^2 + y^2}{x^2 - 3x + 2}$ b. $g(x, y) = \frac{1}{x^2 - y}$

Continuity for Three VariablesAt what points (x, y, z) in space are the functions in Exercises 35–40 continuous?

35. a. $f(x, y, z) = x^2 + y^2 - 2z^2$ b. $f(x, y, z) = \sqrt{x^2 + y^2 - 1}$
36. a. $f(x, y, z) = \ln xyz$ b. $f(x, y, z) = e^{x+y} \cos z$
37. a. $h(x, y, z) = xy \sin \frac{1}{z}$ b. $h(x, y, z) = \frac{1}{x^2 + z^2 - 1}$
38. a. $h(x, y, z) = \frac{1}{|y| + |z|}$ b. $h(x, y, z) = \frac{1}{|xy| + |z|}$
39. a. $h(x, y, z) = \ln(z - x^2 - y^2 - 1)$ b. $h(x, y, z) = \frac{1}{z - \sqrt{x^2 + y^2}}$
40. a. $h(x, y, z) = \sqrt{4 - x^2 - y^2 - z^2}$ b. $h(x, y, z) = \frac{1}{4 - \sqrt{x^2 + y^2 + z^2} - 9}$

No Limit Exists at the OriginBy considering different paths of approach, show that the functions in Exercises 41–48 have no limit as $(x, y) \rightarrow (0, 0)$.

41. $f(x, y) = -\frac{x}{\sqrt{x^2 + y^2}}$ 42. $f(x, y) = \frac{x^4}{x^4 + y^2}$



43. $f(x, y) = \frac{x^4 - y^2}{x^4 + y^2}$ 44. $f(x, y) = \frac{xy}{|xy|}$
45. $g(x, y) = \frac{x - y}{x + y}$ 46. $g(x, y) = \frac{x^2 - y}{x - y}$
47. $h(x, y) = \frac{x^2 + y}{y}$ 48. $h(x, y) = \frac{x^2y}{x^4 + y^2}$

Theory and Examples

In Exercises 49–54, show that the limits do not exist.

49. $\lim_{(x,y) \rightarrow (1,1)} \frac{xy^2 - 1}{y - 1}$ 50. $\lim_{(x,y) \rightarrow (1,-1)} \frac{xy + 1}{x^2 - y^2}$
51. $\lim_{(x,y) \rightarrow (0,1)} \frac{x \ln y}{x^2 + (\ln y)^2}$ 52. $\lim_{(x,y) \rightarrow (1,0)} \frac{xe^y - 1}{xe^y - 1 + y}$
53. $\lim_{(x,y) \rightarrow (0,0)} \frac{y + \sin x}{x + \sin y}$ 54. $\lim_{(x,y) \rightarrow (1,1)} \frac{\tan y - y \tan x}{y - x}$

55. Let $f(x, y) = \begin{cases} 1, & y \geq x^4 \\ 1, & y \leq 0 \\ 0, & \text{otherwise.} \end{cases}$

Find each of the following limits, or explain that the limit does not exist.

a. $\lim_{(x,y) \rightarrow (0,1)} f(x, y)$

b. $\lim_{(x,y) \rightarrow (2,3)} f(x, y)$

c. $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$

56. Let $f(x, y) = \begin{cases} x^2, & x \geq 0 \\ x^3, & x < 0 \end{cases}$.

Find the following limits.

a. $\lim_{(x,y) \rightarrow (3,-2)} f(x, y)$

b. $\lim_{(x,y) \rightarrow (-2,1)} f(x, y)$

c. $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$

57. Show that the function in Example 6 has limit 0 along every straight line approaching (0, 0).

58. If $f(x_0, y_0) = 3$, what can you say about

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$$

if f is continuous at (x_0, y_0) ? If f is not continuous at (x_0, y_0) ? Give reasons for your answers.

The Sandwich Theorem for functions of two variables states that if $g(x, y) \leq f(x, y) \leq h(x, y)$ for all $(x, y) \neq (x_0, y_0)$ in a disk centered at (x_0, y_0) and if g and h have the same finite limit L as $(x, y) \rightarrow (x_0, y_0)$, then

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y) = L.$$

Use this result to support your answers to the questions in Exercises 59–62.

59. Does knowing that

$$1 - \frac{x^2 y^2}{3} < \frac{\tan^{-1} xy}{xy} < 1$$

tell you anything about

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\tan^{-1} xy}{xy}?$$

Give reasons for your answer.

60. Does knowing that

$$2|xy| - \frac{x^2 y^2}{6} < 4 - 4\cos\sqrt{|xy|} < 2|xy|$$

tell you anything about

$$\lim_{(x,y) \rightarrow (0,0)} \frac{4 - 4\cos\sqrt{|xy|}}{|xy|}?$$

Give reasons for your answer.

61. Does knowing that $|\sin(1/x)| \leq 1$ tell you anything about

$$\lim_{(x,y) \rightarrow (0,0)} y \sin \frac{1}{x}?$$

Give reasons for your answer.

62. Does knowing that $|\cos(1/y)| \leq 1$ tell you anything about

$$\lim_{(x,y) \rightarrow (0,0)} x \cos \frac{1}{y}?$$

Give reasons for your answer.

63. (Continuation of Example 5.)

a. Reread Example 5. Then substitute $m = \tan \theta$ into the formula

$$f(x, y) \Big|_{y=mx} = \frac{2m}{1+m^2}$$

and simplify the result to show how the value of f varies with the line's angle of inclination.

b. Use the formula you obtained in part (a) to show that the limit of f as $(x, y) \rightarrow (0, 0)$ along the line $y = mx$ varies from -1 to 1 , depending on the angle of approach.

64. **Continuous extension** Define $f(0, 0)$ in a way that extends

$$f(x, y) = xy \frac{x^2 - y^2}{x^2 + y^2}$$

to be continuous at the origin.

Changing Variables to Polar Coordinates

If you cannot make any headway with $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ in rectangular coordinates, try changing to polar coordinates. Substitute $x = r \cos \theta$, $y = r \sin \theta$, and investigate the limit of the resulting expression as $r \rightarrow 0$. In other words, try to decide whether there exists a number L satisfying the following criterion:

Given $\varepsilon > 0$, there exists a $\delta > 0$ such that for all r and θ ,

$$|r| < \delta \Rightarrow |f(r, \theta) - L| < \varepsilon. \quad (1)$$

If such an L exists, then

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{r \rightarrow 0} f(r \cos \theta, r \sin \theta) = L.$$

For instance,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + y^2} = \lim_{r \rightarrow 0} \frac{r^3 \cos^3 \theta}{r^2} = \lim_{r \rightarrow 0} r \cos^3 \theta = 0.$$

To verify the last of these equalities, we need to show that Equation (1) is satisfied with $f(r, \theta) = r \cos^3 \theta$ and $L = 0$. That is, we need to show that given any $\varepsilon > 0$, there exists a $\delta > 0$ such that for all r and θ ,

$$|r| < \delta \Rightarrow |r \cos^3 \theta - 0| < \varepsilon.$$

Since

$$|r \cos^3 \theta| = |r| |\cos^3 \theta| \leq |r| \cdot 1 = |r|,$$

the implication holds for all r and θ if we take $\delta = \varepsilon$.

In contrast,

$$\frac{x^2}{x^2 + y^2} = \frac{r^2 \cos^2 \theta}{r^2} = \cos^2 \theta$$

takes on all values from 0 to 1 regardless of how small $|r|$ is, so that

$\lim_{(x,y) \rightarrow (0,0)} x^2/(x^2 + y^2)$ does not exist.

In each of these instances, the existence or nonexistence of the limit as $r \rightarrow 0$ is fairly clear. Shifting to polar coordinates does not always help, however, and may even tempt us to false conclusions. For example, the limit may exist along every straight line (or ray) $\theta = \text{constant}$ and yet fail to exist in the broader sense. Example 5 illustrates this point. In polar coordinates, $f(x, y) = (2x^2y)/(x^4 + y^2)$ becomes

$$f(r \cos \theta, r \sin \theta) = \frac{r \cos \theta \sin 2\theta}{r^2 \cos^4 \theta + \sin^2 \theta}$$

for $r \neq 0$. If we hold θ constant and let $r \rightarrow 0$, the limit is 0. On the path $y = x^2$, however, we have $r \sin \theta = r^2 \cos^2 \theta$ and

$$\begin{aligned} f(r \cos \theta, r \sin \theta) &= \frac{r \cos \theta \sin 2\theta}{r^2 \cos^4 \theta + (r \cos^2 \theta)^2} \\ &= \frac{2r \cos^2 \theta \sin \theta}{2r^2 \cos^4 \theta} = \frac{r \sin \theta}{r^2 \cos^2 \theta} = 1. \end{aligned}$$

In Exercises 65–70, find the limit of f as $(x, y) \rightarrow (0, 0)$ or show that the limit does not exist.

$$65. f(x, y) = \frac{x^3 - xy^2}{x^2 + y^2} \quad 66. f(x, y) = \cos\left(\frac{x^3 - y^3}{x^2 + y^2}\right)$$

$$67. f(x, y) = \frac{y^2}{x^2 + y^2} \quad 68. f(x, y) = \frac{2x}{x^2 + x + y^2}$$

$$69. f(x, y) = \tan^{-1}\left(\frac{|x| + |y|}{x^2 + y^2}\right)$$

$$70. f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$$

In Exercises 71 and 72, define $f(0, 0)$ in a way that extends f to be continuous at the origin.

$$71. f(x, y) = \ln\left(\frac{3x^2 - x^2y^2 + 3y^2}{x^2 + y^2}\right)$$

$$72. f(x, y) = \frac{3x^2y}{x^2 + y^2}$$

Using the Limit Definition

Each of Exercises 73–78 gives a function $f(x, y)$ and a positive number ε . In each exercise, show that there exists a $\delta > 0$ such that for all (x, y) ,

$$\sqrt{x^2 + y^2} < \delta \Rightarrow |f(x, y) - f(0, 0)| < \varepsilon.$$

$$73. f(x, y) = x^2 + y^2, \quad \varepsilon = 0.01$$

$$74. f(x, y) = y/(x^2 + 1), \quad \varepsilon = 0.05$$

$$75. f(x, y) = (x + y)/(x^2 + 1), \quad \varepsilon = 0.01$$

$$76. f(x, y) = (x + y)/(2 + \cos x), \quad \varepsilon = 0.02$$

$$77. f(x, y) = \frac{xy^2}{x^2 + y^2} \quad \text{and} \quad f(0, 0) = 0, \quad \varepsilon = 0.04$$

$$78. f(x, y) = \frac{x^3 + y^4}{x^2 + y^2} \quad \text{and} \quad f(0, 0) = 0, \quad \varepsilon = 0.02$$

Each of Exercises 79–82 gives a function $f(x, y, z)$ and a positive number ε . In each exercise, show that there exists a $\delta > 0$ such that for all (x, y, z) ,

$$\sqrt{x^2 + y^2 + z^2} < \delta \Rightarrow |f(x, y, z) - f(0, 0, 0)| < \varepsilon.$$

$$79. f(x, y, z) = x^2 + y^2 + z^2, \quad \varepsilon = 0.015$$

$$80. f(x, y, z) = xyz, \quad \varepsilon = 0.008$$

$$81. f(x, y, z) = \frac{x + y + z}{x^2 + y^2 + z^2 + 1}, \quad \varepsilon = 0.015$$

$$82. f(x, y, z) = \tan^2 x + \tan^2 y + \tan^2 z, \quad \varepsilon = 0.03$$

$$83. \text{ Show that } f(x, y, z) = x + y - z \text{ is continuous at every point } (x_0, y_0, z_0).$$

$$84. \text{ Show that } f(x, y, z) = x^2 + y^2 + z^2 \text{ is continuous at the origin.}$$

13.3 Partial Derivatives

The calculus of several variables is similar to single-variable calculus applied to several variables, one at a time. When we hold all but one of the independent variables of a function constant and differentiate with respect to that one variable, we get a “partial” derivative. This section shows how partial derivatives are defined and interpreted geometrically, and how to calculate them by applying the familiar rules for differentiating functions of a single variable. The idea of *differentiability* for functions of several variables requires more than the existence of the partial derivatives, because a point can be approached from many different directions. However, we will see that differentiable functions of several variables behave similarly to differentiable single-variable functions. In particular, they are continuous and can be well approximated by linear functions.

Partial Derivatives of a Function of Two Variables

If (x_0, y_0) is a point in the domain of a function $f(x, y)$, the vertical plane $y = y_0$ will cut the surface $z = f(x, y)$ in the curve $z = f(x, y_0)$ (Figure 13.16). This curve is the graph

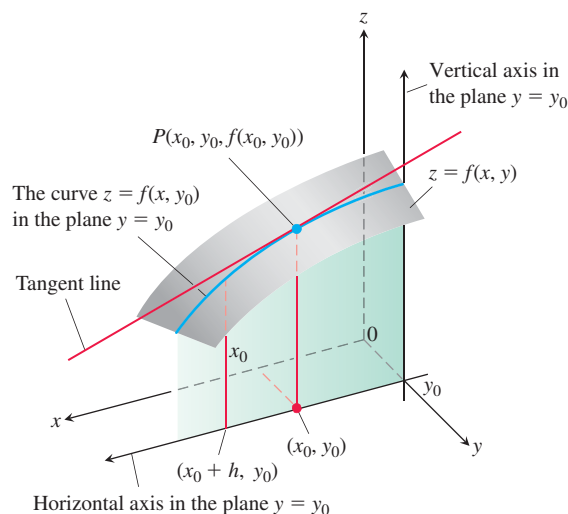


FIGURE 13.16 The intersection of the plane $y = y_0$ with the surface $z = f(x, y)$, viewed from above the first quadrant of the xy -plane.

of the function $z = f(x, y_0)$ in the plane $y = y_0$. The horizontal coordinate in this plane is x ; the vertical coordinate is z . The y -value is held constant at y_0 , so y is not a variable.

We define the partial derivative of f with respect to x at the point (x_0, y_0) as the ordinary derivative of $f(x, y_0)$ with respect to x at the point $x = x_0$. To distinguish partial derivatives from ordinary derivatives, we use the symbol ∂ rather than the d previously used. In the definition, h represents a real number, positive or negative.

DEFINITION The **partial derivative of $f(x, y)$ with respect to x at the point (x_0, y_0)** is

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the limit exists.

The partial derivative of $f(x, y)$ with respect to x at the point (x_0, y_0) is the same as the ordinary derivative of $f(x, y_0)$ at the point x_0 :

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0}.$$

A variety of notations are used to denote the partial derivative at a point (x_0, y_0) , including

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}, \quad f_x(x_0, y_0), \quad \text{and} \quad \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}.$$

When we do not specify a specific point (x_0, y_0) at which the partial derivative is being evaluated, then the partial derivative becomes a *function* whose domain is the points where the partial derivative exists. Notations for this function include

$$\frac{\partial f}{\partial x}, \quad f_x, \quad \text{and} \quad \frac{\partial z}{\partial x}.$$

The slope of the curve $z = f(x, y_0)$ at the point $P(x_0, y_0, f(x_0, y_0))$ in the plane $y = y_0$ is the value of the partial derivative of f with respect to x at (x_0, y_0) . (In Figure 13.16 this slope is negative.) The tangent line to the curve at P is the line in the plane $y = y_0$ that passes through P with this slope. The partial derivative $\partial f / \partial x$ at (x_0, y_0) gives the rate of change of f with respect to x when y is held fixed at the value y_0 .

The definition of the partial derivative of $f(x, y)$ with respect to y at a point (x_0, y_0) is similar to the definition of the partial derivative of f with respect to x . We hold x fixed at the value x_0 and take the ordinary derivative of $f(x_0, y)$ with respect to y at y_0 .

DEFINITION The partial derivative of $f(x, y)$ with respect to y at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h},$$

provided the limit exists.

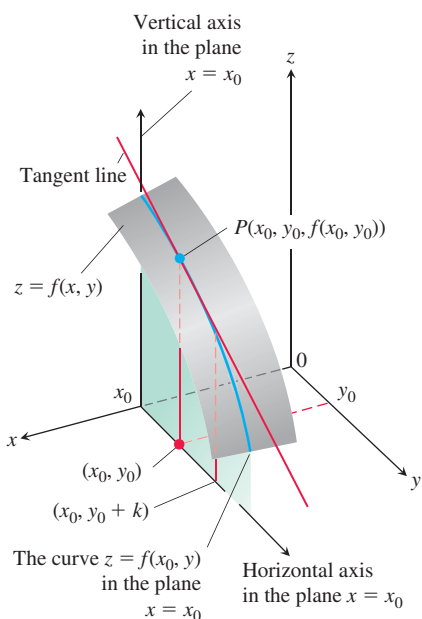


FIGURE 13.17 The intersection of the plane $x = x_0$ with the surface $z = f(x, y)$, viewed from above the first quadrant of the xy -plane.

The slope of the curve $z = f(x_0, y)$ at the point $P(x_0, y_0, f(x_0, y_0))$ in the vertical plane $x = x_0$ (Figure 13.17) is the partial derivative of f with respect to y at (x_0, y_0) . The tangent line to the curve at P is the line in the plane $x = x_0$ that passes through P with this slope. The partial derivative gives the rate of change of f with respect to y at (x_0, y_0) when x is held fixed at the value x_0 .

The partial derivative with respect to y is denoted the same way as the partial derivative with respect to x :

$$\frac{\partial f}{\partial y}(x_0, y_0), \quad f_y(x_0, y_0), \quad \frac{\partial f}{\partial y}, \quad f_y.$$

Notice that we now have two tangent lines associated with the surface $z = f(x, y)$ at the point $P(x_0, y_0, f(x_0, y_0))$ (Figure 13.18). Is the plane they determine tangent to the surface at P ? We will see that it is for the *differentiable* functions defined at the end of this section, and we will learn how to find the tangent plane in Section 13.6. First we have to better understand partial derivatives.

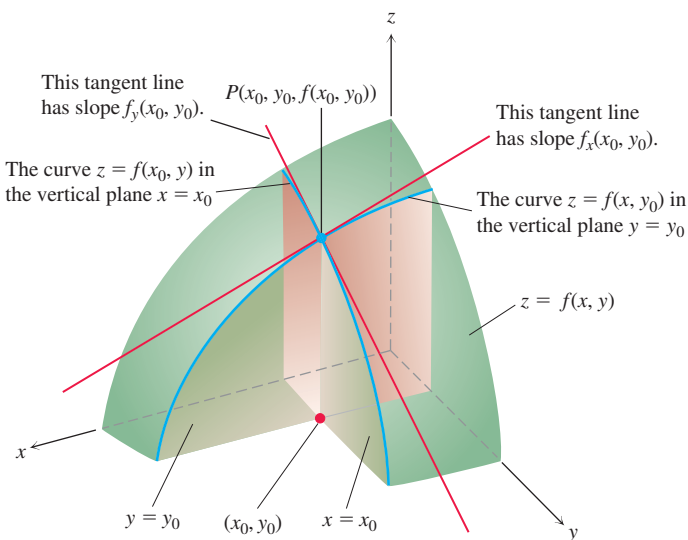


FIGURE 13.18 Figures 13.16 and 13.17 combined. The tangent lines at the point $(x_0, y_0, f(x_0, y_0))$ determine a plane that, in this picture at least, appears to be tangent to the surface.

Calculations

The definitions of $\partial f/\partial x$ and $\partial f/\partial y$ give us two different ways of differentiating f at a point: with respect to x in the usual way while treating y as a constant, and with respect to y in the usual way while treating x as a constant. As the following examples show, the values of these partial derivatives are usually different at a given point (x_0, y_0) .

EXAMPLE 1 Find the values of $\partial f/\partial x$ and $\partial f/\partial y$ at the point $(4, -5)$ if

$$f(x, y) = x^2 + 3xy + y - 1.$$

Solution To find $\partial f/\partial x$, we treat y as a constant and differentiate with respect to x :

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + 3xy + y - 1) = 2x + 3 \cdot 1 \cdot y + 0 - 0 = 2x + 3y.$$

The value of $\partial f/\partial x$ at $(4, -5)$ is $2(4) + 3(-5) = -7$.

To find $\partial f/\partial y$, we treat x as a constant and differentiate with respect to y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2 + 3xy + y - 1) = 0 + 3 \cdot x \cdot 1 + 1 - 0 = 3x + 1.$$

The value of $\partial f/\partial y$ at $(4, -5)$ is $3(4) + 1 = 13$. ■

EXAMPLE 2 Find $\partial f/\partial y$ as a function if $f(x, y) = y \sin xy$.

Solution We treat x as a constant and f as a product of y and $\sin xy$:

$$\begin{aligned} \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(y \sin xy) = y \frac{\partial}{\partial y} \sin xy + (\sin xy) \frac{\partial}{\partial y}(y) \\ &= (y \cos xy) \frac{\partial}{\partial y}(xy) + \sin xy = xy \cos xy + \sin xy. \end{aligned}$$
 ■

EXAMPLE 3 Find f_x and f_y as functions if

$$f(x, y) = \frac{2y}{y + \cos x}.$$

Solution We treat f as a quotient. With y held constant, we use the quotient rule to get

$$\begin{aligned} f_x &= \frac{\partial}{\partial x} \left(\frac{2y}{y + \cos x} \right) = \frac{(y + \cos x) \frac{\partial}{\partial x}(2y) - 2y \frac{\partial}{\partial x}(y + \cos x)}{(y + \cos x)^2} \\ &= \frac{(y + \cos x)(0) - 2y(-\sin x)}{(y + \cos x)^2} = \frac{2y \sin x}{(y + \cos x)^2}. \end{aligned}$$

With x held constant and again applying the quotient rule, we get

$$\begin{aligned} f_y &= \frac{\partial}{\partial y} \left(\frac{2y}{y + \cos x} \right) = \frac{(y + \cos x) \frac{\partial}{\partial y}(2y) - 2y \frac{\partial}{\partial y}(y + \cos x)}{(y + \cos x)^2} \\ &= \frac{(y + \cos x)(2) - 2y(1)}{(y + \cos x)^2} = \frac{2 \cos x}{(y + \cos x)^2}. \end{aligned}$$
 ■

Implicit differentiation works for partial derivatives the way it works for ordinary derivatives, as the next example illustrates.

EXAMPLE 4 Find $\partial z / \partial x$ assuming that the equation

$$yz - \ln z = x + y$$

defines z as a function of the two independent variables x and y and the partial derivative exists.

Solution We differentiate both sides of the equation with respect to x , holding y constant and treating z as a differentiable function of x :

$$\begin{aligned} \frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial x} \ln z &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial x} \\ y \frac{\partial z}{\partial x} - \frac{1}{z} \frac{\partial z}{\partial x} &= 1 + 0 && \text{With } y \text{ constant, } \frac{\partial}{\partial x}(yz) = y \frac{\partial z}{\partial x}. \\ \left(y - \frac{1}{z}\right) \frac{\partial z}{\partial x} &= 1 \\ \frac{\partial z}{\partial x} &= \frac{z}{yz - 1}. \end{aligned}$$

EXAMPLE 5 The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$ in a parabola. Find the slope of the tangent line to the parabola at $(1, 2, 5)$ (Figure 13.19).

Solution The parabola lies in a plane parallel to the yz -plane, and the slope is the value of the partial derivative $\partial z / \partial y$ at $(1, 2)$:

$$\left. \frac{\partial z}{\partial y} \right|_{(1,2)} = \left. \frac{\partial}{\partial y}(x^2 + y^2) \right|_{(1,2)} = 2y \Big|_{(1,2)} = 2(2) = 4.$$

As a check, we can treat the parabola as the graph of the single-variable function $z = (1)^2 + y^2 = 1 + y^2$ in the plane $x = 1$ and ask for the slope at $y = 2$. The slope, calculated now as an ordinary derivative, is

$$\left. \frac{dz}{dy} \right|_{y=2} = \left. \frac{d}{dy}(1 + y^2) \right|_{y=2} = 2y \Big|_{y=2} = 4.$$

Functions of More Than Two Variables

The definitions of the partial derivatives of functions of more than two independent variables are similar to the definitions for functions of two variables. They are ordinary derivatives with respect to one variable, taken while the other independent variables are held constant.

EXAMPLE 6 If x , y , and z are independent variables and

$$f(x, y, z) = x \sin(y + 3z),$$

then

$$\begin{aligned} \frac{\partial f}{\partial z} &= \frac{\partial}{\partial z}[x \sin(y + 3z)] = x \frac{\partial}{\partial z} \sin(y + 3z) && x \text{ held constant} \\ &= x \cos(y + 3z) \frac{\partial}{\partial z}(y + 3z) && \text{Chain rule} \\ &= 3x \cos(y + 3z). && y \text{ held constant} \end{aligned}$$

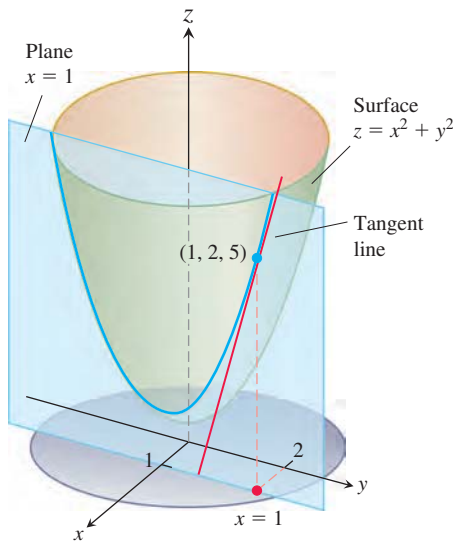


FIGURE 13.19 The tangent line to the curve of intersection of the plane $x = 1$ and the surface $z = x^2 + y^2$ at the point $(1, 2, 5)$ (Example 5).

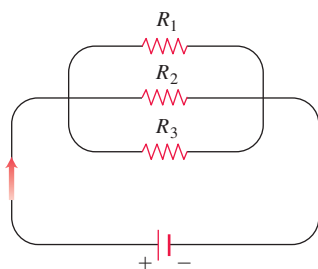


FIGURE 13.20 Resistors arranged this way are said to be connected in parallel (Example 7). Each resistor lets a portion of the current through. Their equivalent resistance R is calculated with the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

EXAMPLE 7 If resistors of R_1 , R_2 , and R_3 ohms are connected in parallel to make an R -ohm resistor, the value of R can be found from the equation

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(Figure 13.20). Find the value of $\partial R / \partial R_2$ when $R_1 = 30$, $R_2 = 45$, and $R_3 = 90$ ohms.

Solution To find $\partial R / \partial R_2$, we treat R_1 and R_3 as constants and, using implicit differentiation, differentiate both sides of the equation with respect to R_2 :

$$\begin{aligned} \frac{\partial}{\partial R_2} \left(\frac{1}{R} \right) &= \frac{\partial}{\partial R_2} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) \\ -\frac{1}{R^2} \frac{\partial R}{\partial R_2} &= 0 - \frac{1}{R_2^2} + 0 \\ \frac{\partial R}{\partial R_2} &= \frac{R^2}{R_2^2} = \left(\frac{R}{R_2} \right)^2. \end{aligned}$$

When $R_1 = 30$, $R_2 = 45$, and $R_3 = 90$,

$$\frac{1}{R} = \frac{1}{30} + \frac{1}{45} + \frac{1}{90} = \frac{3 + 2 + 1}{90} = \frac{6}{90} = \frac{1}{15},$$

so $R = 15$ and

$$\frac{\partial R}{\partial R_2} = \left(\frac{15}{45} \right)^2 = \left(\frac{1}{3} \right)^2 = \frac{1}{9}.$$

Thus at the given values, a small change in the resistance R_2 leads to a change in R about one-ninth as large. ■

Partial Derivatives and Continuity

A function $f(x, y)$ can have partial derivatives with respect to both x and y at a point without the function being continuous there. This is different from functions of a single variable, where the existence of a derivative implies continuity. If the partial derivatives of $f(x, y)$ exist and are continuous throughout a disk centered at (x_0, y_0) , however, then f is continuous at (x_0, y_0) , as we see at the end of this section.

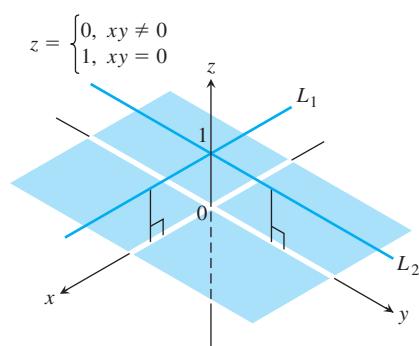


FIGURE 13.21 The graph of

$$f(x, y) = \begin{cases} 0, & xy \neq 0 \\ 1, & xy = 0 \end{cases}$$

consists of the lines L_1 and L_2 (lying 1 unit above the xy -plane) and the four open quadrants of the xy -plane. The function has partial derivatives at the origin but is not continuous there (Example 8).

EXAMPLE 8 Let

$$f(x, y) = \begin{cases} 0, & xy \neq 0 \\ 1, & xy = 0 \end{cases}$$

(Figure 13.21).

- Find the limit of f as (x, y) approaches $(0, 0)$ along the line $y = x$.
- Find the limit of f as (x, y) approaches $(0, 0)$ along the line $y = 0$.
- Prove that f is not continuous at the origin.
- Show that both partial derivatives $\partial f / \partial x$ and $\partial f / \partial y$ exist at the origin.

Solution

- (a) Since $f(x, y)$ is zero at every point on the line $y = x$ (except at the origin), we have

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \Big|_{y=x} = \lim_{(x, y) \rightarrow (0, 0)} 0 = 0.$$

- (b) Since $f(x, y)$ takes the constant value 1 at every point on the line $y = 0$, we have

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \Big|_{y=0} = \lim_{(x, y) \rightarrow (0, 0)} 1 = 1.$$

- (c) By the two-path test, f has no limit as (x, y) approaches $(0, 0)$. Consequently, f is not continuous at $(0, 0)$.
- (d) To find $\partial f / \partial x$ at $(0, 0)$, we hold y fixed at $y = 0$. Then $f(x, y) = 1$ for all x , and the graph of f is the line L_1 in Figure 13.21. The slope of this line at any x is $\partial f / \partial x = 0$. In particular, $\partial f / \partial x = 0$ at $(0, 0)$. Similarly, $\partial f / \partial y$ is the slope of line L_2 at any y , so $\partial f / \partial y = 0$ at $(0, 0)$. ■

What Example 8 suggests is that we need a stronger requirement for differentiability in higher dimensions than the mere existence of the partial derivatives. We define differentiability for functions of two variables (which is somewhat more complicated than for single-variable functions) at the end of this section and then revisit the connection to continuity.

Second-Order Partial Derivatives

When we differentiate a function $f(x, y)$ twice, we produce its second-order derivatives. These derivatives are usually denoted by

$$\frac{\partial^2 f}{\partial x^2} \text{ or } f_{xx}, \quad \frac{\partial^2 f}{\partial y^2} \text{ or } f_{yy},$$

$$\frac{\partial^2 f}{\partial x \partial y} \text{ or } f_{yx}, \quad \text{and} \quad \frac{\partial^2 f}{\partial y \partial x} \text{ or } f_{xy}.$$

The defining equations are

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right),$$

and so on. Notice the order in which the mixed partial derivatives are taken:

$$\frac{\partial^2 f}{\partial x \partial y} \quad \text{Differentiate first with respect to } y, \text{ then with respect to } x.$$

$$f_{yx} = (f_y)_x \quad \text{Means the same thing}$$

HISTORICAL BIOGRAPHY

Pierre-Simon Laplace
(1749–1827)

Mathematician and astronomer, Laplace was born in Normandy, France. He was among the most influential scientists of his time and was called the Newton of France for contributions to the understanding of the solar system's stability. Laplace also generalized the laws of mechanics for their application to the motion and properties of the heavenly bodies.

To know more, visit the companion Website.

EXAMPLE 9 If $f(x, y) = x \cos y + ye^x$, find the second-order derivatives

$$\frac{\partial^2 f}{\partial x^2}, \quad \frac{\partial^2 f}{\partial y \partial x}, \quad \frac{\partial^2 f}{\partial y^2}, \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y}.$$

Solution The first step is to calculate both first partial derivatives.

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(x \cos y + ye^x) & \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(x \cos y + ye^x) \\ &= \cos y + ye^x & &= -x \sin y + e^x \end{aligned}$$

Now we find both partial derivatives of each first partial:

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = -\sin y + e^x & \frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = -\sin y + e^x \\ \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = ye^x & \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = -x \cos y. \end{aligned} \quad \blacksquare$$

The Mixed Derivative Theorem

You may have noticed that the “mixed” second-order partial derivatives

$$\frac{\partial^2 f}{\partial y \partial x} \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y}$$

in Example 9 are equal. This is not a coincidence. They must be equal whenever f , f_x , f_y , f_{xy} , and f_{yx} are continuous, as stated in the following theorem. However, the mixed derivatives can be different when the continuity conditions are not satisfied (see Exercise 82).

THEOREM 2—The Mixed Derivative Theorem

If $f(x, y)$ and its partial derivatives f_x , f_y , f_{xy} , and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

HISTORICAL BIOGRAPHY

Alexis Clairaut (1713–1765)

Alexis Clairaut was a mathematical genius, who was called to visit the Academy of Sciences in Paris when he was only 12 years old. In a study published in 1743, the Clairaut proposition postulates in a simple way the dependency of the geometrical flattening ratio on the relationship between the gravity and the centrifugal force.

To know more, visit the companion Website.

Theorem 2 is also known as Clairaut's Theorem, after the French mathematician Alexis Clairaut, who discovered it. A proof is given in Appendix A.10. Theorem 2 says that to calculate a mixed second-order derivative, we may differentiate in either order, provided the continuity conditions are satisfied. This ability to proceed in different order sometimes simplifies our calculations.

EXAMPLE 10 Find $\frac{\partial^2 w}{\partial x \partial y}$ if

$$w = xy + \frac{e^y}{y^2 + 1}.$$

Solution The symbol $\partial^2 w / \partial x \partial y$ tells us to differentiate first with respect to y and then with respect to x . However, if we interchange the order of differentiation and differentiate first with respect to x , we get the answer more quickly. In two steps,

$$\frac{\partial w}{\partial x} = y \quad \text{and} \quad \frac{\partial^2 w}{\partial y \partial x} = 1.$$

If we differentiate first with respect to y , we obtain $\partial^2 w / \partial x \partial y = 1$ as well, but with more work. We can differentiate in either order because the conditions of Theorem 2 hold for w at all points (x_0, y_0) . ■

Partial Derivatives of Still Higher Order

Although we will deal mostly with first- and second-order partial derivatives, because these appear the most frequently in applications, there is no theoretical limit to how many times we can differentiate a function as long as the derivatives involved exist. Thus, we get third- and fourth-order derivatives denoted by symbols like

$$\begin{aligned} \frac{\partial^3 f}{\partial x \partial y^2} &= f_{yyx}, \\ \frac{\partial^4 f}{\partial x^2 \partial y^2} &= f_{yyxx}, \end{aligned}$$

and so on. As with second-order derivatives, the order of differentiation is immaterial as long as all the derivatives through the order in question are continuous.

EXAMPLE 11 Find f_{yxyz} if $f(x, y, z) = 1 - 2xy^2z + x^2y$.

Solution We first differentiate with respect to the variable y , then x , then y again, and finally with respect to z :

$$\begin{aligned} f_y &= -4xyz + x^2 \\ f_{yx} &= -4yz + 2x \\ f_{yxy} &= -4z \\ f_{yxyz} &= -4. \end{aligned}$$



Differentiability

The concept of differentiability for functions of several variables is more complicated than for single-variable functions, because a point in the domain can be approached from many directions and along any path, not just from the left or from the right. The existence of both partial derivatives at a point (x_0, y_0) is not by itself even enough to show continuity at (x_0, y_0) , as we saw in Example 8. The differentiability of f is instead based on the idea that a linear function gives a good model of a differentiable function near a point.

In Section 3.11, we saw that a differentiable function f can be approximated near a point x_0 by its linearization,

$$L(x) = f(x_0) + f'(x_0)(x - x_0).$$

This formula allows us to find a linear function L , a function whose graph is a straight line, such that L closely approximates f near x_0 . This can be done whenever f is differentiable, even when f itself is described by a very complicated formula. Approximations are much more useful and meaningful when they are accompanied by information on their accuracy. In Section 3.11, Equation (1), we saw that a differentiable function f satisfies

$$f(x) - f(x_0) = f'(x_0)(x - x_0) + \varepsilon(x - x_0),$$

where $\varepsilon \rightarrow 0$ as $x \rightarrow x_0$. Framed in terms of approximating f by L , this becomes

$$f(x) - L(x) = \varepsilon(x - x_0), \quad (1)$$

where again $\varepsilon \rightarrow 0$ as $x \rightarrow x_0$.

Rather than being a consequence of the definition, the differentiability for a function of two variables $f(x, y)$ is defined to mean that f can be approximated by a linear function. The approximating linear function $L(x, y)$ for $f(x, y)$ near the point (x_0, y_0) takes the form

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0),$$

and the graph of L is a plane, called the *tangent plane*, that approximates the graph of f near (x_0, y_0) . Notice that $L(x_0, y_0) = f(x_0, y_0)$, so the functions L and f coincide at (x_0, y_0) . Moreover the partial derivatives of L and f are also equal at (x_0, y_0) . We will study tangent planes in detail in Section 13.6.

We now specify how closely f is approximated by L at (x_0, y_0) . Extending the formula for single variable functions in Equation (1), we require that the difference between f and L satisfies

$$f(x, y) - L(x, y) = \varepsilon_1(x - x_0) + \varepsilon_2(y - y_0), \quad (2)$$

where both $\varepsilon_1 \rightarrow 0$ and $\varepsilon_2 \rightarrow 0$ as $(x, y) \rightarrow (x_0, y_0)$.

If we insert the formula for $L(x, y)$ into Equation (2) we see that

$$f(x, y) - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + \varepsilon_1(x - x_0) + \varepsilon_2(y - y_0).$$

Setting $\Delta x = x - x_0$, $\Delta y = y - y_0$, and $\Delta z = f(x, y) - f(x_0, y_0)$, we get

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y.$$

Based on these ideas, we now state the formal definition of differentiability, which captures the idea that f is well approximated by L .

DEFINITION A function $z = f(x, y)$ is **differentiable at** (x_0, y_0) if both $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist and if $\Delta z = f(x, y) - f(x_0, y_0)$ satisfies

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where $\Delta x = x - x_0$, $\Delta y = y - y_0$, and both $\varepsilon_1 \rightarrow 0$ and $\varepsilon_2 \rightarrow 0$ as $(x, y) \rightarrow (x_0, y_0)$. We call the function f **differentiable** if it is differentiable at every point in its domain, and we then say that its graph is a **smooth surface**.

Differentiability is defined in a similar way for functions of more than two variables.

The following theorem (proved in Appendix A.10) and its accompanying corollary tell us that functions with *continuous* first partial derivatives at (x_0, y_0) are differentiable there, and they are closely approximated locally by a linear function. We study this approximation in Section 13.6.

THEOREM 3—The Increment Theorem for Functions of Two Variables

Suppose that the first partial derivatives of $f(x, y)$ are defined throughout an open region R containing the point (x_0, y_0) and that f_x and f_y are continuous at (x_0, y_0) . Then the change

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

in the value of f that results from moving from (x_0, y_0) to another point $(x_0 + \Delta x, y_0 + \Delta y)$ in R satisfies an equation of the form

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

in which each of $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as both $\Delta x, \Delta y \rightarrow 0$.

In many cases the partial derivatives are defined and continuous at every point in the domain of f . We then have the following Corollary.

Corollary of Theorem 3

If the partial derivatives f_x and f_y of a function $f(x, y)$ are continuous throughout an open region R , then f is differentiable at every point of R .

If $z = f(x, y)$ is differentiable, then the definition of differentiability ensures that $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$ approaches 0 as Δx and Δy approach 0. This tells us that a function of two variables is continuous at every point where it is differentiable.

THEOREM 4—Differentiable Implies Continuous

If a function $f(x, y)$ is differentiable at (x_0, y_0) , then f is continuous at (x_0, y_0) .

As we can see from Corollary 3 and Theorem 4, a function $f(x, y)$ must be continuous at a point (x_0, y_0) if f_x and f_y are continuous throughout an open region containing (x_0, y_0) . Remember, however, that it is still possible for a function of two variables to be discontinuous at a point where its first partial derivatives exist, as we saw in Example 8. Existence alone of the partial derivatives at that point is not enough, but continuity of the partial derivatives guarantees differentiability.

EXERCISES 13.3

Calculating First-Order Partial Derivatives

In Exercises 1–22, find $\partial f/\partial x$ and $\partial f/\partial y$.

1. $f(x, y) = 2x^2 - 3y - 4$
2. $f(x, y) = x^2 - xy + y^2$
3. $f(x, y) = (x^2 - 1)(y + 2)$
4. $f(x, y) = 5xy - 7x^2 - y^2 + 3x - 6y + 2$
5. $f(x, y) = (xy - 1)^2$
6. $f(x, y) = (2x - 3y)^3$
7. $f(x, y) = \sqrt{x^2 + y^2}$
8. $f(x, y) = (x^3 + (y/2))^{2/3}$
9. $f(x, y) = 1/(x + y)$
10. $f(x, y) = x/(x^2 + y^2)$
11. $f(x, y) = (x + y)/(xy - 1)$
12. $f(x, y) = \tan^{-1}(y/x)$
13. $f(x, y) = e^{(x+y+1)}$
14. $f(x, y) = e^{-x} \sin(x + y)$

15. $f(x, y) = \ln(x + y)$
16. $f(x, y) = e^{xy} \ln y$
17. $f(x, y) = \sin^2(x - 3y)$
18. $f(x, y) = \cos^2(3x - y^2)$
19. $f(x, y) = x^y$
20. $f(x, y) = \log_y x$
21. $f(x, y) = \int_x^y g(t) dt$ (g continuous for all t)
22. $f(x, y) = \sum_{n=0}^{\infty} (xy)^n$ ($|xy| < 1$)

In Exercises 23–34, find f_x, f_y , and f_z .

23. $f(x, y, z) = 1 + xy^2 - 2z^2$
24. $f(x, y, z) = xy + yz + xz$
25. $f(x, y, z) = x - \sqrt{y^2 + z^2}$

26. $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$
 27. $f(x, y, z) = \arcsin(xyz)$
 28. $f(x, y, z) = \operatorname{arcsec}(x + yz)$
 29. $f(x, y, z) = \ln(x + 2y + 3z)$
 30. $f(x, y, z) = yz \ln(xy)$
 31. $f(x, y, z) = e^{-(x^2 + y^2 + z^2)}$
 32. $f(x, y, z) = e^{-xyz}$
 33. $f(x, y, z) = \tanh(x + 2y + 3z)$
 34. $f(x, y, z) = \sinh(xy - z^2)$

In Exercises 35–40, find the partial derivative of the function with respect to each variable.

35. $f(t, \alpha) = \cos(2\pi t - \alpha)$
 36. $g(u, v) = v^2 e^{(2u/v)}$
 37. $h(\rho, \phi, \theta) = \rho \sin \phi \cos \theta$
 38. $g(r, \theta, z) = r(1 - \cos \theta) - z$
 39. **Work done by the heart** (Section 3.11, Exercise 59)

$$W(P, V, \delta, v, g) = PV + \frac{V\delta v^2}{2g}$$

40. **Wilson lot size formula** (Section 4.6, Exercise 61)

$$A(c, h, k, m, q) = \frac{km}{q} + cm + \frac{hq}{2}$$

Calculating Second-Order Partial Derivatives

Find all the second-order partial derivatives of the functions in Exercises 41–54.

41. $f(x, y) = x + y + xy$ 42. $f(x, y) = \sin xy$
 43. $g(x, y) = x^2y + \cos y + y \sin x$
 44. $h(x, y) = xe^y + y + 1$ 45. $r(x, y) = \ln(x + y)$
 46. $s(x, y) = \arctan(y/x)$ 47. $w = x^2 \tan(xy)$
 48. $w = ye^{x^2 - y}$ 49. $w = x \sin(x^2y)$
 50. $w = \frac{x - y}{x^2 + y}$ 51. $f(x, y) = x^2y^3 - x^4 + y^5$
 52. $g(x, y) = \cos x^2 - \sin 3y$ 53. $z = x \sin(2x - y^2)$
 54. $z = xe^{x/y^2}$

Mixed Partial Derivatives

In Exercises 55–60, verify that $w_{xy} = w_{yx}$.

55. $w = \ln(2x + 3y)$ 56. $w = e^x + x \ln y + y \ln x$
 57. $w = xy^2 + x^2y^3 + x^3y^4$
 58. $w = x \sin y + y \sin x + xy$
 59. $w = \frac{x^2}{y^3}$ 60. $w = \frac{3x - y}{x + y}$

61. Which order of differentiation enables one to calculate f_{xy} faster: x first or y first? Try to answer without writing anything down.

- a. $f(x, y) = x \sin y + e^y$
 b. $f(x, y) = 1/x$
 c. $f(x, y) = y + (x/y)$
 d. $f(x, y) = y + x^2y + 4y^3 - \ln(y^2 + 1)$
 e. $f(x, y) = x^2 + 5xy + \sin x + 7e^x$
 f. $f(x, y) = x \ln xy$

62. The fifth-order partial derivative $\partial^5 f / \partial x^2 \partial y^3$ is zero for each of the following functions. To show this as quickly as possible, which variable would you differentiate with respect to first: x or y ? Try to answer without writing anything down.

- a. $f(x, y) = y^2x^4e^x + 2$
 b. $f(x, y) = y^2 + y(\sin x - x^4)$
 c. $f(x, y) = x^2 + 5xy + \sin x + 7e^x$
 d. $f(x, y) = xe^{y^2/2}$

Using the Partial Derivative Definition

In Exercises 63–66, use the limit definition of partial derivative to compute the partial derivatives of the functions at the specified points.

63. $f(x, y) = 1 - x + y - 3x^2y$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(1, 2)$
 64. $f(x, y) = 4 + 2x - 3y - xy^2$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(-2, 1)$
 65. $f(x, y) = \sqrt{2x + 3y - 1}$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(-2, 3)$
 66. $f(x, y) = \begin{cases} \frac{\sin(x^3 + y^4)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$
 $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(0, 0)$

67. **Three variables** Let $w = f(x, y, z)$ be a function of three independent variables and write the formal definition of the partial derivative $\partial f / \partial z$ at (x_0, y_0, z_0) . Use this definition to find $\partial f / \partial z$ at $(1, 2, 3)$ for $f(x, y, z) = x^2yz^2$.

68. **Three variables** Let $w = f(x, y, z)$ be a function of three independent variables and write the formal definition of the partial derivative $\partial f / \partial y$ at (x_0, y_0, z_0) . Use this definition to find $\partial f / \partial y$ at $(-1, 0, 3)$ for $f(x, y, z) = -2xy^2 + yz^2$.

Differentiating Implicitly

69. Find the value of $\partial z / \partial x$ at the point $(1, 1, 1)$ if the equation

$$xy + z^3x - 2yz = 0$$

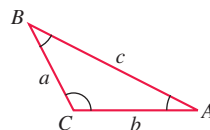
defines z as a function of the two independent variables x and y and the partial derivative exists.

70. Find the value of $\partial x / \partial z$ at the point $(1, -1, -3)$ if the equation

$$xz + y \ln x - x^2 + 4 = 0$$

defines x as a function of the two independent variables y and z and the partial derivative exists.

Exercises 71 and 72 are about the triangle shown here.



71. Express A implicitly as a function of a , b , and c and calculate $\partial A / \partial a$ and $\partial A / \partial b$.
 72. Express a implicitly as a function of A , b , and B and calculate $\partial a / \partial A$ and $\partial a / \partial B$.

73. Two dependent variables Express v_x in terms of u and y if the equations $x = v \ln u$ and $y = u \ln v$ define u and v as functions of the independent variables x and y , and if v_x exists. (Hint: Differentiate both equations with respect to x and solve for v_x by eliminating u_x .)

74. Two dependent variables Find $\partial x/\partial u$ and $\partial y/\partial u$ if the equations $u = x^2 - y^2$ and $v = x^2 + y^2$ define x and y as functions of the independent variables u and v , and the partial derivatives exist. (See the hint in Exercise 73.) Then let $s = x^2 + y^2$ and find $\partial s/\partial u$.

Theory and Examples

75. Let $f(x, y) = 2x + 3y - 4$. Find the slope of the line tangent to this surface at the point $(2, -1)$ and lying in **a.** the plane $x = 2$ **b.** the plane $y = -1$.

76. Let $f(x, y) = x^2 + y^3$. Find the slope of the line tangent to this surface at the point $(-1, 1)$ and lying in **a.** the plane $x = -1$ **b.** the plane $y = 1$.

In Exercises 77–80, find a function $z = f(x, y)$ whose partial derivatives are as given, or explain why this is impossible.

77. $\frac{\partial f}{\partial x} = 3x^2y^2 - 2x$, $\frac{\partial f}{\partial y} = 2x^3y + 6y$

78. $\frac{\partial f}{\partial x} = 2xe^{xy^2} + x^2y^2e^{xy^2} + 3$, $\frac{\partial f}{\partial y} = 2x^3ye^{xy^2} - e^y$

79. $\frac{\partial f}{\partial x} = \frac{2y}{(x+y)^2}$, $\frac{\partial f}{\partial y} = \frac{2x}{(x+y)^2}$

80. $\frac{\partial f}{\partial x} = xy \cos(xy) + \sin(xy)$, $\frac{\partial f}{\partial y} = x \cos(xy)$

81. Let $f(x, y) = \begin{cases} y^3, & y \geq 0 \\ -y^2, & y < 0. \end{cases}$

Find f_x, f_y, f_{xy} , and f_{yx} , and state the domain for each partial derivative.

82. Let $f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$

a. Show that $\frac{\partial f}{\partial y}(x, 0) = x$ for all x , and $\frac{\partial f}{\partial x}(0, y) = -y$ for all y .

b. Show that $\frac{\partial^2 f}{\partial y \partial x}(0, 0) \neq \frac{\partial^2 f}{\partial x \partial y}(0, 0)$.

The three-dimensional Laplace equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

is satisfied by steady-state temperature distributions $T = f(x, y, z)$ in space, by gravitational potentials, and by electrostatic potentials. The

two-dimensional Laplace equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0,$$

obtained by dropping the $\partial^2 f/\partial z^2$ term from the previous equation, describes potentials and steady-state temperature distributions in a plane. The plane may be treated as a thin slice of the solid perpendicular to the z -axis.

Show that each function in Exercises 83–90 satisfies a Laplace equation.

83. $f(x, y, z) = x^2 + y^2 - 2z^2$

84. $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z$

85. $f(x, y) = e^{-2y} \cos 2x$

86. $f(x, y) = \ln \sqrt{x^2 + y^2}$

87. $f(x, y) = 3x + 2y - 4$

88. $f(x, y) = \arctan \frac{x}{y}$

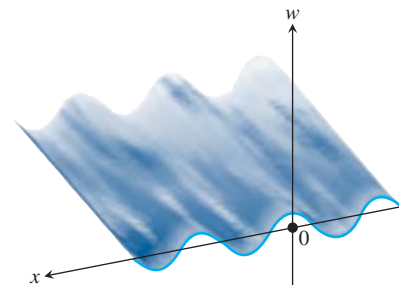
89. $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$

90. $f(x, y, z) = e^{3x+4y} \cos 5z$

The wave equation If we stand on an ocean shore and take a snapshot of the waves, the picture shows a regular pattern of peaks and valleys in an instant of time. We see periodic vertical motion in space, with respect to distance. If we stand in the water, we can feel the rise and fall of the water as the waves go by. We see periodic vertical motion in time. In physics, this beautiful symmetry is expressed by the **one-dimensional wave equation**

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2},$$

where w is the wave height, x is the distance variable, t is the time variable, and c is the velocity with which the waves are propagated.



In our example, x is the distance across the ocean's surface, but in other applications, x might be the distance along a vibrating string, distance through air (sound waves), or distance through space (light waves). The number c varies with the medium and type of wave.

Show that the functions in Exercises 91–97 are all solutions of the wave equation.

91. $w = \sin(x + ct)$

92. $w = \cos(2x + 2ct)$

93. $w = \sin(x + ct) + \cos(2x + 2ct)$

94. $w = \ln(2x + 2ct)$

95. $w = \tan(2x - 2ct)$

96. $w = 5 \cos(3x + 3ct) + e^{x+ct}$

97. $w = f(u)$, where f is a differentiable function of u , and $u = a(x + ct)$, where a is a constant

98. Does a function $f(x, y)$ with continuous first partial derivatives throughout an open region R have to be continuous on R ? Give reasons for your answer.

99. If a function $f(x, y)$ has continuous second partial derivatives throughout an open region R , must the first-order partial derivatives of f be continuous on R ? Give reasons for your answer.

- 100. The heat equation** An important partial differential equation that describes the distribution of heat in a region at time t can be represented by the *one-dimensional heat equation*

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}.$$

Show that $u(x, t) = \sin(\alpha x) \cdot e^{-\beta t}$ satisfies the heat equation for constants α and β . What is the relationship between α and β for this function to be a solution?

101. Let $f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^4}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$

Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but f is not differentiable at $(0, 0)$. (Hint: Use Theorem 4 and show that f is not continuous at $(0, 0)$.)

102. Let $f(x, y) = \begin{cases} 0, & x^2 < y < 2x^2 \\ 1, & \text{otherwise.} \end{cases}$

Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but f is not differentiable at $(0, 0)$.

103. The Korteweg–de Vries equation

This nonlinear differential equation, which describes wave motion on shallow water surfaces, is given by

$$u_t + u_{xxx} + 12uu_x = 0.$$

Show that $u(x, t) = \text{sech}^2(x - t)$ satisfies the Korteweg–de Vries equation.

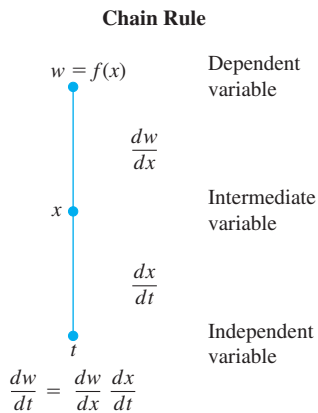
104. Show that $T = \frac{1}{\sqrt{x^2 + y^2}}$ satisfies the equation $T_{xx} + T_{yy} = T^3$.

13.4 The Chain Rule

The Chain Rule for functions of a single variable studied in Section 3.6 says that if $w = f(x)$ is a differentiable function of x , and $x = g(t)$ is a differentiable function of t , then w is a differentiable function of t , and dw/dt can be calculated by the formula

$$\frac{dw}{dt} = \frac{dw}{dx} \frac{dx}{dt}.$$

To find dw/dt , we read down the route from w to t , multiplying derivatives along the way.



For this composite function $w(t) = f(g(t))$, we can think of t as the independent variable and $x = g(t)$ as the “intermediate variable” because t determines the value of x that in turn gives the value of w from the function f . We display the Chain Rule in a “dependency diagram” in the margin. Such diagrams capture which variables depend on which.

For functions of several variables the Chain Rule has more than one form, which depends on how many independent and intermediate variables are involved. However, once the variables are taken into account, the Chain Rule works in the same way we just discussed.

Functions of Two Variables

The Chain Rule formula for a differentiable function $w = f(x, y)$ when $x = x(t)$ and $y = y(t)$ are both differentiable functions of t is given in the following theorem.

THEOREM 5—Chain Rule for Functions of One Independent Variable and Two Intermediate Variables

If $w = f(x, y)$ is differentiable and if $x = x(t)$, $y = y(t)$ are differentiable functions of t , then the composition $w = f(x(t), y(t))$ is a differentiable function of t and

$$\frac{dw}{dt} = f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t),$$

or

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Each of $\frac{\partial f}{\partial x}$, $\frac{\partial w}{\partial x}$, f_x indicates the partial derivative of f with respect to x .

Proof The proof consists of showing that if x and y are differentiable at $t = t_0$, then w is differentiable at t_0 and

$$\frac{dw}{dt}(t_0) = \frac{\partial w}{\partial x}(P_0) \frac{dx}{dt}(t_0) + \frac{\partial w}{\partial y}(P_0) \frac{dy}{dt}(t_0),$$

where $P_0 = (x(t_0), y(t_0))$.

Let Δx , Δy , and Δw be the increments that result from changing t from t_0 to $t_0 + \Delta t$. Since f is differentiable (see the definition in Section 13.3),

$$\Delta w = \frac{\partial w}{\partial x}(P_0) \Delta x + \frac{\partial w}{\partial y}(P_0) \Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y,$$

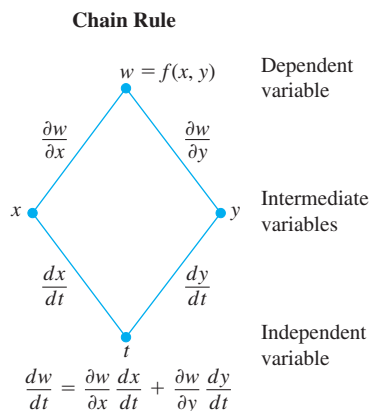
where $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $\Delta x, \Delta y \rightarrow 0$. To find dw/dt , we divide this equation through by Δt and let Δt approach zero (therefore, Δx and Δy approach zero as well since the fact that $x(t)$ and $y(t)$ are differentiable implies that they are continuous). The division gives

$$\frac{\Delta w}{\Delta t} = \frac{\partial w}{\partial x}(P_0) \frac{\Delta x}{\Delta t} + \frac{\partial w}{\partial y}(P_0) \frac{\Delta y}{\Delta t} + \varepsilon_1 \frac{\Delta x}{\Delta t} + \varepsilon_2 \frac{\Delta y}{\Delta t}.$$

Letting Δt approach zero gives

$$\begin{aligned} \frac{dw}{dt}(t_0) &= \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} \\ &= \frac{\partial w}{\partial x}(P_0) \frac{dx}{dt}(t_0) + \frac{\partial w}{\partial y}(P_0) \frac{dy}{dt}(t_0) + 0 \cdot \frac{dx}{dt}(t_0) + 0 \cdot \frac{dy}{dt}(t_0). \end{aligned} \quad \blacksquare$$

To remember the Chain Rule, picture the diagram below. To find dw/dt , start at w and read down each route to t , multiplying derivatives along the way. Then add the products.



Often we write $\partial w/\partial x$ for the partial derivative $\partial f/\partial x$, so we can rewrite the Chain Rule in Theorem 5 in the form

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}.$$

However, the meaning of the dependent variable w is different on each side of the preceding equation. On the left-hand side, it refers to the composite function $w = f(x(t), y(t))$ as a function of the single variable t . On the right-hand side, it refers to the function $w = f(x, y)$ as a function of the two variables x and y . Moreover, the single derivatives dw/dt , dx/dt , and dy/dt are being evaluated at a point t_0 , whereas the partial derivatives $\partial w/\partial x$ and $\partial w/\partial y$ are being evaluated at the point (x_0, y_0) , with $x_0 = x(t_0)$ and $y_0 = y(t_0)$. With that understanding, we will use both of these forms interchangeably throughout the text whenever no confusion will arise.

The **dependency diagram** on the preceding page provides a convenient way to remember the Chain Rule. The “true” independent variable in the composite function is t , whereas x and y are *intermediate variables* (controlled by t) and w is the dependent variable.

A more precise notation for the Chain Rule shows where the various derivatives in Theorem 5 are evaluated:

$$\frac{dw}{dt}(t_0) = \frac{\partial f}{\partial x}(x_0, y_0) \frac{dx}{dt}(t_0) + \frac{\partial f}{\partial y}(x_0, y_0) \frac{dy}{dt}(t_0),$$

or, using another notation,

$$\left. \frac{dw}{dt} \right|_{t_0} = \left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} \left. \frac{dx}{dt} \right|_{t_0} + \left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} \left. \frac{dy}{dt} \right|_{t_0}.$$

EXAMPLE 1 Use the Chain Rule to find the derivative of

$$w = xy$$

with respect to t along the path $x = \cos t$, $y = \sin t$. What is the derivative's value at $t = \pi/2$?

Solution We apply the Chain Rule to find dw/dt as follows:

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} \\ &= \frac{\partial(xy)}{\partial x} \frac{d}{dt}(\cos t) + \frac{\partial(xy)}{\partial y} \frac{d}{dt}(\sin t) \\ &= (y)(-\sin t) + (x)(\cos t) \\ &= (\sin t)(-\sin t) + (\cos t)(\cos t) \\ &= -\sin^2 t + \cos^2 t \\ &= \cos 2t.\end{aligned}$$

In this example, we can check the result with a more direct calculation. As a function of t ,

$$w = xy = \cos t \sin t = \frac{1}{2} \sin 2t,$$

so

$$\frac{dw}{dt} = \frac{d}{dt} \left(\frac{1}{2} \sin 2t \right) = \frac{1}{2} (2 \cos 2t) = \cos 2t.$$

In either case, at the given value of t ,

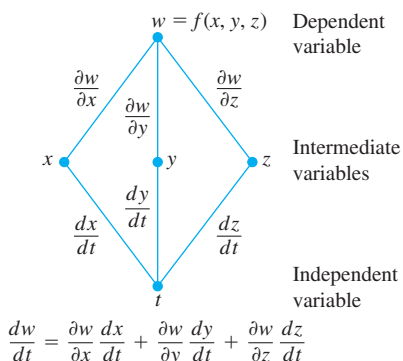
$$\left. \frac{dw}{dt} \right|_{t=\pi/2} = \cos \left(2 \frac{\pi}{2} \right) = \cos \pi = -1. \quad \blacksquare$$

Functions of Three Variables

You can probably predict the Chain Rule for functions of three intermediate variables, as it involves adding the expected third term to the two-variable formula.

Here we have three routes from w to t instead of two, but finding dw/dt is still the same. Read down each route, multiplying derivatives along the way; then add.

Chain Rule



THEOREM 6—Chain Rule for Functions of One Independent Variable and Three Intermediate Variables

If $w = f(x, y, z)$ is differentiable and x, y , and z are differentiable functions of t , then w is a differentiable function of t , and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}.$$

The proof is identical to the proof of Theorem 5, except that there are now three intermediate variables instead of two. The dependency diagram we use for remembering the new equation is similar as well, with three routes from w to t .

EXAMPLE 2 Find dw/dt if

$$w = xy + z, \quad x = \cos t, \quad y = \sin t, \quad z = t.$$

In this example the values of $w(t)$ are changing along the path of a helix (Section 12.1) as t changes. What is the derivative's value at $t = 0$?

Solution Using the Chain Rule for three intermediate variables, we have

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} \\ &= (y)(-\sin t) + (x)(\cos t) + (1)(1) \\ &= (\sin t)(-\sin t) + (\cos t)(\cos t) + 1 && \text{Substitute for intermediate variables.} \\ &= -\sin^2 t + \cos^2 t + 1 = 1 + \cos 2t,\end{aligned}$$

so

$$\left. \frac{dw}{dt} \right|_{t=0} = 1 + \cos(0) = 2. \quad \blacksquare$$

For a physical interpretation of change along a curve, think of an object whose position is changing with time t . If $w = T(x, y, z)$ is the temperature at each point (x, y, z) along a curve C with parametric equations $x = x(t)$, $y = y(t)$, and $z = z(t)$, then the composite function $w = T(x(t), y(t), z(t))$ represents the temperature relative to t along the curve. The derivative dw/dt is then the instantaneous rate of change of temperature due to the motion along the curve, as calculated in Theorem 6.

Functions Defined on Surfaces

If we are interested in the temperature $w = f(x, y, z)$ at points (x, y, z) on Earth's surface, we might prefer to think of x , y , and z as functions of the variables r and s that give the points' longitudes and latitudes. If $x = g(r, s)$, $y = h(r, s)$, and $z = k(r, s)$, we could then express the temperature as a function of r and s with the composite function

$$w = f(g(r, s), h(r, s), k(r, s)).$$

Under the conditions stated below, w has partial derivatives with respect to both r and s that can be calculated in the following way.

THEOREM 7—Chain Rule for Two Independent Variables and Three Intermediate Variables

Suppose that $w = f(x, y, z)$, $x = g(r, s)$, $y = h(r, s)$, and $z = k(r, s)$. If all four functions are differentiable, then w has partial derivatives with respect to r and s , given by the formulas

$$\begin{aligned}\frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r} \\ \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s}.\end{aligned}$$

The first of these equations can be derived from the Chain Rule in Theorem 6 by holding s fixed and treating r as t . The second can be derived in the same way, holding r fixed and treating s as t . The dependency diagrams for both equations are shown in Figure 13.22.

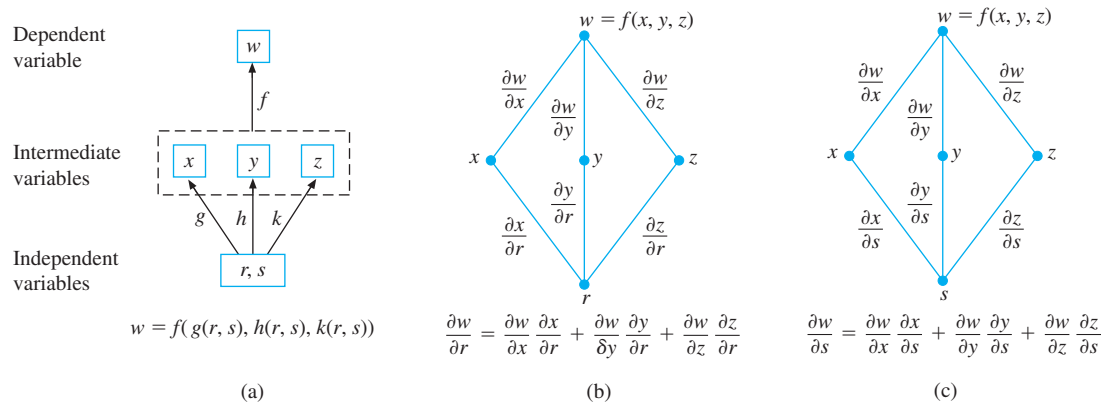


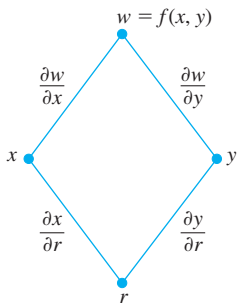
FIGURE 13.22 Composite function and dependency diagrams for Theorem 7.

EXAMPLE 3 Express $\partial w/\partial r$ and $\partial w/\partial s$ in terms of r and s if

$$w = x + 2y + z^2, \quad x = \frac{r}{s}, \quad y = r^2 + \ln s, \quad z = 2r.$$

Solution Using the formulas in Theorem 7, we find

$$\begin{aligned} \frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r} \\ &= (1)\left(\frac{1}{s}\right) + (2)(2r) + (2z)(2) \\ &= \frac{1}{s} + 4r + (4r)(2) = \frac{1}{s} + 12r \quad \text{Substitute for intermediate variable } z. \\ \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s} \\ &= (1)\left(-\frac{r}{s^2}\right) + (2)\left(\frac{1}{s}\right) + (2z)(0) = \frac{2}{s} - \frac{r}{s^2}. \end{aligned}$$

Chain Rule

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}$$

FIGURE 13.23 Dependency diagram for the equation

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}.$$

If f is a function of two intermediate variables instead of three, each equation in Theorem 7 becomes correspondingly one term shorter.

If $w = f(x, y)$, $x = g(r, s)$, and $y = h(r, s)$, then

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \quad \text{and} \quad \frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s}.$$

Figure 13.23 shows the dependency diagram for the first of these equations. The diagram for the second equation is similar; just replace r with s .

EXAMPLE 4 Express $\partial w/\partial r$ and $\partial w/\partial s$ in terms of r and s if

$$w = x^2 + y^2, \quad x = r - s, \quad y = r + s.$$

Solution The preceding discussion gives the following.

$$\begin{aligned}
 \frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} & \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} \\
 &= (2x)(1) + (2y)(1) & &= (2x)(-1) + (2y)(1) \\
 &= 2(r-s) + 2(r+s) & &= -2(r-s) + 2(r+s) \\
 &= 4r & &= 4s
 \end{aligned}$$

Substitute for
the intermediate
variables.

If f is a function of a single intermediate variable x , our equations are even simpler.

If $w = f(x)$ and $x = g(r, s)$, then

$$\frac{\partial w}{\partial r} = \frac{dw}{dx} \frac{\partial x}{\partial r} \quad \text{and} \quad \frac{\partial w}{\partial s} = \frac{dw}{dx} \frac{\partial x}{\partial s}.$$

Chain Rule

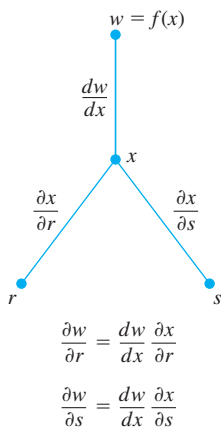


FIGURE 13.24 Dependency diagram for differentiating f as a composite function of r and s with one intermediate variable.

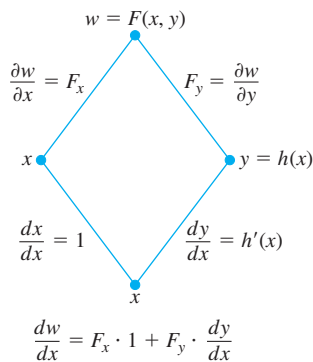


FIGURE 13.25 Dependency diagram for differentiating $w = F(x, y)$ with respect to x . Setting $dw/dx = 0$ leads to a simple computational formula for implicit differentiation (Theorem 8).

In this case, we use the ordinary (single-variable) derivative, dw/dx . The dependency diagram is shown in Figure 13.24.

Implicit Differentiation Revisited

The two-variable Chain Rule in Theorem 5 leads to a formula that takes some of the algebra out of implicit differentiation. Suppose that

1. The function $F(x, y)$ is differentiable and
2. The equation $F(x, h(x)) = 0$ defines y implicitly as a differentiable function of x , say $y = h(x)$.

Since $w = F(x, h(x)) = 0$, the derivative dw/dx must be zero. Computing the derivative from the Chain Rule (dependency diagram in Figure 13.25), we find

$$\begin{aligned}
 0 &= \frac{dw}{dx} = F_x \frac{dx}{dx} + F_y \frac{dy}{dx} && \text{Theorem 5 with } t = x \text{ and } f = F \\
 &= F_x \cdot 1 + F_y \cdot \frac{dy}{dx}.
 \end{aligned}$$

If $F_y = \partial w / \partial y \neq 0$, we can solve this equation for dy/dx to get

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

We state this result formally.

THEOREM 8—A Formula for Implicit Differentiation

Suppose that $F(x, y)$ is differentiable and that the equation $F(x, y) = 0$ defines y as a differentiable function of x . Then, at any point where $F_y \neq 0$,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}. \quad (1)$$

EXAMPLE 5 Use Theorem 8 to find dy/dx if $y^2 - x^2 - \sin xy = 0$.

Solution Take $F(x, y) = y^2 - x^2 - \sin xy$. Then

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{-2x - y \cos xy}{2y - x \cos xy} = \frac{2x + y \cos xy}{2y - x \cos xy}.$$

This calculation is significantly shorter than a single-variable calculation using implicit differentiation. ■

The result in Theorem 8 is easily extended to three variables. Suppose that the equation $F(x, y, z) = 0$ defines the variable z implicitly as a function $z = f(x, y)$. Then, for all (x, y) in the domain of f , we have $F(x, y, f(x, y)) = 0$. Assuming that F and f are differentiable functions, we can use the Chain Rule to differentiate the equation $F(x, y, z) = 0$ with respect to the independent variable x :

$$\begin{aligned} 0 &= \frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} \\ &= F_x \cdot 1 + F_y \cdot 0 + F_z \cdot \frac{\partial z}{\partial x}, \end{aligned} \quad \begin{array}{l} \text{y is constant when} \\ \text{we differentiate} \\ \text{with respect to x.} \end{array}$$

so

$$F_x + F_z \frac{\partial z}{\partial x} = 0.$$

A similar calculation for differentiating with respect to the independent variable y gives

$$F_y + F_z \frac{\partial z}{\partial y} = 0.$$

Whenever $F_z \neq 0$, we can solve these last two equations for the partial derivatives of $z = f(x, y)$ to obtain

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}. \quad (2)$$

An important result from advanced calculus, called the **Implicit Function Theorem**, states the conditions for which our results in Equations (2) are valid. If the partial derivatives F_x , F_y , and F_z are continuous throughout an open region R in space containing the point (x_0, y_0, z_0) , and if for some constant c , $F(x_0, y_0, z_0) = c$ and $F_z(x_0, y_0, z_0) \neq 0$, then the equation $F(x, y, z) = c$ defines z implicitly as a differentiable function of x and y near (x_0, y_0, z_0) , and the partial derivatives of z are given by Equations (2).

EXAMPLE 6 Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(0, 0, 0)$ if $x^3 + z^2 + ye^{xz} + z \cos y = 0$.

Solution Let $F(x, y, z) = x^3 + z^2 + ye^{xz} + z \cos y$. Then

$$F_x = 3x^2 + zye^{xz}, \quad F_y = e^{xz} - z \sin y, \quad \text{and} \quad F_z = 2z + xye^{xz} + \cos y.$$

Since $F(0, 0, 0) = 0$, $F_z(0, 0, 0) = 1 \neq 0$, and all first partial derivatives are continuous, the Implicit Function Theorem says that $F(x, y, z) = 0$ defines z as a differentiable function of x and y near the point $(0, 0, 0)$. From Equations (2),

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{3x^2 + zye^{xz}}{2z + xye^{xz} + \cos y} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{e^{xz} - z \sin y}{2z + xye^{xz} + \cos y}.$$

At $(0, 0, 0)$ we find

$$\frac{\partial z}{\partial x} = -\frac{0}{1} = 0 \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{1}{1} = -1. \quad \blacksquare$$

Functions of Many Variables

We have seen several different forms of the Chain Rule in this section, but each one is just a special case of one general formula. When solving particular problems, it may help to draw the appropriate dependency diagram by placing the dependent variable on top, the intermediate variables in the middle, and the selected independent variable at the bottom. To find the derivative of the dependent variable with respect to the selected independent variable, start at the dependent variable and read down each route of the dependency diagram to the independent variable, calculating and multiplying the derivatives along each route. Then add the products found for the different routes.

In general, suppose that $w = f(x_1, x_2, \dots, x_n)$ is a differentiable function of the intermediate variables $x_1, x_2, \dots, x_n$ (a finite set) and that $x_1, x_2, \dots, x_n$ are differentiable functions of the independent variables $t_1, t_2, \dots, t_m$ (another finite set). Then w is a differentiable function of the variables $t_1, t_2, \dots, t_m$, and the partial derivatives of w with respect to these variables are given by equations of the form

$$\frac{\partial w}{\partial t_i} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial w}{\partial x_n} \frac{\partial x_n}{\partial t_i} \quad \text{for } i = 1, 2, \dots, m.$$

One way to remember this equation is to think of the right-hand side as the dot product of two n -dimensional vectors:

$$\frac{\partial w}{\partial t_i} = \underbrace{\left\langle \frac{\partial w}{\partial x_1}, \frac{\partial w}{\partial x_2}, \dots, \frac{\partial w}{\partial x_n} \right\rangle}_{\substack{\text{Derivatives of } w \text{ with} \\ \text{respect to the} \\ \text{intermediate variables}}} \cdot \underbrace{\left\langle \frac{\partial x_1}{\partial t_i}, \frac{\partial x_2}{\partial t_i}, \dots, \frac{\partial x_n}{\partial t_i} \right\rangle}_{\substack{\text{Derivatives of the} \\ \text{intermediate variables with} \\ \text{respect to the selected} \\ \text{independent variable}}}.$$

The first vector describes how w changes in various directions, while the second vector indicates the velocity vector of $\mathbf{x}(t_i) = \langle x_1(t_i), x_2(t_i), \dots, x_n(t_i) \rangle$. These concepts will be studied further in the next section.

EXERCISES 13.4

Chain Rule: One Independent Variable

In Exercises 1–6, (a) express dw/dt as a function of t , both by using the Chain Rule and by expressing w in terms of t and differentiating directly with respect to t . Then (b) evaluate dw/dt at the given value of t .

- $w = x^2 + y^2$, $x = \cos t$, $y = \sin t$; $t = \pi$
- $w = x^2 + y^2$, $x = \cos t + \sin t$, $y = \cos t - \sin t$; $t = 0$
- $w = \frac{x}{z} + \frac{y}{z}$, $x = \cos^2 t$, $y = \sin^2 t$, $z = 1/t$; $t = 3$
- $w = \ln(x^2 + y^2 + z^2)$, $x = \cos t$, $y = \sin t$, $z = 4\sqrt{t}$; $t = 3$

$$5. w = 2ye^x - \ln z, \quad x = \ln(t^2 + 1), \quad y = \tan^{-1} t, \quad z = e^t; \quad t = 1$$

$$6. w = z - \sin xy, \quad x = t, \quad y = \ln t, \quad z = e^{t-1}; \quad t = 1$$

Chain Rule: Two and Three Independent Variables

In Exercises 7 and 8, (a) express $\partial z/\partial u$ and $\partial z/\partial v$ as functions of u and v both by using the Chain Rule and by expressing z directly in terms of u and v before differentiating. Then (b) evaluate $\partial z/\partial u$ and $\partial z/\partial v$ at the given point (u, v) .

$$7. z = 4e^x \ln y, \quad x = \ln(u \cos v), \quad y = u \sin v; \quad (u, v) = (2, \pi/4)$$

8. $z = \tan^{-1}(x/y)$, $x = u \cos v$, $y = u \sin v$;
 $(u, v) = (1.3, \pi/6)$

In Exercises 9 and 10, (a) express $\partial w/\partial u$ and $\partial w/\partial v$ as functions of u and v both by using the Chain Rule and by expressing w directly in terms of u and v before differentiating. Then (b) evaluate $\partial w/\partial u$ and $\partial w/\partial v$ at the given point (u, v) .

9. $w = xy + yz + xz$, $x = u + v$, $y = u - v$, $z = uv$;
 $(u, v) = (1/2, 1)$

10. $w = \ln(x^2 + y^2 + z^2)$, $x = ue^v \sin u$, $y = ue^v \cos u$,
 $z = ue^v$; $(u, v) = (-2, 0)$

In Exercises 11 and 12, (a) express $\partial u/\partial x$, $\partial u/\partial y$, and $\partial u/\partial z$ as functions of x , y , and z both by using the Chain Rule and by expressing u directly in terms of x , y , and z before differentiating. Then (b) evaluate $\partial u/\partial x$, $\partial u/\partial y$, and $\partial u/\partial z$ at the given point (x, y, z) .

11. $u = \frac{p-q}{q-r}$, $p = x + y + z$, $q = x - y + z$,
 $r = x + y - z$; $(x, y, z) = (\sqrt{3}, 2, 1)$

12. $u = e^{qr} \sin^{-1} p$, $p = \sin x$, $q = z^2 \ln y$, $r = 1/z$;
 $(x, y, z) = (\pi/4, 1/2, -1/2)$

Using a Dependency Diagram

In Exercises 13–24, draw a dependency diagram and write a Chain Rule formula for each derivative.

13. $\frac{dz}{dt}$ for $z = f(x, y)$, $x = g(t)$, $y = h(t)$

14. $\frac{dz}{dt}$ for $z = f(u, v, w)$, $u = g(t)$, $v = h(t)$, $w = k(t)$

15. $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ for $w = h(x, y, z)$, $x = f(u, v)$, $y = g(u, v)$,
 $z = k(u, v)$

16. $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$ for $w = f(r, s, t)$, $r = g(x, y)$, $s = h(x, y)$,
 $t = k(x, y)$

17. $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ for $w = g(x, y)$, $x = h(u, v)$, $y = k(u, v)$

18. $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$ for $w = g(u, v)$, $u = h(x, y)$, $v = k(x, y)$

19. $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial s}$ for $z = f(x, y)$, $x = g(t, s)$, $y = h(t, s)$

20. $\frac{\partial y}{\partial r}$ for $y = f(u)$, $u = g(r, s)$

21. $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$ for $w = g(u)$, $u = h(s, t)$

22. $\frac{\partial w}{\partial p}$ for $w = f(x, y, z, v)$, $x = g(p, q)$, $y = h(p, q)$,
 $z = j(p, q)$, $v = k(p, q)$

23. $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ for $w = f(x, y)$, $x = g(r)$, $y = h(s)$

24. $\frac{\partial w}{\partial s}$ for $w = g(x, y)$, $x = h(r, s, t)$, $y = k(r, s, t)$

Implicit Differentiation

Assuming that the equations in Exercises 25–30 define y as a differentiable function of x , use Theorem 8 to find the value of dy/dx at the given point.

25. $x^3 - 2y^2 + xy = 0$, $(1, 1)$

26. $xy + y^2 - 3x - 3 = 0$, $(-1, 1)$

27. $x^2 + xy + y^2 - 7 = 0$, $(1, 2)$

28. $xe^y + \sin xy + y - \ln 2 = 0$, $(0, \ln 2)$

29. $(x^3 - y^4)^6 + \ln(x^2 + y) = 1$, $(-1, 0)$

30. $xe^{x^2y} - ye^x = x + y - 2$, $(1, 1)$

Find the values of $\partial z/\partial x$ and $\partial z/\partial y$ at the points in Exercises 31–34.

31. $z^3 - xy + yz + y^3 - 2 = 0$, $(1, 1, 1)$

32. $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1 = 0$, $(2, 3, 6)$

33. $\sin(x + y) + \sin(y + z) + \sin(x + z) = 0$, (π, π, π)

34. $xe^y + ye^z + 2 \ln x - 2 - 3 \ln 2 = 0$, $(1, \ln 2, \ln 3)$

Finding Partial Derivatives at Specified Points

35. Find $\partial w/\partial r$ when $r = 1, s = -1$ if $w = (x + y + z)^2$,
 $x = r - s, y = \cos(r + s), z = \sin(r + s)$.

36. Find $\partial w/\partial v$ when $u = -1, v = 2$ if $w = xy + \ln z$,
 $x = v^2/u, y = u + v, z = \cos u$.

37. Find $\partial w/\partial v$ when $u = 0, v = 0$ if $w = x^2 + (y/x)$,
 $x = u - 2v + 1, y = 2u + v - 2$.

38. Find $\partial z/\partial u$ when $u = 0, v = 1$ if $z = \sin xy + x \sin y$,
 $x = u^2 + v^2, y = uv$.

39. Find $\partial z/\partial u$ and $\partial z/\partial v$ when $u = \ln 2, v = 1$ if $z = 5 \tan^{-1} x$
and $x = e^u + \ln v$.

40. Find $\partial z/\partial u$ and $\partial z/\partial v$ when $u = 1, v = -2$ if $z = \ln q$ and
 $q = \sqrt{v} + 3 \tan^{-1} u$.

Theory and Examples

41. Assume that $w = f(s^3 + t^2)$ and $f'(x) = e^x$. Find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$.

42. Assume that $w = f\left(ts^2, \frac{s}{t}\right)$, $\frac{\partial f}{\partial x}(x, y) = xy$, and
 $\frac{\partial f}{\partial y}(x, y) = \frac{x^2}{2}$. Find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$.

43. Assume that $z = f(x, y)$, $x = g(t)$, $y = h(t)$, $f_x(2, -1) = 3$,
and $f_y(2, -1) = -2$. If $g(0) = 2$, $h(0) = -1$, $g'(0) = 5$, and
 $h'(0) = -4$, find $\frac{dz}{dt}\bigg|_{t=0}$.

44. Assume that $z = f(x, y)^2$, $x = g(t)$, $y = h(t)$, $f_x(1, 0) = -1$,
 $f_y(1, 0) = 1$, and $f(1, 0) = 2$. If $g(3) = 1$, $h(3) = 0$,
 $g'(3) = -3$, and $h'(3) = 4$, find $\frac{dz}{dt}\bigg|_{t=3}$.

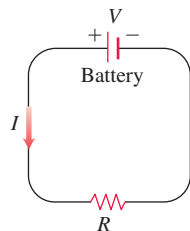
45. Assume that $z = f(w)$, $w = g(x, y)$, $x = 2r^3 - s^2$, and
 $y = re^s$. If $g_x(2, 1) = -3$, $g_y(2, 1) = 2$, $f'(7) = -1$, and
 $g(2, 1) = 7$, find $\frac{\partial z}{\partial r}\bigg|_{r=1, s=0}$ and $\frac{\partial z}{\partial s}\bigg|_{r=1, s=0}$.

46. Assume that $z = \ln(f(w))$, $w = g(x, y)$, $x = \sqrt{r - s}$, and
 $y = r^2s$. If $g_x(2, -9) = -1$, $g_y(2, -9) = 3$, $f'(-2) = 2$,
 $f(-2) = 5$, and $g(2, -9) = -2$, find $\frac{\partial z}{\partial r}\bigg|_{r=3, s=-1}$ and
 $\frac{\partial z}{\partial s}\bigg|_{r=3, s=-1}$.

47. **Changing voltage in a circuit** The voltage V in a circuit that satisfies the law $V = IR$ is slowly dropping as the battery wears out. At the same time, the resistance R is increasing as the resistor heats up. Use the equation

$$\frac{dV}{dt} = \frac{\partial V}{\partial I} \frac{dI}{dt} + \frac{\partial V}{\partial R} \frac{dR}{dt}$$

to find how the current is changing at the instant when $R = 600$ ohms, $I = 0.04$ amp, $dR/dt = 0.5$ ohm/s, and $dV/dt = -0.01$ volt/s.



- 48. Changing dimensions in a box** The lengths a , b , and c of the edges of a rectangular box are changing with time. At the instant in question, $a = 1$ m, $b = 2$ m, $c = 3$ m, $da/dt = db/dt = 1$ m/s, and $dc/dt = -3$ m/s. At what rates are the box's volume V and surface area S changing at that instant? Are the box's interior diagonals increasing in length or decreasing?

- 49.** If $f(u, v, w)$ is differentiable and $u = x - y$, $v = y - z$, and $w = z - x$, show that

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0.$$

- 50. Polar coordinates** Suppose that we substitute polar coordinates $x = r \cos \theta$ and $y = r \sin \theta$ in a differentiable function $w = f(x, y)$.

- a. Show that

$$\frac{\partial w}{\partial r} = f_x \cos \theta + f_y \sin \theta$$

and

$$\frac{1}{r} \frac{\partial w}{\partial \theta} = -f_x \sin \theta + f_y \cos \theta.$$

- b. Solve the equations in part (a) to express f_x and f_y in terms of $\partial w / \partial r$ and $\partial w / \partial \theta$.

- c. Show that

$$(f_x)^2 + (f_y)^2 = \left(\frac{\partial w}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta} \right)^2.$$

- 51. Laplace equations** Show that if $w = f(u, v)$ satisfies the Laplace equation $f_{uu} + f_{vv} = 0$ and if $u = (x^2 - y^2)/2$ and $v = xy$, then w satisfies the Laplace equation $w_{xx} + w_{yy} = 0$.

- 52. Laplace equations** Let $w = f(u) + g(v)$, where $u = x + iy$, $v = x - iy$, and $i = \sqrt{-1}$. Show that w satisfies the Laplace equation $w_{xx} + w_{yy} = 0$ if all the necessary functions are differentiable.

- 53. Extreme values on a helix** Suppose that the partial derivatives of a function $f(x, y, z)$ at points on the helix $x = \cos t$, $y = \sin t$, $z = t$ are

$$f_x = \cos t, \quad f_y = \sin t, \quad f_z = t^2 + t - 2.$$

At what points on the curve, if any, can f take on extreme values?

- 54. A space curve** Let $w = x^2 e^{2y} \cos 3z$. Find the value of dw/dt at the point $(1, \ln 2, 0)$ on the curve $x = \cos t$, $y = \ln(t + 2)$, $z = t$.

- 55. Temperature on a circle** Let $T = f(x, y)$ be the temperature at the point (x, y) on the circle $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, and suppose that

$$\frac{\partial T}{\partial x} = 8x - 4y, \quad \frac{\partial T}{\partial y} = 8y - 4x.$$

- a. Find where the maximum and minimum temperatures on the circle occur by examining the derivatives dT/dt and d^2T/dt^2 .
b. Suppose that $T = 4x^2 - 4xy + 4y^2$. Find the maximum and minimum values of T on the circle.

- 56. Temperature on an ellipse** Let $T = g(x, y)$ be the temperature at the point (x, y) on the ellipse

$$x = 2\sqrt{2} \cos t, \quad y = \sqrt{2} \sin t, \quad 0 \leq t \leq 2\pi,$$

and suppose that

$$\frac{\partial T}{\partial x} = y, \quad \frac{\partial T}{\partial y} = x.$$

- a. Locate the maximum and minimum temperatures on the ellipse by examining dT/dt and d^2T/dt^2 .
b. Suppose that $T = xy - 2$. Find the maximum and minimum values of T on the ellipse.

- 57.** The temperature $T = T(x, y)$ in $^\circ\text{C}$ at point (x, y) satisfies $T_x(1, 2) = 3$ and $T_y(1, 2) = -1$. If $x = e^{2t-2}$ cm and $y = 2 + \ln t$ cm, find the rate at which the temperature T changes when $t = 1$ s.

- 58.** A bug crawls on the surface $z = x^2 - y^2$ directly above a path in the xy -plane given by $x = f(t)$ and $y = g(t)$. If $f(2) = 4$, $f'(2) = -1$, $g(2) = -2$, and $g'(2) = -3$, then at what rate is the bug's elevation z changing when $t = 2$?

Differentiating Integrals Under mild continuity restrictions, it is true that if

$$F(x) = \int_a^b g(t, x) dt,$$

then $F'(x) = \int_a^b g_x(t, x) dt$. Using this fact and the Chain Rule, we can find the derivative of

$$F(x) = \int_a^{f(x)} g(t, x) dt$$

by letting

$$G(u, x) = \int_a^u g(t, x) dt,$$

where $u = f(x)$. Find the derivatives of the functions in Exercises 59 and 60.

59. $F(x) = \int_0^{x^2} \sqrt{t^4 + x^3} dt$

60. $F(x) = \int_{x^2}^1 \sqrt{t^3 + x^2} dt$

- 61.** Water is flowing into a tank in the form of a right-circular cylinder at the rate of $(4/5)\pi \text{ m}^3/\text{min}$. The tank is stretching in such a way that even though it remains cylindrical, its radius is increasing at the rate of 0.002 m/min. How fast is the surface of the water rising when the radius is 2 m and the volume of water in the tank is $20\pi \text{ m}^3$?

- 62.** Suppose f is a differentiable function of x , y , and z and $u = f(x, y, z)$. Then if $x = r \sin \phi \cos \theta$, $y = r \sin \phi \sin \theta$, and $z = r \cos \phi$, express $\partial u / \partial r$, $\partial u / \partial \phi$, and $\partial u / \partial \theta$ in terms of $\partial u / \partial x$, $\partial u / \partial y$, and $\partial u / \partial z$.

- 63.** At a given instant, the length of one leg of a right triangle is 10 m, and it is increasing at the rate of 1 m/min, and the length of the other leg of the right triangle is 12 m, and it is decreasing at the rate of 2 m/min. Find the rate of change of the measure of the acute angle opposite the leg of length 12 m at the given instant.

13.5 Directional Derivatives and Gradient Vectors

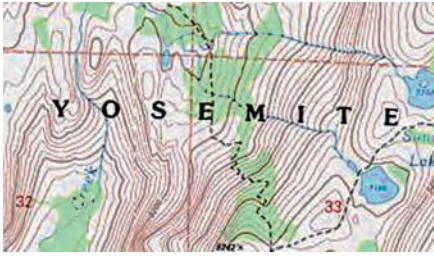


FIGURE 13.26 Contours within Yosemite National Park in California show streams, which follow paths of steepest descent, running perpendicular to the contours. (Source: Yosemite National Park Map from U.S. Geological Survey, <http://www.usgs.gov>)

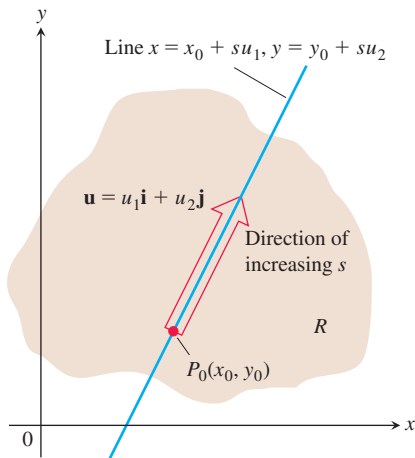


FIGURE 13.27 The rate of change of f in the direction of $\mathbf{u}$ at a point P_0 is the rate at which f changes along this line at P_0 .

If you look at the map (Figure 13.26) showing contours within Yosemite National Park in California, you will notice that the streams flow perpendicular to the contours. The streams are following paths of steepest descent so the waters reach lower elevations as quickly as possible. Therefore, the fastest instantaneous rate of change in a stream's elevation above sea level has a particular direction. In this section, you will see why this direction, called the “downhill” direction, is perpendicular to the contours.

Directional Derivatives in the Plane

We know from Section 13.4 that if $f(x, y)$ is differentiable, then the rate at which f changes with respect to t along a differentiable curve $x = g(t)$, $y = h(t)$ is

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

At any point $P_0(x_0, y_0) = P_0(g(t_0), h(t_0))$, this equation gives the rate of change of f with respect to increasing t and therefore depends, among other things, on the direction of motion along the curve. If the curve is a straight line and t is the arc length parameter along the line measured from P_0 in the direction of a given unit vector $\mathbf{u}$, then df/dt is the rate of change of f with respect to distance in its domain in the direction of $\mathbf{u}$. By varying $\mathbf{u}$, we find the rates at which f changes with respect to distance as we move through P_0 in different directions. We now define this idea more precisely.

Suppose that the function $f(x, y)$ is defined throughout a region R in the xy -plane, that $P_0(x_0, y_0)$ is a point in R , and that $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector. Then the equations

$$x = x_0 + su_1, \quad y = y_0 + su_2$$

parametrize the line through P_0 parallel to $\mathbf{u}$. If the parameter s measures arc length from P_0 in the direction of $\mathbf{u}$, we find the rate of change of f at P_0 in the direction of $\mathbf{u}$ by calculating df/ds at P_0 (Figure 13.27).

DEFINITION The **derivative of f at $P_0(x_0, y_0)$ in the direction of the unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$** is the number

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s}, \quad (1)$$

provided the limit exists.

The **directional derivative** defined by Equation (1) is also denoted by

$$D_{\mathbf{u}}f(P_0) \quad \text{or} \quad D_{\mathbf{u}}f|_{P_0}.$$

“The derivative of f in the direction of $\mathbf{u}$, evaluated at P_0 ”

The partial derivatives $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ are the directional derivatives of f at P_0 in the $\mathbf{i}$ and $\mathbf{j}$ directions. This observation can be seen by comparing Equation (1) to the definitions of the two partial derivatives given in Section 13.3.

EXAMPLE 1 Using the definition, find the derivative of

$$f(x, y) = x^2 + xy$$

at $P_0(1, 2)$ in the direction of the unit vector $\mathbf{u} = (1/\sqrt{2})\mathbf{i} + (1/\sqrt{2})\mathbf{j}$.

Solution Applying the definition in Equation (1), we obtain

$$\begin{aligned}
 \left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} &= \lim_{s \rightarrow 0} \frac{f(x_0 + su_1, y_0 + su_2) - f(x_0, y_0)}{s} && \text{Eq. (1)} \\
 &= \lim_{s \rightarrow 0} \frac{f\left(1 + s \cdot \frac{1}{\sqrt{2}}, 2 + s \cdot \frac{1}{\sqrt{2}}\right) - f(1, 2)}{s} && \text{Substitute.} \\
 &= \lim_{s \rightarrow 0} \frac{\left(1 + \frac{s}{\sqrt{2}}\right)^2 + \left(1 + \frac{s}{\sqrt{2}}\right)\left(2 + \frac{s}{\sqrt{2}}\right) - (1^2 + 1 \cdot 2)}{s} \\
 &= \lim_{s \rightarrow 0} \frac{\left(1 + \frac{2s}{\sqrt{2}} + \frac{s^2}{2}\right) + \left(2 + \frac{3s}{\sqrt{2}} + \frac{s^2}{2}\right) - 3}{s} \\
 &= \lim_{s \rightarrow 0} \frac{\frac{5s}{\sqrt{2}} + s^2}{s} = \lim_{s \rightarrow 0} \left(\frac{5}{\sqrt{2}} + s\right) = \frac{5}{\sqrt{2}}.
 \end{aligned}$$

The rate of change of $f(x, y) = x^2 + xy$ at $P_0(1, 2)$ in the direction $\mathbf{u}$ is $5/\sqrt{2}$. ■

Interpretation of the Directional Derivative

The equation $z = f(x, y)$ represents a surface S in space. If $z_0 = f(x_0, y_0)$, then the point $P(x_0, y_0, z_0)$ lies on S . The vertical plane that passes through P and $P_0(x_0, y_0)$ parallel to $\mathbf{u}$ intersects S in a curve C (Figure 13.28). The rate of change of f in the direction of $\mathbf{u}$ is the slope of the tangent to C at P in the right-handed system formed by the vectors $\mathbf{u}$ and $\mathbf{k}$.

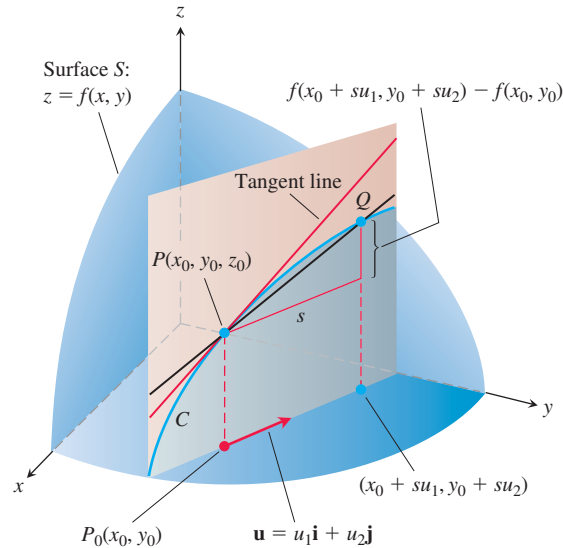


FIGURE 13.28 The slope of the trace curve C at P_0 is $\lim_{Q \rightarrow P} \text{slope}(PQ)$; this is the directional derivative

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = D_{\mathbf{u}}f \Big|_{P_0}.$$

When $\mathbf{u} = \mathbf{i}$, the directional derivative at P_0 is $\partial f / \partial x$ evaluated at (x_0, y_0) . When $\mathbf{u} = \mathbf{j}$, the directional derivative at P_0 is $\partial f / \partial y$ evaluated at (x_0, y_0) . The directional derivative generalizes the two partial derivatives. We can now ask for the rate of change of f in any direction $\mathbf{u}$, not just in the directions $\mathbf{i}$ and $\mathbf{j}$.

For a physical interpretation of the directional derivative, suppose that $T = f(x, y)$ is the temperature at each point (x, y) over a region in the plane. Then $f(x_0, y_0)$ is the

temperature at the point $P_0(x_0, y_0)$, and $D_{\mathbf{u}}f|_{P_0}$ is the instantaneous rate of change of the temperature at P_0 stepping off in the direction $\mathbf{u}$.

Calculation and Gradients

We now develop an efficient formula to calculate the directional derivative for a differentiable function f . We begin with the line

$$x = x_0 + su_1, \quad y = y_0 + su_2, \quad (2)$$

through $P_0(x_0, y_0)$, parametrized with the arc length parameter s increasing in the direction of the unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$. Then, by the Chain Rule we find

$$\begin{aligned} \left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} &= \left.\frac{\partial f}{\partial x}\right|_{P_0} \frac{dx}{ds} + \left.\frac{\partial f}{\partial y}\right|_{P_0} \frac{dy}{ds} && \text{Chain Rule for differentiable } f \\ &= \left.\frac{\partial f}{\partial x}\right|_{P_0} u_1 + \left.\frac{\partial f}{\partial y}\right|_{P_0} u_2 && \text{From Eqs. (2), } dx/ds = u_1 \\ &&& \text{and } dy/ds = u_2 \\ &= \underbrace{\left[\left.\frac{\partial f}{\partial x}\right|_{P_0} \mathbf{i} + \left.\frac{\partial f}{\partial y}\right|_{P_0} \mathbf{j}\right]}_{\text{Gradient of } f \text{ at } P_0} \cdot \underbrace{\left[u_1\mathbf{i} + u_2\mathbf{j}\right]}_{\text{Direction } \mathbf{u}}. \end{aligned} \quad (3)$$

Equation (3) says that the derivative of a differentiable function f in the direction of $\mathbf{u}$ at P_0 is the dot product of $\mathbf{u}$ with a special vector, which we now define.

DEFINITION The **gradient vector** (or **gradient**) of $f(x, y)$ is the vector

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}.$$

The value of the gradient vector obtained by evaluating the partial derivatives at a point $P_0(x_0, y_0)$ is written

$$\nabla f|_{P_0} \quad \text{or} \quad \nabla f(x_0, y_0).$$

The notation ∇f is read “grad f ” as well as “gradient of f ” and “del f .” The symbol ∇ by itself is read “del.” Another notation for the gradient is $\text{grad } f$. Using the gradient notation, we restate Equation (3) as a theorem.

THEOREM 9—The Directional Derivative Is a Dot Product

If $f(x, y)$ is differentiable in an open region containing $P_0(x_0, y_0)$, then

$$\left(\frac{df}{ds}\right)_{\mathbf{u}, P_0} = \nabla f|_{P_0} \cdot \mathbf{u}, \quad (4)$$

the dot product of the gradient ∇f at P_0 with the vector $\mathbf{u}$. In brief, $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

EXAMPLE 2 Find the derivative of $f(x, y) = xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$.

Solution Recall that the direction of a vector $\mathbf{v}$ is the unit vector obtained by dividing $\mathbf{v}$ by its length:

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\mathbf{v}}{5} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}.$$

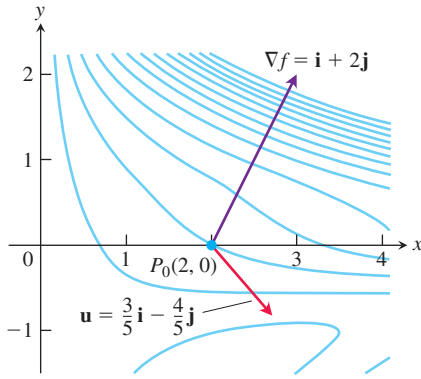


FIGURE 13.29 Picture ∇f as a vector in the domain of f . The figure shows a number of level curves of f . The rate at which f changes at $(2, 0)$ in the direction $\mathbf{u}$ is $\nabla f \cdot \mathbf{u} = -1$, which is the component of ∇f in the direction of unit vector $\mathbf{u}$ (Example 2).

The partial derivatives of f are everywhere continuous and at $(2, 0)$ are given by

$$f_x(2, 0) = (e^y - y \sin(xy)) \Big|_{(2,0)} = e^0 - 0 = 1$$

$$f_y(2, 0) = (xe^y - x \sin(xy)) \Big|_{(2,0)} = 2e^0 - 2 \cdot 0 = 2.$$

The gradient of f at $(2, 0)$ is

$$\nabla f|_{(2,0)} = f_x(2, 0)\mathbf{i} + f_y(2, 0)\mathbf{j} = \mathbf{i} + 2\mathbf{j}$$

(Figure 13.29). The derivative of f at $(2, 0)$ in the direction of $\mathbf{v}$ is therefore

$$\begin{aligned} D_{\mathbf{u}}f|_{(2,0)} &= \nabla f|_{(2,0)} \cdot \mathbf{u} && \text{Eq. (4) with the } D_{\mathbf{u}}f|_{P_0} \text{ notation} \\ &= (\mathbf{i} + 2\mathbf{j}) \cdot \left(\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right) = \frac{3}{5} - \frac{8}{5} = -1. \end{aligned}$$

Evaluating the dot product in the brief version of Equation (4) gives

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f||\mathbf{u}| \cos \theta = |\nabla f| \cos \theta,$$

where θ is the angle between the vectors $\mathbf{u}$ and ∇f , and reveals the following properties.

Properties of the Directional Derivative $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos \theta$

1. The function f increases most rapidly when $\cos \theta = 1$, which means that $\theta = 0$ and $\mathbf{u}$ is the direction of ∇f . That is, at each point P in its domain, f increases most rapidly in the direction of the gradient vector ∇f at P . The derivative in this direction is

$$D_{\mathbf{u}}f = |\nabla f| \cos(0) = |\nabla f|.$$

2. Similarly, f decreases most rapidly in the direction of $-\nabla f$. The derivative in this direction is $D_{\mathbf{u}}f = |\nabla f| \cos(\pi) = -|\nabla f|$.

3. Any direction $\mathbf{u}$ orthogonal to a gradient $\nabla f \neq 0$ is a direction of zero change in f because θ then equals $\pi/2$ and

$$D_{\mathbf{u}}f = |\nabla f| \cos(\pi/2) = |\nabla f| \cdot 0 = 0.$$

As we discuss later, these properties hold in three dimensions as well as two.

EXAMPLE 3 Find the directions in which $f(x, y) = (x^2/2) + (y^2/2)$

- (a) increases most rapidly at the point $(1, 1)$, and
- (b) decreases most rapidly at $(1, 1)$.
- (c) What are the directions of zero change in f at $(1, 1)$?

Solution

- (a) The function increases most rapidly in the direction of ∇f at $(1, 1)$. The gradient there is

$$\nabla f|_{(1,1)} = (x\mathbf{i} + y\mathbf{j}) \Big|_{(1,1)} = \mathbf{i} + \mathbf{j}.$$

Its direction is

$$\mathbf{u} = \frac{\mathbf{i} + \mathbf{j}}{|\mathbf{i} + \mathbf{j}|} = \frac{\mathbf{i} + \mathbf{j}}{\sqrt{(1)^2 + (1)^2}} = \frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}.$$

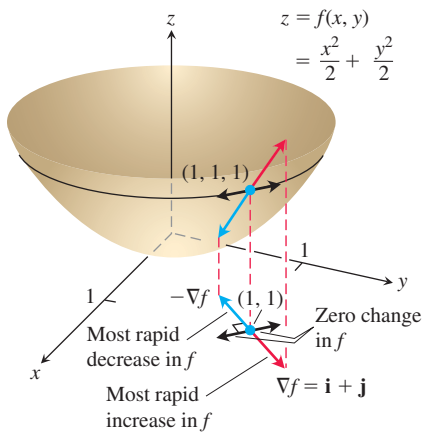


FIGURE 13.30 The direction in which $f(x, y)$ increases most rapidly at $(1, 1)$ is the direction of $\nabla f|_{(1,1)} = \mathbf{i} + \mathbf{j}$. It corresponds to the direction of steepest ascent on the surface at $(1, 1, 1)$ (Example 3).

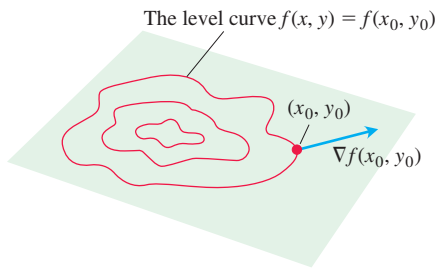


FIGURE 13.31 When it is nonzero, the gradient of a differentiable function of two variables at a point is always normal to the function's level curve through that point.

(b) The function decreases most rapidly in the direction of $-\nabla f$ at $(1, 1)$, which is

$$-\mathbf{u} = -\frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}.$$

(c) The directions of zero change at $(1, 1)$ are the directions orthogonal to ∇f :

$$\mathbf{n} = -\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j} \quad \text{and} \quad -\mathbf{n} = \frac{1}{\sqrt{2}}\mathbf{i} - \frac{1}{\sqrt{2}}\mathbf{j}.$$

See Figure 13.30.

Gradients and Tangents to Level Curves

If a differentiable function $f(x, y)$ has a constant value c along a smooth curve $\mathbf{r} = g(t)\mathbf{i} + h(t)\mathbf{j}$ (making the curve part of a level curve of f), then $f(g(t), h(t)) = c$. Differentiating both sides of this equation with respect to t leads to the equations

$$\begin{aligned} \frac{d}{dt}f(g(t), h(t)) &= \frac{d}{dt}(c) \\ \frac{\partial f}{\partial x} \frac{dg}{dt} + \frac{\partial f}{\partial y} \frac{dh}{dt} &= 0 && \text{Chain Rule} \\ \underbrace{\left(\frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} \right)}_{\nabla f} \cdot \underbrace{\left(\frac{dg}{dt} \mathbf{i} + \frac{dh}{dt} \mathbf{j} \right)}_{\frac{d\mathbf{r}}{dt}} &= 0. \end{aligned} \quad (5)$$

Assuming the gradient of f is a nonzero vector, Equation (5) says that ∇f is normal to the tangent vector $d\mathbf{r}/dt$, so it is normal to the curve. This is seen in Figure 13.31.

At every point (x_0, y_0) in the domain of a differentiable function $f(x, y)$ where the gradient of f is a nonzero vector, this vector is normal to the level curve through (x_0, y_0) (Figure 13.31).

Equation (5) validates our observation that streams flow perpendicular to the contours in topographical maps (see Figure 13.26). Since the downflowing stream will reach its destination in the fastest way, it must flow in the direction of the negative gradient vectors from Property 2 for the directional derivative. Equation (5) tells us these directions are perpendicular to the level curves.

This observation also enables us to find equations for tangent lines to level curves. They are the lines normal to the gradients. The line through a point $P_0(x_0, y_0)$ normal to a nonzero vector $\mathbf{N} = A\mathbf{i} + B\mathbf{j}$ has the equation

$$A(x - x_0) + B(y - y_0) = 0$$

(Exercise 39). If $\mathbf{N}$ is the gradient $\nabla f|_{(x_0, y_0)} = f_x(x_0, y_0)\mathbf{i} + f_y(x_0, y_0)\mathbf{j}$, and this gradient is not the zero vector, then this equation gives the following formula.

Equation for the Tangent Line to a Level Curve

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 0 \quad (6)$$

EXAMPLE 4 Find an equation for the tangent to the ellipse

$$\frac{x^2}{4} + y^2 = 2$$

(Figure 13.32) at the point $(-2, 1)$.

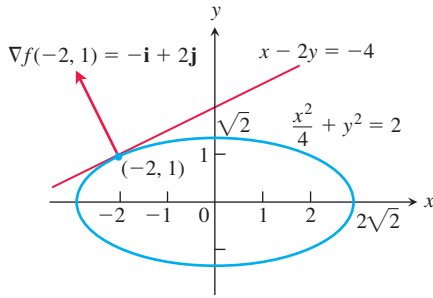


FIGURE 13.32 We can find the tangent to the ellipse $(x^2/4) + y^2 = 2$ by treating the ellipse as a level curve of the function $f(x, y) = (x^2/4) + y^2$ (Example 4).

Solution The ellipse is a level curve of the function

$$f(x, y) = \frac{x^2}{4} + y^2.$$

The gradient of f at $(-2, 1)$ is

$$\nabla f|_{(-2,1)} = \left(\frac{x}{2}\mathbf{i} + 2y\mathbf{j} \right) \Big|_{(-2,1)} = -\mathbf{i} + 2\mathbf{j}.$$

Because this gradient vector is nonzero, the tangent to the ellipse at $(-2, 1)$ is the line

$$(-1)(x + 2) + (2)(y - 1) = 0 \quad \text{Eq. (6)}$$

$$x - 2y = -4. \quad \text{Simplify.}$$

If we know the gradients of two functions f and g , we automatically know the gradients of their sum, difference, constant multiples, product, and quotient. You are asked to establish the following rules in Exercise 40. Notice that these rules have the same form as the corresponding rules for derivatives of single-variable functions.

Algebra Rules for Gradients

- | | | |
|-----------------------------------|--|---|
| 1. <i>Sum Rule:</i> | $\nabla(f + g) = \nabla f + \nabla g$ | |
| 2. <i>Difference Rule:</i> | $\nabla(f - g) = \nabla f - \nabla g$ | |
| 3. <i>Constant Multiple Rule:</i> | $\nabla(kf) = k\nabla f$ | (any number k) |
| 4. <i>Product Rule:</i> | $\nabla(fg) = f\nabla g + g\nabla f$ | |
| 5. <i>Quotient Rule:</i> | $\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$ | Scalar multipliers on left of gradients |

EXAMPLE 5 We illustrate two of the rules with

$$\begin{aligned} f(x, y) &= x - y & g(x, y) &= 3y \\ \nabla f &= \mathbf{i} - \mathbf{j} & \nabla g &= 3\mathbf{j}. \end{aligned}$$

We have

$$1. \nabla(f - g) = \nabla(x - 4y) = \mathbf{i} - 4\mathbf{j} = \nabla f - \nabla g \quad \text{Rule 2}$$

$$2. \nabla(fg) = \nabla(3xy - 3y^2) = 3y\mathbf{i} + (3x - 6y)\mathbf{j}$$

and

$$f\nabla g + g\nabla f = (x - y)3\mathbf{j} + 3y(\mathbf{i} - \mathbf{j}) \quad \text{Substitute.}$$

$$= 3y\mathbf{i} + (3x - 6y)\mathbf{j}. \quad \text{Simplify.}$$

We have therefore verified that for this example, $\nabla(fg) = f\nabla g + g\nabla f$.

Functions of Three Variables

For a differentiable function $f(x, y, z)$ and a unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ in space, we have

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

and

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \frac{\partial f}{\partial x}u_1 + \frac{\partial f}{\partial y}u_2 + \frac{\partial f}{\partial z}u_3.$$

The directional derivative can once again be written in the form

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| |\mathbf{u}| \cos \theta = |\nabla f| \cos \theta,$$

so the properties listed earlier for functions of two variables extend to three variables. At any given point, f increases most rapidly in the direction of ∇f and decreases most rapidly in the direction of $-\nabla f$. In any direction orthogonal to ∇f , the derivative is zero.

EXAMPLE 6

- (a) Find the derivative of $f(x, y, z) = x^3 - xy^2 - z$ at $P_0(1, 1, 0)$ in the direction of $\mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$.
 (b) In what directions does f change most rapidly at P_0 , and what are the rates of change in these directions?

Solution

- (a) The direction of $\mathbf{v}$ is obtained by dividing $\mathbf{v}$ by its length:

$$|\mathbf{v}| = \sqrt{(2)^2 + (-3)^2 + (6)^2} = \sqrt{49} = 7$$

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}.$$

The partial derivatives of f at P_0 are

$$f_x = (3x^2 - y^2) \Big|_{(1,1,0)} = 2, \quad f_y = -2xy \Big|_{(1,1,0)} = -2, \quad f_z = -1 \Big|_{(1,1,0)} = -1.$$

The gradient of f at P_0 is

$$\nabla f|_{(1,1,0)} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}.$$

The derivative of f at P_0 in the direction of $\mathbf{v}$ is therefore

$$\begin{aligned} D_{\mathbf{u}}f|_{(1,1,0)} &= \nabla f|_{(1,1,0)} \cdot \mathbf{u} = (2\mathbf{i} - 2\mathbf{j} - \mathbf{k}) \cdot \left(\frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k} \right) \\ &= \frac{4}{7} + \frac{6}{7} - \frac{6}{7} = \frac{4}{7}. \end{aligned}$$

- (b) The function increases most rapidly in the direction of $\nabla f = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$ and decreases most rapidly in the direction of $-\nabla f$. The rates of change in the directions are, respectively,

$$|\nabla f| = \sqrt{(2)^2 + (-2)^2 + (-1)^2} = \sqrt{9} = 3 \quad \text{and} \quad -|\nabla f| = -3. \quad \blacksquare$$

Functions of More Than Three Variables

The gradient of a differentiable function of n variables $f(x_1, x_2, \dots, x_n)$ is

$$\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

If $\mathbf{u} = \langle u_1, u_2, \dots, u_n \rangle$ is an n -dimensional vector such that $u_1^2 + u_2^2 + \dots + u_n^2 = 1$ (so $\mathbf{u}$ is a unit vector since $|\mathbf{u}| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2} = 1$), then the directional derivative of f in the direction of $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \frac{\partial f}{\partial x_1} u_1 + \frac{\partial f}{\partial x_2} u_2 + \dots + \frac{\partial f}{\partial x_n} u_n.$$

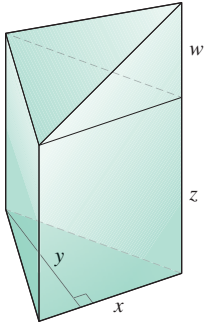


FIGURE 13.33 A tetrahedron on top of a triangular prism (Example 7).

EXAMPLE 7 The volume of the solid shown in Figure 13.33 consisting of a tetrahedron on top of a triangular prism is given by $f(x, y, z, w) = \frac{xy}{2}\left(z + \frac{w}{3}\right)$.

- (a) Calculate the derivative of $f(x, y, z, w)$ at the point $P_0(6, 5, 8, 4)$ in the direction of $\mathbf{v} = \langle 1, -1, -1, 1 \rangle$.
 (b) What is the geometric significance of the value obtained in part (a)?

Solution

- (a) The direction of $\mathbf{v}$ is the unit vector

$$\begin{aligned}\mathbf{u} &= \frac{1}{|\mathbf{v}|} \mathbf{v} \\ &= \frac{1}{\sqrt{1^2 + (-1)^2 + (-1)^2 + 1^2}} \langle 1, -1, -1, 1 \rangle \\ &= \frac{1}{2} \langle 1, -1, -1, 1 \rangle \\ &= \left\langle \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right\rangle.\end{aligned}$$

The four partial derivatives of f at the point P_0 are

$$\begin{aligned}f_x &= \frac{y}{2} \left(z + \frac{w}{3} \right) \Big|_{(6,5,8,4)} = \frac{70}{3}, \\ f_y &= \frac{x}{2} \left(z + \frac{w}{3} \right) \Big|_{(6,5,8,4)} = 28, \\ f_z &= \frac{xy}{2} \Big|_{(6,5,8,4)} = 15, \\ f_w &= \frac{xy}{6} \Big|_{(6,5,8,4)} = 5.\end{aligned}$$

The gradient of f at P_0 is

$$\nabla f|_{(6,5,8,4)} = \left\langle \frac{70}{3}, 28, 15, 5 \right\rangle.$$

The derivative of f at $(6, 5, 8, 4)$ in the direction of $\mathbf{v}$ is

$$\begin{aligned}D_{\mathbf{u}}f|_{P_0} &= \nabla f|_{P_0} \cdot \mathbf{u} \\ &= \left\langle \frac{70}{3}, 28, 15, 5 \right\rangle \cdot \left\langle \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right\rangle \\ &= \left(\frac{70}{3} \right) \left(\frac{1}{2} \right) + (28) \left(-\frac{1}{2} \right) + (15) \left(-\frac{1}{2} \right) + (5) \left(\frac{1}{2} \right) \\ &= -\frac{22}{3}.\end{aligned}$$

- (b) Geometrically, this means that if the dimensions are $x = 6, y = 5, z = 8$, and $w = 4$, and the dimensions are changed by moving at unit speed so that x and w increase at the same rate while both y and z decrease at that rate, then the volume of the solid decreases at the rate of $7 \frac{1}{3}$. ■

The Chain Rule for Paths

If $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is a smooth path C , and $w = f(\mathbf{r}(t))$ is a scalar function evaluated along C , then according to the Chain Rule, Theorem 6 in Section 13.4,

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}.$$

The partial derivatives on the right-hand side of the above equation are evaluated along the curve $\mathbf{r}(t)$, and the derivatives of the intermediate variables are evaluated at t . If we express this equation using vector notation, we have

The Derivative Along a Path

$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t). \quad (7)$$

What Equation (7) says is that the derivative of the composite function $f(\mathbf{r}(t))$ is the “derivative” (gradient) of the outside function f , evaluated at $\mathbf{r}(t)$, “times” (dot product) the derivative of the inside function $\mathbf{r}$. This is analogous to the “Outside-Inside” Rule for derivatives of composite functions studied in Section 3.6. That is, the multivariable Chain Rule for paths has exactly *the same form* as the rule for single-variable differential calculus when appropriate interpretations are given to the meanings of the terms and operations involved.

EXERCISES 13.5

Calculating Gradients

In Exercises 1–6, find the gradient of the function at the given point. Then sketch the gradient, together with the level curve that passes through the point.

1. $f(x, y) = y - x$, $(2, 1)$
2. $f(x, y) = \ln(x^2 + y^2)$, $(1, 1)$
3. $g(x, y) = xy^2$, $(2, -1)$
4. $g(x, y) = \frac{x^2}{2} - \frac{y^2}{2}$, $(\sqrt{2}, 1)$
5. $f(x, y) = \sqrt{2x + 3y}$, $(-1, 2)$
6. $f(x, y) = \tan^{-1} \frac{\sqrt{x}}{y}$, $(4, -2)$

In Exercises 7–10, find ∇f at the given point.

7. $f(x, y, z) = x^2 + y^2 - 2z^2 + z \ln x$, $(1, 1, 1)$
8. $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z + \arctan xz$, $(1, 1, 1)$
9. $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2} + \ln(xyz)$, $(-1, 2, -2)$
10. $f(x, y, z) = e^{x+y} \cos z + (y + 1) \arcsin x$, $(0, 0, \pi/6)$

Finding Directional Derivatives

In Exercises 11–18, find the derivative of the function at P_0 in the direction of $\mathbf{v}$.

11. $f(x, y) = 2xy - 3y^2$, $P_0(5, 5)$, $\mathbf{v} = 4\mathbf{i} + 3\mathbf{j}$
12. $f(x, y) = 2x^2 + y^2$, $P_0(-1, 1)$, $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$
13. $g(x, y) = \frac{x - y}{xy + 2}$, $P_0(1, -1)$, $\mathbf{v} = 12\mathbf{i} + 5\mathbf{j}$
14. $h(x, y) = \arctan(y/x) + \sqrt{3} \arcsin(xy/2)$, $P_0(1, 1)$, $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j}$
15. $f(x, y, z) = xy + yz + zx$, $P_0(1, -1, 2)$, $\mathbf{v} = 3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$
16. $f(x, y, z) = x^2 + 2y^2 - 3z^2$, $P_0(1, 1, 1)$, $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
17. $g(x, y, z) = 3e^x \cos yz$, $P_0(0, 0, 0)$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$
18. $h(x, y, z) = \cos xy + e^{yz} + \ln xz$, $P_0(1, 0, 1/2)$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$

In Exercises 19–24, find the directions in which the functions increase most rapidly, and the directions in which they decrease most rapidly, at P_0 . Then find the derivatives of the functions in these directions.

19. $f(x, y) = x^2 + xy + y^2$, $P_0(-1, 1)$

20. $f(x, y) = x^2y + e^{xy} \sin y$, $P_0(1, 0)$
21. $f(x, y, z) = (x/y) - yz$, $P_0(4, 1, 1)$
22. $g(x, y, z) = xe^y + z^2$, $P_0(1, \ln 2, 1/2)$
23. $f(x, y, z) = \ln xy + \ln yz + \ln xz$, $P_0(1, 1, 1)$
24. $h(x, y, z) = \ln(x^2 + y^2 - 1) + y + 6z$, $P_0(1, 1, 0)$

Tangent Lines to Level Curves

In Exercises 25–28, sketch the curve $f(x, y) = c$, together with ∇f and the tangent line at the given point. Then write an equation for the tangent line.

25. $x^2 + y^2 = 4$, $(\sqrt{2}, \sqrt{2})$
26. $x^2 - y = 1$, $(\sqrt{2}, 1)$
27. $xy = -4$, $(2, -2)$
28. $x^2 - xy + y^2 = 7$, $(-1, 2)$

Theory and Examples

29. Let $f(x, y) = x^2 - xy + y^2 - y$. Find the directions $\mathbf{u}$ and the values of $D_{\mathbf{u}}f(1, -1)$ for which
 - a. $D_{\mathbf{u}}f(1, -1)$ is largest
 - b. $D_{\mathbf{u}}f(1, -1)$ is smallest
 - c. $D_{\mathbf{u}}f(1, -1) = 0$
 - d. $D_{\mathbf{u}}f(1, -1) = 4$
 - e. $D_{\mathbf{u}}f(1, -1) = -3$
30. Let $f(x, y) = \frac{(x - y)}{(x + y)}$. Find the directions $\mathbf{u}$ and the values of $D_{\mathbf{u}}f\left(-\frac{1}{2}, \frac{3}{2}\right)$ for which
 - a. $D_{\mathbf{u}}f\left(-\frac{1}{2}, \frac{3}{2}\right)$ is largest
 - b. $D_{\mathbf{u}}f\left(-\frac{1}{2}, \frac{3}{2}\right)$ is smallest
 - c. $D_{\mathbf{u}}f\left(-\frac{1}{2}, \frac{3}{2}\right) = 0$
 - d. $D_{\mathbf{u}}f\left(-\frac{1}{2}, \frac{3}{2}\right) = -2$
 - e. $D_{\mathbf{u}}f\left(-\frac{1}{2}, \frac{3}{2}\right) = 1$
31. **Zero directional derivative** In what direction is the derivative of $f(x, y) = xy + y^2$ at $P(3, 2)$ equal to zero?
32. **Zero directional derivative** In what directions is the derivative of $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$ at $P(1, 1)$ equal to zero?
33. Is there a direction $\mathbf{u}$ in which the rate of change of $f(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals 14? Give reasons for your answer.

- 34. Changing temperature along a circle** Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y, z) = 2xy - yz$ (temperature in degrees Celsius, distance in meters) at $P(1, -1, 1)$ is $-3^\circ\text{C}/\text{m}$? Give reasons for your answer.
- 35.** The derivative of $f(x, y)$ at $P_0(1, 2)$ in the direction of $\mathbf{i} + \mathbf{j}$ is $2\sqrt{2}$ and in the direction of $-2\mathbf{j}$ is -3 . What is the derivative of f in the direction of $-\mathbf{i} - 2\mathbf{j}$? Give reasons for your answer.
- 36.** The derivative of $f(x, y, z)$ at a point P is greatest in the direction of $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$. In this direction, the value of the derivative is $2\sqrt{3}$.
- a. What is ∇f at P ? Give reasons for your answer.
b. What is the derivative of f at P in the direction of $\mathbf{i} + \mathbf{j}$?
- 37. Directional derivatives and scalar components** How is the derivative of a differentiable function $f(x, y, z)$ at a point P_0 in the direction of a unit vector $\mathbf{u}$ related to the scalar component of $\nabla f|_{P_0}$ in the direction of $\mathbf{u}$? Give reasons for your answer.
- 38. Directional derivatives and partial derivatives** Assuming that the necessary derivatives of $f(x, y, z)$ are defined, how are $D_{\mathbf{i}}f$, $D_{\mathbf{j}}f$, and $D_{\mathbf{k}}f$ related to f_x , f_y , and f_z ? Give reasons for your answer.
- 39. Lines in the xy -plane** Show that $A(x - x_0) + B(y - y_0) = 0$ is an equation for the line in the xy -plane through the point (x_0, y_0) normal to the vector $\mathbf{N} = A\mathbf{i} + B\mathbf{j}$.
- 40. The algebra rules for gradients** Given a constant k and the gradients

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k},$$

$$\nabla g = \frac{\partial g}{\partial x}\mathbf{i} + \frac{\partial g}{\partial y}\mathbf{j} + \frac{\partial g}{\partial z}\mathbf{k},$$

establish the algebra rules for gradients.

In Exercises 41–44, find a parametric equation for the line that is perpendicular to the graph of the given equation at the given point.

- 41.** $x^2 + y^2 = 25$, $(-3, 4)$
42. $x^2 + xy + y^2 = 3$, $(2, -1)$
43. $x^2 + y^2 + z^2 = 14$, $(3, -2, 1)$
44. $z = x^3 - xy^2$, $(-1, 1, 0)$

Gradients and Directional Derivatives for Functions of More Than Three Variables

In Exercises 45–48, find ∇f at the given point.

- 45.** $f(x, y, z, w) = \frac{x\sqrt{y}}{w} - x^2z^3$, $(2, 4, -1, 3)$
46. $f(x, y, z, w) = x^3 \sin y + w^2 \cos z$, $(-2, \pi, 0, 3)$
47. $f(x, y, z, s, t) = e^y \ln s + x^2 t \tan z$, $(3, 0, \frac{\pi}{4}, e, 5)$
48. $f(x, y, z, s, t) = \frac{(x^2 + y^2) \arctan t}{z^2 s}$, $(-2, 1, -1, 2, 1)$

In Exercises 49–52, find the derivative of the function at P_0 in the direction of $\mathbf{v}$.

- 49.** $f(x, y, z, w) = \frac{w \ln x}{y^2 z^3}$, $P_0(e^2, -2, 1, -3)$, $\mathbf{v} = \langle -1, 2, -2, 4 \rangle$
50. $f(x, y, z, w) = (x - y)^2 + e^{z-w}$, $P_0(4, 2, 3, 1)$, $\mathbf{v} = \langle 1, 0, -2, 2 \rangle$
51. $f(x, y, z, s, t) = s \arcsin(x + y) - t^2 \arctan(x - z)$, $P_0(0, \frac{1}{2}, -1, 1, -1)$, $\mathbf{v} = \langle -1, 1, 0, 3, 5 \rangle$
52. $f(x, y, z, s, t) = \sin tx + \cos sy - \frac{st}{z}$, $P_0(\frac{\pi}{4}, \frac{\pi}{6}, 2, 5, 1)$, $\mathbf{v} = \langle -3, 2, -2, 2, 2 \rangle$

13.6 Tangent Planes and Differentials

In single-variable differential calculus, we saw how the derivative defined the tangent line to the graph of a differentiable function at a point on the graph. The tangent line then provided for a linearization of the function at the point. In this section, we will see analogously how the gradient defines the *tangent plane* to the level surface of a function $w = f(x, y, z)$ at a point on the surface. The tangent plane then provides for a linearization of f at the point and defines the total differential of the function.

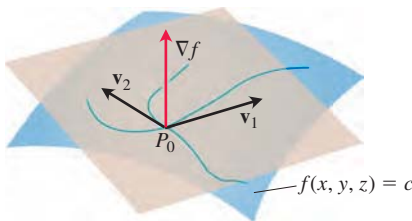


FIGURE 13.34 The gradient ∇f is orthogonal to the velocity vector of every smooth curve in the surface through P_0 . The velocity vectors at P_0 therefore lie in a common plane, which we call the tangent plane at P_0 .

Tangent Planes and Normal Lines

If $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ is a smooth curve on the level surface $f(x, y, z) = c$ of a differentiable function f , we found in Equation (7) of the last section that

$$\frac{d}{dt}f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t).$$

Since f is constant along the curve $\mathbf{r}$, the derivative on the left-hand side of the equation is 0, so the gradient ∇f is orthogonal to the curve's velocity vector $\mathbf{r}'$.

Now let us restrict our attention to the curves that pass through a point P_0 (Figure 13.34). All the velocity vectors at P_0 are orthogonal to ∇f at P_0 , so the curves' tangent lines all lie in the plane through P_0 normal to ∇f . (assuming it is a nonzero vector). We now define this plane.

DEFINITIONS The **tangent plane** to the level surface $f(x, y, z) = c$ of a differentiable function f at a point P_0 where the gradient is not zero is the plane through P_0 normal to $\nabla f|_{P_0}$.

The **normal line** of the surface at P_0 is the line through P_0 parallel to $\nabla f|_{P_0}$.

The results of Section 11.5 imply that the tangent plane and normal line satisfy the following equations, as long as the gradient at the point P_0 is not the zero vector.

Tangent Plane to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0 \quad (1)$$

Normal Line to $f(x, y, z) = c$ at $P_0(x_0, y_0, z_0)$

$$x = x_0 + f_x(P_0)t, \quad y = y_0 + f_y(P_0)t, \quad z = z_0 + f_z(P_0)t \quad (2)$$

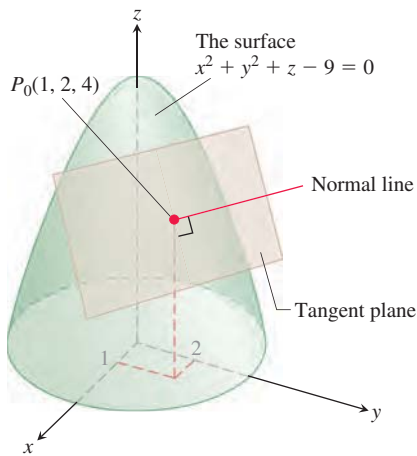


FIGURE 13.35 The tangent plane and normal line to this level surface at P_0 (Example 1).

EXAMPLE 1 Find the tangent plane and normal line of the level surface

$$f(x, y, z) = x^2 + y^2 + z - 9 = 0 \quad \text{A circular paraboloid}$$

at the point $P_0(1, 2, 4)$.

Solution The surface is shown in Figure 13.35.

The tangent plane is the plane through P_0 perpendicular to the gradient of f at P_0 . The gradient is

$$\nabla f|_{P_0} = (2x\mathbf{i} + 2y\mathbf{j} + \mathbf{k}) \Big|_{(1,2,4)} = 2\mathbf{i} + 4\mathbf{j} + \mathbf{k}.$$

The tangent plane is therefore the plane

$$2(x - 1) + 4(y - 2) + (z - 4) = 0, \quad \text{or} \quad 2x + 4y + z = 14.$$

The line normal to the surface at P_0 is

$$x = 1 + 2t, \quad y = 2 + 4t, \quad z = 4 + t. \quad \blacksquare$$

To find an equation for the plane tangent to a smooth surface $z = f(x, y)$ at a point $P_0(x_0, y_0, z_0)$ where $z_0 = f(x_0, y_0)$, we first observe that the equation $z = f(x, y)$ is equivalent to $f(x, y) - z = 0$. The surface $z = f(x, y)$ is therefore the zero level surface of the function $F(x, y, z) = f(x, y) - z$. The partial derivatives of F are

$$F_x = \frac{\partial}{\partial x}(f(x, y) - z) = f_x - 0 = f_x$$

$$F_y = \frac{\partial}{\partial y}(f(x, y) - z) = f_y - 0 = f_y$$

$$F_z = \frac{\partial}{\partial z}(f(x, y) - z) = 0 - 1 = -1.$$

The formula

$$F_x(P_0)(x - x_0) + F_y(P_0)(y - y_0) + F_z(P_0)(z - z_0) = 0$$

for the plane tangent to the level surface at P_0 therefore reduces to

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0.$$

Plane Tangent to a Surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$

The plane tangent to the surface $z = f(x, y)$ of a differentiable function f at the point $P_0(x_0, y_0, z_0) = (x_0, y_0, f(x_0, y_0))$ is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0. \quad (3)$$

EXAMPLE 2 Find the plane tangent to the surface $z = x \cos y - ye^x$ at $(0, 0, 0)$.

Solution We calculate the partial derivatives of $f(x, y) = x \cos y - ye^x$ and use Equation (3):

$$f_x(0, 0) = (\cos y - ye^x) \Big|_{(0,0)} = 1 - 0 \cdot 1 = 1$$

$$f_y(0, 0) = (-x \sin y - e^x) \Big|_{(0,0)} = 0 - 1 = -1.$$

The tangent plane is therefore

$$1 \cdot (x - 0) - 1 \cdot (y - 0) - (z - 0) = 0, \quad \text{Eq. (3)}$$

or

$$x - y - z = 0.$$

EXAMPLE 3 The surfaces

$$f(x, y, z) = x^2 + y^2 - 2 = 0 \quad \text{A cylinder}$$

and

$$g(x, y, z) = x + z - 4 = 0 \quad \text{A plane}$$

meet in an ellipse E (Figure 13.36). Find parametric equations for the line tangent to E at the point $P_0(1, 1, 3)$.

Solution The tangent line is orthogonal to both ∇f and ∇g at P_0 , and therefore parallel to $\mathbf{v} = \nabla f \times \nabla g$. The components of $\mathbf{v}$ and the coordinates of P_0 give us equations for the line. We have

$$\nabla f|_{(1,1,3)} = (2x\mathbf{i} + 2y\mathbf{j}) \Big|_{(1,1,3)} = 2\mathbf{i} + 2\mathbf{j}$$

$$\nabla g|_{(1,1,3)} = (\mathbf{i} + \mathbf{k}) \Big|_{(1,1,3)} = \mathbf{i} + \mathbf{k}$$

$$\mathbf{v} = (2\mathbf{i} + 2\mathbf{j}) \times (\mathbf{i} + \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 2\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}.$$

The tangent line to the ellipse of intersection is

$$x = 1 + 2t, \quad y = 1 - 2t, \quad z = 3 - 2t.$$

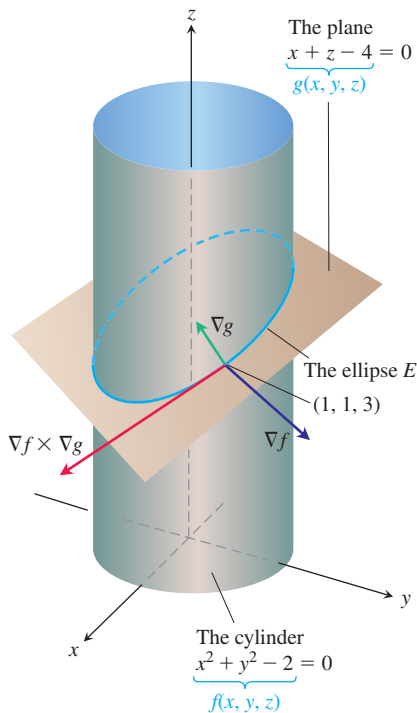


FIGURE 13.36 This cylinder and plane intersect in an ellipse E (Example 3).

Estimating Change in a Specific Direction

The directional derivative plays a role similar to that of an ordinary derivative when we want to estimate how much the value of a function f changes if we move a small distance ds from a point P_0 to another point nearby. If f were a function of a single variable, we would have

$$df = f'(P_0) ds. \quad \text{Ordinary derivative} \times \text{increment}$$

For a function of two or more variables, we use the formula

$$df = (\nabla f|_{P_0} \cdot \mathbf{u}) ds, \quad \text{Directional derivative} \times \text{increment}$$

where $\mathbf{u}$ is the direction of the motion away from P_0 .

Estimating the Change in f in a Direction $\mathbf{u}$

To estimate the change in the value of a differentiable function f when we move a small distance ds from a point P_0 in a particular direction $\mathbf{u}$, use this formula:

$$df = \underbrace{(\nabla f|_{P_0} \cdot \mathbf{u})}_{\text{Directional derivative}} \underbrace{ds}_{\text{Distance increment}}$$

EXAMPLE 4 Estimate how much the value of

$$f(x, y, z) = y \sin x + 2yz$$

will change if the point $P(x, y, z)$ moves 0.1 unit from $P_0(0, 1, 0)$ straight toward $P_1(2, 2, -2)$.

Solution We first find the derivative of f at P_0 in the direction of the vector $\overrightarrow{P_0P_1} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$. The direction of this vector is

$$\mathbf{u} = \frac{\overrightarrow{P_0P_1}}{|\overrightarrow{P_0P_1}|} = \frac{\overrightarrow{P_0P_1}}{3} = \frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}.$$

The gradient of f at P_0 is

$$\nabla f|_{(0,1,0)} = ((y \cos x)\mathbf{i} + (\sin x + 2z)\mathbf{j} + 2y\mathbf{k}) \Big|_{(0,1,0)} = \mathbf{i} + 2\mathbf{k}.$$

Therefore,

$$\nabla f|_{P_0} \cdot \mathbf{u} = (\mathbf{i} + 2\mathbf{k}) \cdot \left(\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k} \right) = \frac{2}{3} - \frac{4}{3} = -\frac{2}{3}.$$

The change df in f that results from moving $ds = 0.1$ unit away from P_0 in the direction of $\mathbf{u}$ is approximately

$$df = (\nabla f|_{P_0} \cdot \mathbf{u})(ds) = \left(-\frac{2}{3}\right)(0.1) \approx -0.067 \text{ unit.}$$

See Figure 13.37. ■

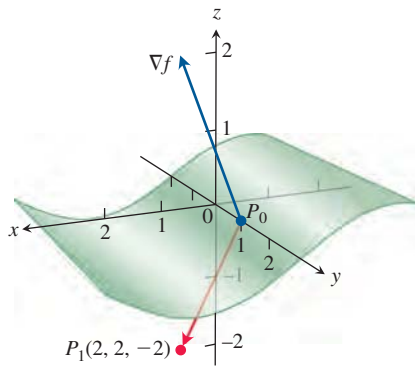


FIGURE 13.37 As $P(x, y, z)$ moves off the level surface at P_0 by 0.1 unit directly toward P_1 , the function f changes value by approximately -0.067 unit (Example 4).

How to Linearize a Function of Two Variables

Functions of two variables can be quite complicated, and we sometimes need to approximate them with simpler ones that give the accuracy required for specific applications without being so difficult to work with. We do this in a way that is similar to the way we find linear replacements for functions of a single variable (Section 3.11).

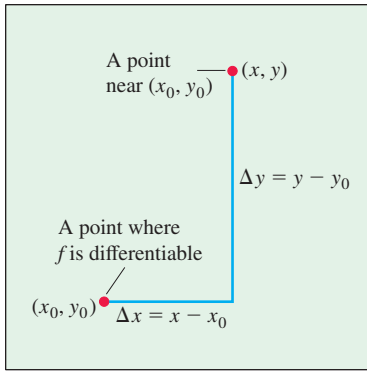


FIGURE 13.38 If f is differentiable at (x_0, y_0) , then the value of f at point (x, y) nearby is approximately $f(x_0, y_0) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$.

Suppose the function we wish to approximate is $z = f(x, y)$ near a point (x_0, y_0) at which we know the values of f , f_x , and f_y , and at which f is differentiable. If we move from (x_0, y_0) to a nearby point (x, y) by increments $\Delta x = x - x_0$ and $\Delta y = y - y_0$ (see Figure 13.38), then the definition of differentiability from Section 13.3 shows that the change

$$f(x, y) - f(x_0, y_0) = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where $\varepsilon_1, \varepsilon_2 \rightarrow 0$ as $\Delta x, \Delta y \rightarrow 0$. If the increments Δx and Δy are small, the products $\varepsilon_1\Delta x$ and $\varepsilon_2\Delta y$ will eventually be smaller still, and we have the approximation

$$f(x, y) \approx \underbrace{f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}_{L(x, y)}.$$

In other words, as long as Δx and Δy are small, f will have approximately the same value as the linear function L .

DEFINITIONS The **linearization** of a function $f(x, y)$ at a point (x_0, y_0) where f is differentiable is the function

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The approximation

$$f(x, y) \approx L(x, y)$$

is the **standard linear approximation** of f at (x_0, y_0) .

From Equation (3), we find that the plane $z = L(x, y)$ is tangent to the surface $z = f(x, y)$ at the point (x_0, y_0) . Thus, the linearization of a function of two variables is a **tangent-plane** approximation in the same way that the linearization of a function of a single variable is a **tangent-line** approximation. (See Exercise 57.)

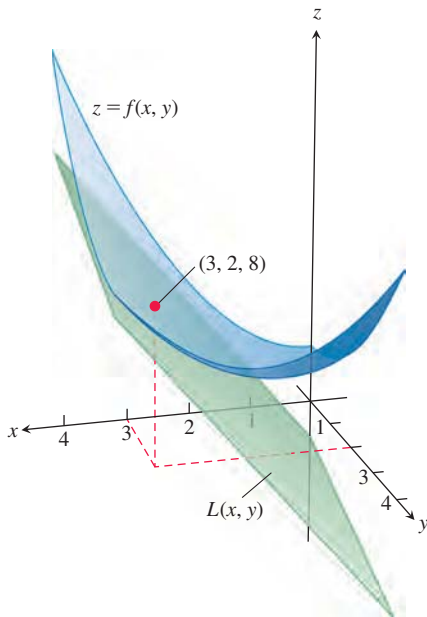


FIGURE 13.39 The tangent plane $L(x, y)$ represents the linearization of $f(x, y)$ in Example 5.

EXAMPLE 5 Find the linearization of

$$f(x, y) = x^2 - xy + \frac{1}{2}y^2 + 3$$

at the point $(3, 2)$.

Solution We first evaluate f , f_x , and f_y at the point $(x_0, y_0) = (3, 2)$:

$$f(3, 2) = \left(x^2 - xy + \frac{1}{2}y^2 + 3\right)\Big|_{(3, 2)} = 8$$

$$f_x(3, 2) = \frac{\partial}{\partial x}\left(x^2 - xy + \frac{1}{2}y^2 + 3\right)\Big|_{(3, 2)} = (2x - y)\Big|_{(3, 2)} = 4$$

$$f_y(3, 2) = \frac{\partial}{\partial y}\left(x^2 - xy + \frac{1}{2}y^2 + 3\right)\Big|_{(3, 2)} = (-x + y)\Big|_{(3, 2)} = -1,$$

which yields

$$\begin{aligned} L(x, y) &= f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \\ &= 8 + (4)(x - 3) + (-1)(y - 2) = 4x - y - 2. \end{aligned}$$

The linearization of f at $(3, 2)$ is $L(x, y) = 4x - y - 2$ (see Figure 13.39). ■

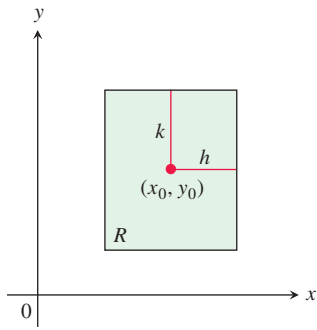


FIGURE 13.40 The rectangular region R : $|x - x_0| \leq h, |y - y_0| \leq k$ in the xy -plane.

When we approximate a differentiable function $f(x, y)$ by its linearization $L(x, y)$ at (x_0, y_0) , an important question is how accurate the approximation might be.

If we can find a common upper bound M for $|f_{xx}|$, $|f_{yy}|$, and $|f_{xy}|$ on a rectangle R centered at (x_0, y_0) (Figure 13.40), then we can bound the error E throughout R by using a simple formula. The **error** is defined by $E(x, y) = f(x, y) - L(x, y)$.

The Error in the Standard Linear Approximation

If f has continuous first and second partial derivatives throughout an open set containing a rectangle R centered at (x_0, y_0) , and if M is any upper bound for the values of $|f_{xx}|$, $|f_{yy}|$, and $|f_{xy}|$ on R , then the error $E(x, y)$ incurred in replacing $f(x, y)$ on R by its linearization

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

satisfies the inequality

$$|E(x, y)| \leq \frac{1}{2}M(|x - x_0| + |y - y_0|)^2.$$

To make $|E(x, y)|$ small for a given M , we just make $|x - x_0|$ and $|y - y_0|$ small.

Differentials

Recall from Section 3.11 that for a function of a single variable, $y = f(x)$, we defined the change in f as x changes from a to $a + \Delta x$ by

$$\Delta f = f(a + \Delta x) - f(a)$$

and the differential of f as

$$df = f'(a)\Delta x.$$

We now consider the differential of a function of two variables.

Suppose a differentiable function $f(x, y)$ and its partial derivatives exist at a point (x_0, y_0) . If we move to a nearby point $(x_0 + \Delta x, y_0 + \Delta y)$, the change in f is

$$\Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0).$$

A straightforward calculation based on the definition of $L(x, y)$, using the notation $x - x_0 = \Delta x$ and $y - y_0 = \Delta y$, shows that the corresponding change in L is

$$\Delta L = L(x_0 + \Delta x, y_0 + \Delta y) - L(x_0, y_0) = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y.$$

The **differentials** dx and dy are independent variables, so they can be assigned any values. Often we take $dx = \Delta x = x - x_0$, and $dy = \Delta y = y - y_0$. We then have the following definition of the differential or *total differential* of f .

DEFINITION If we move from (x_0, y_0) to a point $(x_0 + dx, y_0 + dy)$ nearby, the resulting change

$$df = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$$

in the linearization of f is called the **total differential** of f .

EXAMPLE 6 Suppose that a cylindrical can is designed to have a radius of 1 cm and a height of 5 cm, but that the radius and height are off by the amounts $dr = +0.03$ and $dh = -0.1$. Estimate the resulting absolute change in the volume of the can.

Solution To estimate the absolute change in $V = \pi r^2 h$, we use

$$\Delta V \approx dV = V_r(r_0, h_0) dr + V_h(r_0, h_0) dh.$$

With $V_r = 2\pi rh$ and $V_h = \pi r^2$, we get

$$\begin{aligned} dV &= 2\pi r_0 h_0 dr + \pi r_0^2 dh = 2\pi(1)(5)(0.03) + \pi(1)^2(-0.1) \\ &= 0.3\pi - 0.1\pi = 0.2\pi \approx 0.63 \text{ cm}^3. \end{aligned}$$

EXAMPLE 7 Your company manufactures stainless steel right circular cylindrical molasses storage tanks that are 2.5 m high with a radius of 0.5 m. How sensitive are the tanks' volumes to small variations in height and radius?

Solution With $V = \pi r^2 h$, the total differential gives the approximation for the change in volume as

$$\begin{aligned} dV &= V_r(0.5, 2.5) dr + V_h(0.5, 2.5) dh \\ &= (2\pi rh) \Big|_{(0.5, 2.5)} dr + (\pi r^2) \Big|_{(0.5, 2.5)} dh \\ &= 2.5\pi dr + 0.25\pi dh. \end{aligned}$$

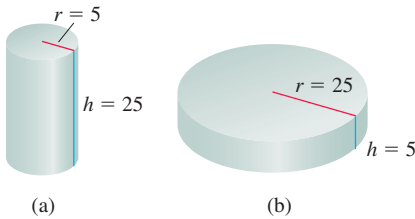


FIGURE 13.41 The volume of cylinder (a) is more sensitive to a small change in r than it is to an equally small change in h . The volume of cylinder (b) is more sensitive to small changes in h than it is to small changes in r (Example 7).

Thus, a 1-unit change in r will change V by about 2.5π units. A 1-unit change in h will change V by about 0.25π units. The tank's volume is 10 times more sensitive to a small change in r than it is to a small change of equal size in h . As a quality control engineer concerned with being sure the tanks have the correct volume, you would want to pay special attention to their radii.

In contrast, if the values of r and h are reversed to make $r = 2.5$ and $h = 0.5$, then the total differential in V becomes

$$dV = (2\pi rh) \Big|_{(2.5, 0.5)} dr + (\pi r^2) \Big|_{(2.5, 0.5)} dh = 2.5\pi dr + 6.25\pi dh.$$

Now the volume is more sensitive to changes in h than to changes in r (Figure 13.41).

The general rule is that functions are most sensitive to small changes in the variables that generate the largest partial derivatives.

Functions of More Than Two Variables

Analogous results hold for differentiable functions of more than two variables.

1. The **linearization** of $f(x, y, z)$ at a point $P_0(x_0, y_0, z_0)$ is

$$L(x, y, z) = f(P_0) + f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0).$$

2. Suppose that R is a closed rectangular solid centered at P_0 and lying in an open region on which the second partial derivatives of f are continuous. Suppose also that $|f_{xx}|, |f_{yy}|, |f_{zz}|, |f_{xy}|, |f_{xz}|$, and $|f_{yz}|$ are all less than or equal to M throughout R . Then the **error** $E(x, y, z) = f(x, y, z) - L(x, y, z)$ in the approximation of f by L is bounded throughout R by the inequality

$$|E| \leq \frac{1}{2} M(|x - x_0| + |y - y_0| + |z - z_0|)^2.$$

3. If the second partial derivatives of f are continuous and if x , y , and z change from x_0 , y_0 , and z_0 by small amounts dx , dy , and dz , the **total differential**

$$df = f_x(P_0) dx + f_y(P_0) dy + f_z(P_0) dz$$

gives a good approximation of the resulting change in f .

EXAMPLE 8 Find the linearization $L(x, y, z)$ of

$$f(x, y, z) = x^2 - xy + 3 \sin z$$

at the point $(x_0, y_0, z_0) = (2, 1, 0)$. Find an upper bound for the error incurred in replacing f by L on the rectangular region

$$R: |x - 2| \leq 0.01, \quad |y - 1| \leq 0.02, \quad |z| \leq 0.01.$$

Solution Routine calculations give

$$f(2, 1, 0) = 2, \quad f_x(2, 1, 0) = 3, \quad f_y(2, 1, 0) = -2, \quad f_z(2, 1, 0) = 3.$$

Thus,

$$L(x, y, z) = 2 + 3(x - 2) + (-2)(y - 1) + 3(z - 0) = 3x - 2y + 3z - 2.$$

Since

$$f_{xx} = 2, \quad f_{yy} = 0, \quad f_{zz} = -3 \sin z, \quad f_{xy} = -1, \quad f_{xz} = 0, \quad f_{yz} = 0,$$

and $|-3 \sin z| \leq 3 \sin 0.01 \approx 0.03$, we may take $M = 2$ as a bound on the second partials. Hence, the error incurred by replacing f by L on R satisfies

$$|E| \leq \frac{1}{2}(2)(0.01 + 0.02 + 0.01)^2 = 0.0016. \quad \blacksquare$$

EXERCISES 13.6

Tangent Planes and Normal Lines to Surfaces

In Exercises 1–10, find equations for the

(a) tangent plane and

(b) normal line at the point P_0 on the given surface.

1. $x^2 + y^2 + z^2 = 3$, $P_0(1, 1, 1)$
2. $x^2 + y^2 - z^2 = 18$, $P_0(3, 5, -4)$
3. $2z - x^2 = 0$, $P_0(2, 0, 2)$
4. $x^2 + 2xy - y^2 + z^2 = 7$, $P_0(1, -1, 3)$
5. $\cos \pi x - x^2 y + e^{xz} + yz = 4$, $P_0(0, 1, 2)$
6. $x^2 - xy - y^2 - z = 0$, $P_0(1, 1, -1)$
7. $x + y + z = 1$, $P_0(0, 1, 0)$
8. $x^2 + y^2 - 2xy - x + 3y - z = -4$, $P_0(2, -3, 18)$
9. $x \ln y + y \ln z = x$, $P_0(1, 1, e)$
10. $ye^x + ze^{y^2} = z$, $P_0(0, 0, 1)$

In Exercises 11–14, find an equation for the plane that is tangent to the given surface at the given point.

11. $z = \ln(x^2 + y^2)$, $(1, 0, 0)$
12. $z = e^{-(x^2 + y^2)}$, $(0, 0, 1)$

13. $z = \sqrt{y - x}$, $(1, 2, 1)$

14. $z = 4x^2 + y^2$, $(1, 1, 5)$

Tangent Lines to Intersecting Surfaces

In Exercises 15–20, find parametric equations for the line tangent to the curve of intersection of the surfaces at the given point.

15. Surfaces: $x + y^2 + 2z = 4$, $x = 1$

Point: $(1, 1, 1)$

16. Surfaces: $xyz = 1$, $x^2 + 2y^2 + 3z^2 = 6$

Point: $(1, 1, 1)$

17. Surfaces: $x^2 + 2y + 2z = 4$, $y = 1$

Point: $(1, 1, 1/2)$

18. Surfaces: $x + y^2 + z = 2$, $y = 1$

Point: $(1/2, 1, 1/2)$

19. Surfaces: $x^3 + 3x^2y^2 + y^3 + 4xy - z^2 = 0$,

$$x^2 + y^2 + z^2 = 11$$

Point: $(1, 1, 3)$

20. Surfaces: $x^2 + y^2 = 4$, $x^2 + y^2 - z = 0$

Point: $(\sqrt{2}, \sqrt{2}, 4)$

Estimating Change

21. By about how much will

$$f(x, y, z) = \ln \sqrt{x^2 + y^2 + z^2}$$

change if the point $P(x, y, z)$ moves from $P_0(3, 4, 12)$ a distance of $ds = 0.1$ unit in the direction of $3\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$?

22. By about how much will

$$f(x, y, z) = e^x \cos yz$$

change as the point $P(x, y, z)$ moves from the origin a distance of $ds = 0.1$ unit in the direction of $2\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$?

23. By about how much will

$$g(x, y, z) = x + x \cos z - y \sin z + y$$

change if the point $P(x, y, z)$ moves from $P_0(2, -1, 0)$ a distance of $ds = 0.2$ unit toward the point $P_1(0, 1, 2)$?

24. By about how much will

$$h(x, y, z) = \cos(\pi xy) + xz^2$$

change if the point $P(x, y, z)$ moves from $P_0(-1, -1, -1)$ a distance of $ds = 0.1$ unit toward the origin?

25. **Temperature change along a circle** Suppose that the Celsius temperature at the point (x, y) in the xy -plane is $T(x, y) = x \sin 2y$ and that distance in the xy -plane is measured in meters. A particle is moving *clockwise* around the circle of radius 1 m centered at the origin at the constant rate of 2 m/s.

- How fast is the temperature experienced by the particle changing in degrees Celsius per meter at the point $P(1/2, \sqrt{3}/2)$?
- How fast is the temperature experienced by the particle changing in degrees Celsius per second at P ?

26. **Changing temperature along a space curve** The Celsius temperature in a region in space is given by $T(x, y, z) = 2x^2 - xyz$. A particle is moving in this region and its position at time t is given by $x = 2t^2$, $y = 3t$, $z = -t^2$, where time is measured in seconds and distance in meters.

- How fast is the temperature experienced by the particle changing in degrees Celsius per meter when the particle is at the point $P(8, 6, -4)$?
- How fast is the temperature experienced by the particle changing in degrees Celsius per second at P ?

Finding Linearizations

In Exercises 27–32, find the linearization $L(x, y)$ of the function at each point.

- $f(x, y) = x^2 + y^2 + 1$ at $\mathbf{a. (0, 0), \quad b. (1, 1)}$
- $f(x, y) = (x + y + 2)^2$ at $\mathbf{a. (0, 0), \quad b. (1, 2)}$
- $f(x, y) = 3x - 4y + 5$ at $\mathbf{a. (0, 0), \quad b. (1, 1)}$
- $f(x, y) = x^3 y^4$ at $\mathbf{a. (1, 1), \quad b. (0, 0)}$
- $f(x, y) = e^x \cos y$ at $\mathbf{a. (0, 0), \quad b. (0, \pi/2)}$
- $f(x, y) = e^{2y-x}$ at $\mathbf{a. (0, 0), \quad b. (1, 2)}$

33. **Wind chill factor** Wind chill, a measure of the apparent temperature felt on exposed skin, is a function of air temperature and

wind speed. The precise formula, updated by the National Weather Service in 2001 and based on modern heat transfer theory, a human face model, and skin tissue resistance, is (after unit conversion)

$$W = W(v, T) = 13.13 + 0.6215T - 11.36v^{0.16} + 0.396T \cdot v^{0.16},$$

where T is air temperature in $^{\circ}\text{C}$ and v is wind speed in km/h. A partial wind chill chart is given.

		$T(^{\circ}\text{C})$						
		5	0	-5	-10	-15	-20	-25
v (km/h)	10	2.7	-3.3	-9.3	-15.2	-21.2	-27.2	-33.1
	20	1.1	-5.2	-11.5	-17.8	-24.1	-30.4	-36.7
	30	0.1	-6.4	-13.0	-19.5	-26.0	-32.5	-39.0
	40	-0.7	-7.4	-14.0	-20.7	-27.4	-34.1	-40.8
	50	-1.3	-8.1	-14.9	-21.7	-28.5	-35.4	-42.2
	60	-1.8	-8.7	-15.7	-22.6	-29.5	-36.4	-43.3

- Use the table to find $W(30, -5)$, $W(50, -25)$, and $W(30, -10)$.
 - Use the formula to find $W(15, -40)$, $W(80, -40)$, and $W(90, 0)$.
 - Find the linearization $L(v, T)$ of the function $W(v, T)$ at the point $(40, -10)$.
 - Use $L(v, T)$ in part (c) to estimate the following wind chill values.
 - $W(39, -9)$
 - $W(42, -12)$
 - $W(10, -25)$ (Explain why this value is much different from the value found in the table.)
34. Find the linearization $L(v, T)$ of the function $W(v, T)$ in Exercise 31 at the point $(50, -20)$. Use it to estimate the following wind chill values.
- $W(49, -22)$
 - $W(53, -19)$
 - $W(60, -30)$

Bounding the Error in Linear Approximations

In Exercises 35–40, find the linearization $L(x, y)$ of the function $f(x, y)$ at P_0 . Then find an upper bound for the magnitude $|E|$ of the error in the approximation $f(x, y) \approx L(x, y)$ over the rectangle R .

- $f(x, y) = x^2 - 3xy + 5$ at $P_0(2, 1)$,
 $R: |x - 2| \leq 0.1, |y - 1| \leq 0.1$
- $f(x, y) = (1/2)x^2 + xy + (1/4)y^2 + 3x - 3y + 4$ at $P_0(2, 2)$,
 $R: |x - 2| \leq 0.1, |y - 2| \leq 0.1$
- $f(x, y) = 1 + y + x \cos y$ at $P_0(0, 0)$,
 $R: |x| \leq 0.2, |y| \leq 0.2$
 (Use $|\cos y| \leq 1$ and $|\sin y| \leq 1$ in estimating E .)
- $f(x, y) = xy^2 + y \cos(x - 1)$ at $P_0(1, 2)$,
 $R: |x - 1| \leq 0.1, |y - 2| \leq 0.1$

39. $f(x, y) = e^x \cos y$ at $P_0(0, 0)$,
 $R: |x| \leq 0.1, |y| \leq 0.1$
 (Use $e^x \leq 1.11$ and $|\cos y| \leq 1$ in estimating E .)
40. $f(x, y) = \ln x + \ln y$ at $P_0(1, 1)$,
 $R: |x - 1| \leq 0.2, |y - 1| \leq 0.2$

Linearizations for Three Variables

Find the linearizations $L(x, y, z)$ of the functions in Exercises 41–46 at the given points.

41. $f(x, y, z) = xy + yz + xz$ at
 a. $(1, 1, 1)$ b. $(1, 0, 0)$ c. $(0, 0, 0)$
42. $f(x, y, z) = x^2 + y^2 + z^2$ at
 a. $(1, 1, 1)$ b. $(0, 1, 0)$ c. $(1, 0, 0)$
43. $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at
 a. $(1, 0, 0)$ b. $(1, 1, 0)$ c. $(1, 2, 2)$
44. $f(x, y, z) = (\sin xy)/z$ at
 a. $(\pi/2, 1, 1)$ b. $(2, 0, 1)$
45. $f(x, y, z) = e^x + \cos(y + z)$ at
 a. $(0, 0, 0)$ b. $(0, \frac{\pi}{2}, 0)$ c. $(0, \frac{\pi}{4}, \frac{\pi}{4})$
46. $f(x, y, z) = \tan^{-1}(xyz)$ at
 a. $(1, 0, 0)$ b. $(1, 1, 0)$ c. $(1, 1, 1)$

In Exercises 47–50, find the linearization $L(x, y, z)$ of the function $f(x, y, z)$ at P_0 . Then find an upper bound for the magnitude of the error E in the approximation $f(x, y, z) \approx L(x, y, z)$ over the region R .

47. $f(x, y, z) = xz - 3yz + 2$ at $P_0(1, 1, 2)$,
 $R: |x - 1| \leq 0.01, |y - 1| \leq 0.01, |z - 2| \leq 0.02$
48. $f(x, y, z) = x^2 + xy + yz + (1/4)z^2$ at $P_0(1, 1, 2)$,
 $R: |x - 1| \leq 0.01, |y - 1| \leq 0.01, |z - 2| \leq 0.08$
49. $f(x, y, z) = xy + 2yz - 3xz$ at $P_0(1, 1, 0)$,
 $R: |x - 1| \leq 0.01, |y - 1| \leq 0.01, |z| \leq 0.01$
50. $f(x, y, z) = \sqrt{2} \cos x \sin(y + z)$ at $P_0(0, 0, \pi/4)$,
 $R: |x| \leq 0.01, |y| \leq 0.01, |z - \pi/4| \leq 0.01$

Estimating Error; Sensitivity to Change

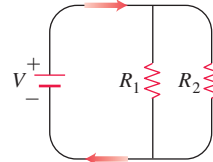
51. **Estimating maximum error** Suppose that T is to be found from the formula $T = x(e^y + e^{-y})$, where x and y are found to be 2 and $\ln 2$ with maximum possible errors of $|dx| = 0.1$ and $|dy| = 0.02$. Estimate the maximum possible error in the computed value of T .
52. **Variation in electrical resistance** The resistance R produced by wiring resistors of R_1 and R_2 ohms in parallel (see accompanying figure) can be calculated from the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

- a. Show that

$$dR = \left(\frac{R}{R_1}\right)^2 dR_1 + \left(\frac{R}{R_2}\right)^2 dR_2.$$

- b. You have designed a two-resistor circuit, like the one shown, to have resistances of $R_1 = 100$ ohms and $R_2 = 400$ ohms, but there is always some variation in manufacturing, and the resistors received by your firm will probably not have these exact values. Will the value of R be more sensitive to variation in R_1 or to variation in R_2 ? Give reasons for your answer.



- c. In another circuit like the one shown, you plan to change R_1 from 20 to 20.1 ohms and R_2 from 25 to 24.9 ohms. By about what percentage will this change R ?
53. You plan to calculate the area of a long, thin rectangle from measurements of its length and width. Which dimension should you measure more carefully? Give reasons for your answer.
54. a. Around the point $(1, 0)$, is $f(x, y) = x^2(y + 1)$ more sensitive to changes in x or to changes in y ? Give reasons for your answer.
 b. What ratio of dx to dy will make df equal zero at $(1, 0)$?
55. **Value of a 2×2 determinant** If $|a|$ is much greater than $|b|, |c|$, and $|d|$, to which of a, b, c , and d is the value of the determinant

$$f(a, b, c, d) = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

most sensitive? Give reasons for your answer.

56. **The Wilson lot size formula** The Wilson lot size formula in economics says that the most economical quantity Q of goods (radios, shoes, brooms, whatever) for a store to order is given by the formula $Q = \sqrt{2KM/h}$, where K is the cost of placing the order, M is the number of items sold per week, and h is the weekly holding cost for each item (cost of space, utilities, security, and so on). To which of the variables K, M , and h is Q most sensitive near the point $(K_0, M_0, h_0) = (2, 20, 0.05)$? Give reasons for your answer.

Theory and Examples

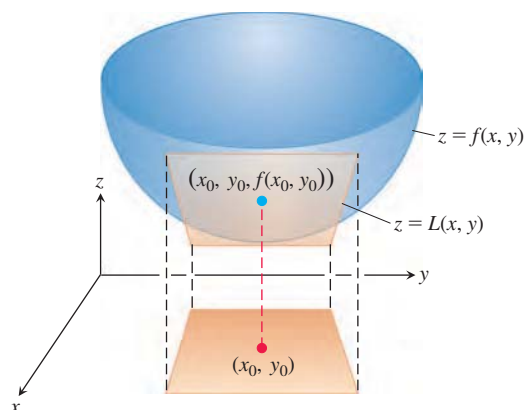
57. **The linearization of $f(x, y)$ is a tangent-plane approximation** Show that the tangent plane at the point $P_0(x_0, y_0, f(x_0, y_0))$ on the surface $z = f(x, y)$ defined by a differentiable function f is the plane

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - f(x_0, y_0)) = 0,$$

or

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Thus, the tangent plane at P_0 is the graph of the linearization of f at P_0 (see accompanying figure).



- 58. Change along the involute of a circle** Find the derivative of $f(x, y) = x^2 + y^2$ in the direction of the unit tangent vector of the curve

$$\mathbf{r}(t) = (\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}, \quad t > 0.$$

- 59. Tangent curves** A smooth curve is *tangent* to the surface at a point of intersection if its velocity vector is orthogonal to ∇f there.

Show that the curve

$$\mathbf{r}(t) = \sqrt{t}\mathbf{i} + \sqrt{t}\mathbf{j} + (2t - 1)\mathbf{k}$$

is tangent to the surface $x^2 + y^2 - z = 1$ when $t = 1$.

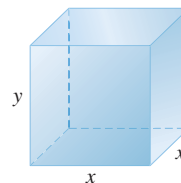
- 60. Normal curves** A smooth curve is *normal* to a surface $f(x, y, z) = c$ at a point of intersection if the curve's velocity vector is a nonzero scalar multiple of ∇f at the point.

Show that the curve

$$\mathbf{r}(t) = \sqrt{t}\mathbf{i} + \sqrt{t}\mathbf{j} - \frac{1}{4}(t + 3)\mathbf{k}$$

is normal to the surface $x^2 + y^2 - z = 3$ when $t = 1$.

- 61.** Consider a closed rectangular box with a square base, as shown in the figure. Assume x is measured with an error of at most 0.5% and y is measured with an error of at most 0.75%, so we have $|dx|/x < 0.005$ and $|dy|/y < 0.0075$.



- Use a differential to estimate the relative error $|dV|/V$ in computing the box's volume V .
- Use a differential to estimate the relative error $|dS|/S$ in computing the box's surface area S .

Hint for b: $\frac{4x^2 + 4xy}{2x^2 + 4xy} \leq \frac{4x^2 + 8xy}{2x^2 + 4xy} = 2$ and $\frac{4xy}{2x^2 + 4xy} \leq \frac{2x^2 + 4xy}{2x^2 + 4xy} = 1$.

13.7 Extreme Values and Saddle Points

HISTORICAL BIOGRAPHY

Siméon-Denis Poisson
(1781–1840)

French mathematician Poisson studied with Lagrange and Laplace at the École polytechnique and did so well that he was made an assistant professor upon his graduation. In 1806, he replaced Fourier as the professor of mathematics. Poisson's early work in mechanics appeared in his first volume of *Traité de mécanique* (1811), where he applied mathematics to applications in physics and mechanics, including elasticity and vibrations.

To know more, visit the companion Website.

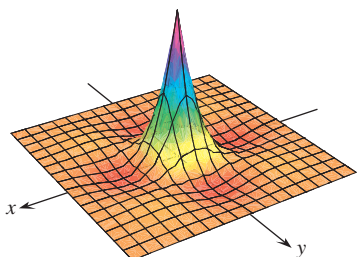


FIGURE 13.42 The function

$$z = (\cos x)(\cos y)e^{-\sqrt{x^2+y^2}}$$

has a maximum value of 1 and a minimum value of about -0.067 on the square region $|x| \leq 3\pi/2$, $|y| \leq 3\pi/2$.

Continuous functions of two variables assume extreme values on closed, bounded domains (see Figures 13.42 and 13.43). We see in this section that we can narrow the search for these extreme values by examining the functions' first partial derivatives. A function of two variables can assume extreme values only at boundary points of the domain or at interior domain points where both first partial derivatives are zero or where one or both of the first partial derivatives fail to exist. However, the vanishing of derivatives at an interior point (a, b) does not always signal the presence of an extreme value. The surface that is the graph of the function might be shaped like a saddle right above (a, b) and cross its tangent plane there.

Local Extreme Values for Functions of Two Variables

To find the local extreme values of a function of a single variable, we look for points where the graph has a horizontal tangent line. At such points, we then look for local maxima, local minima, and points of inflection. For a function $f(x, y)$ of two variables, we look for points where the surface $z = f(x, y)$ has a horizontal tangent plane. At such points, we then look for local maxima, local minima, and saddle points. We begin by defining maxima and minima.

DEFINITIONS Let $f(x, y)$ be defined on a region R containing the point (a, b) . Then

- $f(a, b)$ is a **local maximum** value of f if $f(a, b) \geq f(x, y)$ for all domain points (x, y) in an open disk centered at (a, b) . $f(a, b)$ is an **absolute maximum** value of f on R if $f(a, b) \geq f(x, y)$ for all domain points (x, y) in R .
- $f(a, b)$ is a **local minimum** value of f if $f(a, b) \leq f(x, y)$ for all domain points (x, y) in an open disk centered at (a, b) . $f(a, b)$ is an **absolute minimum** value of f on R if $f(a, b) \leq f(x, y)$ for all domain points (x, y) in R .

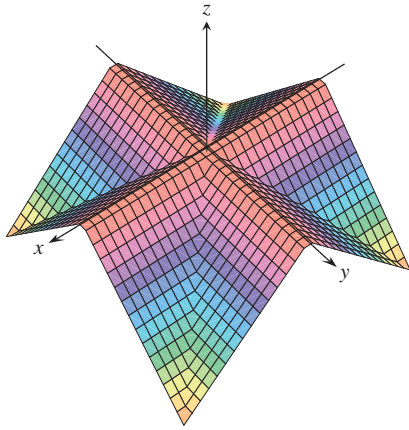


FIGURE 13.43 The “roof surface”

$$z = \frac{1}{2}(|x| - |y|) - |x| - |y|$$

has a maximum value of 0 and a minimum value of $-a$ on the square region $|x| \leq a$, $|y| \leq a$.

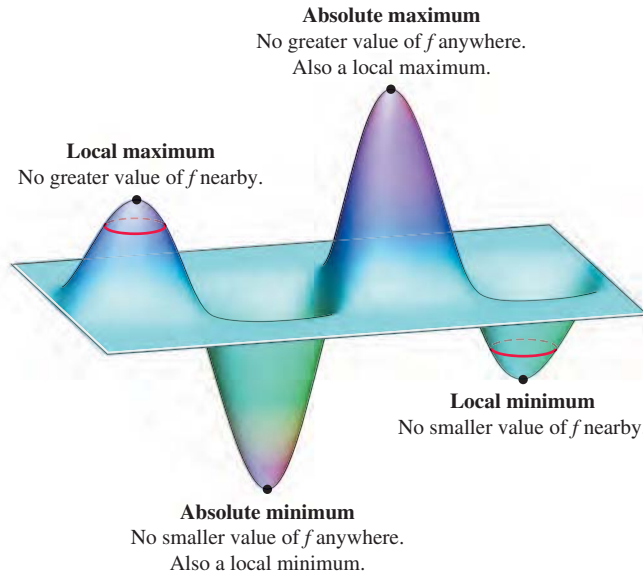


FIGURE 13.44 A local maximum occurs at a mountain peak, and a local minimum occurs at a valley low point.

THEOREM 10—First Derivative Theorem for Local Extreme Values

If $f(x, y)$ has a local maximum or minimum value at an interior point (a, b) of its domain and if the first partial derivatives exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

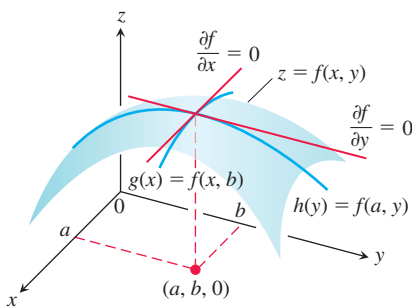


FIGURE 13.45 If a local maximum of f occurs at $x = a$, $y = b$, then the first partial derivatives $f_x(a, b)$ and $f_y(a, b)$ are both zero.

Proof If f has a local extremum at (a, b) , then the function $g(x) = f(x, b)$ has a local extremum at $x = a$ (Figure 13.45). Therefore, $g'(a) = 0$ (Chapter 4, Theorem 2). Now $g'(a) = f_x(a, b)$, so $f_x(a, b) = 0$. A similar argument with the function $h(y) = f(a, y)$ shows that $f_y(a, b) = 0$. ■

If we substitute the values $f_x(a, b) = 0$ and $f_y(a, b) = 0$ into the equation

$$f_x(a, b)(x - a) + f_y(a, b)(y - b) - (z - f(a, b)) = 0$$

for the tangent plane to the surface $z = f(x, y)$ at (a, b) , the equation reduces to

$$0 \cdot (x - a) + 0 \cdot (y - b) - z + f(a, b) = 0,$$

or

$$z = f(a, b).$$

Thus, Theorem 10 says that the surface does indeed have a horizontal tangent plane at a local extremum, provided there is a tangent plane there.

DEFINITION An interior point of the domain of a function $f(x, y)$ where both f_x and f_y are zero or where one or both of f_x and f_y do not exist is a **critical point** of f .

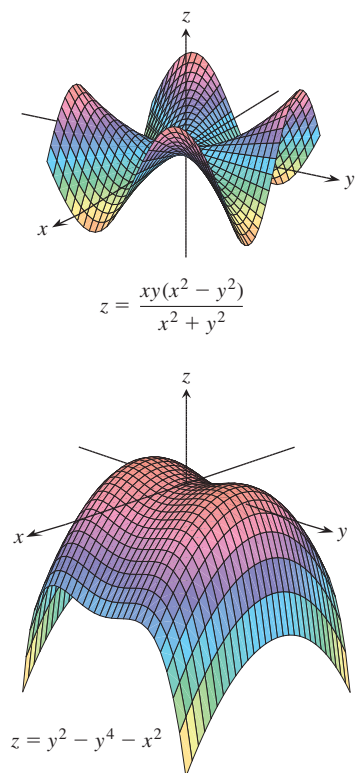


FIGURE 13.46 Saddle points at the origin.

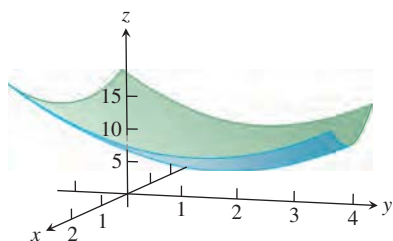


FIGURE 13.47 The graph of the function $f(x, y) = x^2 + y^2 - 4y + 9$ is a paraboloid which has a local minimum value of 5 at the point $(0, 2)$ (Example 1).

Theorem 10 says that the only points where a function $f(x, y)$ can assume extreme values are critical points and boundary points. As with differentiable functions of a single variable, not every critical point gives rise to a local extremum. A differentiable function of a single variable might have a point of inflection. A differentiable function of two variables might have a *saddle point*, with the graph of f crossing the tangent plane defined there.

DEFINITION A differentiable function $f(x, y)$ has a **saddle point** at a critical point (a, b) if in every open disk centered at (a, b) there are domain points (x, y) where $f(x, y) > f(a, b)$ and domain points (x, y) where $f(x, y) < f(a, b)$. The corresponding point $(a, b, f(a, b))$ on the surface $z = f(x, y)$ is called a saddle point of the surface (Figure 13.46).

EXAMPLE 1 Find the local extreme values of $f(x, y) = x^2 + y^2 - 4y + 9$.

Solution The domain of f is the entire plane (so there are no boundary points) and the partial derivatives $f_x = 2x$ and $f_y = 2y - 4$ exist everywhere. Therefore, local extreme values can occur only where

$$f_x = 2x = 0 \quad \text{and} \quad f_y = 2y - 4 = 0.$$

The only possibility is the point $(0, 2)$, where the value of f is 5. Since $f(x, y) = x^2 + (y - 2)^2 + 5$ is never less than 5, we see that the critical point $(0, 2)$ gives a local minimum (Figure 13.47). ■

EXAMPLE 2 Find the local extreme values (if any) of $f(x, y) = y^2 - x^2$.

Solution The domain of f is the entire plane (so there are no boundary points) and the partial derivatives $f_x = -2x$ and $f_y = 2y$ exist everywhere. Therefore, local extrema can occur only at the origin $(0, 0)$, where $f_x = 0$ and $f_y = 0$. The value of f at the origin is 0. However, away from the origin along the positive x -axis, f has the value $f(x, 0) = -x^2 < 0$; along the positive y -axis, f has the value $f(0, y) = y^2 > 0$. Therefore, every open disk in the xy -plane centered at $(0, 0)$ contains points where the function is positive and points where it is negative. The function has a saddle point at the origin and no local extreme values (Figure 13.48a). Figure 13.48b displays the level curves (they are hyperbolas) of f and shows the function decreasing and increasing in an alternating fashion among the groupings of hyperbolas. ■

That $f_x = f_y = 0$ at an interior point (a, b) of R does not guarantee that f has a local extreme value there. If f and its first and second partial derivatives are continuous on R , however, we may be able to learn more from the following theorem.

THEOREM 11—Second Derivative Test for Local Extreme Values

Suppose that $f(x, y)$ and its first and second partial derivatives are continuous throughout a disk centered at (a, b) and that $f_x(a, b) = f_y(a, b) = 0$. Then

- i) f has a **local maximum** at (a, b) if $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) .
- ii) f has a **local minimum** at (a, b) if $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) .
- iii) f has a **saddle point** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a, b) .
- iv) **the test is inconclusive** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a, b) . In this case, we must find some other way to determine the behavior of f at (a, b) .

The expression $f_{xx}f_{yy} - f_{xy}^2$ is called the **discriminant** or **Hessian** of f . It is sometimes easier to remember it in determinant form,

$$f_{xx}f_{yy} - f_{xy}^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}.$$

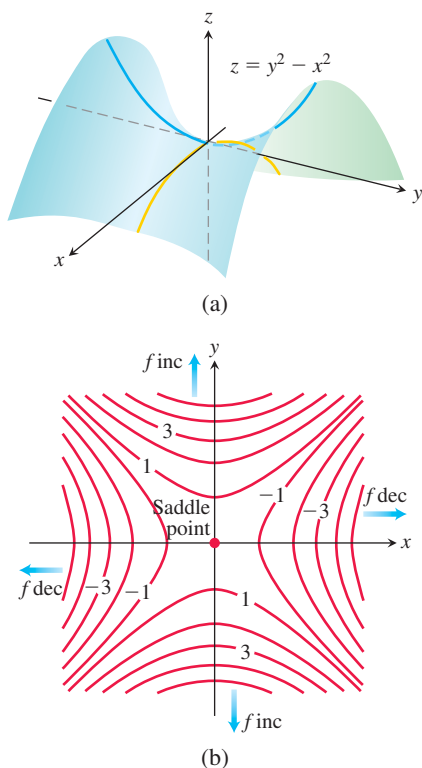


FIGURE 13.48 (a) The origin is a saddle point of the function $f(x, y) = y^2 - x^2$. There are no local extreme values (Example 2). (b) Level curves for the function f in Example 2.

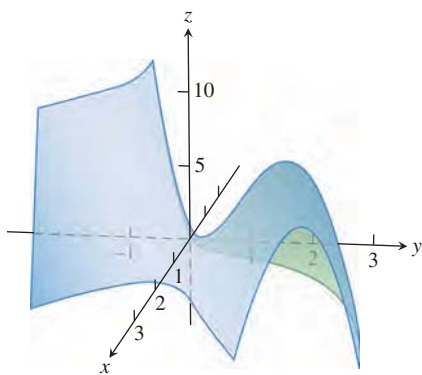


FIGURE 13.49 The surface $z = 3y^2 - 2y^3 - 3x^2 + 6xy$ has a saddle point at the origin and a local maximum at the point $(2, 2)$ (Example 4).

The discriminant is the determinant of the **Hessian matrix** of f ,

$$Hf(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Note that by Theorem 2, we have $f_{xy}(a, b) = f_{yx}(a, b)$ at any point (a, b) satisfying the assumptions of Theorem 11.

Theorem 11 says that if the discriminant is positive at the point (a, b) , then the surface curves the same way in all directions: downward if $f_{xx} < 0$, giving rise to a local maximum, and upward if $f_{xx} > 0$, giving a local minimum. On the other hand, if the discriminant is negative at (a, b) , then the surface curves up in some directions and down in others, so we have a saddle point.

EXAMPLE 3 Find the local extreme values of the function

$$f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4.$$

Solution The function is defined and differentiable for all x and y , and its domain has no boundary points. The function therefore has extreme values only at the points where f_x and f_y are simultaneously zero. This leads to

$$f_x = y - 2x - 2 = 0, \quad f_y = x - 2y - 2 = 0,$$

or

$$x = y = -2.$$

Therefore, the point $(-2, -2)$ is the only point where f may take on an extreme value. To see whether it does so, we calculate

$$f_{xx} = -2, \quad f_{yy} = -2, \quad f_{xy} = 1.$$

The discriminant of f at $(a, b) = (-2, -2)$ is

$$f_{xx}f_{yy} - f_{xy}^2 = (-2)(-2) - (1)^2 = 4 - 1 = 3.$$

The combination

$$f_{xx} < 0 \quad \text{and} \quad f_{xx}f_{yy} - f_{xy}^2 > 0$$

tells us that f has a local maximum at $(-2, -2)$. The value of f at this point is $f(-2, -2) = 8$. ■

EXAMPLE 4 Find the local extreme values of $f(x, y) = 3y^2 - 2y^3 - 3x^2 + 6xy$.

Solution Since f is differentiable everywhere, it can assume extreme values only where

$$f_x = 6y - 6x = 0 \quad \text{and} \quad f_y = 6y - 6y^2 + 6x = 0.$$

From the first of these equations we find $x = y$, and substitution for y into the second equation then gives

$$6x - 6x^2 + 6x = 0 \quad \text{or} \quad 6x(2 - x) = 0.$$

The two critical points are therefore $(0, 0)$ and $(2, 2)$.

To classify the critical points, we calculate the second derivatives:

$$f_{xx} = -6, \quad f_{yy} = 6 - 12y, \quad f_{xy} = 6.$$

The discriminant is given by

$$f_{xx}f_{yy} - f_{xy}^2 = (-36 + 72y) - 36 = 72(y - 1).$$

At the critical point $(0, 0)$ we see that the value of the discriminant is the negative number -72 , so the function has a saddle point at the origin. At the critical point $(2, 2)$ we see that the discriminant has the positive value 72 . Combining this result with the negative value of the second partial $f_{xx} = -6$, Theorem 11 says that the critical point $(2, 2)$ gives a local maximum value of $f(2, 2) = 12 - 16 - 12 + 24 = 8$. A graph of the surface is shown in Figure 13.49. ■

EXAMPLE 5 Find the critical points of the function $f(x, y) = 10xye^{-(x^2+y^2)}$ and use the Second Derivative Test to classify each point as one where a saddle, local minimum, or local maximum occurs.

Solution First we find the partial derivatives f_x and f_y and set them simultaneously to zero in seeking the critical points:

$$f_x = 10ye^{-(x^2+y^2)} - 20x^2ye^{-(x^2+y^2)} = 10y(1 - 2x^2)e^{-(x^2+y^2)} = 0 \Rightarrow y = 0 \text{ or } 1 - 2x^2 = 0,$$

$$f_y = 10xe^{-(x^2+y^2)} - 20xy^2e^{-(x^2+y^2)} = 10x(1 - 2y^2)e^{-(x^2+y^2)} = 0 \Rightarrow x = 0 \text{ or } 1 - 2y^2 = 0.$$

Since both partial derivatives are continuous everywhere, the only critical points are

$$(0, 0), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \text{ and } \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right).$$

Next we calculate the second partial derivatives in order to evaluate the discriminant at each critical point:

$$f_{xx} = -20xy(1 - 2x^2)e^{-(x^2+y^2)} - 40xye^{-(x^2+y^2)} = -20xy(3 - 2x^2)e^{-(x^2+y^2)},$$

$$f_{xy} = f_{yx} = 10(1 - 2x^2)e^{-(x^2+y^2)} - 20y^2(1 - 2x^2)e^{-(x^2+y^2)} = 10(1 - 2x^2)(1 - 2y^2)e^{-(x^2+y^2)},$$

$$f_{yy} = -20xy(1 - 2y^2)e^{-(x^2+y^2)} - 40xye^{-(x^2+y^2)} = -20xy(3 - 2y^2)e^{-(x^2+y^2)}.$$

The following table summarizes the values needed by the Second Derivative Test.

Critical Point	f_{xx}	f_{xy}	f_{yy}	Discriminant D
$(0, 0)$	0	10	0	-100
$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$-\frac{20}{e}$	0	$-\frac{20}{e}$	$\frac{400}{e^2}$
$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$	$\frac{20}{e}$	0	$\frac{20}{e}$	$\frac{400}{e^2}$
$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$	$\frac{20}{e}$	0	$\frac{20}{e}$	$\frac{400}{e^2}$
$\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$	$-\frac{20}{e}$	0	$-\frac{20}{e}$	$\frac{400}{e^2}$

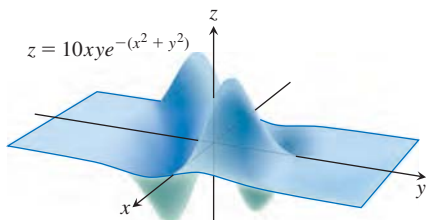


FIGURE 13.50 A graph of the function in Example 5.

From the table we find that $D < 0$ at the critical point $(0, 0)$, giving a saddle; $D > 0$ and $f_{xx} < 0$ at the critical points $(1/\sqrt{2}, 1/\sqrt{2})$ and $(-1/\sqrt{2}, -1/\sqrt{2})$, giving local maximum values there; and $D > 0$ and $f_{xx} > 0$ at the critical points $(-1/\sqrt{2}, 1/\sqrt{2})$ and $(1/\sqrt{2}, -1/\sqrt{2})$, each giving local minimum values. A graph of the surface is shown in Figure 13.50. ■

Absolute Maxima and Minima on Closed Bounded Regions

We organize the search for the absolute extrema of a continuous function $f(x, y)$ on a closed and bounded region R into three steps.

1. List the interior points of R where f may have local maxima and minima and evaluate f at these points. These are the critical points of f .
2. List the boundary points of R where f has local maxima and minima and evaluate f at these points. We show how to do this in the next example.
3. Look through the lists for the maximum and minimum values of f . These will be the absolute maximum and minimum values of f on R .

EXAMPLE 6 Find the absolute maximum and minimum values of

$$f(x, y) = 2 + 2x + 4y - x^2 - y^2$$

on the triangular region in the first quadrant bounded by the lines $x = 0$, $y = 0$, and $y = 9 - x$.

Solution Since f is differentiable, the only places where f can assume these values are points inside the triangle where $f_x = f_y = 0$ and points on the boundary (Figure 13.51a).

(a) Interior points. For these we have

$$f_x = 2 - 2x = 0, \quad f_y = 4 - 2y = 0,$$

yielding the single point $(x, y) = (1, 2)$. The value of f there is

$$f(1, 2) = 7.$$

(b) Boundary points. We take the triangle one side at a time:

i) On the segment OA we always have $y = 0$. Therefore, we can regard $f(x, y)$ as being solely a function of x on this segment. That is, on this segment we want to consider the function

$$g(x) = f(x, 0) = 2 + 2x - x^2$$

for $0 \leq x \leq 9$. Its extreme values (as we know from Chapter 4) may occur at the endpoints

$$x = 0 \quad \text{where} \quad g(0) = f(0, 0) = 2$$

$$x = 9 \quad \text{where} \quad g(9) = f(9, 0) = 2 + 18 - 81 = -61$$

or at the interior points where $g'(x) = 2 - 2x = 0$. The only interior point where $g'(x) = 0$ is $x = 1$, where

$$g(1) = f(1, 0) = 3.$$

ii) On the segment OB we always have $x = 0$. Therefore, on this segment we can regard $f(x, y)$ as being solely a function of y , and so we consider the function

$$h(y) = f(0, y) = 2 + 4y - y^2$$

on the closed interval $[0, 9]$. Its extreme values can occur at the endpoints or at interior points where $h'(y) = 0$. Since $h'(y) = 4 - 2y$, the only interior point where $h'(y) = 0$ occurs at $(0, 2)$, with $h(2) = 6$. So the candidates for this segment are

$$h(0) = f(0, 0) = 2, \quad h(9) = f(0, 9) = -43, \quad \text{and} \quad h(2) = f(0, 2) = 6.$$

iii) We have already accounted for the values of f at the endpoints of AB , so we need only look at the interior points of the line segment AB . On this segment we have $y = 9 - x$, so we consider the function

$$k(x) = f(x, 9 - x) = 2 + 2x + 4(9 - x) - x^2 - (9 - x)^2 = -43 + 16x - 2x^2.$$

Setting $k'(x) = 16 - 4x = 0$ gives

$$x = 4.$$

At this value of x ,

$$y = 9 - 4 = 5 \quad \text{and} \quad k(4) = f(4, 5) = -11.$$

Summary We list all the function value candidates: 7, 2, -61, 3, -43, 6, -11. The maximum is 7, which f assumes at $(1, 2)$. The minimum is -61, which f assumes at $(9, 0)$. See Figure 13.51b. ■

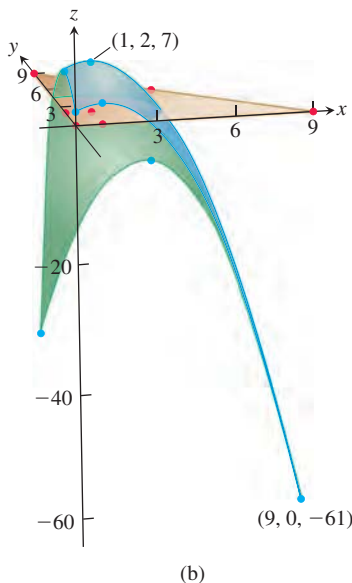
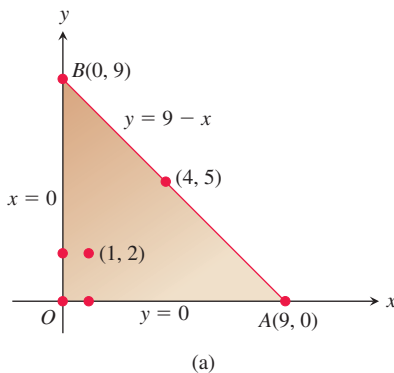


FIGURE 13.51 (a) This triangular region is the domain of the function in Example 6. (b) The graph of the function in Example 6. The blue points are the candidates for maxima or minima.

Solving extreme value problems with algebraic constraints on the variables usually requires the method of Lagrange multipliers, which is introduced in the next section. But sometimes we can solve such problems directly, as in the next example.

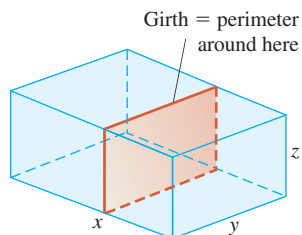


FIGURE 13.52 The box in Example 7.

EXAMPLE 7 A delivery company accepts only rectangular boxes the sum of whose length and girth (perimeter of a cross-section) does not exceed 270 cm. Find the dimensions of an acceptable box of largest volume.

Solution Let x , y , and z represent the length, width, and height of the rectangular box, respectively. Then the girth is $2y + 2z$. We want to maximize the volume $V = xyz$ of the box (Figure 13.52) satisfying $x + 2y + 2z = 270$ (the largest box accepted by the delivery company). Thus, we can write the volume of the box as a function of two variables:

$$\begin{aligned} V(y, z) &= (270 - 2y - 2z)yz & V &= xyz \text{ and} \\ &= 270yz - 2y^2z - 2yz^2. & x &= 270 - 2y - 2z \end{aligned}$$

Setting the first partial derivatives equal to zero,

$$V_y(y, z) = 270z - 4yz - 2z^2 = (270 - 4y - 2z)z = 0$$

$$V_z(y, z) = 270y - 2y^2 - 4yz = (270 - 2y - 4z)y = 0,$$

gives the critical points $(0, 0)$, $(0, 135)$, $(135, 0)$, and $(45, 45)$. The volume is zero at $(0, 0)$, $(0, 135)$, and $(135, 0)$, which are not maximum values. At the point $(45, 45)$, we apply the Second Derivative Test (Theorem 11):

$$V_{yy} = -4z, \quad V_{zz} = -4y, \quad V_{yz} = 270 - 4y - 4z.$$

Then

$$V_{yy}V_{zz} - V_{yz}^2 = 16yz - 4(135 - 2y - 2z)^2.$$

Thus,

$$V_{yy}(45, 45) = -4(45) < 0$$

and

$$(V_{yy}V_{zz} - V_{yz}^2) \Big|_{(45, 45)} = 16(45)(45) - 4(-45)^2 > 0,$$

so $(45, 45)$ gives a maximum volume. The dimensions of the package are $x = 270 - 2(45) - 2(45) = 90$ cm, $y = 45$ cm, and $z = 45$ cm. The maximum volume is $V = (90)(45)(45) = 182,250$ cm³, or 182.25 liters. ■

Despite the power of Theorem 11, we urge you to remember its limitations. It does not apply to boundary points of a function's domain, where it is possible for a function to have extreme values along with nonzero derivatives. Also, it does not apply to points where either f_x or f_y fails to exist.

Summary of Max-Min Tests

The extreme values of $f(x, y)$ can occur only at

- i) **boundary points** of the domain of f
- ii) **critical points** (interior points where $f_x = f_y = 0$ or points where f_x or f_y fails to exist)

If the first- and second-order partial derivatives of f are continuous throughout a disk centered at a point (a, b) and if $f_x(a, b) = f_y(a, b) = 0$, then the nature of $f(a, b)$ can be tested with the **Second Derivative Test**:

- i) $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at $(a, b) \Rightarrow$ **local maximum**
- ii) $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at $(a, b) \Rightarrow$ **local minimum**
- iii) $f_{xx}f_{yy} - f_{xy}^2 < 0$ at $(a, b) \Rightarrow$ **saddle point**
- iv) $f_{xx}f_{yy} - f_{xy}^2 = 0$ at $(a, b) \Rightarrow$ **test is inconclusive**

Finding maximum and minimum values for functions of more than two variables is an important problem with many important applications, from machine learning to making economic predictions. The problem becomes much harder as the number of variables increases. The process of finding extrema for functions with high-dimensional domains is discussed in Appendices B.2 and B.3.

EXERCISES 13.7

Finding Local Extrema

Find all the local maxima, local minima, and saddle points of the functions in Exercises 1–30.

- $f(x, y) = x^2 + xy + y^2 + 3x - 3y + 4$
- $f(x, y) = 2xy - 5x^2 - 2y^2 + 4x + 4y - 4$
- $f(x, y) = x^2 + xy + 3x + 2y + 5$
- $f(x, y) = 5xy - 7x^2 + 3x - 6y + 2$
- $f(x, y) = 2xy - x^2 - 2y^2 + 3x + 4$
- $f(x, y) = x^2 - 4xy + y^2 + 6y + 2$
- $f(x, y) = 2x^2 + 3xy + 4y^2 - 5x + 2y$
- $f(x, y) = x^2 - 2xy + 2y^2 - 2x + 2y + 1$
- $f(x, y) = x^2 - y^2 - 2x + 4y + 6$
- $f(x, y) = x^2 + 2xy$
- $f(x, y) = \sqrt{56x^2 - 8y^2 - 16x - 31} + 1 - 8x$
- $f(x, y) = 1 - \sqrt[3]{x^2 + y^2}$
- $f(x, y) = x^3 - y^3 - 2xy + 6$
- $f(x, y) = x^3 + 3xy + y^3$
- $f(x, y) = 6x^2 - 2x^3 + 3y^2 + 6xy$
- $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2 - 8$
- $f(x, y) = x^3 + 3xy^2 - 15x + y^3 - 15y$
- $f(x, y) = 2x^3 + 2y^3 - 9x^2 + 3y^2 - 12y$
- $f(x, y) = 4xy - x^4 - y^4$
- $f(x, y) = x^4 + y^4 + 4xy$
- $f(x, y) = \frac{1}{x^2 + y^2 - 1}$
- $f(x, y) = \frac{1}{x} + xy + \frac{1}{y}$
- $f(x, y) = y \sin x$
- $f(x, y) = e^{2x} \cos y$
- $f(x, y) = e^{x^2 + y^2 - 4x}$
- $f(x, y) = e^y - ye^x$
- $f(x, y) = e^{-y}(x^2 + y^2)$
- $f(x, y) = e^x(x^2 - y^2)$
- $f(x, y) = 2 \ln x + \ln y - 4x - y$
- $f(x, y) = \ln(x + y) + x^2 - y$

Finding Absolute Extrema

In Exercises 31–38, find the absolute maxima and minima of the functions on the given domains.

- $f(x, y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0$, $y = 2$, $y = 2x$ in the first quadrant
- $D(x, y) = x^2 - xy + y^2 + 1$ on the closed triangular plate in the first quadrant bounded by the lines $x = 0$, $y = 4$, $y = x$
- $f(x, y) = x^2 + y^2$ on the closed triangular plate bounded by the lines $x = 0$, $y = 0$, $y + 2x = 2$ in the first quadrant
- $T(x, y) = x^2 + xy + y^2 - 6x$ on the rectangular plate $0 \leq x \leq 5$, $-3 \leq y \leq 3$

- $T(x, y) = x^2 + xy + y^2 - 6x + 2$ on the rectangular plate $0 \leq x \leq 5$, $-3 \leq y \leq 0$
- $f(x, y) = 48xy - 32x^3 - 24y^2$ on the rectangular plate $0 \leq x \leq 1$, $0 \leq y \leq 1$
- $f(x, y) = (4x - x^2)\cos y$ on the rectangular plate $1 \leq x \leq 3$, $-\pi/4 \leq y \leq \pi/4$
- $f(x, y) = 4x - 8xy + 2y + 1$ on the triangular plate bounded by the lines $x = 0$, $y = 0$, $x + y = 1$ in the first quadrant
- Find two numbers a and b with $a \leq b$ such that

$$\int_a^b (6 - x - x^2) dx$$

has its largest value.

- Find two numbers a and b with $a \leq b$ such that

$$\int_a^b (24 - 2x - x^2)^{1/3} dx$$

has its largest value.

- Temperatures** A flat circular plate has the shape of the region $x^2 + y^2 \leq 1$. The plate, including the boundary where $x^2 + y^2 = 1$, is heated so that the temperature at the point (x, y) is

$$T(x, y) = x^2 + 2y^2 - x.$$

Find the temperatures at the hottest and coldest points on the plate.

- Find the critical point of

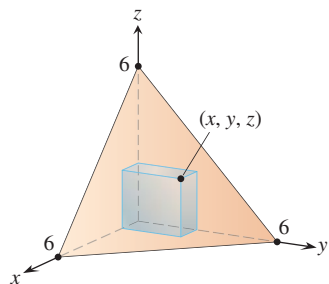
$$f(x, y) = xy + 2x - \ln x^2 y$$

in the open first quadrant ($x > 0$, $y > 0$) and show that f takes on a minimum there.

Theory and Examples

- Find the maxima, minima, and saddle points of $f(x, y)$, if any, given that
 - $f_x = 2x - 4y$ and $f_y = 2y - 4x$
 - $f_x = 2x - 2$ and $f_y = 2y - 4$
 - $f_x = 9x^2 - 9$ and $f_y = 2y + 4$
 Describe your reasoning in each case.
- The discriminant $f_{xx}f_{yy} - f_{xy}^2$ is zero at the origin for each of the following functions, so the Second Derivative Test fails there. Determine whether the function has a maximum, a minimum, or neither at the origin by imagining what the surface $z = f(x, y)$ looks like. Describe your reasoning in each case.
 - $f(x, y) = x^2 y^2$
 - $f(x, y) = 1 - x^2 y^2$
 - $f(x, y) = xy^2$
 - $f(x, y) = x^3 y^2$
 - $f(x, y) = x^3 y^3$
 - $f(x, y) = x^4 y^4$
- Show that $(0, 0)$ is a critical point of $f(x, y) = x^2 + kxy + y^2$ no matter what value the constant k has. (*Hint:* Consider two cases: $k = 0$ and $k \neq 0$.)

46. For what values of the constant k does the Second Derivative Test guarantee that $f(x, y) = x^2 + kxy + y^2$ will have a saddle point at $(0, 0)$? A local minimum at $(0, 0)$? For what values of k is the Second Derivative Test inconclusive? Give reasons for your answers.
47. If $f_x(a, b) = f_y(a, b) = 0$, must f have a local maximum or minimum value at (a, b) ? Give reasons for your answer.
48. Can you conclude anything about $f(a, b)$ if f and its first and second partial derivatives are continuous throughout a disk centered at the critical point (a, b) and $f_{xx}(a, b)$ and $f_{yy}(a, b)$ differ in sign? Give reasons for your answer.
49. Among all the points on the graph of $z = 10 - x^2 - y^2$ that lie above the plane $x + 2y + 3z = 0$, find the point farthest from the plane.
50. Find the point on the graph of $z = x^2 + y^2 + 10$ nearest the plane $x + 2y - z = 0$.
51. Find the point on the plane $3x + 2y + z = 6$ that is nearest the origin.
52. Find the minimum distance from the point $(2, -1, 1)$ to the plane $x + y - z = 2$.
53. Find three numbers whose sum is 9 and whose sum of squares is a minimum.
54. Find three positive numbers whose sum is 3 and whose product is a maximum.
55. Find the maximum value of $s = xy + yz + xz$ where $x + y + z = 6$.
56. Find the minimum distance from the cone $z = \sqrt{x^2 + y^2}$ to the point $(-6, 4, 0)$.
57. Find the dimensions of the rectangular box of maximum volume that can be inscribed inside the sphere $x^2 + y^2 + z^2 = 4$.
58. Among all closed rectangular boxes of volume 27 cm^3 , what is the smallest surface area?
59. You are to construct an open rectangular box from 12 m^2 of material. What dimensions will result in a box of maximum volume?
60. Consider the function $f(x, y) = x^2 + y^2 + 2xy - x - y + 1$ over the square $0 \leq x \leq 1$ and $0 \leq y \leq 1$.
- Show that f has an absolute minimum along the line segment $2x + 2y = 1$ in this square. What is the absolute minimum value?
 - Find the absolute maximum value of f over the square.
61. Find the point on the graph of $y^2 - xz^2 = 4$ nearest the origin.
62. A rectangular box is inscribed in the region in the first octant bounded above by the plane with x -intercept 6, y -intercept 6, and z -intercept 6.



- Find an equation for the plane.
- Find the dimensions of the box of maximum volume.

Extreme Values on Parametrized Curves To find the extreme values of a function $f(x, y)$ on a curve $x = x(t)$, $y = y(t)$, we treat f as a function of the single variable t and use the Chain Rule to find where df/dt is zero. As in any other single-variable case, the extreme values of f are then found among the values at

- The critical points (points where df/dt is zero or fails to exist), and
- The endpoints of the parameter domain.

In Exercises 63–66, find the absolute maximum and minimum values of the following functions on the given curves.

63. Functions:

- $f(x, y) = x + y$
- $g(x, y) = xy$
- $h(x, y) = 2x^2 + y^2$

Curves:

- The semicircle $x^2 + y^2 = 4$, $y \geq 0$
- The quarter circle $x^2 + y^2 = 4$, $x \geq 0$, $y \geq 0$

Use the parametric equations $x = 2 \cos t$, $y = 2 \sin t$.

64. Functions:

- $f(x, y) = 2x + 3y$
- $g(x, y) = xy$
- $h(x, y) = x^2 + 3y^2$

Curves:

- The semiellipse $(x^2/9) + (y^2/4) = 1$, $y \geq 0$
- The quarter ellipse $(x^2/9) + (y^2/4) = 1$, $x \geq 0$, $y \geq 0$

Use the parametric equations $x = 3 \cos t$, $y = 2 \sin t$.

65. Function: $f(x, y) = xy$

Curves:

- The line $x = 2t$, $y = t + 1$
- The line segment $x = 2t$, $y = t + 1$, $-1 \leq t \leq 0$
- The line segment $x = 2t$, $y = t + 1$, $0 \leq t \leq 1$

66. Functions:

- $f(x, y) = x^2 + y^2$
- $g(x, y) = 1/(x^2 + y^2)$

Curves:

- The line $x = t$, $y = 2 - 2t$
- The line segment $x = t$, $y = 2 - 2t$, $0 \leq t \leq 1$

67. Least squares and regression lines When we try to fit a line $y = mx + b$ to a set of numerical data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we usually choose the line that minimizes the sum of the squares of the vertical distances from the points to the line. In theory, this means finding the values of m and b that minimize the value of the function

$$w = (mx_1 + b - y_1)^2 + \cdots + (mx_n + b - y_n)^2. \quad (1)$$

(See the accompanying figure.) Show that the values of m and b that do this are

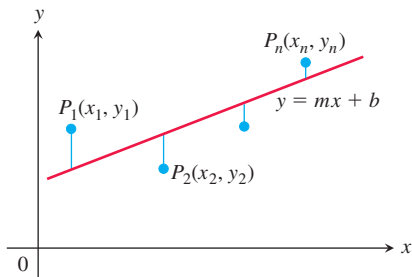
$$m = \frac{\left(\sum x_k\right)\left(\sum y_k\right) - n \sum x_k y_k}{\left(\sum x_k\right)^2 - n \sum x_k^2}, \quad (2)$$

$$b = \frac{1}{n} \left(\sum y_k - m \sum x_k \right), \quad (3)$$

with all sums running from $k = 1$ to $k = n$. Many scientific calculators have these formulas built in, enabling you to find m and b with only a few keystrokes after you have entered the data.

The line $y = mx + b$ determined by these values of m and b is called the **least squares line**, **regression line**, or **trend line** for the data under study. Finding a least squares line lets you

1. summarize data with a simple expression,
2. predict values of y for other, experimentally untried values of x ,
3. handle data analytically.



In Exercises 68–70, use Equations (2) and (3) to find the least squares line for each set of data points. Then use the linear equation you obtain to predict the value of y that would correspond to $x = 4$.

68. $(-2, 0), (0, 2), (2, 3)$

69. $(-1, 2), (0, 1), (3, -4)$

70. $(0, 0), (1, 2), (2, 3)$

COMPUTER EXPLORATIONS

In Exercises 71–76, you will explore functions to identify their local extrema. Use a CAS to perform the following steps:

- a. Plot the function over the given rectangle.
- b. Plot some level curves in the rectangle.
- c. Calculate the function's first partial derivatives and use the CAS equation solver to find the critical points. How are the critical points related to the level curves plotted in part (b)? Which critical points, if any, appear to give a saddle point? Give reasons for your answer.
- d. Calculate the function's second partial derivatives and find the discriminant $f_{xx}f_{yy} - f_{xy}^2$.
- e. Using the max-min tests, classify the critical points found in part (c). Are your findings consistent with your discussion in part (c)?

71. $f(x, y) = x^2 + y^3 - 3xy, \quad -5 \leq x \leq 5, \quad -5 \leq y \leq 5$

72. $f(x, y) = x^3 - 3xy^2 + y^2, \quad -2 \leq x \leq 2, \quad -2 \leq y \leq 2$

73. $f(x, y) = x^4 + y^2 - 8x^2 - 6y + 16, \quad -3 \leq x \leq 3, \quad -6 \leq y \leq 6$

74. $f(x, y) = 2x^4 + y^4 - 2x^2 - 2y^2 + 3, \quad -3/2 \leq x \leq 3/2, \quad -3/2 \leq y \leq 3/2$

75. $f(x, y) = 5x^6 + 18x^5 - 30x^4 + 30xy^2 - 120x^3, \quad -4 \leq x \leq 3, \quad -2 \leq y \leq 2$

76. $f(x, y) = \begin{cases} x^5 \ln(x^2 + y^2), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}, \quad -2 \leq x \leq 2, \quad -2 \leq y \leq 2$

13.8 Lagrange Multipliers

HISTORICAL BIOGRAPHY

Joseph Louis Lagrange
(1736–1813)

Lagrange was born in Turin, Italy. He enjoyed studying mathematics, despite his father's wish that he study law. Lagrange's mathematical contributions began as early as 1754 with the discovery of the calculus of variations and continued with applications to mechanics in 1756.

To know more, visit the companion Website.

Sometimes we need to find the extreme values of a function whose domain is constrained to lie within some particular subset of the plane—for example, a disk, a closed triangular region, or along a curve. We saw an instance of this situation in Example 6 of the previous section. Here we explore a powerful method for finding extreme values of constrained functions: the method of *Lagrange multipliers*.

Constrained Maxima and Minima

To gain some insight, we first consider a problem where a constrained minimum can be found by eliminating a variable.

EXAMPLE 1 Find the point $p(x, y, z)$ on the plane $2x + y - z - 5 = 0$ that is closest to the origin.

Solution The problem asks us to find the minimum value of the function

$$|\overline{OP}| = \sqrt{(x-0)^2 + (y-0)^2 + (z-0)^2} = \sqrt{x^2 + y^2 + z^2}$$

subject to the constraint that

$$2x + y - z - 5 = 0.$$

Since $|\overline{OP}|$ has a minimum value wherever the function

$$f(x, y, z) = x^2 + y^2 + z^2$$

has a minimum value, we may solve the problem by finding the minimum value of $f(x, y, z)$ subject to the constraint $2x + y - z - 5 = 0$ (thus avoiding square roots). If we regard x and y as the independent variables in this equation and write z as

$$z = 2x + y - 5,$$

our problem reduces to finding the points (x, y) at which the function

$$h(x, y) = f(x, y, 2x + y - 5) = x^2 + y^2 + (2x + y - 5)^2$$

has its minimum value or values. Since the domain of h is the entire xy -plane, the First Derivative Theorem of Section 13.7 tells us that any minima that h might have must occur at points where

$$h_x = 2x + 2(2x + y - 5)(2) = 0, \quad h_y = 2y + 2(2x + y - 5) = 0.$$

This leads to

$$10x + 4y = 20, \quad 4x + 4y = 10,$$

which has the solution

$$x = \frac{5}{3}, \quad y = \frac{5}{6}.$$

We may apply a geometric argument together with the Second Derivative Test to show that these values minimize h . The z -coordinate of the corresponding point on the plane $z = 2x + y - 5$ is

$$z = 2\left(\frac{5}{3}\right) + \frac{5}{6} - 5 = -\frac{5}{6}.$$

Therefore, the point we seek is

$$\text{Closest point: } P\left(\frac{5}{3}, \frac{5}{6}, -\frac{5}{6}\right).$$

The distance from P to the origin is $5/\sqrt{6} \approx 2.04$. ■

Attempts to solve a constrained maximum or minimum problem by substitution, as we might call the method of Example 1, do not always go smoothly.

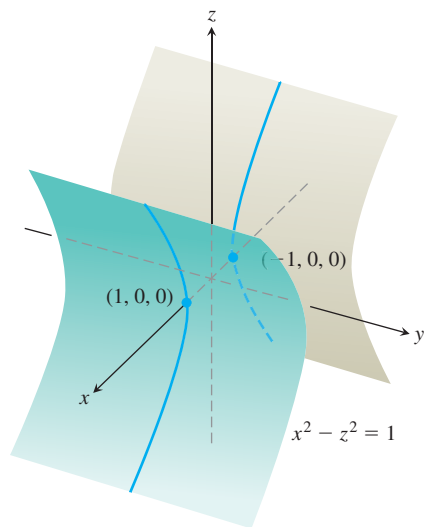


FIGURE 13.53 The hyperbolic cylinder $x^2 - z^2 - 1 = 0$ in Example 2.

EXAMPLE 2 Find the points on the hyperbolic cylinder $x^2 - z^2 - 1 = 0$ that are closest to the origin.

Solution 1 The cylinder is shown in Figure 13.53. We seek the points on the cylinder closest to the origin. These are the points whose coordinates minimize the value of the function

$$f(x, y, z) = x^2 + y^2 + z^2 \quad \text{Square of the distance}$$

subject to the constraint that $x^2 - z^2 - 1 = 0$. If we regard x and y as independent variables in the constraint equation, then

$$z^2 = x^2 - 1,$$

and the values of $f(x, y, z) = x^2 + y^2 + z^2$ on the cylinder are given by the function

$$h(x, y) = x^2 + y^2 + (x^2 - 1) = 2x^2 + y^2 - 1.$$

To find the points on the cylinder whose coordinates minimize f , we look for the points in the xy -plane whose coordinates minimize h . The only extreme value of h occurs where

$$h_x = 4x = 0 \quad \text{and} \quad h_y = 2y = 0,$$

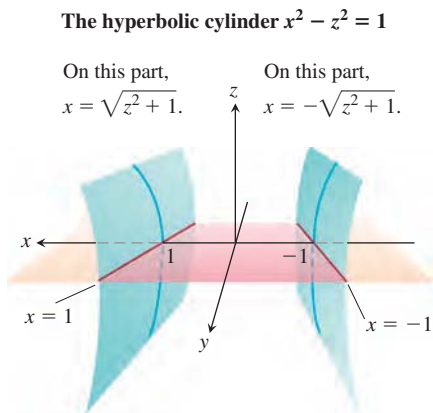


FIGURE 13.54 The region in the xy -plane from which the first two coordinates of the points (x, y, z) on the hyperbolic cylinder $x^2 - z^2 = 1$ are selected excludes the band $-1 < x < 1$ in the xy -plane (Example 2).

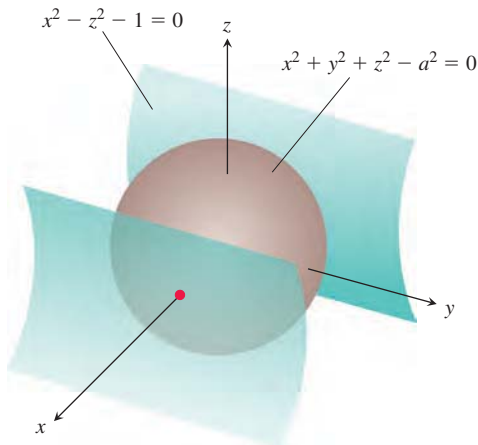


FIGURE 13.55 A sphere expanding like a soap bubble centered at the origin until it just touches the hyperbolic cylinder $x^2 - z^2 - 1 = 0$ (Example 2).

λ is the Greek letter lambda.

that is, at the point $(0, 0)$. But there are no points on the cylinder where both x and y are zero. What went wrong?

What happened is that the First Derivative Theorem found (as it should have) the point *in the domain of h* where h has a minimum value. We, on the other hand, want the points *on the cylinder* where h has a minimum value. Although the domain of h is the entire xy -plane, the domain from which we can select the first two coordinates of the points (x, y, z) on the cylinder is restricted to the projection, or “shadow” of the cylinder on the xy -plane; it does not include the band between the lines $x = -1$ and $x = 1$ (Figure 13.54).

We can avoid this problem if we treat y and z as independent variables (instead of x and y) and express x in terms of y and z as

$$x^2 = z^2 + 1.$$

With this substitution, $f(x, y, z) = x^2 + y^2 + z^2$ becomes

$$k(y, z) = (z^2 + 1) + y^2 + z^2 = 1 + y^2 + 2z^2$$

and we look for the points where k takes on its smallest value. The domain of k in the yz -plane now matches the domain from which we select the y - and z -coordinates of the points (x, y, z) on the cylinder. Hence, the points that minimize k in the plane will have corresponding points on the cylinder. The smallest values of k occur where

$$k_y = 2y = 0 \quad \text{and} \quad k_z = 4z = 0,$$

or where $y = z = 0$. This leads to

$$x^2 = z^2 + 1 = 1, \quad x = \pm 1.$$

The corresponding points on the cylinder are $(\pm 1, 0, 0)$. We can see from the inequality

$$k(y, z) = 1 + y^2 + 2z^2 \geq 1$$

that the points $(\pm 1, 0, 0)$ give a minimum value for k . We can also see that the minimum distance from the origin to a point on the cylinder is 1 unit.

Solution 2 Another way to find the points on the cylinder closest to the origin is to imagine a small sphere centered at the origin expanding like a soap bubble until it just touches the cylinder (Figure 13.55). At each point of contact, the cylinder and sphere have the same tangent plane and normal line. Therefore, if the sphere and cylinder are represented as the level surfaces obtained by setting

$$f(x, y, z) = x^2 + y^2 + z^2 - a^2 \quad \text{and} \quad g(x, y, z) = x^2 - z^2 - 1$$

equal to 0, then the gradients ∇f and ∇g will be parallel where the surfaces touch. At any point of contact, we should therefore be able to find a scalar λ (“lambda”) such that

$$\nabla f = \lambda \nabla g,$$

or

$$2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} = \lambda(2x\mathbf{i} - 2z\mathbf{k}).$$

Thus, the coordinates x , y , and z of any point of tangency will have to satisfy the three scalar equations

$$2x = 2\lambda x, \quad 2y = 0, \quad 2z = -2\lambda z.$$

For what values of λ will a point (x, y, z) whose coordinates satisfy these scalar equations also lie on the surface $x^2 - z^2 - 1 = 0$? To answer this question, we use our knowledge that no point on the surface has a zero x -coordinate to conclude that $x \neq 0$. Hence, $2x = 2\lambda x$ only if

$$2 = 2\lambda, \quad \text{or} \quad \lambda = 1.$$

For $\lambda = 1$, the equation $2z = -2\lambda z$ becomes $2z = -2z$. If this equation is to be satisfied as well, z must be zero. Since $y = 0$ also (from the equation $2y = 0$), we conclude that the points we seek all have coordinates of the form

$$(x, 0, 0).$$

What points on the surface $x^2 - z^2 = 1$ have coordinates of this form? The answer is the points $(x, 0, 0)$ for which

$$x^2 - (0)^2 = 1, \quad x^2 = 1, \quad \text{or} \quad x = \pm 1.$$

The points on the cylinder closest to the origin are the points $(\pm 1, 0, 0)$. ■

The Method of Lagrange Multipliers

In Solution 2 of Example 2, we used the **method of Lagrange multipliers**. The method says that the local extreme values of a function $f(x, y, z)$ whose variables are subject to a constraint $g(x, y, z) = 0$ are to be found on the surface $g = 0$ among the points where

$$\nabla f = \lambda \nabla g$$

for some scalar λ (called a **Lagrange multiplier**).

To explore the method further and see why it works, we first make the following observation, which we state as a theorem.

THEOREM 12—The Orthogonal Gradient Theorem

Suppose that $f(x, y, z)$ is differentiable in a region whose interior contains a smooth curve

$$C: \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}.$$

If P_0 is a point on C where f has a local maximum or minimum relative to its values on C , then ∇f is orthogonal to the curve's tangent vector $\mathbf{r}'$ at P_0 .

Proof The values of f on C are given by the composition $f(x(t), y(t), z(t))$, whose derivative with respect to t is

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = \nabla f \cdot \mathbf{r}'.$$

At any point P_0 where f has a local maximum or minimum relative to its values on the curve, $df/dt = 0$, so

$$\nabla f \cdot \mathbf{r}' = 0. \quad \blacksquare$$

By dropping the z -terms in Theorem 12, we obtain a similar result for functions of two variables.

COROLLARY At the points on a smooth curve $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ where a differentiable function $f(x, y)$ takes on its local maxima or minima relative to its values on the curve, we have $\nabla f \cdot \mathbf{r}' = 0$.

Theorem 12 is the key to the method of Lagrange multipliers. Suppose that $f(x, y, z)$ and $g(x, y, z)$ are differentiable and that P_0 is a point on the surface $g(x, y, z) = 0$ where f has a local maximum or minimum value relative to its other values on the surface. We assume also that $\nabla g \neq \mathbf{0}$ at points on the surface $g(x, y, z) = 0$. Then f takes on a local maximum or minimum at P_0 relative to its values on every differentiable curve through P_0 on the surface $g(x, y, z) = 0$. Therefore, ∇f is orthogonal to the tangent vector of every

such differentiable curve through P_0 . Moreover, so is ∇g (because ∇g is perpendicular to the level surface $g = 0$, as we saw in Section 13.5). Therefore, at P_0 , ∇f is some scalar multiple λ of ∇g .

The Method of Lagrange Multipliers

Suppose that $f(x, y, z)$ and $g(x, y, z)$ are differentiable and $\nabla g \neq \mathbf{0}$ when $g(x, y, z) = 0$. To find the local maximum and minimum values of f subject to the constraint $g(x, y, z) = 0$ (if these exist), find the values of x, y, z , and λ that simultaneously satisfy the equations

$$\nabla f = \lambda \nabla g \quad \text{and} \quad g(x, y, z) = 0. \quad (1)$$

If they exist, absolute extrema can be found by comparing these values of f at each critical point satisfying Equation (1). For functions of two independent variables, the condition is similar, but without the variable z .

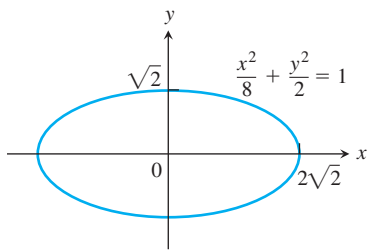


FIGURE 13.56 Example 3 shows how to find the largest and smallest values of the product xy on this ellipse.

Some care must be used in applying this method. An extreme value may not actually exist (Exercise 45).

EXAMPLE 3 Find the largest and smallest values that the function

$$f(x, y) = xy$$

takes on the ellipse (Figure 13.56)

$$\frac{x^2}{8} + \frac{y^2}{2} = 1.$$

Solution We want to find the extreme values of $f(x, y) = xy$ subject to the constraint

$$g(x, y) = \frac{x^2}{8} + \frac{y^2}{2} - 1 = 0.$$

To do so, we first find the values of x, y , and λ for which

$$\nabla f = \lambda \nabla g \quad \text{and} \quad g(x, y) = 0.$$

The gradient equation in Equations (1) gives

$$y\mathbf{i} + x\mathbf{j} = \frac{\lambda}{4}x\mathbf{i} + \lambda y\mathbf{j},$$

from which we find

$$y = \frac{\lambda}{4}x, \quad x = \lambda y,$$

and

$$y = \frac{\lambda}{4}(\lambda y) = \frac{\lambda^2}{4}y, \quad \text{Caution: Don't cancel } y \text{ without considering the case where } y = 0.$$

so that

$$y = 0 \text{ or } \lambda = \pm 2.$$

We now consider these two cases.

Case 1: If $y = 0$, then $x = y = 0$. But $(0, 0)$ is not on the ellipse. Hence, $y \neq 0$.

Case 2: If $y \neq 0$, then $\lambda = \pm 2$ and $x = \pm 2y$. Substituting this in the equation $g(x, y) = 0$ gives

$$\frac{(\pm 2y)^2}{8} + \frac{y^2}{2} = 1, \quad 4y^2 + 4y^2 = 8 \quad \text{and} \quad y = \pm 1.$$

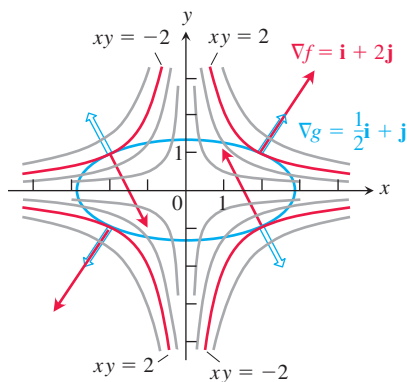


FIGURE 13.57 When subjected to the constraint $g(x, y) = x^2/8 + y^2/2 - 1 = 0$, the function $f(x, y) = xy$ takes on extreme values at the four points $(\pm 2, \pm 1)$. These are the points on the ellipse where ∇f (red) is a scalar multiple of ∇g (blue) (Example 3).

The function $f(x, y) = xy$ therefore has critical points on the ellipse at the four points $(\pm 2, 1)$, $(\pm 2, -1)$. The extreme values are found by examining the values of f at these four points. The absolute maximum is $f(2, 1) = f(-2, -1) = 2$, and the absolute minimum is $f(-2, 1) = f(2, -1) = -2$.

The Geometry of the Solution The level curves of the function $f(x, y) = xy$ are the hyperbolas $xy = c$ (Figure 13.57). The farther the hyperbolas lie from the origin, the larger the absolute value of f . We want to find the extreme values of $f(x, y)$, given that the point (x, y) also lies on the ellipse $x^2 + 4y^2 = 8$. Which hyperbolas intersecting the ellipse lie farthest from the origin? The hyperbolas that just graze the ellipse, the ones that are tangent to it, are farthest. At these points, any vector normal to the hyperbola is normal to the ellipse, so $\nabla f = y\mathbf{i} + x\mathbf{j}$ is a multiple ($\lambda = \pm 2$) of $\nabla g = (x/4)\mathbf{i} + y\mathbf{j}$. At the point $(2, 1)$, for example,

$$\nabla f = \mathbf{i} + 2\mathbf{j}, \quad \nabla g = \frac{1}{2}\mathbf{i} + \mathbf{j}, \quad \text{and} \quad \nabla f = 2\nabla g.$$

At the point $(-2, 1)$,

$$\nabla f = \mathbf{i} - 2\mathbf{j}, \quad \nabla g = -\frac{1}{2}\mathbf{i} + \mathbf{j}, \quad \text{and} \quad \nabla f = -2\nabla g. \quad \blacksquare$$

EXAMPLE 4 Find the maximum and minimum values of the function $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$.

Solution We model this as a Lagrange multiplier problem with

$$f(x, y) = 3x + 4y, \quad g(x, y) = x^2 + y^2 - 1$$

and look for the values of x , y , and λ that satisfy the equations

$$\begin{aligned} \nabla f &= \lambda \nabla g: & 3\mathbf{i} + 4\mathbf{j} &= 2x\lambda\mathbf{i} + 2y\lambda\mathbf{j} \\ g(x, y) &= 0: & x^2 + y^2 - 1 &= 0. \end{aligned}$$

The gradient equation implies that $\lambda \neq 0$ and gives

$$x = \frac{3}{2\lambda}, \quad y = \frac{2}{\lambda}.$$

These equations tell us, among other things, that x and y have the same sign. With these values for x and y , the equation $g(x, y) = 0$ gives

$$\left(\frac{3}{2\lambda}\right)^2 + \left(\frac{2}{\lambda}\right)^2 - 1 = 0,$$

so

$$\frac{9}{4\lambda^2} + \frac{4}{\lambda^2} = 1, \quad 9 + 16 = 4\lambda^2, \quad 4\lambda^2 = 25, \quad \text{and} \quad \lambda = \pm \frac{5}{2}.$$

Thus,

$$x = \frac{3}{2\lambda} = \pm \frac{3}{5}, \quad y = \frac{2}{\lambda} = \pm \frac{4}{5},$$

and $f(x, y) = 3x + 4y$ has critical points at $(x, y) = \pm(3/5, 4/5)$.

By calculating the value of $3x + 4y$ at the points $\pm(3/5, 4/5)$, we see that its maximum and minimum values on the circle $x^2 + y^2 = 1$ are

$$3\left(\frac{3}{5}\right) + 4\left(\frac{4}{5}\right) = \frac{25}{5} = 5 \quad \text{and} \quad 3\left(-\frac{3}{5}\right) + 4\left(-\frac{4}{5}\right) = -\frac{25}{5} = -5.$$

The Geometry of the Solution The level curves of $f(x, y) = 3x + 4y$ are the lines $3x + 4y = c$ (Figure 13.58). The farther the lines lie from the origin, the larger the absolute value of f . We want to find the extreme values of $f(x, y)$ given that the point (x, y)

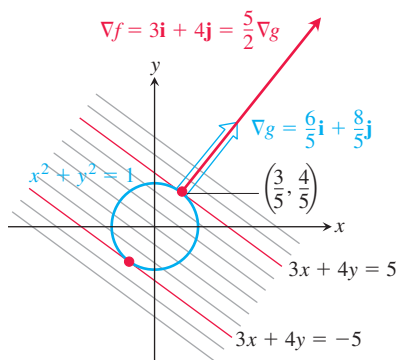


FIGURE 13.58 The function $f(x, y) = 3x + 4y$ takes on its largest value on the unit circle $g(x, y) = x^2 + y^2 - 1 = 0$ at the point $(3/5, 4/5)$ and its smallest value at the point $(-3/5, -4/5)$ (Example 4). At each of these points, ∇f is a scalar multiple of ∇g . The figure shows the gradients at the first point but not at the second.

also lies on the circle $x^2 + y^2 = 1$. Which lines intersecting the circle lie farthest from the origin? The lines tangent to the circle are farthest. At the points of tangency, any vector normal to the line is normal to the circle, so the gradient $\nabla f = 3\mathbf{i} + 4\mathbf{j}$ is a multiple ($\lambda = \pm 5/2$) of the gradient $\nabla g = 2x\mathbf{i} + 2y\mathbf{j}$. At the point $(3/5, 4/5)$, for example,

$$\nabla f = 3\mathbf{i} + 4\mathbf{j}, \quad \nabla g = \frac{6}{5}\mathbf{i} + \frac{8}{5}\mathbf{j}, \quad \text{and} \quad \nabla f = \frac{5}{2}\nabla g. \quad \blacksquare$$

Lagrange Multipliers with Two Constraints

Many problems require us to find the extreme values of a differentiable function $f(x, y, z)$ whose variables are subject to two constraints. If the constraints are

$$g_1(x, y, z) = 0 \quad \text{and} \quad g_2(x, y, z) = 0$$

and g_1 and g_2 are differentiable, with ∇g_1 not parallel to ∇g_2 , we find the constrained local maxima and minima of f by introducing two Lagrange multipliers λ and μ (mu, pronounced “mew”). That is, we locate the points $P(x, y, z)$ where f takes on its constrained extreme values by finding the values of x, y, z, λ , and μ that simultaneously satisfy the three equations

$$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2, \quad g_1(x, y, z) = 0, \quad g_2(x, y, z) = 0 \quad (2)$$

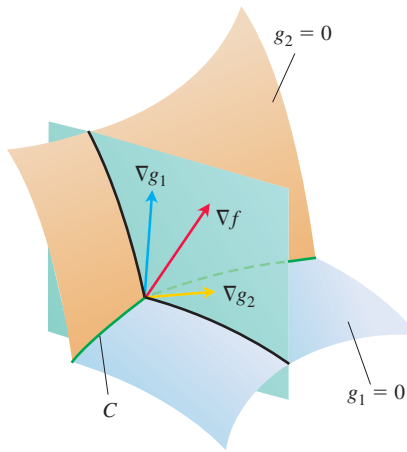


FIGURE 13.59 The vectors ∇g_1 and ∇g_2 lie in a plane perpendicular to the curve C , because ∇g_1 is normal to the surface $g_1 = 0$ and ∇g_2 is normal to the surface $g_2 = 0$.

μ is the Greek letter mu, pronounced “mew”.

Equations (2) have a nice geometric interpretation. The surfaces $g_1 = 0$ and $g_2 = 0$ (usually) intersect in a smooth curve, say C (Figure 13.59). Along this curve we seek the points where f has local maximum and minimum values relative to its other values on the curve. These are the points where ∇f is normal to C , as we saw in Theorem 12. But ∇g_1 and ∇g_2 are also normal to C at these points because C lies in the surfaces $g_1 = 0$ and $g_2 = 0$. Therefore, ∇f lies in the plane determined by ∇g_1 and ∇g_2 , which means that $\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$ for some λ and μ . Since the points we seek also lie in both surfaces, their coordinates must satisfy the equations $g_1(x, y, z) = 0$ and $g_2(x, y, z) = 0$, which are the remaining requirements in Equations (2).

EXAMPLE 5 The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse (Figure 13.60). Find the points on the ellipse that lie closest to and farthest from the origin.

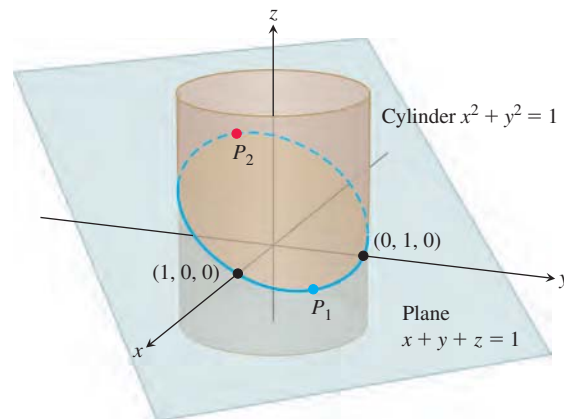


FIGURE 13.60 On the ellipse where the plane and cylinder meet, we find the points closest to and farthest from the origin (Example 5).

Solution We find the extreme values of

$$f(x, y, z) = x^2 + y^2 + z^2$$

(the square of the distance from (x, y, z) to the origin) subject to the constraints

$$g_1(x, y, z) = x^2 + y^2 - 1 = 0 \quad (3)$$

$$g_2(x, y, z) = x + y + z - 1 = 0. \quad (4)$$

The gradient equation in Equations (2) then gives

$$\begin{aligned} \nabla f &= \lambda \nabla g_1 + \mu \nabla g_2 \\ 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} &= \lambda(2x\mathbf{i} + 2y\mathbf{j}) + \mu(\mathbf{i} + \mathbf{j} + \mathbf{k}) \\ 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} &= (2\lambda x + \mu)\mathbf{i} + (2\lambda y + \mu)\mathbf{j} + \mu\mathbf{k}, \end{aligned}$$

or

$$2x = 2\lambda x + \mu, \quad 2y = 2\lambda y + \mu, \quad 2z = \mu. \quad (5)$$

The scalar equations in Equations (5) yield

$$\begin{aligned} 2x &= 2\lambda x + 2z \Rightarrow (1 - \lambda)x = z, \\ 2y &= 2\lambda y + 2z \Rightarrow (1 - \lambda)y = z. \end{aligned} \quad (6)$$

Equations (6) are satisfied simultaneously if either $\lambda = 1$ and $z = 0$ or $\lambda \neq 1$ and $x = y = z/(1 - \lambda)$.

In the first case, where $z = 0$, solving Equations (3) and (4) simultaneously to find the corresponding points on the ellipse gives the two points $(1, 0, 0)$ and $(0, 1, 0)$. This makes sense when you look at Figure 13.60.

In the second case, where $x = y$, Equations (3) and (4) give

$$\begin{aligned} x^2 + x^2 - 1 &= 0 & x + x + z - 1 &= 0 \\ 2x^2 &= 1 & z &= 1 - 2x \\ x &= \pm \frac{\sqrt{2}}{2} & z &= 1 \mp \sqrt{2}. \end{aligned}$$

The corresponding points on the ellipse are

$$P_1 = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1 - \sqrt{2} \right) \quad \text{and} \quad P_2 = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 1 + \sqrt{2} \right).$$

To find the points at maximum and minimum distance from the origin, we evaluate f at the four critical points $(1, 0, 0)$, $(0, 1, 0)$, P_1 , and P_2 . We see that

$$f(1, 0, 0) = f(0, 1, 0) = 1, \quad f(P_1) = 4 - 2\sqrt{2}, \quad \text{and} \quad f(P_2) = 4 + 2\sqrt{2}.$$

The largest and smallest of these give the absolute extrema. Since

$$1 < 4 - 2\sqrt{2} < 4 + 2\sqrt{2},$$

we see that the absolute minimum value of f is 1 and is attained when f is evaluated at either $(1, 0, 0)$ or $(0, 1, 0)$. The absolute maximum value of f is $4 + 2\sqrt{2}$ and occurs when f is evaluated at P_2 . The value $f(P_1) = 4 - 2\sqrt{2}$ is neither the largest nor the smallest among the values of f at the critical points, so f does not have an absolute extremum at P_1 .

The points on the ellipse closest to the origin are $(1, 0, 0)$ and $(0, 1, 0)$. The point on the ellipse farthest from the origin is P_2 . (See Figure 13.60.) ■

EXERCISES 13.8

Two Independent Variables with One Constraint

- 1. Extrema on an ellipse** Find the points on the ellipse $x^2 + 2y^2 = 1$ where $f(x, y) = xy$ has its extreme values.
- 2. Extrema on a circle** Find the extreme values of $f(x, y) = xy$ subject to the constraint $g(x, y) = x^2 + y^2 - 10 = 0$.

- 3. Maximum on a line** Find the maximum value of $f(x, y) = 49 - x^2 - y^2$ on the line $x + 3y = 10$.
- 4. Extrema on a line** Find the local extreme values of $f(x, y) = x^2y$ on the line $x + y = 3$.

- 5. Constrained minimum** Find the points on the curve $xy^2 = 54$ nearest the origin.
- 6. Constrained minimum** Find the points on the curve $x^2y = 2$ nearest the origin.
- 7. Use the method of Lagrange multipliers to find**
- Minimum on a hyperbola** The minimum value of $x + y$, subject to the constraints $xy = 16$, $x > 0$, $y > 0$.
 - Maximum on a line** The maximum value of xy , subject to the constraint $x + y = 16$.
- Comment on the geometry of each solution.
- 8. Extrema on a curve** Find the points on the curve $x^2 + xy + y^2 = 1$ in the xy -plane that are nearest to and farthest from the origin.
- 9. Minimum surface area with fixed volume** Find the dimensions of the closed right circular cylindrical can of smallest surface area whose volume is $16\pi \text{ cm}^3$.
- 10. Cylinder in a sphere** Find the radius and height of the open right circular cylinder of largest surface area that can be inscribed in a sphere of radius a . What is the largest surface area?
- 11. Rectangle of greatest area in an ellipse** Use the method of Lagrange multipliers to find the dimensions of the rectangle of greatest area that can be inscribed in the ellipse $x^2/16 + y^2/9 = 1$ with sides parallel to the coordinate axes.
- 12. Rectangle of longest perimeter in an ellipse** Find the dimensions of the rectangle of largest perimeter that can be inscribed in the ellipse $x^2/a^2 + y^2/b^2 = 1$ with sides parallel to the coordinate axes. What is the largest perimeter?
- 13. Extrema on a circle** Find the maximum and minimum values of $x^2 + y^2$ subject to the constraint $x^2 - 2x + y^2 - 4y = 0$.
- 14. Extrema on a circle** Find the maximum and minimum values of $3x - y + 6$ subject to the constraint $x^2 + y^2 = 4$.
- 15. Ant on a metal plate** The temperature at a point (x, y) on a metal plate is $T(x, y) = 4x^2 - 4xy + y^2$. An ant on the plate walks around the circle of radius 5 centered at the origin. What are the highest and lowest temperatures encountered by the ant?
- 16. Cheapest storage tank** Your firm has been asked to design a storage tank for liquid petroleum gas. The customer's specifications call for a cylindrical tank with hemispherical ends, and the tank is to hold 8000 m^3 of gas. The customer also wants to use the smallest amount of material possible in building the tank. What radius and height do you recommend for the cylindrical portion of the tank?

Three Independent Variables with One Constraint

- 17. Minimum distance to a point** Find the point on the plane $x + 2y + 3z = 13$ closest to the point $(1, 1, 1)$.
- 18. Maximum distance to a point** Find the point on the sphere $x^2 + y^2 + z^2 = 4$ farthest from the point $(1, -1, 1)$.
- 19. Minimum distance to the origin** Find the minimum distance from the surface $x^2 - y^2 - z^2 = 1$ to the origin.
- 20. Minimum distance to the origin** Find the point on the surface $z = xy + 1$ nearest the origin.
- 21. Minimum distance to the origin** Find the points on the surface $z^2 = xy + 4$ closest to the origin.
- 22. Minimum distance to the origin** Find the point(s) on the surface $xyz = 1$ closest to the origin.
- 23. Extrema on a sphere** Find the maximum and minimum values of
- $$f(x, y, z) = x - 2y + 5z$$
- on the sphere $x^2 + y^2 + z^2 = 30$.
- 24. Extrema on a sphere** Find the points on the sphere $x^2 + y^2 + z^2 = 25$ where $f(x, y, z) = x + 2y + 3z$ has its maximum and minimum values.
- 25. Minimizing a sum of squares** Find three real numbers whose sum is 9 and the sum of whose squares is as small as possible.
- 26. Maximizing a product** Find the largest product the positive numbers x , y , and z can have if $x + y + z^2 = 16$.
- 27. Rectangular box of largest volume in a sphere** Find the dimensions of the closed rectangular box with maximum volume that can be inscribed in the unit sphere.
- 28. Box with vertex on a plane** Find the volume of the largest closed rectangular box in the first octant having three faces in the coordinate planes and a vertex on the plane $x/a + y/b + z/c = 1$, where $a > 0$, $b > 0$, and $c > 0$.
- 29. Hottest point on a space probe** A space probe in the shape of the ellipsoid
- $$4x^2 + y^2 + 4z^2 = 16$$
- enters Earth's atmosphere and its surface begins to heat. After 1 hour, the temperature at the point (x, y, z) on the probe's surface is
- $$T(x, y, z) = 8x^2 + 4yz - 16z + 600.$$
- Find the hottest point on the probe's surface.
- 30. Extreme temperatures on a sphere** Suppose that the Celsius temperature at the point (x, y, z) on the sphere $x^2 + y^2 + z^2 = 1$ is $T = 400xyz^2$. Locate the highest and lowest temperatures on the sphere.
- 31. Cobb–Douglas production function** During the 1920s, Charles Cobb and Paul Douglas modeled total production output P (of a firm, industry, or entire economy) as a function of labor hours involved x and capital invested y (which includes the monetary worth of all buildings and equipment). The Cobb–Douglas production function is given by
- $$P(x, y) = kx^\alpha y^{1-\alpha},$$
- where k and α are constants representative of a particular firm or economy.
- Show that a doubling of both labor and capital results in a doubling of production P .
 - Suppose a particular firm has the production function for $k = 120$ and $\alpha = 3/4$. Assume that each unit of labor costs \$250 and each unit of capital costs \$400, and that the total expenses for all costs cannot exceed \$100,000. Find the maximum production level for the firm.
- 32. (Continuation of Exercise 31.)** If the cost of a unit of labor is c_1 and the cost of a unit of capital is c_2 , and if the firm can spend only B dollars as its total budget, then production P is constrained by $c_1x + c_2y = B$. Show that the maximum production level subject to the constraint occurs at the point
- $$x = \frac{\alpha B}{c_1} \quad \text{and} \quad y = \frac{(1 - \alpha)B}{c_2}.$$

33. Maximizing a utility function: an example from economics

In economics, the usefulness or *utility* of amounts x and y of two capital goods G_1 and G_2 is sometimes measured by a function $U(x, y)$. For example, G_1 and G_2 might be two chemicals a pharmaceutical company needs to have on hand, and $U(x, y)$ might be the gain from manufacturing a product whose synthesis requires different amounts of the chemicals depending on the process used. If G_1 costs a dollars per kilogram, G_2 costs b dollars per kilogram, and the total amount allocated for the purchase of G_1 and G_2 together is c dollars, then the company's managers want to maximize $U(x, y)$ given that $ax + by = c$. Thus, they need to solve a typical Lagrange multiplier problem.

Suppose that

$$U(x, y) = xy + 2x$$

and that the equation $ax + by = c$ simplifies to

$$2x + y = 30.$$

Find the maximum value of U and the corresponding values of x and y subject to this latter constraint.

34. Blood types Human blood types are classified by three gene forms A, B, and O. Blood types AA, BB, and OO are *homozygous*, and blood types AB, AO, and BO are *heterozygous*. If p , q , and r represent the proportions of the three gene forms to the population, respectively, then the *Hardy-Weinberg Law* asserts that the proportion Q of heterozygous persons in any specific population is modeled by

$$Q(p, q, r) = 2(pq + pr + qr),$$

subject to $p + q + r = 1$. Find the maximum value of Q .

35. Length of a beam In Section 4.6, Exercise 47, we posed a problem of finding the length L of the shortest beam that can reach over a wall of height h to a tall building located k units from the wall. Use Lagrange multipliers to show that

$$L = (h^{2/3} + k^{2/3})^{3/2}.$$

36. Locating a radio telescope You are in charge of erecting a radio telescope on a newly discovered planet. To minimize interference, you want to place it where the magnetic field of the planet is weakest. The planet is spherical, with a radius of 6 units. Based on a coordinate system whose origin is at the center of the planet, the strength of the magnetic field is given by $M(x, y, z) = 6x - y^2 + xz + 60$. Where should you locate the radio telescope?**Extreme Values Subject to Two Constraints****37.** Maximize the function $f(x, y, z) = x^2 + 2y - z^2$ subject to the constraints $2x - y = 0$ and $y + z = 0$.**38.** Minimize the function $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraints $x + 2y + 3z = 6$ and $x + 3y + 9z = 9$.**39. Minimum distance to the origin** Find the point closest to the origin on the line of intersection of the planes $y + 2z = 12$ and $x + y = 6$.**40.** Find the extreme values of $f(x, y, z) = 2x^2 + yz$ on the intersection of the cylinder $x^2 + z^2 = 9$ and the plane $y - z = 4$.**41. Extrema on a curve of intersection** Find the extreme values of $f(x, y, z) = x^2yz + 1$ on the intersection of the plane $z = 1$ with the sphere $x^2 + y^2 + z^2 = 10$.**42. a. Maximum on line of intersection** Find the maximum value of $w = xyz$ on the line of intersection of the two planes $x + y + z = 40$ and $x + y - z = 0$.

b. Give a geometric argument to support your claim that you have found a maximum, and not a minimum, value of w .

43. Extrema on a circle of intersection Find the extreme values of the function $f(x, y, z) = xy + z^2$ on the circle in which the plane $y - x = 0$ intersects the sphere $x^2 + y^2 + z^2 = 4$.**44. Minimum distance to the origin** Find the point closest to the origin on the curve of intersection of the plane $2y + 4z = 5$ and the cone $z^2 = 4x^2 + 4y^2$.**Theory and Examples****45. The condition $\nabla f = \lambda \nabla g$ is not sufficient** Even though $\nabla f = \lambda \nabla g$ is a necessary condition for the occurrence of an extreme value of $f(x, y)$ subject to the conditions $g(x, y) = 0$ and $\nabla g \neq \mathbf{0}$, it does not in itself guarantee that one exists. As a case in point, try using the method of Lagrange multipliers to find a maximum value of $f(x, y) = x + y$ subject to the constraint that $xy = 16$. The method will identify the two points $(4, 4)$ and $(-4, -4)$ as candidates for the location of extreme values. Yet the sum $x + y$ has no maximum value on the hyperbola $xy = 16$. The farther you go from the origin on this hyperbola in the first quadrant, the larger the sum $f(x, y) = x + y$ becomes.**46. A least squares plane** The plane $z = Ax + By + C$ is to be “fitted” to the following points (x_k, y_k, z_k) :

$$(0, 0, 0), \quad (0, 1, 1), \quad (1, 1, 1), \quad (1, 0, -1).$$

Find the values of A , B , and C that minimize

$$\sum_{k=1}^4 (Ax_k + By_k + C - z_k)^2,$$

the sum of the squares of the deviations.

47. a. Maximum on a sphere Show that the maximum value of $a^2b^2c^2$ on a sphere of radius r centered at the origin of a Cartesian abc -coordinate system is $(r^2/3)^3$.**b. Geometric and arithmetic means** Using part (a), show that for nonnegative numbers a , b , and c ,

$$(abc)^{1/3} \leq \frac{a + b + c}{3};$$

that is, the *geometric mean* of three nonnegative numbers is less than or equal to their *arithmetic mean*.

48. Sum of products Let $a_1, a_2, \dots, a_n$ be n positive numbers. Find the maximum of $\sum_{i=1}^n a_i x_i$ subject to the constraint $\sum_{i=1}^n x_i^2 = 1$.**COMPUTER EXPLORATIONS**

In Exercises 49–54, use a CAS to perform the following steps implementing the method of Lagrange multipliers for finding constrained extrema:

- Form the function $h = f - \lambda_1 g_1 - \lambda_2 g_2$, where f is the function to optimize subject to the constraints $g_1 = 0$ and $g_2 = 0$.
- Determine all the first partial derivatives of h , including the partials with respect to λ_1 and λ_2 , and set them equal to 0.
- Solve the system of equations found in part (b) for all the unknowns, including λ_1 and λ_2 .
- Evaluate f at each of the solution points found in part (c), and select the extreme value subject to the constraints asked for in the exercise.

49. Minimize $f(x, y, z) = xy + yz$ subject to the constraints $x^2 + y^2 - 2 = 0$ and $x^2 + z^2 - 2 = 0$.
50. Minimize $f(x, y, z) = xyz$ subject to the constraints $x^2 + y^2 - 1 = 0$ and $x - z = 0$.
51. Maximize $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraints $2y + 4z - 5 = 0$ and $4x^2 + 4y^2 - z^2 = 0$.
52. Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraints $x^2 - xy + y^2 - z^2 - 1 = 0$ and $x^2 + y^2 - 1 = 0$.

53. Minimize $f(x, y, z, w) = x^2 + y^2 + z^2 + w^2$ subject to the constraints $2x - y + z - w - 1 = 0$ and $x + y - z + w - 1 = 0$.
54. Determine the distance from the line $y = x + 1$ to the parabola $y^2 = x$. (Hint: Let (x, y) be a point on the line and (w, z) a point on the parabola. You want to minimize $(x - w)^2 + (y - z)^2$.)

13.9 Taylor's Formula for Two Variables

In this section we use Taylor's formula to derive the Second Derivative Test for local extreme values (Section 13.7) and the error formula for linearizations of functions of two independent variables (Section 13.6). The use of Taylor's formula in these derivations leads to an extension of the formula that provides polynomial approximations of all orders for functions of two independent variables.

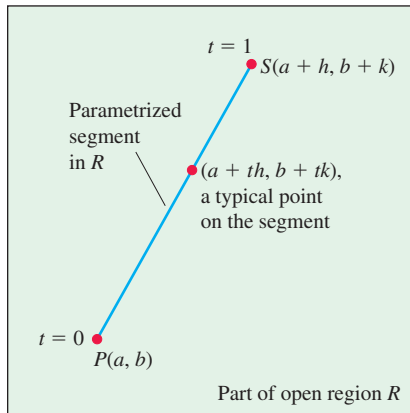


FIGURE 13.61 We begin the derivation of the Second Derivative Test at $P(a, b)$ by parametrizing a typical line segment from P to a point S nearby.

Derivation of the Second Derivative Test

Let $f(x, y)$ have continuous first and second partial derivatives in an open region R containing a point $P(a, b)$ where $f_x = f_y = 0$ (Figure 13.61). Let h and k be increments small enough to put the point $S(a + h, b + k)$ and the line segment joining it to P inside R . We parametrize the segment PS as

$$x = a + th, \quad y = b + tk, \quad 0 \leq t \leq 1.$$

If $F(t) = f(a + th, b + tk)$, the Chain Rule gives

$$F'(t) = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = hf_x + kf_y.$$

Since f_x and f_y are differentiable (because they have continuous partial derivatives), F' is a differentiable function of t and

$$\begin{aligned} F'' &= \frac{\partial F'}{\partial x} \frac{dx}{dt} + \frac{\partial F'}{\partial y} \frac{dy}{dt} = \frac{\partial}{\partial x}(hf_x + kf_y) \cdot h + \frac{\partial}{\partial y}(hf_x + kf_y) \cdot k \\ &= h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}. \quad f_{xy} = f_{yx} \end{aligned}$$

Since F and F' are continuous on $[0, 1]$ and F' is differentiable on $(0, 1)$, we can apply Taylor's formula with $n = 2$ and $a = 0$ to obtain

$$\begin{aligned} F(1) &= F(0) + F'(0)(1 - 0) + F''(c) \frac{(1 - 0)^2}{2} \\ &= F(0) + F'(0) + \frac{1}{2} F''(c) \end{aligned} \quad (1)$$

for some c between 0 and 1. Writing Equation (1) in terms of f gives

$$\begin{aligned} f(a + h, b + k) &= f(a, b) + hf_x(a, b) + kf_y(a, b) \\ &\quad + \frac{1}{2} (h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}) \Big|_{(a+ch, b+ck)}. \end{aligned} \quad (2)$$

Since $f_x(a, b) = f_y(a, b) = 0$, this reduces to

$$f(a + h, b + k) - f(a, b) = \frac{1}{2} (h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}) \Big|_{(a+ch, b+ck)}. \quad (3)$$

To determine whether f has an extremum at (a, b) , we examine the sign of the difference $f(a + h, b + k) - f(a, b)$. By Equation (3), this is the same as the sign of

$$Q(c) = (h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}) \Big|_{(a+ch, b+ck)}.$$

Now, if $Q(0) \neq 0$, the sign of $Q(c)$ will be the same as the sign of $Q(0)$ for sufficiently small values of h and k . We can predict the sign of

$$Q(0) = h^2 f_{xx}(a, b) + 2hkf_{xy}(a, b) + k^2 f_{yy}(a, b) \quad (4)$$

from the signs of f_{xx} and $f_{xx}f_{yy} - f_{xy}^2$ at (a, b) . Multiply both sides of Equation (4) by f_{xx} and rearrange the right-hand side to get

$$f_{xx}Q(0) = (hf_{xx} + kf_{xy})^2 + (f_{xx}f_{yy} - f_{xy}^2)k^2. \quad (5)$$

From Equation (5) we see that

1. If $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) , then $Q(0) < 0$ for all sufficiently small nonzero values of h and k , and f has a *local maximum* value at (a, b) .
2. If $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) , then $Q(0) > 0$ for all sufficiently small nonzero values of h and k , and f has a *local minimum* value at (a, b) .
3. If $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a, b) , there are combinations of arbitrarily small nonzero values of h and k for which $Q(0) > 0$, and other values for which $Q(0) < 0$. Arbitrarily close to the point $P_0(a, b, f(a, b))$ on the surface $z = f(x, y)$ there are points above P_0 and points below P_0 , so f has a *saddle point* at (a, b) .
4. If $f_{xx}f_{yy} - f_{xy}^2 = 0$, another test is needed. The possibility that $Q(0)$ equals zero prevents us from drawing conclusions about the sign of $Q(c)$.

The Error Formula for Linear Approximations

We want to show that the difference $E(x, y)$ between the values of a function $f(x, y)$ and its linearization $L(x, y)$ at (x_0, y_0) satisfies the inequality

$$|E(x, y)| \leq \frac{1}{2}M(|x - x_0| + |y - y_0|)^2.$$

The function f is assumed to have continuous second partial derivatives throughout an open set containing a closed rectangular region R centered at (x_0, y_0) . The number M is an upper bound for $|f_{xx}|$, $|f_{yy}|$, and $|f_{xy}|$ on R .

The inequality we want comes from Equation (2). We substitute x_0 and y_0 for a and b , and $x - x_0$ and $y - y_0$ for h and k , respectively, and rearrange the result as

$$\begin{aligned} f(x, y) &= \underbrace{f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}_{\text{linearization } L(x, y)} \\ &\quad + \underbrace{\frac{1}{2}((x - x_0)^2 f_{xx} + 2(x - x_0)(y - y_0)f_{xy} + (y - y_0)^2 f_{yy})}_{\text{error } E(x, y)} \Big|_{(x_0 + c(x - x_0), y_0 + c(y - y_0))}. \end{aligned}$$

This equation reveals that

$$|E| \leq \frac{1}{2}(|x - x_0|^2 |f_{xx}| + 2|x - x_0||y - y_0||f_{xy}| + |y - y_0|^2 |f_{yy}|).$$

Hence, if M is an upper bound for the values of $|f_{xx}|$, $|f_{xy}|$, and $|f_{yy}|$ on R , then

$$\begin{aligned} |E| &\leq \frac{1}{2}(|x - x_0|^2 M + 2|x - x_0||y - y_0| M + |y - y_0|^2 M) \\ &= \frac{1}{2}M(|x - x_0| + |y - y_0|)^2. \end{aligned}$$

Taylor's Formula for Functions of Two Variables

The formulas derived earlier for F' and F'' can be obtained by applying to $f(x, y)$ the differentiation operators

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) \quad \text{and} \quad \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 = h^2 \frac{\partial^2}{\partial x^2} + 2hk \frac{\partial^2}{\partial x \partial y} + k^2 \frac{\partial^2}{\partial y^2}.$$

These are the first two instances of a more general formula,

$$F^{(n)}(t) = \frac{d^n}{dt^n} F(t) = \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^n f(x, y), \quad (6)$$

which says that applying d^n/dt^n to $F(t)$ gives the same result as applying the operator

$$\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^n$$

to $f(x, y)$ after expanding it by the Binomial Theorem.

If the partial derivatives of f through order $n + 1$ are continuous throughout a rectangular region centered at (a, b) , we may extend the Taylor formula for $F(t)$ to

$$F(t) = F(0) + F'(0)t + \frac{F''(0)}{2!}t^2 + \cdots + \frac{F^{(n)}(0)}{n!}t^n + \text{remainder},$$

and take $t = 1$ to obtain

$$F(1) = F(0) + F'(0) + \frac{F''(0)}{2!} + \cdots + \frac{F^{(n)}(0)}{n!} + \text{remainder}.$$

When we replace the first n derivatives on the right of this last series by their equivalent expressions from Equation (6) evaluated at $t = 0$ and add the appropriate remainder term, we arrive at the following formula.

Taylor's Formula for $f(x, y)$ at the Point (a, b)

Suppose $f(x, y)$ and its partial derivatives through order $n + 1$ are continuous throughout an open rectangular region R centered at a point (a, b) . Then, throughout R ,

$$\begin{aligned} f(a + h, b + k) &= f(a, b) + \left(hf_x + kf_y\right)\Big|_{(a,b)} + \frac{1}{2!}\left(h^2f_{xx} + 2hkf_{xy} + k^2f_{yy}\right)\Big|_{(a,b)} \\ &\quad + \frac{1}{3!}\left(h^3f_{xxx} + 3h^2kf_{xxy} + 3hk^2f_{xyy} + k^3f_{yyy}\right)\Big|_{(a,b)} + \cdots + \frac{1}{n!}\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^n f\Big|_{(a,b)} \\ &\quad + \frac{1}{(n+1)!}\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^{n+1} f\Big|_{(a+ch, b+ck)}. \end{aligned} \quad (7)$$

The first n derivative terms are evaluated at (a, b) . The last term is evaluated at some point $(a + ch, b + ck)$ on the line segment joining (a, b) and $(a + h, b + k)$.

If $(a, b) = (0, 0)$ and we treat h and k as independent variables (denoting them now by x and y), then Equation (7) assumes the following form.

Taylor's Formula for $f(x, y)$ at the Origin

$$\begin{aligned}
 f(x, y) = & f(0, 0) + xf_x + yf_y + \frac{1}{2!} \left(x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} \right) \\
 & + \frac{1}{3!} \left(x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy} \right) + \cdots + \frac{1}{n!} \left(x^n \frac{\partial^n f}{\partial x^n} + nx^{n-1} y \frac{\partial^n f}{\partial x^{n-1} \partial y} + \cdots + y^n \frac{\partial^n f}{\partial y^n} \right) \\
 & + \frac{1}{(n+1)!} \left(x^{n+1} \frac{\partial^{n+1} f}{\partial x^{n+1}} + (n+1)x^n y \frac{\partial^{n+1} f}{\partial x^n \partial y} + \cdots + y^{n+1} \frac{\partial^{n+1} f}{\partial y^{n+1}} \right) \Bigg|_{(cx, cy)} \quad (8)
 \end{aligned}$$

The first n derivative terms are evaluated at $(0, 0)$. The last term is evaluated at a point on the line segment joining the origin and (x, y) .

Taylor's formula provides polynomial approximations of two-variable functions. The first n derivative terms give the polynomial; the last term gives the approximation error. The first three terms of Taylor's formula give the function's linearization. To improve on the linearization, we add higher-power terms.

EXAMPLE 1 Find a quadratic approximation to $f(x, y) = \sin x \sin y$ near the origin. How accurate is the approximation if $|x| \leq 0.1$ and $|y| \leq 0.1$?

Solution We take $n = 2$ in Equation (8):

$$\begin{aligned}
 f(x, y) = & f(0, 0) + (xf_x + yf_y) + \frac{1}{2}(x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy}) \\
 & + \frac{1}{6}(x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy}) \Bigg|_{(cx, cy)}.
 \end{aligned}$$

Calculating the values of the partial derivatives,

$$\begin{aligned}
 f(0, 0) &= \sin x \sin y \Big|_{(0,0)} = 0, & f_{xx}(0, 0) &= -\sin x \sin y \Big|_{(0,0)} = 0, \\
 f_x(0, 0) &= \cos x \sin y \Big|_{(0,0)} = 0, & f_{xy}(0, 0) &= \cos x \cos y \Big|_{(0,0)} = 1, \\
 f_y(0, 0) &= \sin x \cos y \Big|_{(0,0)} = 0, & f_{yy}(0, 0) &= -\sin x \sin y \Big|_{(0,0)} = 0,
 \end{aligned}$$

we have the result

$$\sin x \sin y \approx 0 + 0 + 0 + \frac{1}{2}(x^2(0) + 2xy(1) + y^2(0)), \text{ or } \sin x \sin y \approx xy.$$

The error in the approximation is

$$E(x, y) = \frac{1}{6}(x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy}) \Bigg|_{(cx, cy)}.$$

The third derivatives never exceed 1 in absolute value because they are products of sines and cosines. Also, $|x| \leq 0.1$ and $|y| \leq 0.1$. Hence

$$|E(x, y)| \leq \frac{1}{6}((0.1)^3 + 3(0.1)^3 + 3(0.1)^3 + (0.1)^3) = \frac{8}{6}(0.1)^3 \leq 0.00134$$

(rounded up). The error will not exceed 0.00134 if $|x| \leq 0.1$ and $|y| \leq 0.1$. ■

EXERCISES 13.9

Finding Quadratic and Cubic Approximations

In Exercises 1–10, use Taylor's formula for $f(x, y)$ at the origin to find quadratic and cubic approximations of f near the origin.

1. $f(x, y) = xe^y$
2. $f(x, y) = e^x \cos y$
3. $f(x, y) = y \sin x$
4. $f(x, y) = \sin x \cos y$
5. $f(x, y) = e^x \ln(1 + y)$
6. $f(x, y) = \ln(2x + y + 1)$
7. $f(x, y) = \sin(x^2 + y^2)$
8. $f(x, y) = \cos(x^2 + y^2)$

$$9. f(x, y) = \frac{1}{1 - x - y} \quad 10. f(x, y) = \frac{1}{1 - x - y + xy}$$

11. Use Taylor's formula to find a quadratic approximation of $f(x, y) = \cos x \cos y$ at the origin. Estimate the error in the approximation if $|x| \leq 0.1$ and $|y| \leq 0.1$.
12. Use Taylor's formula to find a quadratic approximation of $e^x \sin y$ at the origin. Estimate the error in the approximation if $|x| \leq 0.1$ and $|y| \leq 0.1$.

13.10 Partial Derivatives with Constrained Variables

In finding partial derivatives of functions like $w = f(x, y)$, we have assumed x and y to be independent. In many applications, however, this is not the case. For example, the internal energy U of a gas may be expressed as a function $U = f(P, V, T)$ of pressure P , volume V , and temperature T . If the individual molecules of the gas do not interact, however, P , V , and T obey (and are constrained by) the ideal gas law

$$PV = nRT \quad (n \text{ and } R \text{ constant}),$$

and fail to be independent. In this section we learn how to find partial derivatives in situations like this, which occur in economics, engineering, and physics.

Decide Which Variables Are Dependent and Which Are Independent

If the variables in a function $w = f(x, y, z)$ are constrained by a relation like the one imposed on x , y , and z by the equation $z = x^2 + y^2$, the geometric meanings and the numerical values of the partial derivatives of f will depend on which variables are chosen to be dependent and which are chosen to be independent. To see how this choice can affect the outcome, we consider the calculation of $\partial w / \partial x$ when $w = x^2 + y^2 + z^2$ and $z = x^2 + y^2$.

EXAMPLE 1 Find $\partial w / \partial x$ if $w = x^2 + y^2 + z^2$ and $z = x^2 + y^2$.

Solution We are given two equations in the four unknowns x , y , z , and w . Like many such systems, this one can be solved for two of the unknowns (the dependent variables) in terms of the others (the independent variables). In being asked for $\partial w / \partial x$, we are told that w is to be a dependent variable and x an independent variable. The possible choices for the other variables come down to

	Dependent	Independent
Choice 1:	w, z	x, y
Choice 2:	w, y	x, z

In either case, we can express w explicitly in terms of the selected independent variables. We do this by using the second equation $z = x^2 + y^2$ to eliminate the remaining dependent variable in the first equation.

In the first case, the remaining dependent variable is z . We eliminate it from the first equation by replacing it by $x^2 + y^2$. The resulting expression for w is

$$\begin{aligned} w &= x^2 + y^2 + z^2 = x^2 + y^2 + (x^2 + y^2)^2 \\ &= x^2 + y^2 + x^4 + 2x^2y^2 + y^4 \end{aligned}$$

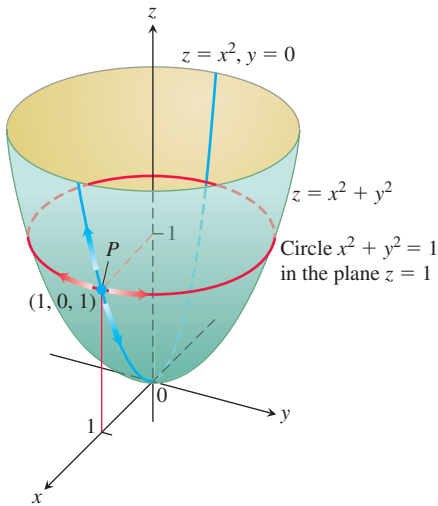


FIGURE 13.62 If P is constrained to lie on the paraboloid $z = x^2 + y^2$, the value of the partial derivative of $w = x^2 + y^2 + z^2$ with respect to x at P depends on the direction of motion (Example 1). (1) As x changes, with $y = 0$, P moves up or down the surface on the parabola $z = x^2$ in the xz -plane with $\partial w/\partial x = 2x + 4x^3$. (2) As x changes, with $z = 1$, P moves on the circle $x^2 + y^2 = 1$, $z = 1$, and $\partial w/\partial x = 0$.

and therefore

$$\frac{\partial w}{\partial x} = 2x + 4x^3 + 4xy^2. \quad (1)$$

This is the formula for $\partial w/\partial x$ when x and y are the independent variables.

In the second case, where the independent variables are x and z and the remaining dependent variable is y , we eliminate the dependent variable y in the expression for w by replacing y^2 in the second equation by $z - x^2$. This gives

$$w = x^2 + y^2 + z^2 = x^2 + (z - x^2) + z^2 = z + z^2$$

and therefore

$$\frac{\partial w}{\partial x} = 0. \quad (2)$$

This is the formula for $\partial w/\partial x$ when x and z are the independent variables.

The formulas for $\partial w/\partial x$ in Equations (1) and (2) are genuinely different. We cannot change either formula into the other by using the relation $z = x^2 + y^2$. There is not just one $\partial w/\partial x$, there are two, and we see that the original instruction to find $\partial w/\partial x$ was incomplete. *Which $\partial w/\partial x$?* we ask.

The geometric interpretations of Equations (1) and (2) help to explain why the equations differ. The function $w = x^2 + y^2 + z^2$ measures the square of the distance from the point (x, y, z) to the origin. The condition $z = x^2 + y^2$ says that the point (x, y, z) lies on the paraboloid of revolution shown in Figure 13.62. What does it mean to calculate $\partial w/\partial x$ at a point $P(x, y, z)$ that can move only on this surface? What is the value of $\partial w/\partial x$ when the coordinates of P are, say, $(1, 0, 1)$?

If we take x and y to be independent, then we find $\partial w/\partial x$ by holding y fixed (at $y = 0$ in this case) and letting x vary. Hence, P moves along the parabola $z = x^2$ in the xz -plane. As P moves on this parabola, w , which is the square of the distance from P to the origin, changes. We calculate $\partial w/\partial x$ in this case (our first solution above) to be

$$\frac{\partial w}{\partial x} = 2x + 4x^3 + 4xy^2.$$

At the point $P(1, 0, 1)$, the value of this derivative is

$$\frac{\partial w}{\partial x} = 2 + 4 + 0 = 6.$$

If we take x and z to be independent, then we find $\partial w/\partial x$ by holding z fixed while x varies. Since the z -coordinate of P is 1, varying x moves P along a circle in the plane $z = 1$. As P moves along this circle, its distance from the origin remains constant, and w , being the square of this distance, does not change. That is,

$$\frac{\partial w}{\partial x} = 0,$$

as we found in our second solution. ■

How to Find $\partial w/\partial x$ When the Variables in $w = f(x, y, z)$ Are Constrained by Another Equation

As we saw in Example 1, a typical routine for finding $\partial w/\partial x$ when the variables in the function $w = f(x, y, z)$ are related by another equation has three steps. These steps apply to finding $\partial w/\partial y$ and $\partial w/\partial z$ as well.

1. *Decide* which variables are to be dependent and which are to be independent. (In practice, the decision is based on the physical or theoretical context of our work. In the exercises at the end of this section, we say which variables are which.)
2. *Eliminate* the other dependent variable(s) in the expression for w .
3. *Differentiate* as usual.

If we cannot carry out Step 2 after deciding which variables are dependent, we differentiate the equations as they are and try to solve for $\partial w/\partial x$ afterward. The next example shows how this is done.

EXAMPLE 2 Find $\partial w/\partial x$ at the point $(x, y, z) = (2, -1, 1)$ if

$$w = x^2 + y^2 + z^2, \quad z^3 - xy + yz + y^3 = 1,$$

and x and y are the independent variables.

Solution It is not convenient to eliminate z in the expression for w . We therefore differentiate both equations implicitly with respect to x , treating x and y as independent variables and w and z as dependent variables. This gives

$$\frac{\partial w}{\partial x} = 2x + 2z \frac{\partial z}{\partial x} \quad (3)$$

and

$$3z^2 \frac{\partial z}{\partial x} - y + y \frac{\partial z}{\partial x} + 0 = 0. \quad (4)$$

These equations may now be combined to express $\partial w/\partial x$ in terms of x , y , and z . We solve Equation (4) for $\partial z/\partial x$ to get

$$\frac{\partial z}{\partial x} = \frac{y}{y + 3z^2}$$

and substitute into Equation (3) to get

$$\frac{\partial w}{\partial x} = 2x + \frac{2yz}{y + 3z^2}.$$

The value of this derivative at $(x, y, z) = (2, -1, 1)$ is

$$\left. \frac{\partial w}{\partial x} \right|_{(2, -1, 1)} = 2(2) + \frac{2(-1)(1)}{-1 + 3(1)^2} = 4 + \frac{-2}{2} = 3. \quad \blacksquare$$

HISTORICAL BIOGRAPHY

Sonya Kovalevsky
(1850–1891)

Kovalevsky, a Russian mathematician, primarily worked on the theory of partial differential equations, and a central result on the existence of solutions still bears her name. She published numerous papers on partial differential equations, eventually gaining recognition as the first woman to be elected a member of the Russian Imperial Academy of Sciences in 1889.

To know more, visit the companion Website.

Notation

To show what variables are assumed to be independent in calculating a derivative, we can use the following notation:

$$\left(\frac{\partial w}{\partial x} \right)_y \quad \partial w/\partial x \text{ with } x \text{ and } y \text{ independent}$$

$$\left(\frac{\partial f}{\partial y} \right)_{x, t} \quad \partial f/\partial y \text{ with } y, x, \text{ and } t \text{ independent.}$$

EXAMPLE 3 Find $(\partial w / \partial x)_{y,z}$ if $w = x^2 + y - z + \sin t$ and $x + y = t$.

Solution With x, y, z independent, we have

$$\begin{aligned} t &= x + y, & w &= x^2 + y - z + \sin(x + y) \\ \left(\frac{\partial w}{\partial x}\right)_{y,z} &= 2x + 0 - 0 + \cos(x + y) \frac{\partial}{\partial x}(x + y) \\ &= 2x + \cos(x + y). \end{aligned}$$

Arrow Diagrams

In solving problems like the one in Example 3, it often helps to start with an arrow diagram that shows how the variables and functions are related. If

$$w = x^2 + y - z + \sin t \quad \text{and} \quad x + y = t$$

and we are asked to find $\partial w / \partial x$ when x, y , and z are independent, the appropriate diagram is one like this:

$$\begin{array}{ccc} \begin{pmatrix} x \\ y \\ z \end{pmatrix} & \rightarrow & \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \rightarrow w \end{array} \quad (5)$$

Independent variables Intermediate variables Dependent variable

To avoid confusion between the independent and intermediate variables with the same symbolic names in the diagram, it is helpful to rename the intermediate variables (so they are seen as *functions* of the independent variables). Thus, let $u = x$, $v = y$, and $s = z$ denote the renamed intermediate variables. With this notation, the arrow diagram becomes

$$\begin{array}{ccc} \begin{pmatrix} x \\ y \\ z \end{pmatrix} & \rightarrow & \begin{pmatrix} u \\ v \\ s \\ t \end{pmatrix} \rightarrow w \end{array} \quad (6)$$

Independent variables Intermediate variables and relations Dependent variable

$u = x$
 $v = y$
 $s = z$
 $t = x + y$

The diagram shows the independent variables on the left, the intermediate variables and their relation to the independent variables in the middle, and the dependent variable on the right. The function w now becomes

$$w = u^2 + v - s + \sin t,$$

where

$$u = x, \quad v = y, \quad s = z, \quad \text{and} \quad t = x + y.$$

To find $\partial w / \partial x$, we apply the four-variable form of the Chain Rule to w , guided by the arrow diagram in Equation (6):

$$\begin{aligned} \frac{\partial w}{\partial x} &= \frac{\partial w}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial w}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial w}{\partial t} \frac{\partial t}{\partial x} \\ &= (2u)(1) + (1)(0) + (-1)(0) + (\cos t)(1) \\ &= 2u + \cos t \\ &= 2x + \cos(x + y). \end{aligned}$$

Substitute the original independent variables $u = x$ and $t = x + y$

EXERCISES 13.10

Finding Partial Derivatives with Constrained Variables

In Exercises 1–3, begin by drawing a diagram that shows the relations among the variables.

1. If $w = x^2 + y^2 + z^2$ and $z = x^2 + y^2$, find

a. $\left(\frac{\partial w}{\partial y}\right)_z$ b. $\left(\frac{\partial w}{\partial z}\right)_x$ c. $\left(\frac{\partial w}{\partial z}\right)_y$.

2. If $w = x^2 + y - z + \sin t$ and $x + y = t$, find

a. $\left(\frac{\partial w}{\partial y}\right)_{x,z}$ b. $\left(\frac{\partial w}{\partial y}\right)_{z,t}$ c. $\left(\frac{\partial w}{\partial z}\right)_{x,y}$

d. $\left(\frac{\partial w}{\partial z}\right)_{y,t}$ e. $\left(\frac{\partial w}{\partial t}\right)_{x,z}$ f. $\left(\frac{\partial w}{\partial t}\right)_{y,z}$.

3. Let $U = f(P, V, T)$ be the internal energy of a gas that obeys the ideal gas law $PV = nRT$ (n and R constant). Find

a. $\left(\frac{\partial U}{\partial P}\right)_V$ b. $\left(\frac{\partial U}{\partial T}\right)_V$.

4. Find

a. $\left(\frac{\partial w}{\partial x}\right)_y$ b. $\left(\frac{\partial w}{\partial z}\right)_y$

at the point $(x, y, z) = (0, 1, \pi)$ if

$$w = x^2 + y^2 + z^2 \quad \text{and} \quad y \sin z + z \sin x = 0.$$

5. Find

a. $\left(\frac{\partial w}{\partial y}\right)_x$ b. $\left(\frac{\partial w}{\partial y}\right)_z$

at the point $(w, x, y, z) = (4, 2, 1, -1)$ if

$$w = x^2y^2 + yz - z^3 \quad \text{and} \quad x^2 + y^2 + z^2 = 6.$$

6. Find $(\partial u / \partial y)_x$ at the point $(u, v) = (\sqrt{2}, 1)$ if $x = u^2 + v^2$ and $y = uv$.

7. Suppose that $x^2 + y^2 = r^2$ and $x = r \cos \theta$, as in polar coordinates. Find

$$\left(\frac{\partial x}{\partial r}\right)_\theta \quad \text{and} \quad \left(\frac{\partial r}{\partial x}\right)_y.$$

8. Suppose that

$$w = x^2 - y^2 + 4z + t \quad \text{and} \quad x + 2z + t = 25.$$

Show that the equations

$$\frac{\partial w}{\partial x} = 2x - 1 \quad \text{and} \quad \frac{\partial w}{\partial x} = 2x - 2$$

each give $\partial w / \partial x$, depending on which variables are chosen to be dependent and which variables are chosen to be independent. Identify the independent variables in each case.

Theory and Examples

9. Establish the fact, widely used in hydrodynamics, that if $f(x, y, z) = 0$, then

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1.$$

(Hint: Express all the derivatives in terms of the formal partial derivatives $\partial f / \partial x$, $\partial f / \partial y$, and $\partial f / \partial z$.)

10. If $z = x + f(u)$, where $u = xy$, show that

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = x.$$

11. Suppose that the equation $g(x, y, z) = 0$ determines z as a differentiable function of the independent variables x and y and that $g_z \neq 0$. Show that

$$\left(\frac{\partial z}{\partial y}\right)_x = -\frac{\partial g / \partial y}{\partial g / \partial z}.$$

12. Suppose that $f(x, y, z, w) = 0$ and $g(x, y, z, w) = 0$ determine z and w as differentiable functions of the independent variables x and y , and suppose that

$$\frac{\partial f}{\partial z} \frac{\partial g}{\partial w} - \frac{\partial f}{\partial w} \frac{\partial g}{\partial z} \neq 0.$$

Show that

$$\left(\frac{\partial z}{\partial x}\right)_y = -\frac{\frac{\partial f}{\partial x} \frac{\partial g}{\partial w} - \frac{\partial f}{\partial w} \frac{\partial g}{\partial x}}{\frac{\partial f}{\partial z} \frac{\partial g}{\partial w} - \frac{\partial f}{\partial w} \frac{\partial g}{\partial z}}$$

and

$$\left(\frac{\partial w}{\partial y}\right)_x = -\frac{\frac{\partial f}{\partial z} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial z}}{\frac{\partial f}{\partial z} \frac{\partial g}{\partial w} - \frac{\partial f}{\partial w} \frac{\partial g}{\partial z}}.$$

CHAPTER 13 Questions to Guide Your Review

- What is a real-valued function of two independent variables? Three independent variables? Give examples.
- What does it mean for sets in the plane or in space to be open? Closed? Give examples. Give examples of sets that are neither open nor closed.
- How can you display the values of a function $f(x, y)$ of two independent variables graphically? How do you do the same for a function $f(x, y, z)$ of three independent variables?
- What does it mean for a function $f(x, y)$ to have limit L as $(x, y) \rightarrow (x_0, y_0)$? What are the basic properties of limits of functions of two independent variables?
- When is a function of two (three) independent variables continuous at a point in its domain? Give examples of functions that are continuous at some points but not others.
- What can be said about algebraic combinations and compositions of continuous functions?

7. Explain the two-path test for nonexistence of limits.
8. How are the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ of a function $f(x, y)$ defined? How are they interpreted and calculated?
9. How does the relation between first partial derivatives and continuity of functions of two independent variables differ from the relation between first derivatives and continuity for real-valued functions of a single independent variable? Give an example.
10. What is the Mixed Derivative Theorem for mixed second-order partial derivatives? How can it help in calculating partial derivatives of second and higher orders? Give examples.
11. What does it mean for a function $f(x, y)$ to be differentiable? What does the Increment Theorem say about differentiability?
12. How can you sometimes decide from examining f_x and f_y that a function $f(x, y)$ is differentiable? What is the relation between the differentiability of f and the continuity of f at a point?
13. What is the general Chain Rule? What form does it take for functions of two independent variables? Three independent variables? Functions defined on surfaces? How do you diagram these different forms? Give examples. What pattern enables one to remember all the different forms?
14. What is the derivative of a function $f(x, y)$ at a point P_0 in the direction of a unit vector $\mathbf{u}$? What rate does it describe? What geometric interpretation does it have? Give examples.
15. What is the gradient vector of a differentiable function $f(x, y)$? How is it related to the function's directional derivatives? State the analogous results for functions of three independent variables.
16. How do you find the tangent line at a point on a level curve of a differentiable function $f(x, y)$? How do you find the tangent plane and normal line at a point on a level surface of a differentiable function $f(x, y, z)$? Give examples.
17. How can you use directional derivatives to estimate change?
18. How do you linearize a function $f(x, y)$ of two independent variables at a point (x_0, y_0) ? Why might you want to do this? How do you linearize a function of three independent variables?
19. What can you say about the accuracy of linear approximations of functions of two (three) independent variables?
20. If (x, y) moves from (x_0, y_0) to a point $(x_0 + dx, y_0 + dy)$ nearby, how can you estimate the resulting change in the value of a differentiable function $f(x, y)$? Give an example.
21. How do you define local maxima, local minima, and saddle points for a differentiable function $f(x, y)$? Give examples.
22. What derivative tests are available for determining the local extreme values of a function $f(x, y)$? How do they enable you to narrow your search for these values? Give examples.
23. How do you find the extrema of a continuous function $f(x, y)$ on a closed bounded region of the xy -plane? Give an example.
24. Describe the method of Lagrange multipliers and give examples.
25. How does Taylor's formula for a function $f(x, y)$ generate polynomial approximations and error estimates?
26. If $w = f(x, y, z)$, where the variables x, y , and z are constrained by an equation $g(x, y, z) = 0$, what is the meaning of the notation $(\partial w/\partial x)_y$? How can an arrow diagram help you calculate this partial derivative with constrained variables? Give examples.

CHAPTER 13 Practice Exercises

Domain, Range, and Level Curves

In Exercises 1–4, find the domain and range of the given function and identify its level curves. Sketch a typical level curve.

1. $f(x, y) = 9x^2 + y^2$
2. $f(x, y) = e^{x+y}$
3. $g(x, y) = 1/xy$
4. $g(x, y) = \sqrt{x^2 - y}$

In Exercises 5–8, find the domain and range of the given function and identify its level surfaces. Sketch a typical level surface.

5. $f(x, y, z) = x^2 + y^2 - z$
6. $g(x, y, z) = x^2 + 4y^2 + 9z^2$
7. $h(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$
8. $k(x, y, z) = \frac{1}{x^2 + y^2 + z^2 + 1}$

Evaluating Limits

Find the limits in Exercises 9–14.

9. $\lim_{(x, y) \rightarrow (\pi, \ln 2)} e^y \cos x$
10. $\lim_{(x, y) \rightarrow (0, 0)} \frac{2 + y}{x + \cos y}$
11. $\lim_{(x, y) \rightarrow (1, 1)} \frac{x - y}{x^2 - y^2}$
12. $\lim_{(x, y) \rightarrow (1, 1)} \frac{x^3 y^3 - 1}{xy - 1}$
13. $\lim_{P \rightarrow (1, -1, e)} \ln|x + y + z|$
14. $\lim_{P \rightarrow (1, -1, -1)} \arctan(x + y + z)$

By considering different paths of approach, show that the limits in Exercises 15 and 16 do not exist.

15. $\lim_{\substack{(x, y) \rightarrow (0, 0) \\ y \neq x^2}} \frac{y}{x^2 - y}$
16. $\lim_{\substack{(x, y) \rightarrow (0, 0) \\ xy \neq 0}} \frac{x^2 + y^2}{xy}$

17. Continuous extension Let $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$ for $(x, y) \neq (0, 0)$. Is it possible to define $f(0, 0)$ in a way that makes f continuous at the origin? Why?

18. Continuous extension Let

$$f(x, y) = \begin{cases} \frac{\sin(x - y)}{|x| + |y|}, & |x| + |y| \neq 0 \\ 0, & (x, y) = (0, 0). \end{cases}$$

Is f continuous at the origin? Why?

Partial Derivatives

In Exercises 19–24, find the partial derivative of the function with respect to each variable.

19. $g(r, \theta) = r \cos \theta + r \sin \theta$
20. $f(x, y) = \frac{1}{2} \ln(x^2 + y^2) + \arctan \frac{y}{x}$

21. $f(R_1, R_2, R_3) = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
22. $h(x, y, z) = \sin(2\pi x + y - 3z)$
23. $P(n, R, T, V) = \frac{nRT}{V}$ (the ideal gas law)
24. $f(r, l, T, w) = \frac{1}{2rl} \sqrt{\frac{T}{\pi w}}$

Second-Order Partial Derivatives

Find the second-order partial derivatives of the functions in Exercises 25–28.

25. $g(x, y) = y + \frac{x}{y}$ 26. $g(x, y) = e^x + y \sin x$
27. $f(x, y) = x + xy - 5x^3 + \ln(x^2 + 1)$
28. $f(x, y) = y^2 - 3xy + \cos y + 7e^y$

Chain Rule Calculations

29. Find dw/dt at $t = 0$ if $w = \sin(xy + \pi)$, $x = e^t$, and $y = \ln(t + 1)$.
30. Find dw/dt at $t = 1$ if $w = xe^y + y \sin z - \cos z$, $x = 2\sqrt{t}$, $y = t - 1 + \ln t$, and $z = \pi t$.
31. Find $\partial w/\partial r$ and $\partial w/\partial s$ when $r = \pi$ and $s = 0$ if $w = \sin(2x - y)$, $x = r + \sin s$, $y = rs$.
32. Find $\partial w/\partial u$ and $\partial w/\partial v$ when $u = v = 0$ if $w = \ln\sqrt{1 + x^2} - \tan^{-1}x$ and $x = 2e^u \cos v$.
33. Find the value of the derivative of $f(x, y, z) = xy + yz + xz$ with respect to t on the curve $x = \cos t$, $y = \sin t$, $z = \cos 2t$ at $t = 1$.
34. Show that if $w = f(s)$ is any differentiable function of s and if $s = y + 5x$, then

$$\frac{\partial w}{\partial x} - 5 \frac{\partial w}{\partial y} = 0.$$

Implicit Differentiation

Assuming that the equations in Exercises 35 and 36 define y as a differentiable function of x , find the value of dy/dx at point P .

35. $1 - x - y^2 - \sin xy = 0$, $P(0, 1)$
36. $2xy + e^{x+y} - 2 = 0$, $P(0, \ln 2)$

Directional Derivatives

In Exercises 37–40, find the directions in which f increases and decreases most rapidly at P_0 and find the derivative of f in each direction. Also, find the derivative of f at P_0 in the direction of the vector $\mathbf{v}$.

37. $f(x, y) = \cos x \cos y$, $P_0(\pi/4, \pi/4)$, $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$
38. $f(x, y) = x^2 e^{-2y}$, $P_0(1, 0)$, $\mathbf{v} = \mathbf{i} + \mathbf{j}$
39. $f(x, y, z) = \ln(2x + 3y + 6z)$, $P_0(-1, -1, 1)$, $\mathbf{v} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$
40. $f(x, y, z) = x^2 + 3xy - z^2 + 2y + z + 4$, $P_0(0, 0, 0)$, $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{k}$
41. **Derivative in velocity direction** Find the derivative of $f(x, y, z) = xyz$ in the direction of the velocity vector of the helix

$$\mathbf{r}(t) = (\cos 3t)\mathbf{i} + (\sin 3t)\mathbf{j} + 3t\mathbf{k}$$

at $t = \pi/3$.

42. **Maximum directional derivative** What is the largest value that the directional derivative of $f(x, y, z) = xyz$ can have at the point $(1, 1, 1)$?

43. **Directional derivatives with given values** At the point $(1, 2)$, the function $f(x, y)$ has a derivative of 2 in the direction toward $(2, 2)$ and a derivative of -2 in the direction toward $(1, 1)$.

- a. Find $f_x(1, 2)$ and $f_y(1, 2)$.
- b. Find the derivative of f at $(1, 2)$ in the direction toward the point $(4, 6)$.

44. Which of the following statements are true if $f(x, y)$ is differentiable at (x_0, y_0) ? Give reasons for your answers.

- a. If $\mathbf{u}$ is a unit vector, the derivative of f at (x_0, y_0) in the direction of $\mathbf{u}$ is $(f_x(x_0, y_0)\mathbf{i} + f_y(x_0, y_0)\mathbf{j}) \cdot \mathbf{u}$.
- b. The derivative of f at (x_0, y_0) in the direction of $\mathbf{u}$ is a vector.
- c. The directional derivative of f at (x_0, y_0) has its greatest value in the direction of ∇f .
- d. At (x_0, y_0) , vector ∇f is normal to the curve $f(x, y) = f(x_0, y_0)$.

Gradients, Tangent Planes, and Normal Lines

In Exercises 45 and 46, sketch the surface $f(x, y, z) = c$ together with ∇f at the given points.

45. $x^2 + y + z^2 = 0$; $(0, -1, \pm 1)$, $(0, 0, 0)$
46. $y^2 + z^2 = 4$; $(2, \pm 2, 0)$, $(2, 0, \pm 2)$

In Exercises 47 and 48, find an equation for the plane tangent to the level surface $f(x, y, z) = c$ at the point P_0 . Also, find parametric equations for the line that is normal to the surface at P_0 .

47. $x^2 - y - 5z = 0$, $P_0(2, -1, 1)$
48. $x^2 + y^2 + z = 4$, $P_0(1, 1, 2)$

In Exercises 49 and 50, find an equation for the plane tangent to the surface $z = f(x, y)$ at the given point.

49. $z = \ln(x^2 + y^2)$, $(0, 1, 0)$
50. $z = 1/(x^2 + y^2)$, $(1, 1, 1/2)$

In Exercises 51 and 52, find equations for the lines that are tangent and normal to the level curve $f(x, y) = c$ at the point P_0 . Then sketch the lines and level curve together with ∇f at P_0 .

51. $y - \sin x = 1$, $P_0(\pi, 1)$
52. $\frac{y^2}{2} - \frac{x^2}{2} = \frac{3}{2}$, $P_0(1, 2)$

Tangent Lines to Curves

In Exercises 53 and 54, find parametric equations for the line that is tangent to the curve of intersection of the surfaces at the given point.

53. Surfaces: $x^2 + 2y + 2z = 4$, $y = 1$
Point: $(1, 1, 1/2)$
54. Surfaces: $x + y^2 + z = 2$, $y = 1$
Point: $(1/2, 1, 1/2)$

Linearizations

In Exercises 55 and 56, find the linearization $L(x, y)$ of the function $f(x, y)$ at the point P_0 . Then find an upper bound for the magnitude of the error E in the approximation $f(x, y) \approx L(x, y)$ over the rectangle R .

55. $f(x, y) = \sin x \cos y$, $P_0(\pi/4, \pi/4)$
 $R: \left|x - \frac{\pi}{4}\right| \leq 0.1, \left|y - \frac{\pi}{4}\right| \leq 0.1$
56. $f(x, y) = xy - 3y^2 + 2$, $P_0(1, 1)$
 $R: |x - 1| \leq 0.1, |y - 1| \leq 0.2$

Find the linearizations of the functions in Exercises 57 and 58 at the given points.

57. $f(x, y, z) = xy + 2yz - 3xz$ at $(1, 0, 0)$ and $(1, 1, 0)$
 58. $f(x, y, z) = \sqrt{2} \cos x \sin(y + z)$ at $(0, 0, \pi/4)$ and $(\pi/4, \pi/4, 0)$

Estimates and Sensitivity to Change

59. **Measuring the volume of a pipeline** You plan to calculate the volume inside a stretch of pipeline that is about 36 cm in diameter and 1 km long. With which measurement should you be more careful, the length or the diameter? Why?
60. **Sensitivity to change** Is $f(x, y) = x^2 - xy + y^2 - 3$ more sensitive to changes in x or to changes in y when it is near the point $(1, 2)$? How do you know?
61. **Change in an electrical circuit** Suppose that the current I (amperes) in an electrical circuit is related to the voltage V (volts) and the resistance R (ohms) by the equation $I = V/R$. If the voltage drops from 24 to 23 volts and the resistance drops from 100 to 80 ohms, will I increase or decrease? By about how much? Is the change in I more sensitive to change in the voltage or to change in the resistance? How do you know?
62. **Maximum error in estimating the area of an ellipse** If $a = 10$ cm and $b = 16$ cm to the nearest millimeter, what should you expect the maximum percentage error to be in the calculated area $A = \pi ab$ of the ellipse $x^2/a^2 + y^2/b^2 = 1$?
63. **Error in estimating a product** Let $y = uv$ and $z = u + v$, where u and v are positive independent variables.
- If u is measured with an error of 2% and v with an error of 3%, about what is the percentage error in the calculated value of y ?
 - Show that the percentage error in the calculated value of z is less than the percentage error in the value of y .
64. **Cardiac index** To make different people comparable in studies of cardiac output, researchers divide the measured cardiac output by the body surface area to find the *cardiac index* C :

$$C = \frac{\text{cardiac output}}{\text{body surface area}}.$$

The body surface area B of a person with weight w and height h is approximated by the formula

$$B = 71.84w^{0.425}h^{0.725},$$

which gives B in square centimeters when w is measured in kilograms and h in centimeters. You are about to calculate the cardiac index of a person 180 cm tall, weighing 70 kg, with cardiac output of 7 L/min. Which will have a greater effect on the calculation, a 1-kg error in measuring the weight or a 1-cm error in measuring the height?

Local Extrema

Test the functions in Exercises 65–70 for local maxima and minima and saddle points. Find each function's value at these points.

65. $f(x, y) = x^2 - xy + y^2 + 2x + 2y - 4$
 66. $f(x, y) = 5x^2 + 4xy - 2y^2 + 4x - 4y$
 67. $f(x, y) = 2x^3 + 3xy + 2y^3$
 68. $f(x, y) = x^3 + y^3 - 3xy + 15$
 69. $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2$
 70. $f(x, y) = x^4 - 8x^2 + 3y^2 - 6y$

Absolute Extrema

In Exercises 71–78, find the absolute maximum and minimum values of f on the region R .

71. $f(x, y) = x^2 + xy + y^2 - 3x + 3y$
 R : The triangular region cut from the first quadrant by the line $x + y = 4$
72. $f(x, y) = x^2 - y^2 - 2x + 4y + 1$
 R : The rectangular region in the first quadrant bounded by the coordinate axes and the lines $x = 4$ and $y = 2$
73. $f(x, y) = y^2 - xy - 3y + 2x$
 R : The square region enclosed by the lines $x = \pm 2$ and $y = \pm 2$
74. $f(x, y) = 2x + 2y - x^2 - y^2$
 R : The square region bounded by the coordinate axes and the lines $x = 2, y = 2$ in the first quadrant
75. $f(x, y) = x^2 - y^2 - 2x + 4y$
 R : The triangular region bounded below by the x -axis, above by the line $y = x + 2$, and on the right by the line $x = 2$
76. $f(x, y) = 4xy - x^4 - y^4 + 16$
 R : The triangular region bounded below by the line $y = -2$, above by the line $y = x$, and on the right by the line $x = 2$
77. $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2$
 R : The square region enclosed by the lines $x = \pm 1$ and $y = \pm 1$
78. $f(x, y) = x^3 + 3xy + y^3 + 1$
 R : The square region enclosed by the lines $x = \pm 1$ and $y = \pm 1$

Lagrange Multipliers

79. **Extrema on a circle** Find the extreme values of $f(x, y) = x^3 + y^2$ on the circle $x^2 + y^2 = 1$.
80. **Extrema on a circle** Find the extreme values of $f(x, y) = xy$ on the circle $x^2 + y^2 = 1$.
81. **Extrema in a disk** Find the extreme values of $f(x, y) = x^2 + 3y^2 + 2y$ on the unit disk $x^2 + y^2 \leq 1$.
82. **Extrema in a disk** Find the extreme values of $f(x, y) = x^2 + y^2 - 3x - xy$ on the disk $x^2 + y^2 \leq 9$.
83. **Extrema on a sphere** Find the extreme values of $f(x, y, z) = x - y + z$ on the unit sphere $x^2 + y^2 + z^2 = 1$.
84. **Minimum distance to origin** Find the points on the surface $x^2 - zy = 4$ closest to the origin.
85. **Minimizing cost of a box** A closed rectangular box is to have volume V cm³. The cost of the material used in the box is a cents/cm² for top and bottom, b cents/cm² for front and back, and c cents/cm² for the remaining sides. What dimensions minimize the total cost of materials?
86. **Least volume** Find the plane $x/a + y/b + z/c = 1$ that passes through the point $(2, 1, 2)$ and cuts off the least volume from the first octant.
87. **Extrema on curve of intersecting surfaces** Find the extreme values of $f(x, y, z) = x(y + z)$ on the curve of intersection of the right circular cylinder $x^2 + y^2 = 1$ and the hyperbolic cylinder $xz = 1$.
88. **Minimum distance to origin on curve of intersecting plane and cone** Find the point closest to the origin on the curve of intersection of the plane $x + y + z = 1$ and the cone $z^2 = 2x^2 + 2y^2$.

Theory and Examples

89. Let $w = f(r, \theta)$, $r = \sqrt{x^2 + y^2}$, and $\theta = \tan^{-1}(y/x)$. Find $\partial w / \partial x$ and $\partial w / \partial y$, and express your answers in terms of r and θ .
90. Let $z = f(u, v)$, $u = ax + by$, and $v = ax - by$. Express z_x and z_y in terms of f_u , f_v , and the constants a and b .
91. If a and b are constants, $w = u^3 + \tanh u + \cos u$, and $u = ax + by$, show that

$$a \frac{\partial w}{\partial y} = b \frac{\partial w}{\partial x}.$$

92. **Using the Chain Rule** If $w = \ln(x^2 + y^2 + 2z)$, $x = r + s$, $y = r - s$, and $z = 2rs$, find w_r and w_s by the Chain Rule. Then check your answer another way.
93. **Angle between vectors** The equations $e^u \cos v - x = 0$ and $e^u \sin v - y = 0$ define u and v as differentiable functions of x and y . Show that the angle between the vectors

$$\frac{\partial u}{\partial x} \mathbf{i} + \frac{\partial u}{\partial y} \mathbf{j} \quad \text{and} \quad \frac{\partial v}{\partial x} \mathbf{i} + \frac{\partial v}{\partial y} \mathbf{j}$$

is constant.

94. **Polar coordinates and second derivatives** Introducing polar coordinates $x = r \cos \theta$ and $y = r \sin \theta$ changes $f(x, y)$ to $g(r, \theta)$. Find the value of $\partial^2 g / \partial \theta^2$ at the point $(r, \theta) = (2, \pi/2)$, given that

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2} = 1$$

at that point.

95. **Normal line parallel to a plane** Find the points on the surface

$$(y + z)^2 + (z - x)^2 = 16$$

where the normal line is parallel to the yz -plane.

96. **Tangent plane parallel to xy -plane** Find the points on the surface

$$xy + yz + zx - x - z^2 = 0$$

where the tangent plane is parallel to the xy -plane.

97. **When gradient is parallel to position vector** Suppose that $\nabla f(x, y, z)$ is always parallel to the position vector $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Show that $f(0, 0, a) = f(0, 0, -a)$ for any a .

98. **One-sided directional derivative in all directions, but no gradient** The one-sided directional derivative of f at $P(x_0, y_0, z_0)$ in the direction $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is the number

$$\lim_{s \rightarrow 0^+} \frac{f(x_0 + su_1, y_0 + su_2, z_0 + su_3) - f(x_0, y_0, z_0)}{s}.$$

Show that the one-sided directional derivative of

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

at the origin equals 1 in any direction but that f has no gradient vector at the origin.

99. **Normal line through origin** Show that the line normal to the surface $xy + z = 2$ at the point $(1, 1, 1)$ passes through the origin.

100. **Tangent plane and normal line**

- a. Sketch the surface $x^2 - y^2 + z^2 = 4$.
- b. Find a vector normal to the surface at $(2, -3, 3)$. Add the vector to your sketch.
- c. Find equations for the tangent plane and the normal line at $(2, -3, 3)$.

Partial Derivatives with Constrained Variables

In Exercises 101 and 102, begin by drawing a diagram that shows the relations among the variables.

101. If $w = x^2 e^{yz}$ and $z = x^2 - y^2$ find

$$\text{a. } \left(\frac{\partial w}{\partial y} \right)_z \quad \text{b. } \left(\frac{\partial w}{\partial z} \right)_x \quad \text{c. } \left(\frac{\partial w}{\partial z} \right)_y.$$

102. Let $U = f(P, V, T)$ be the internal energy of a gas that obeys the ideal gas law $PV = nRT$ (n and R constant). Find

$$\text{a. } \left(\frac{\partial U}{\partial T} \right)_P \quad \text{b. } \left(\frac{\partial U}{\partial V} \right)_T.$$

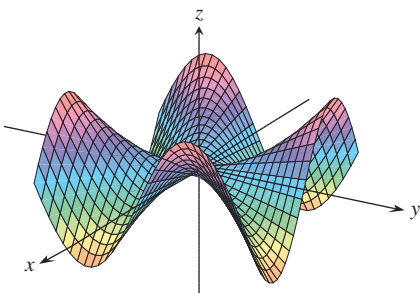
CHAPTER 13 Additional and Advanced Exercises

Partial Derivatives

1. **Function with saddle at the origin** If you did Exercise 64 in Section 13.2, you know that the function

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

(see the accompanying figure) is continuous at $(0, 0)$. Find $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$.



2. **Finding a function from second partials** Find a function $w = f(x, y)$ whose first partial derivatives are $\partial w / \partial x = 1 + e^x \cos y$ and $\partial w / \partial y = 2y - e^x \sin y$ and whose value at the point $(\ln 2, 0)$ is $\ln 2$.
3. **A proof of Leibniz's Rule** Leibniz's Rule says that if f is continuous on $[a, b]$ and if $u(x)$ and $v(x)$ are differentiable functions of x whose values lie in $[a, b]$, then

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x)) \frac{dv}{dx} - f(u(x)) \frac{du}{dx}.$$

Prove the rule by setting

$$g(u, v) = \int_u^v f(t) dt, \quad u = u(x), \quad v = v(x)$$

and calculating dg/dx with the Chain Rule.

4. **Finding a function with constrained second partials** Suppose that f is a twice-differentiable function of r , that $r = \sqrt{x^2 + y^2 + z^2}$, and that

$$f_{xx} + f_{yy} + f_{zz} = 0.$$

Show that for some constants a and b ,

$$f(r) = \frac{a}{r} + b.$$

5. Homogeneous functions A function $f(x, y)$ is *homogeneous of degree n* (n a nonnegative integer) if $f(tx, ty) = t^n f(x, y)$ for all t , x , and y . For such a function (sufficiently differentiable), prove that

$$\text{a. } x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf(x, y)$$

$$\text{b. } x^2 \left(\frac{\partial^2 f}{\partial x^2} \right) + 2xy \left(\frac{\partial^2 f}{\partial x \partial y} \right) + y^2 \left(\frac{\partial^2 f}{\partial y^2} \right) = n(n-1)f.$$

6. Surface in polar coordinates Let

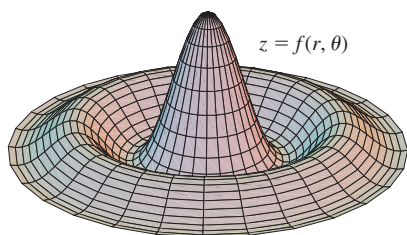
$$f(r, \theta) = \begin{cases} \frac{\sin 6r}{6r}, & r \neq 0 \\ 1, & r = 0, \end{cases}$$

where r and θ are polar coordinates. Find

$$\text{a. } \lim_{r \rightarrow 0} f(r, \theta)$$

$$\text{b. } f_r(0, 0)$$

$$\text{c. } f_\theta(r, \theta), \quad r \neq 0.$$



Gradients and Tangents

7. Properties of position vectors Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and let $r = |\mathbf{r}|$.

$$\text{a. Show that } \nabla r = \mathbf{r}/r.$$

$$\text{b. Show that } \nabla(r^n) = nr^{n-2}\mathbf{r}.$$

$$\text{c. Find a function whose gradient equals } \mathbf{r}.$$

$$\text{d. Show that } \mathbf{r} \cdot d\mathbf{r} = r dr.$$

$$\text{e. Show that } \nabla(\mathbf{A} \cdot \mathbf{r}) = \mathbf{A} \text{ for any constant vector } \mathbf{A}.$$

8. Gradient orthogonal to tangent Suppose that a differentiable function $f(x, y)$ has the constant value c along the differentiable curve $x = g(t)$, $y = h(t)$; that is,

$$f(g(t), h(t)) = c$$

for all values of t . Differentiate both sides of this equation with respect to t to show that ∇f is orthogonal to the curve's tangent vector at every point on the curve.

9. Curve tangent to a surface Show that the curve

$$\mathbf{r}(t) = (\ln t)\mathbf{i} + (t \ln t)\mathbf{j} + t\mathbf{k}$$

is tangent to the surface

$$xz^2 - yz + \cos xy = 1$$

at $(0, 0, 1)$.

10. Curve tangent to a surface Show that the curve

$$\mathbf{r}(t) = \left(\frac{t^3}{4} - 2 \right) \mathbf{i} + \left(\frac{4}{t} - 3 \right) \mathbf{j} + \cos(t-2) \mathbf{k}$$

is tangent to the surface

$$x^3 + y^3 + z^3 - xyz = 0$$

at $(0, -1, 1)$.

Extreme Values

11. Extrema on a surface Show that the only possible maxima and minima of z on the surface $z = x^3 + y^3 - 9xy + 27$ occur at $(0, 0)$ and $(3, 3)$. Show that neither a maximum nor a minimum occurs at $(0, 0)$. Determine whether z has a maximum or a minimum at $(3, 3)$.

12. Maximum in closed first quadrant Find the maximum value of $f(x, y) = 6xye^{-(2x+3y)}$ in the closed first quadrant (includes the nonnegative axes).

13. Minimum volume cut from first octant Find the minimum volume for a region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and a plane tangent to the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

at a point in the first octant.

14. Minimum distance from a line to a parabola in xy -plane By minimizing the function $f(x, y, u, v) = (x - u)^2 + (y - v)^2$ subject to the constraints $y = x + 1$ and $u = v^2$, find the minimum distance in the xy -plane from the line $y = x + 1$ to the parabola $y^2 = x$.

Theory and Examples

15. Boundedness of first partials implies continuity Prove the following theorem: If $f(x, y)$ is defined in an open region R of the xy -plane and if f_x and f_y are bounded on R , then $f(x, y)$ is continuous on R . (The assumption of boundedness is essential.)

16. Suppose that $\mathbf{r}(t) = g(t)\mathbf{i} + h(t)\mathbf{j} + k(t)\mathbf{k}$ is a smooth curve in the domain of a differentiable function $f(x, y, z)$. Describe the relation among df/dt , ∇f , and $\mathbf{v} = d\mathbf{r}/dt$. What can be said about ∇f and $\mathbf{v}$ at interior points of the curve where f has extreme values relative to its other values on the curve? Give reasons for your answer.

17. Finding functions from partial derivatives Suppose that f and g are functions of x and y such that

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x} \quad \text{and} \quad \frac{\partial f}{\partial x} = \frac{\partial g}{\partial y},$$

and suppose that

$$\frac{\partial f}{\partial x} = 0, \quad f(1, 2) = g(1, 2) = 5, \quad \text{and} \quad f(0, 0) = 4.$$

Find $f(x, y)$ and $g(x, y)$.

18. Rate of change of the rate of change We know that if $f(x, y)$ is a function of two variables and if $\mathbf{u} = a\mathbf{i} + b\mathbf{j}$ is a unit vector, then $D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b$ is the rate of change of $f(x, y)$ at (x, y) in the direction of $\mathbf{u}$. Give a similar formula for the rate of change of the rate of change of $f(x, y)$ at (x, y) in the direction $\mathbf{u}$.

19. Path of a heat-seeking particle A heat-seeking particle has the property that at any point (x, y) in the plane, it moves in the direction of maximum temperature increase. If the temperature at (x, y) is $T(x, y) = -e^{-2y} \cos x$, find an equation $y = f(x)$ for the path of a heat-seeking particle at the point $(\pi/4, 0)$.

- 20. Velocity after a ricochet** A particle traveling in a straight line with constant velocity $\mathbf{i} + \mathbf{j} - 5\mathbf{k}$ passes through the point $(0, 0, 30)$ and hits the surface $z = 2x^2 + 3y^2$. The particle ricochets off the surface, the angle of reflection being equal to the angle of incidence. Assuming no loss of speed, what is the velocity of the particle after the ricochet? Simplify your answer.
- 21. Directional derivatives tangent to a surface** Let S be the surface that is the graph of $f(x, y) = 10 - x^2 - y^2$. Suppose that the temperature in space at each point (x, y, z) is $T(x, y, z) = x^2y + y^2z + 4x + 14y + z$.
- Among all the possible directions tangential to the surface S at the point $(0, 0, 10)$, which direction will make the rate of change of temperature at $(0, 0, 10)$ a maximum?
 - Which direction tangential to S at the point $(1, 1, 8)$ will make the rate of change of temperature a maximum?
- 22. Drilling another borehole** On a flat surface of land, geologists drilled a borehole straight down and hit a mineral deposit at 300 m. They drilled a second borehole 30 m to the north of the first and hit the mineral deposit at 285 m. A third borehole 30 m east of the first borehole struck the mineral deposit at 307.5 m. The geologists have reasons to believe that the mineral deposit is in the

shape of a dome, and for the sake of economy, they would like to find where the deposit is closest to the surface. Assuming the surface to be the xy -plane, in what direction from the first borehole would you suggest the geologists drill their fourth borehole?

The one-dimensional heat equation If $w(x, t)$ represents the temperature at position x at time t in a uniform wire with perfectly insulated sides, then the partial derivatives w_{xx} and w_t satisfy a differential equation of the form

$$w_{xx} = \frac{1}{c^2} w_t.$$

This equation is called the *one-dimensional heat equation*. The value of the positive constant c^2 is determined by the material from which the wire is made.

- 23.** Find all solutions of the one-dimensional heat equation of the form $w = e^{rt} \sin \pi x$, where r is a constant.
- 24.** Find all solutions of the one-dimensional heat equation that have the form $w = e^{rt} \sin kx$ and satisfy the conditions that $w(0, t) = 0$ and $w(L, t) = 0$. What happens to these solutions as $t \rightarrow \infty$?

CHAPTER 13 Technology Application Projects

Mathematica/Maple Projects

Projects can be found within MyLab Math.

- **Plotting Surfaces**
Efficiently generate plots of surfaces, contours, and level curves.
- **Exploring the Mathematics Behind Skateboarding: Analysis of the Directional Derivative**
The path of a skateboarder is introduced, first on a level plane, then on a ramp, and finally on a paraboloid. Compute, plot, and analyze the directional derivative in terms of the skateboarder.
- **Looking for Patterns and Applying the Method of Least Squares to Real Data**
Fit a line to a set of numerical data points by choosing the line that minimizes the sum of the squares of the vertical distances from the points to the line.
- **Lagrange Goes Skateboarding: How High Does He Go?**
Revisit and analyze the skateboarders' adventures for maximum and minimum heights from both a graphical and analytic perspective using Lagrange multipliers.